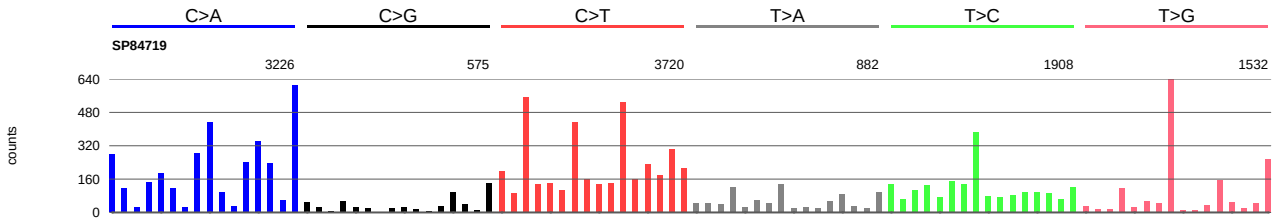
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SP84719
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575
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882
1908
1532
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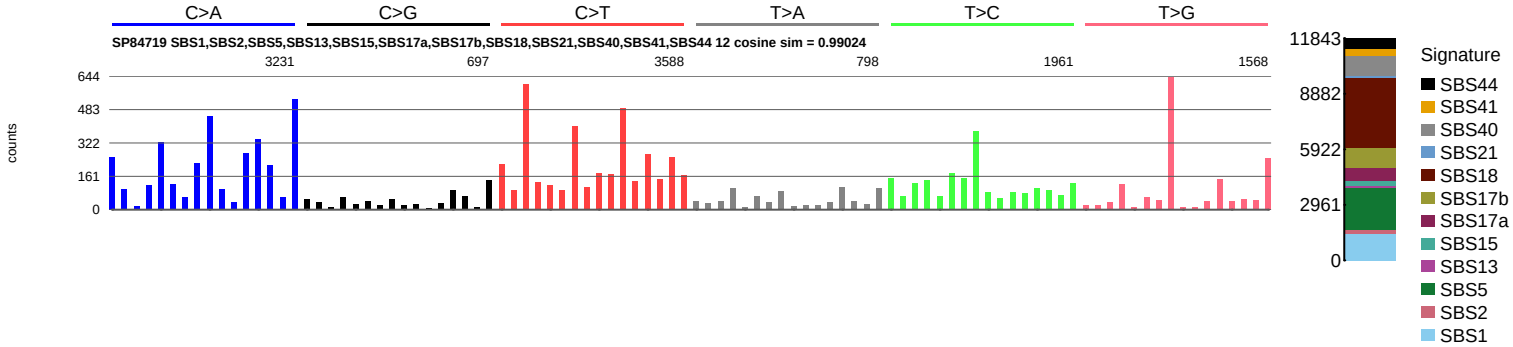
counts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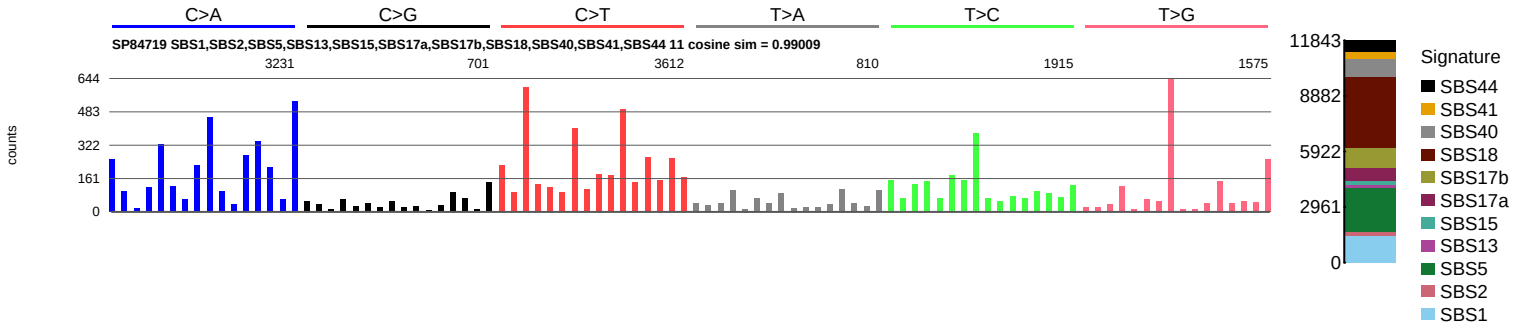
3231
697
3588
798
1961
1568
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99009
3231
701
3612
810
1915
1575
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
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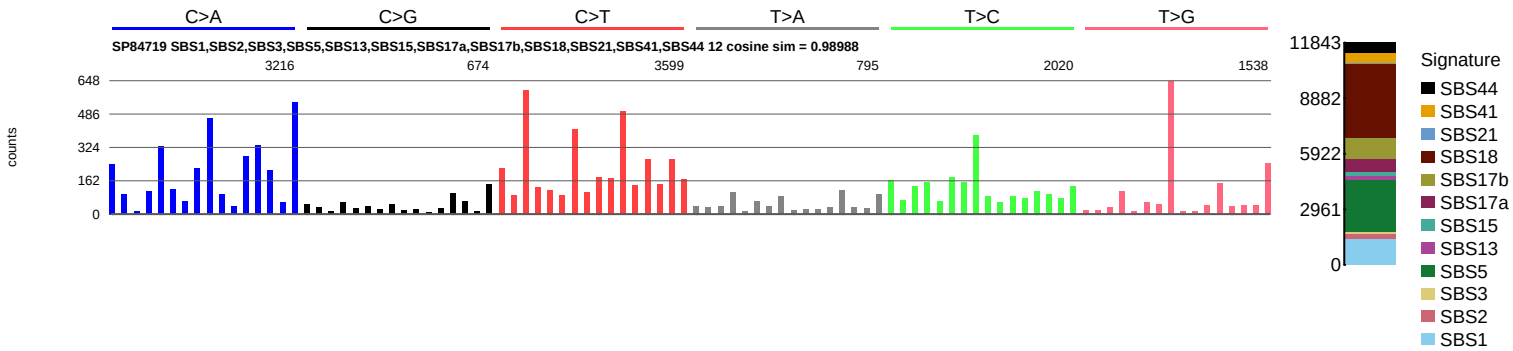
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 12 cosine sim = 0.98988
3216
674
3599
795
2020
1538
648
486
324
162
0
counts
11843
8882
5922
2961
0
Signature
SBS44
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SBS17b
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SBS13
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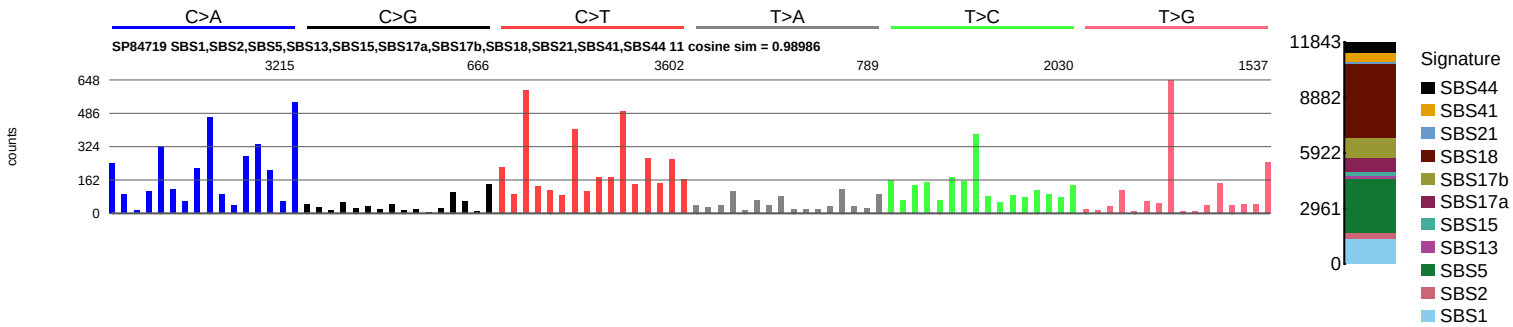
SBS2
SBS1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SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98986
3215
666
3602
789
2030
1537
648
486
324
162
0
counts
11843
8882
5922
2961
0
Signature
SBS44
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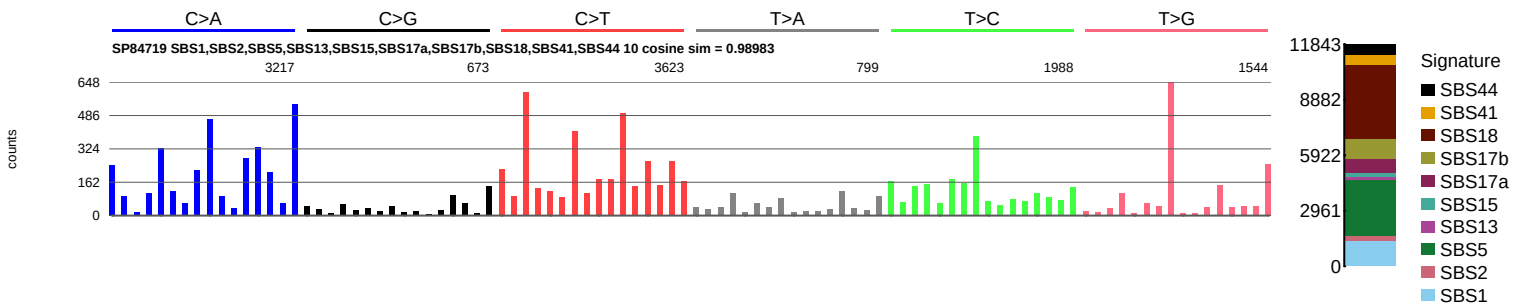
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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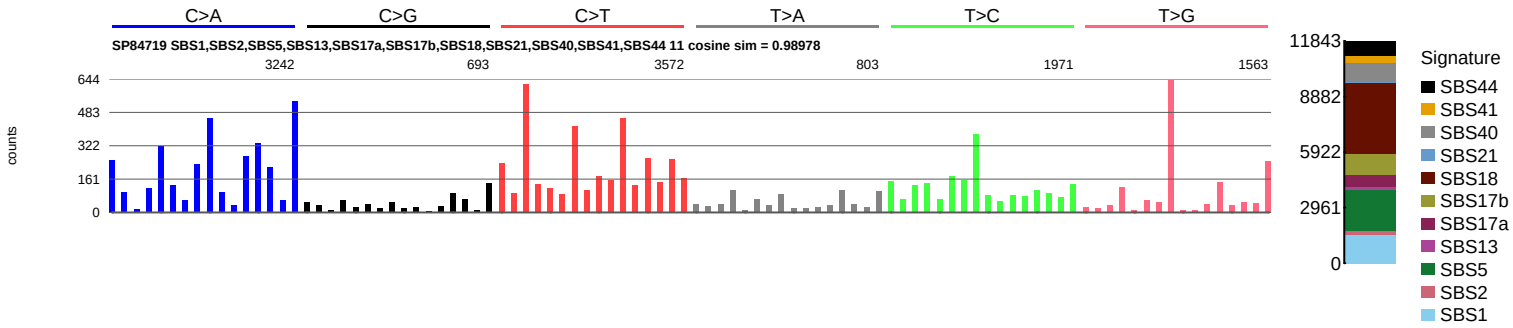
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98983
3217
673
3623
799
1988
1544
648
486
324
162
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>T
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T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98978
3242
693
3572
803
1971
1563
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
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SBS21
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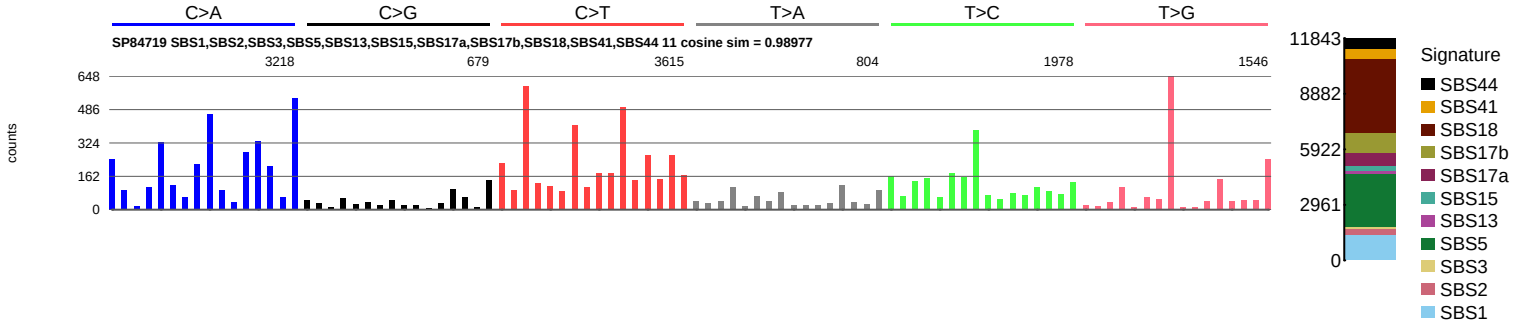
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
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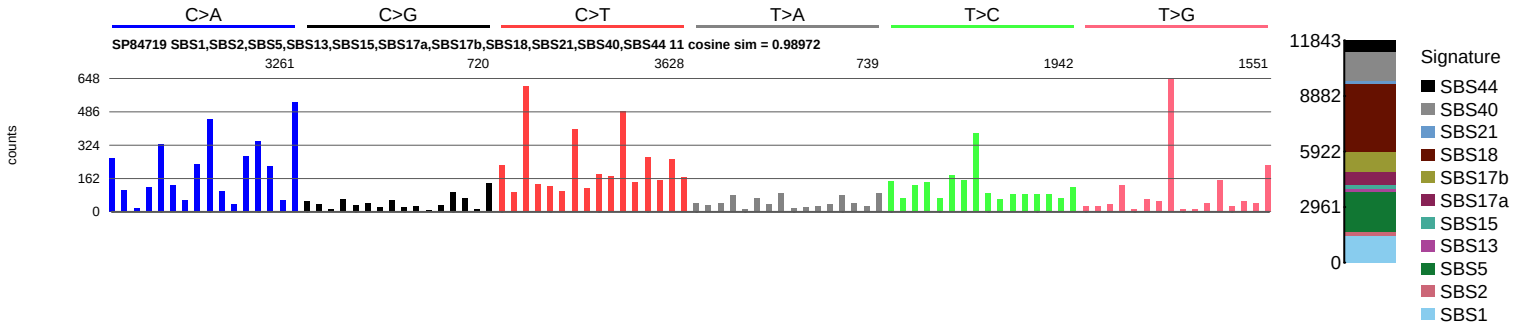
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.98977
3218
679
3615
804
1978
1546
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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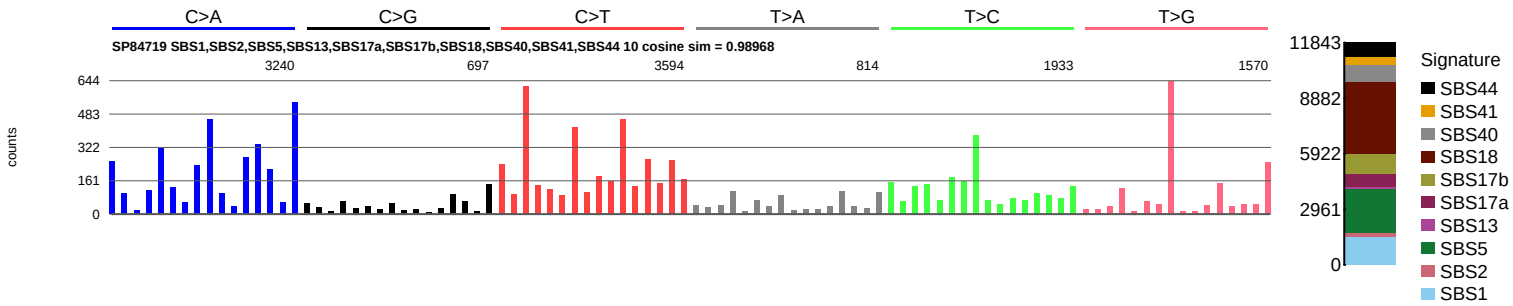
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98972
3261
720
3628
739
1942
1551
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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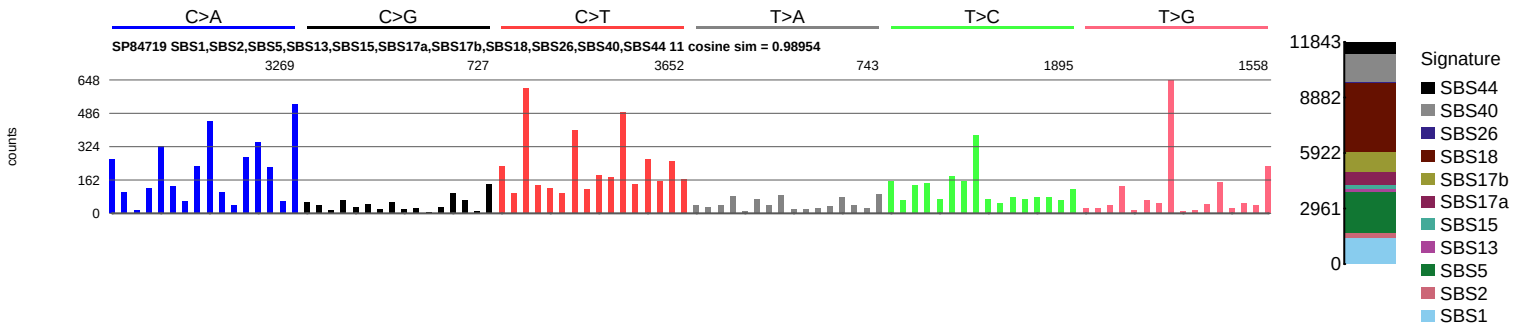
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98968
3240
697
3594
814
1933
1570
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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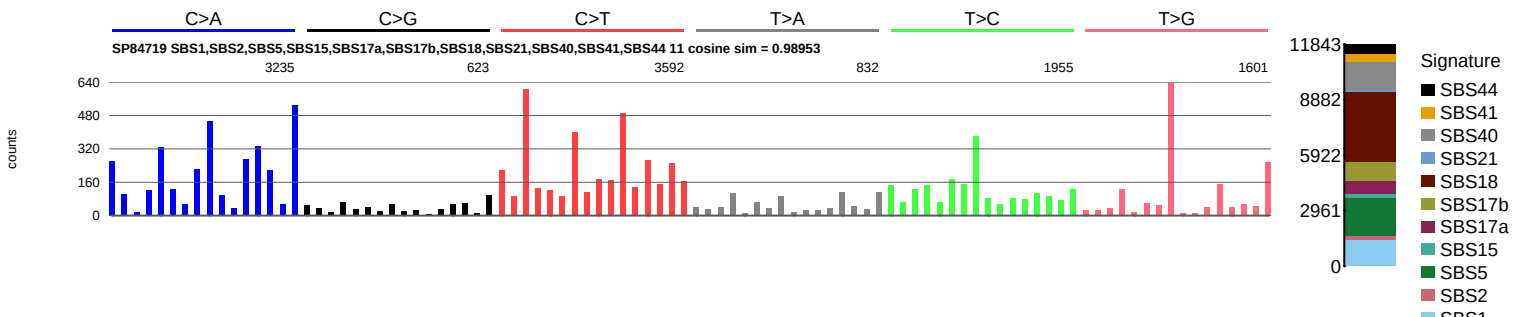
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98954
3269
727
3652
743
1895
1558
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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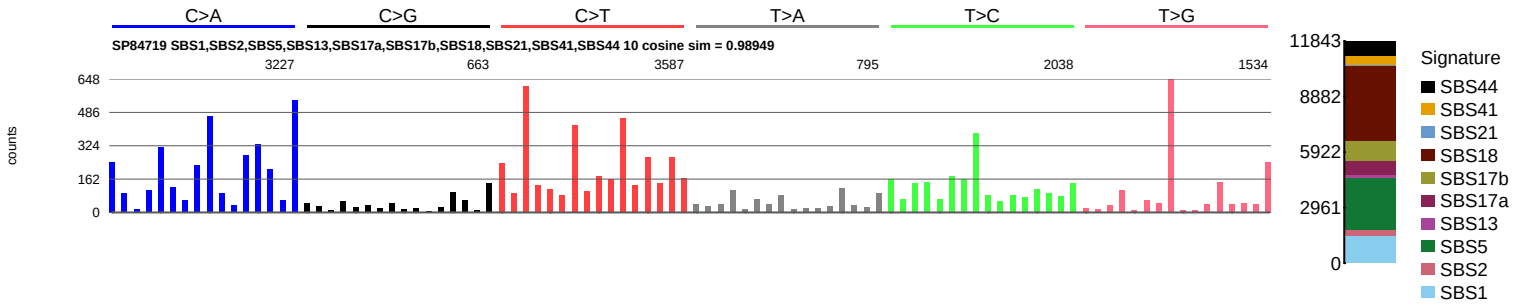
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98953
3235
623
3592
832
1955
1601
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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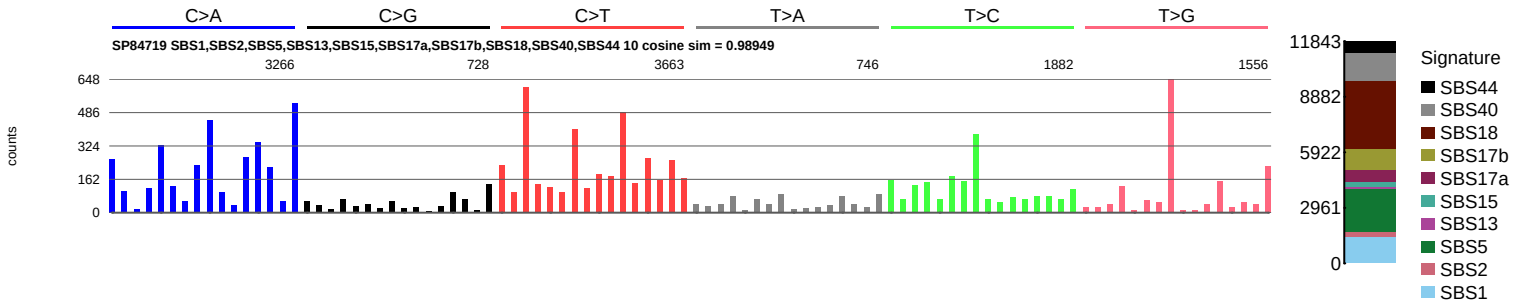
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98949
3227
663
3587
795
2038
1534
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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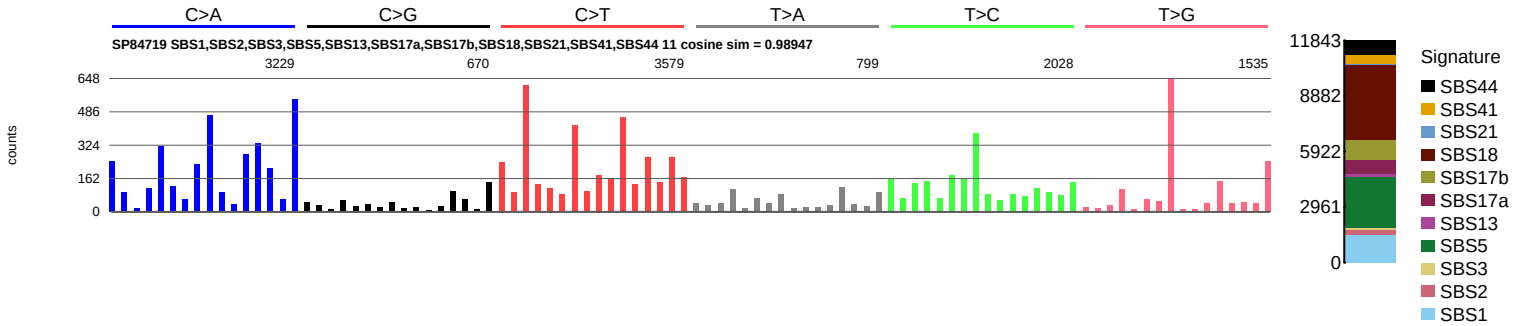
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98949
3266
728
3663
746
1882
1556
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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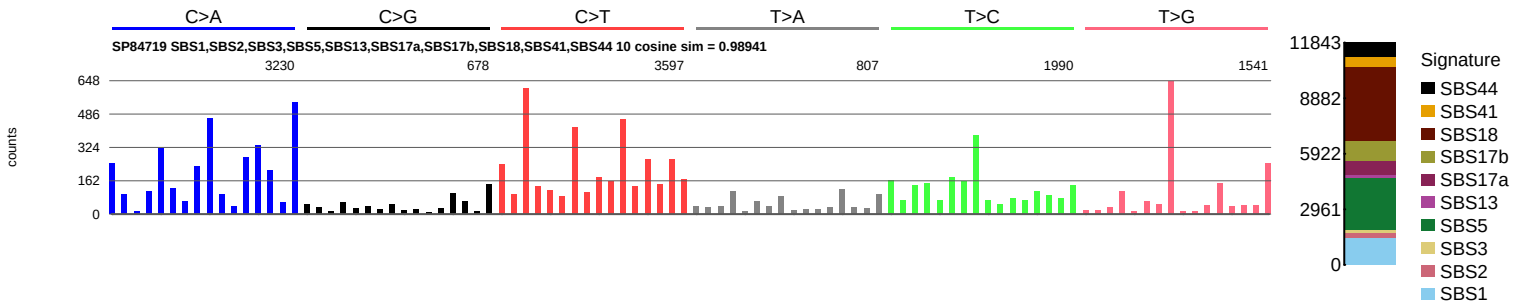
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98947
3229
670
3579
799
2028
1535
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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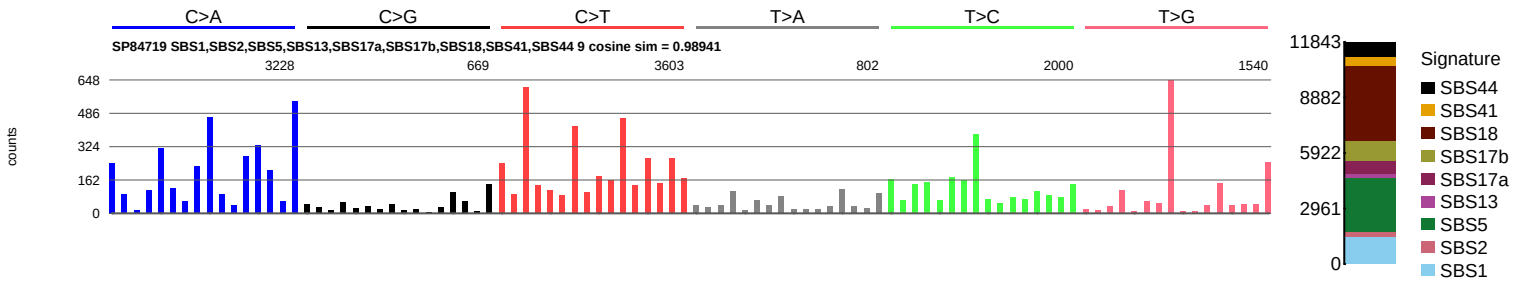
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98941
3230
678
3597
807
1990
1541
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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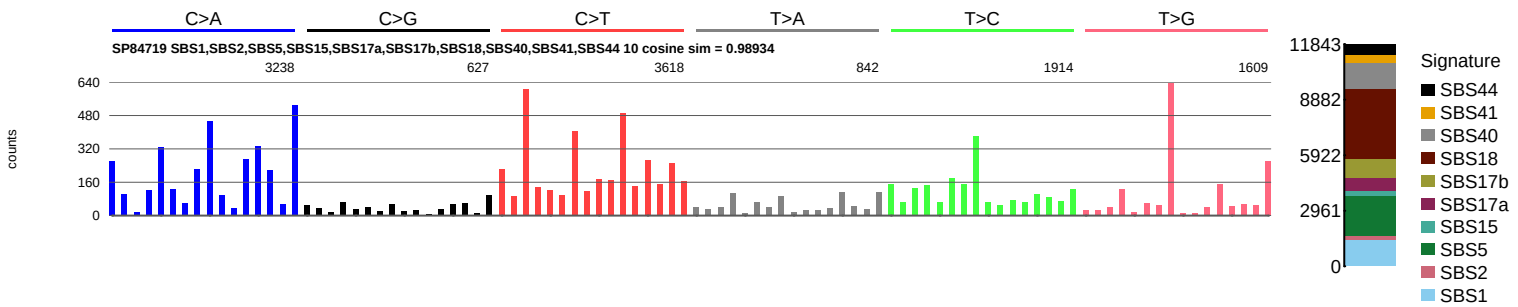
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98941
3228
669
3603
802
2000
1540
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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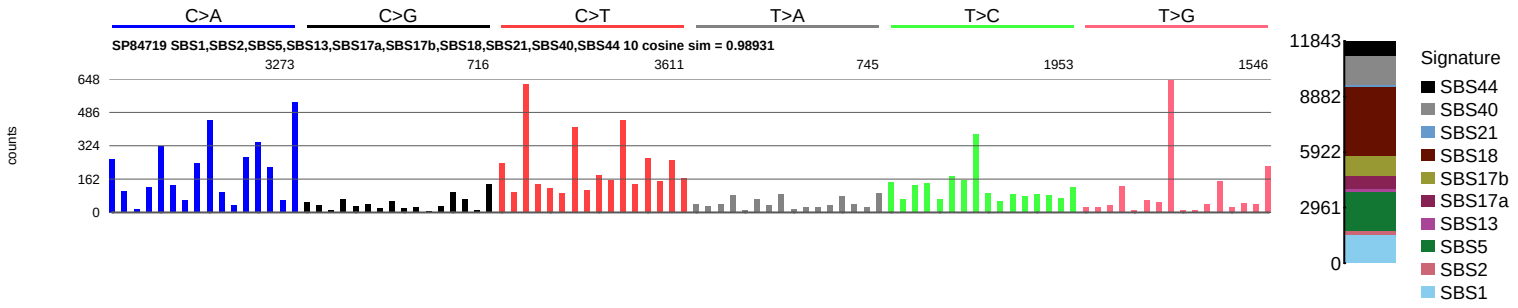
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98934
3238
627
3618
842
1914
1609
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98931
3273
716
3611
745
1953
1546
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
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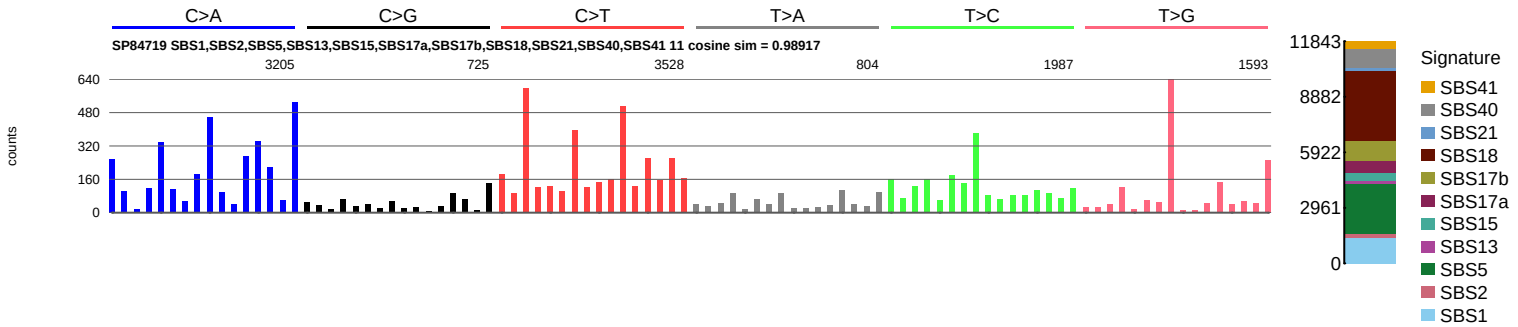
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98917
3205
725
3528
804
1987
1593
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
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SBS18
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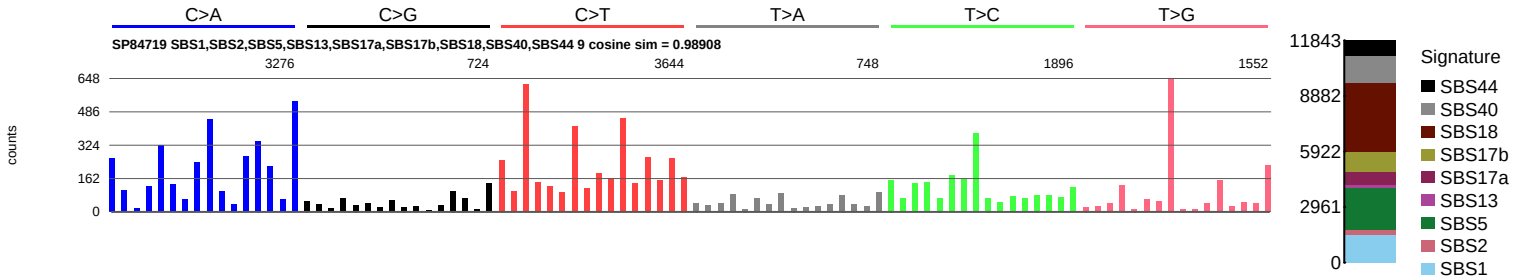
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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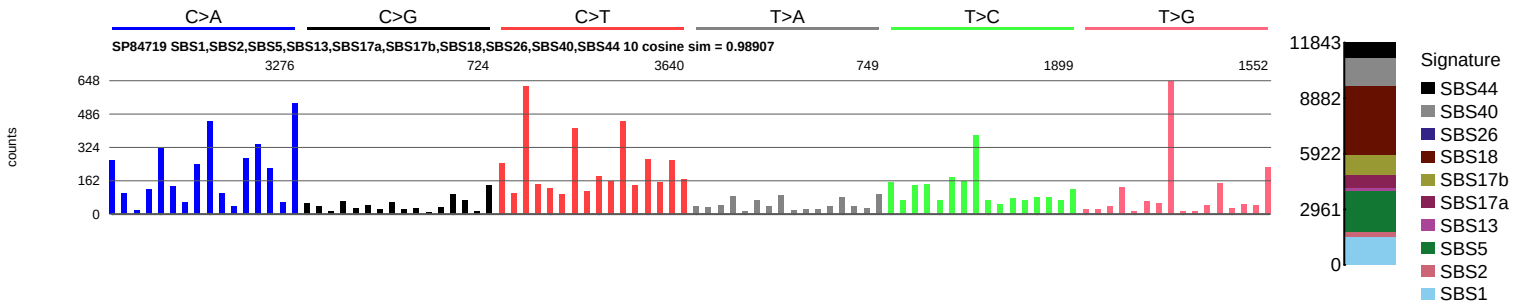
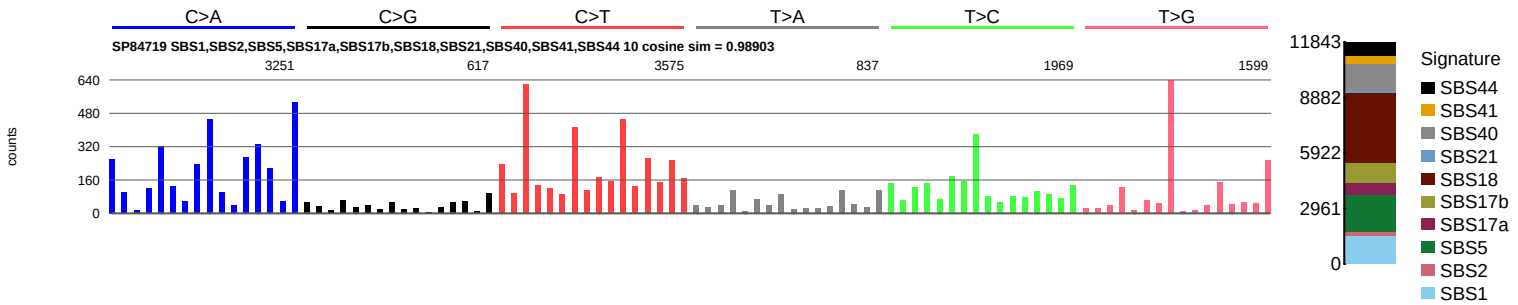
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98908
3276
724
3644
748
1896
1552
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98907
3276
724
3640
749
1899
1552
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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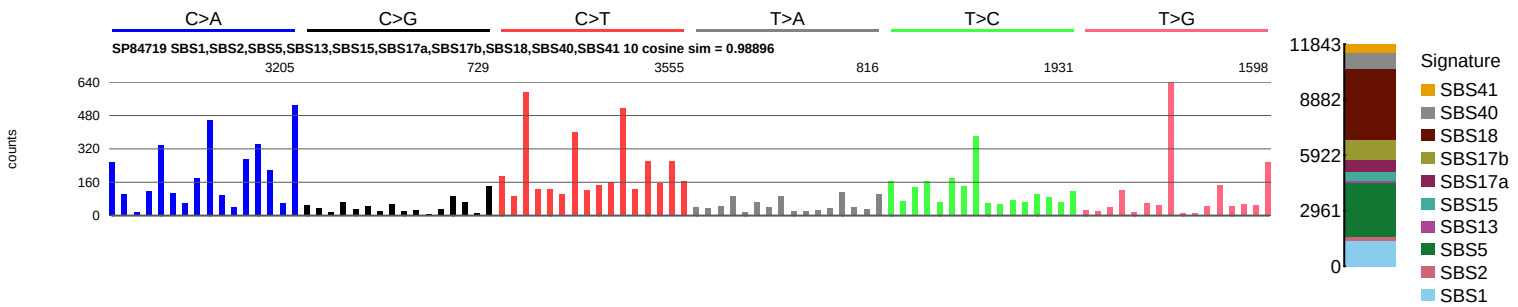
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98903
3251
617
3575
837
1969
1599
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98896
3205
729
3555
816
1931
1598
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS41
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SBS18
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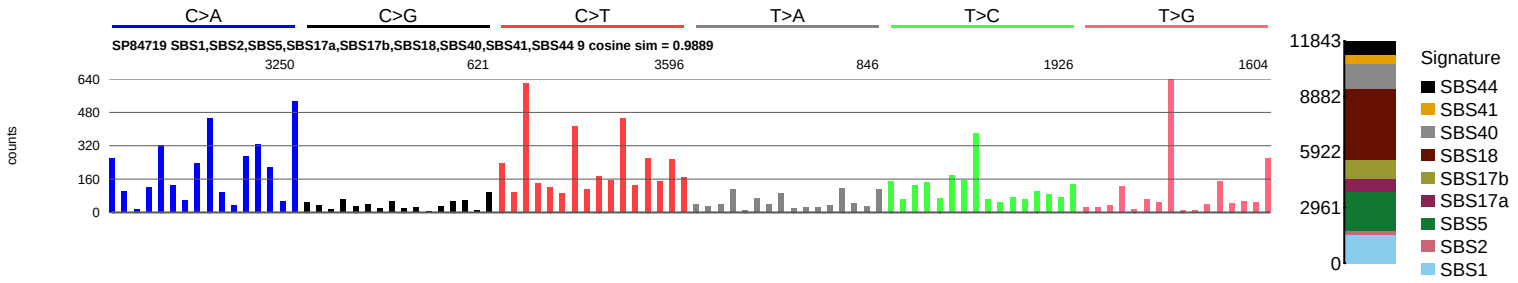
SBS1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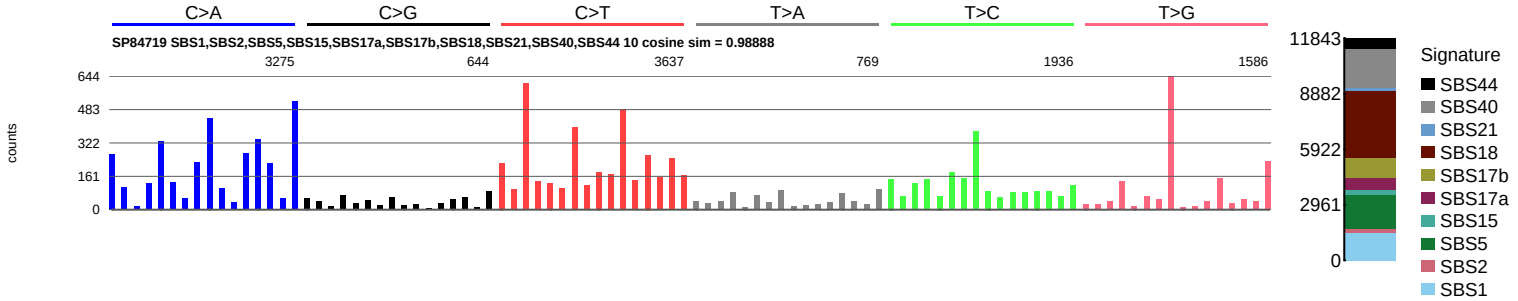
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.9889
3250
621
3596
846
1926
1604
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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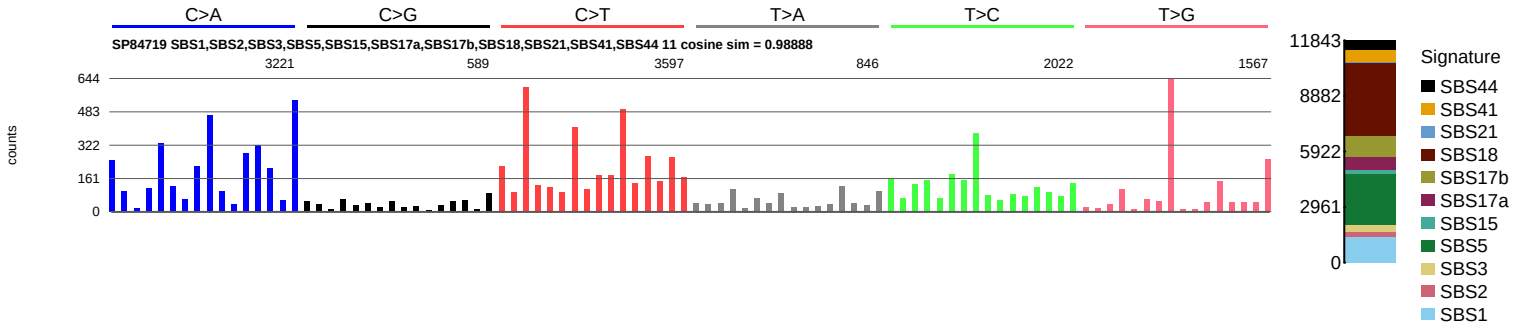
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98888
3275
644
3637
769
1936
1586
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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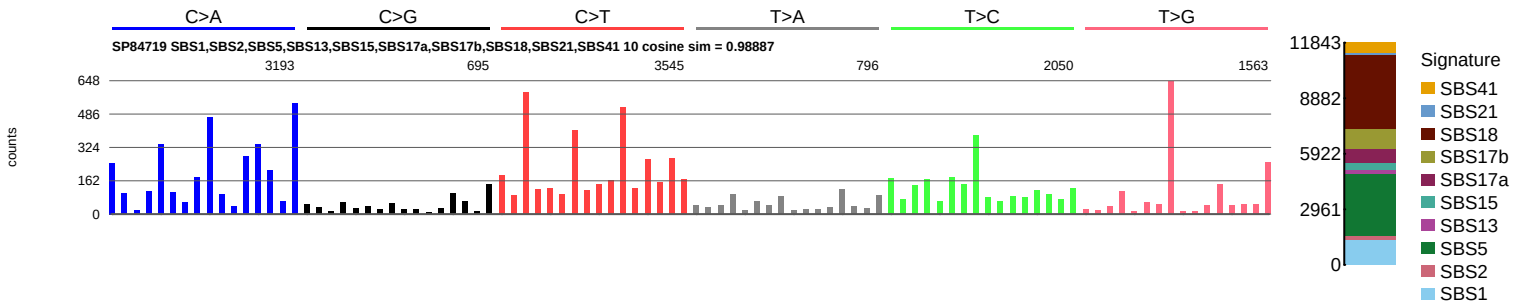
SP84719 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98888
3221
589
3597
846
2022
1567
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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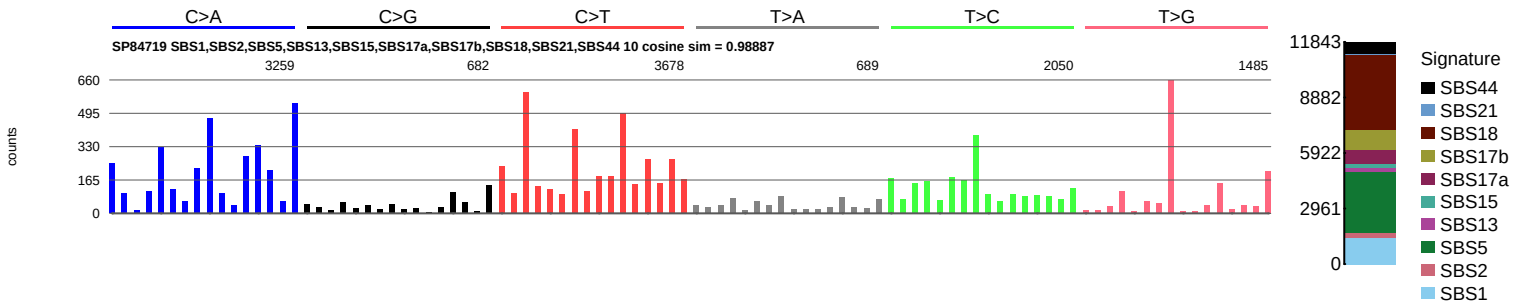
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98887
3193
695
3545
796
2050
1563
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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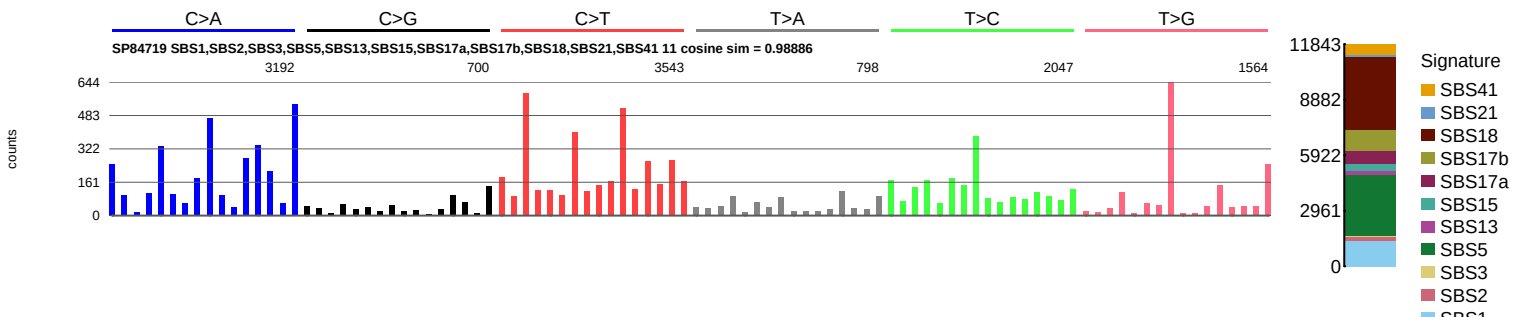
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98887
3259
682
3678
689
2050
1485
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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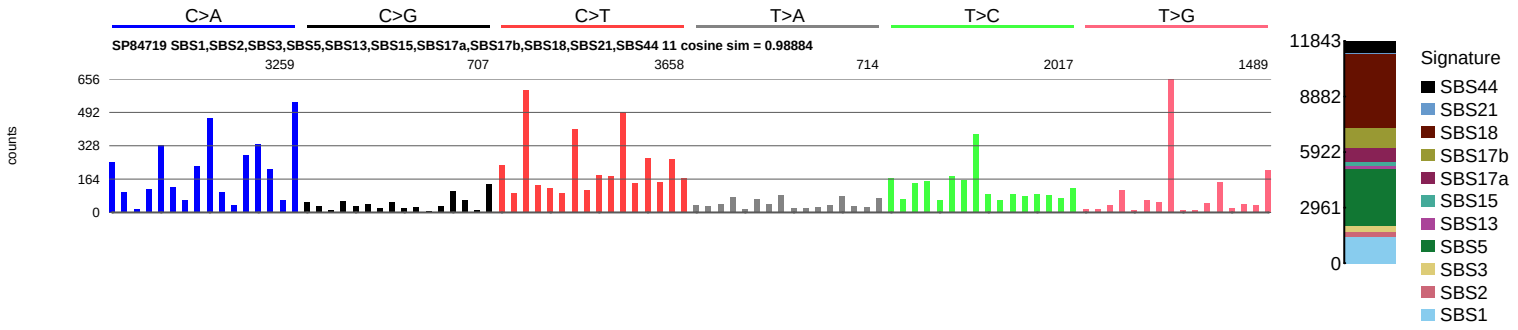
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.98886
3192
700
3543
798
2047
1564
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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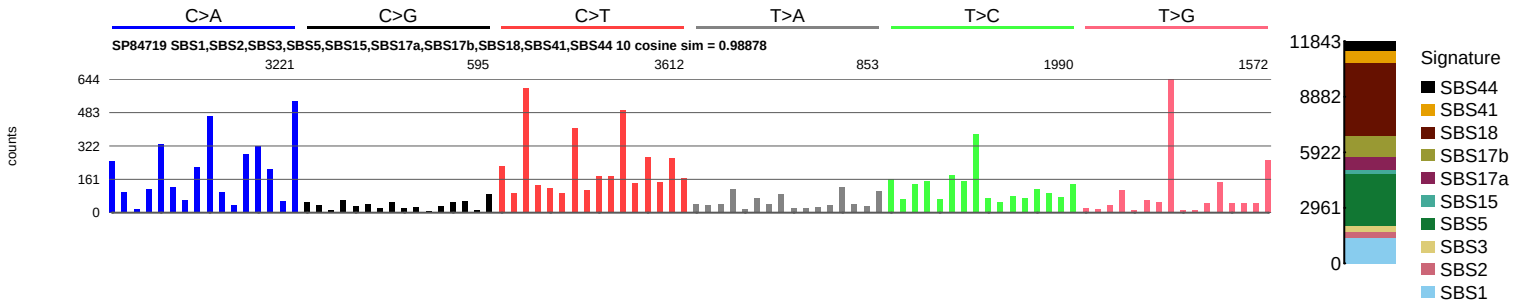
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98884
3259
707
3658
714
2017
1489
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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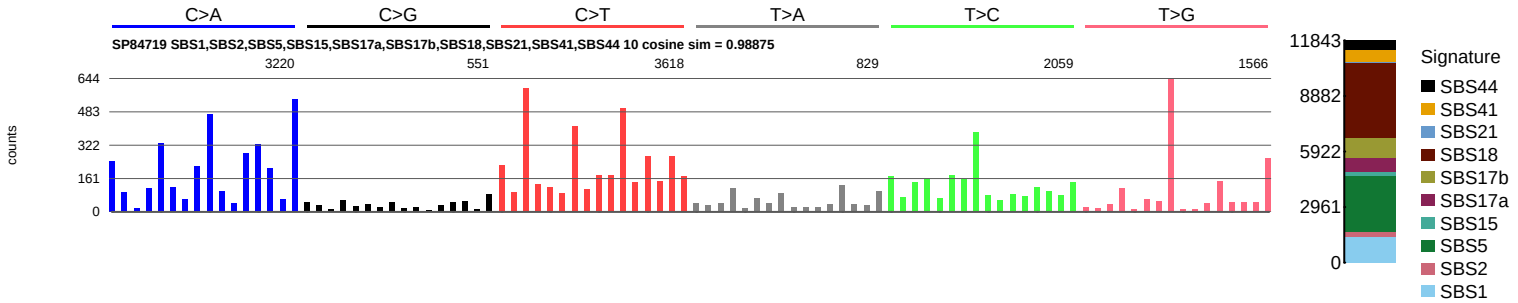
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98878
3221
595
3612
853
1990
1572
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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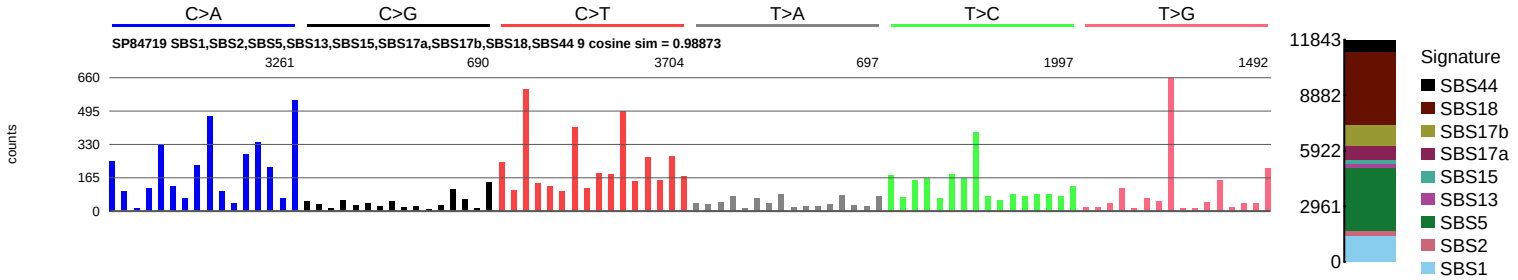
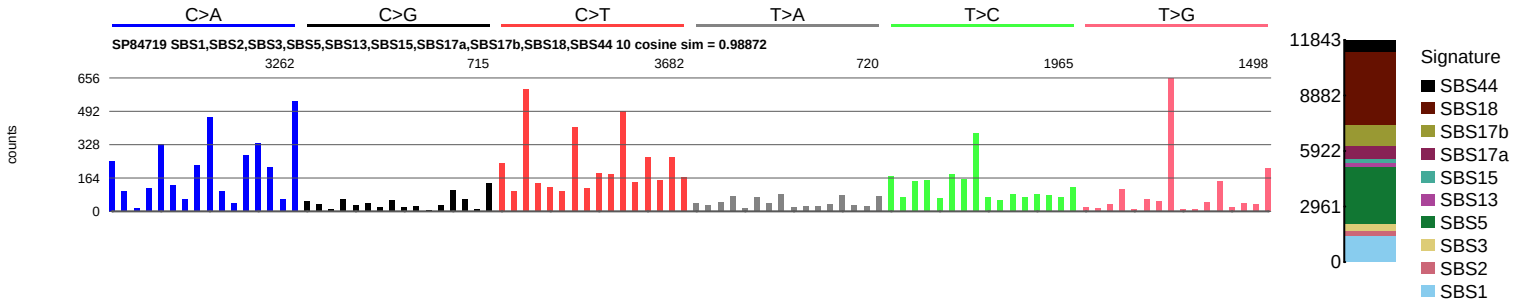
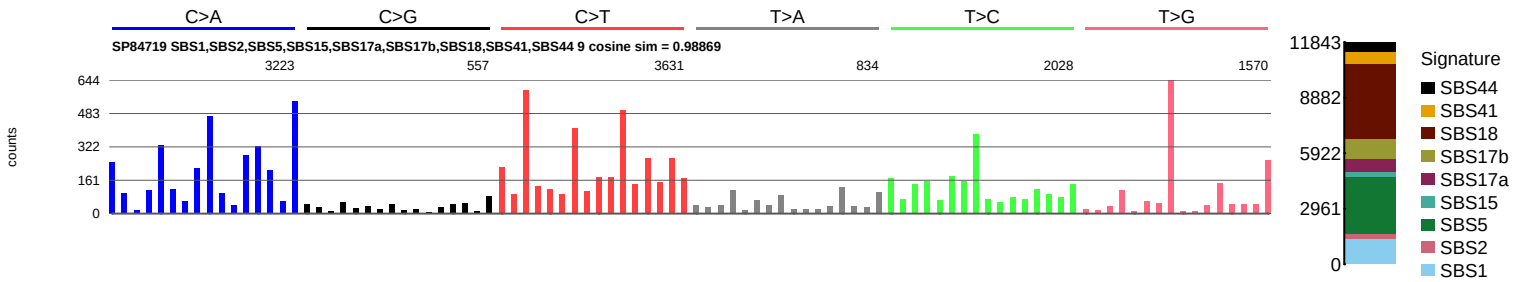
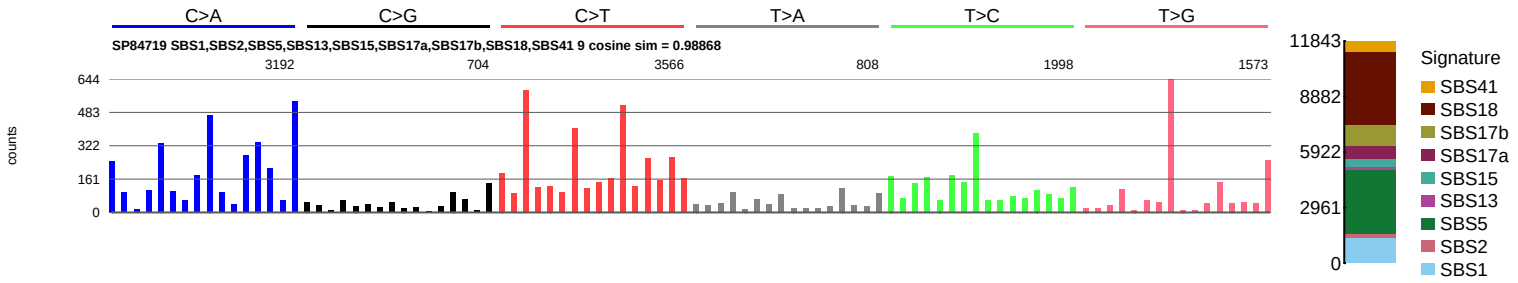
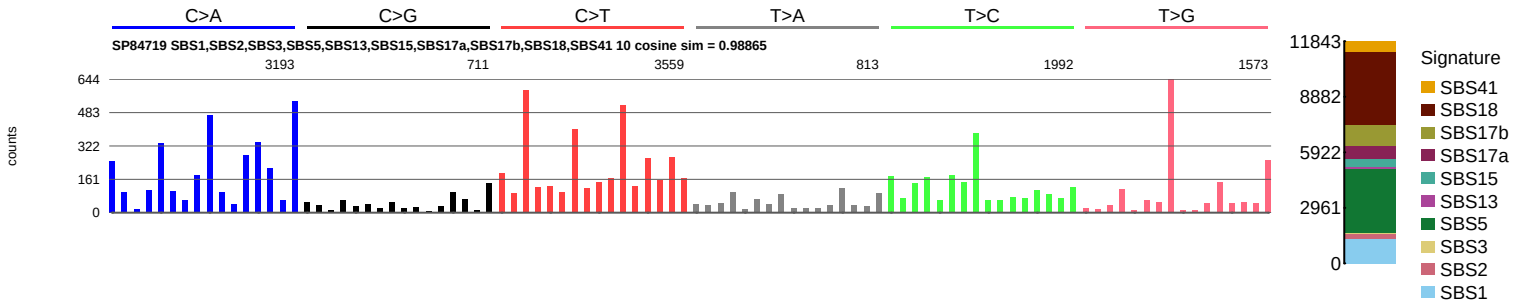
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98875
3220
551
3618
829
2059
1566
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98873
3261
690
3704
697
1997
1492
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98872
3262
715
3682
720
1965
1498
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98869
3223
557
3631
834
2028
1570
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98868
3192
704
3566
808
1998
1573
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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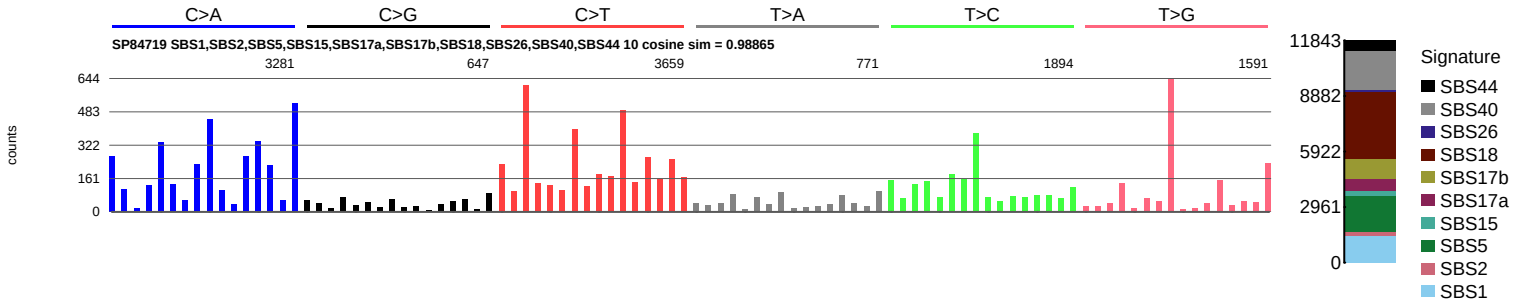
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98865
3193
711
3559
813
1992
1573
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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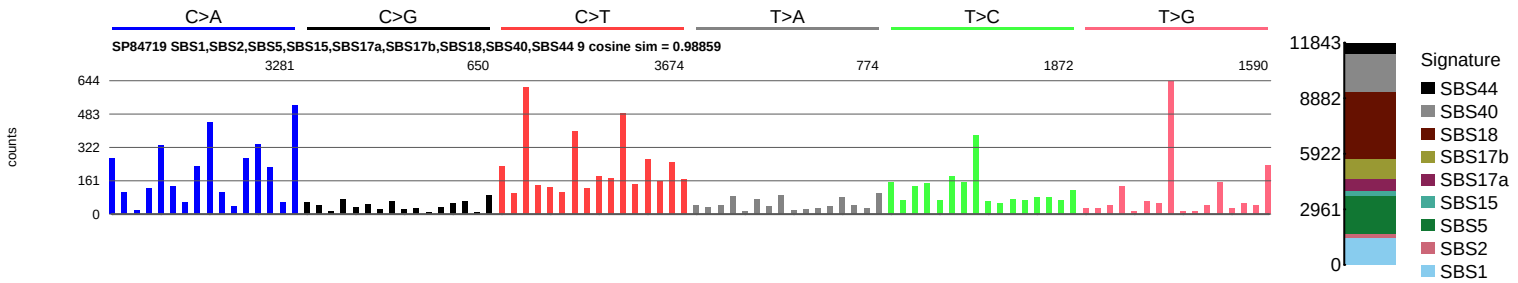
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98865
3281
647
3659
771
1894
1591
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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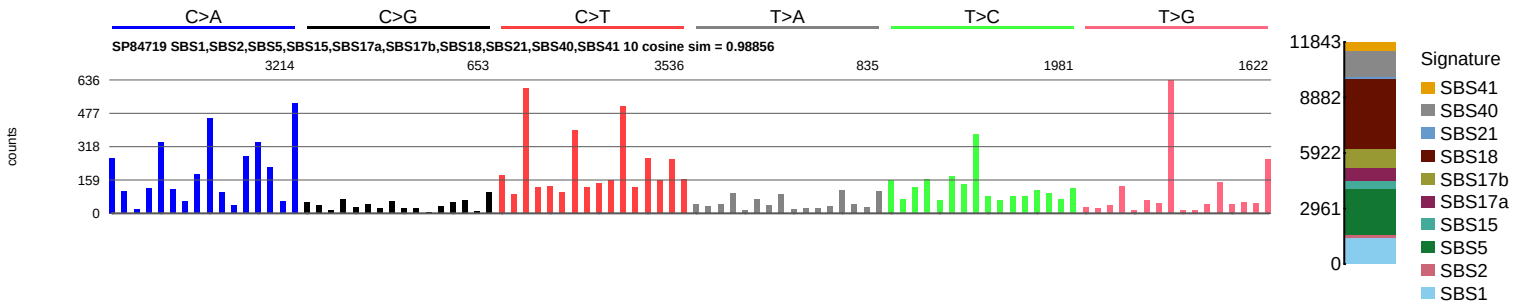
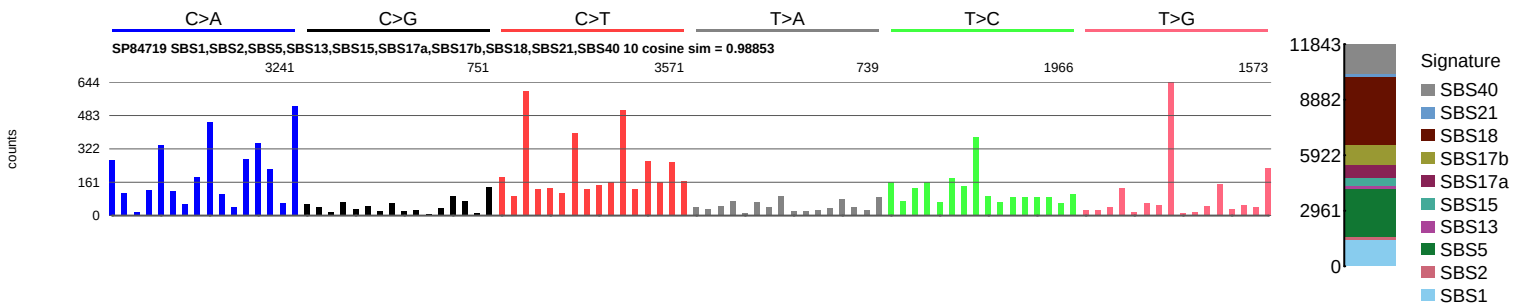
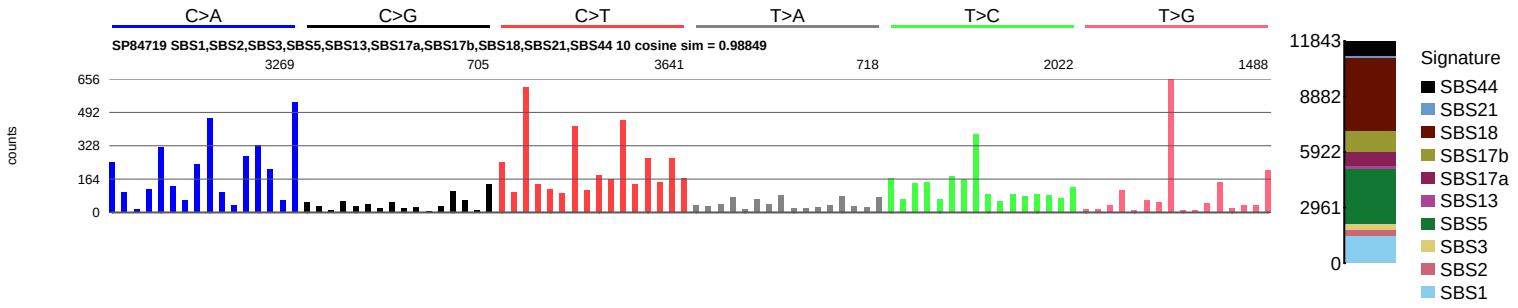
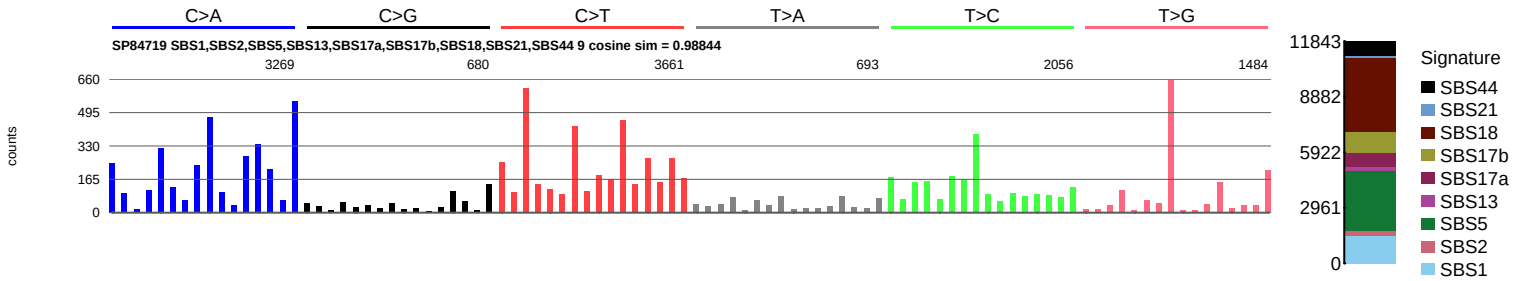
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98859
3281
650
3674
774
1872
1590
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98856
3214
653
3536
835
1981
1622
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98853
3241
751
3571
739
1966
1573
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98849
3269
705
3641
718
2022
1488
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98844
3269
680
3661
693
2056
1484
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
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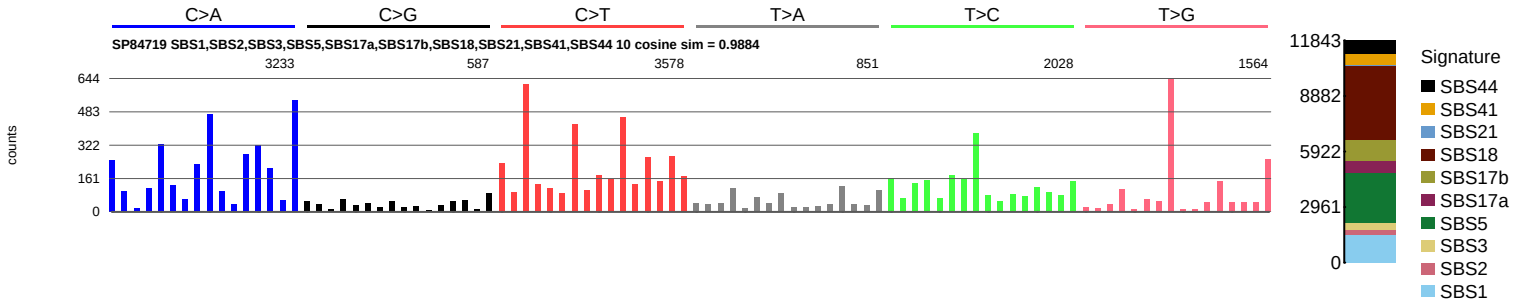
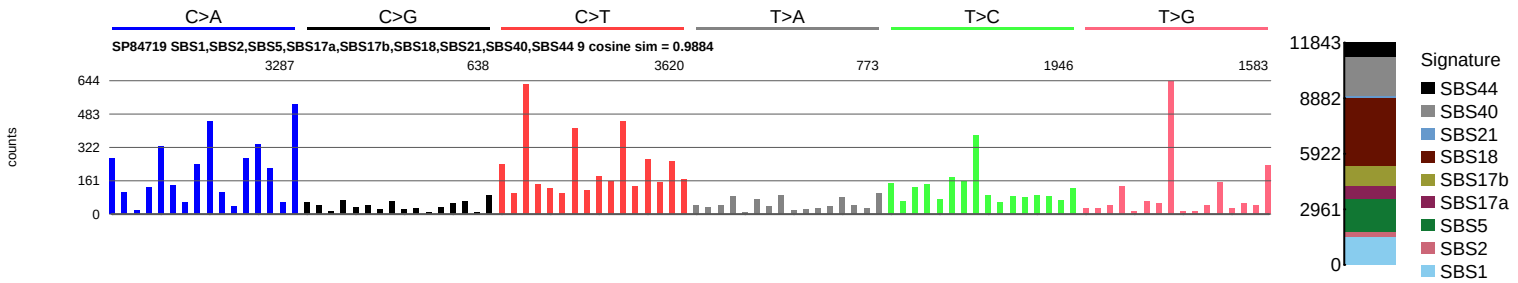
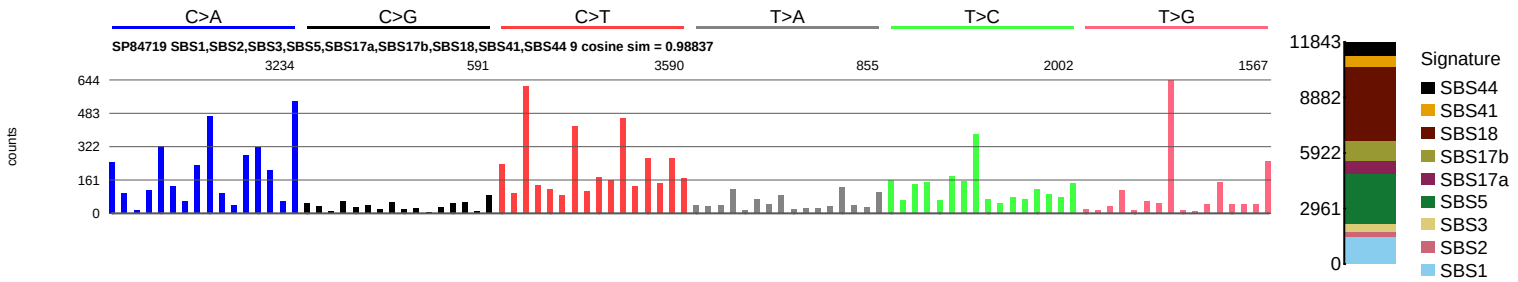
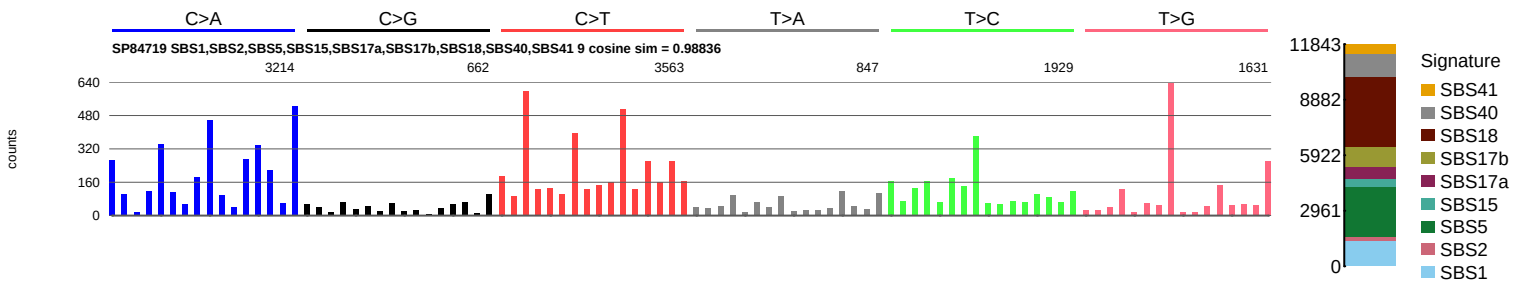
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9884
3233
587
3578
851
2028
1564
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9884
3287
638
3620
773
1946
1583
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98837
3234
591
3590
855
2002
1567
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98836
3214
662
3563
847
1929
1631
counts
640
480
320
160
0
11843
8882
5922
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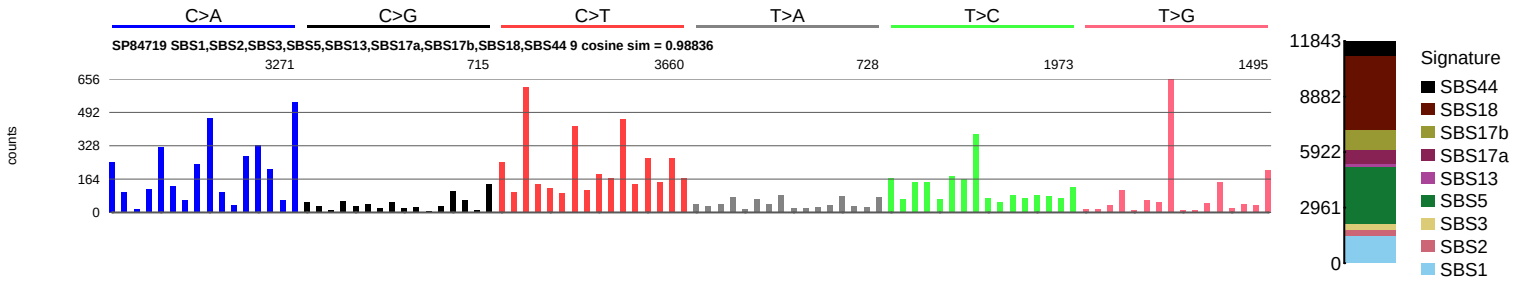
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SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98836
3271
715
3660
728
1973
1495
counts
656
492
328
164
0
11843
8882
5922
2961
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Signature
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SBS17b
SBS17a
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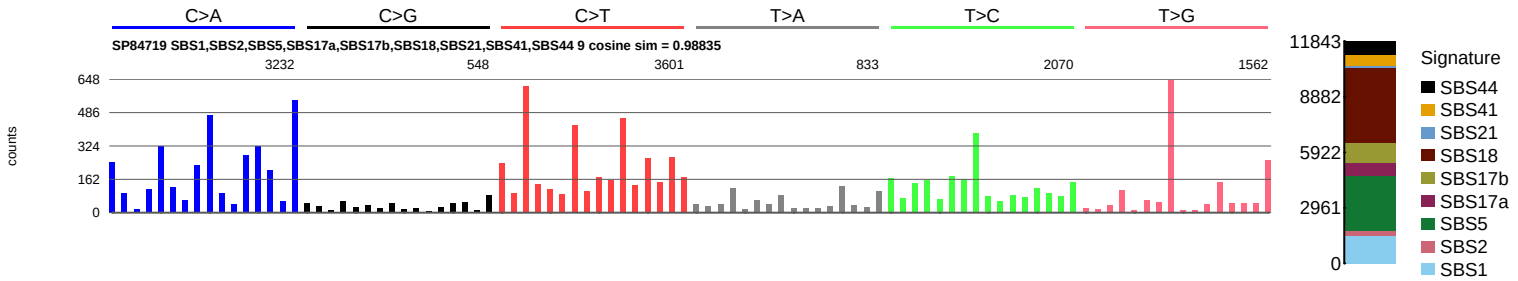
SBS2
SBS1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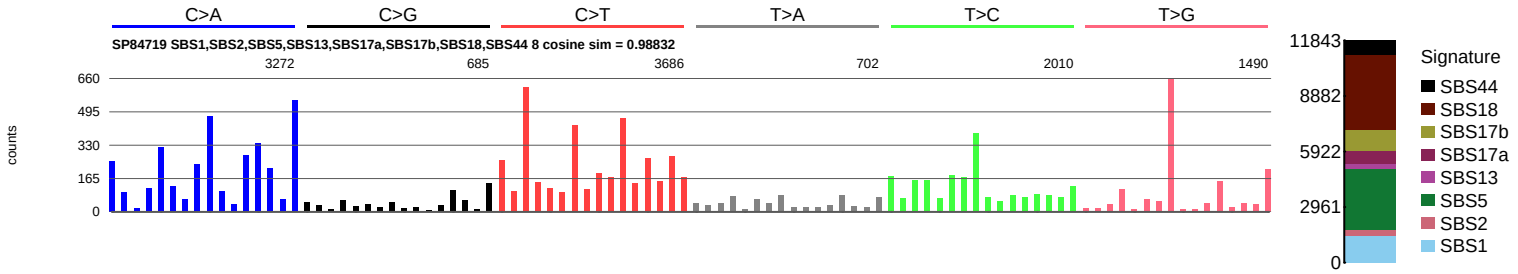
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98835
3232
548
3601
833
2070
1562
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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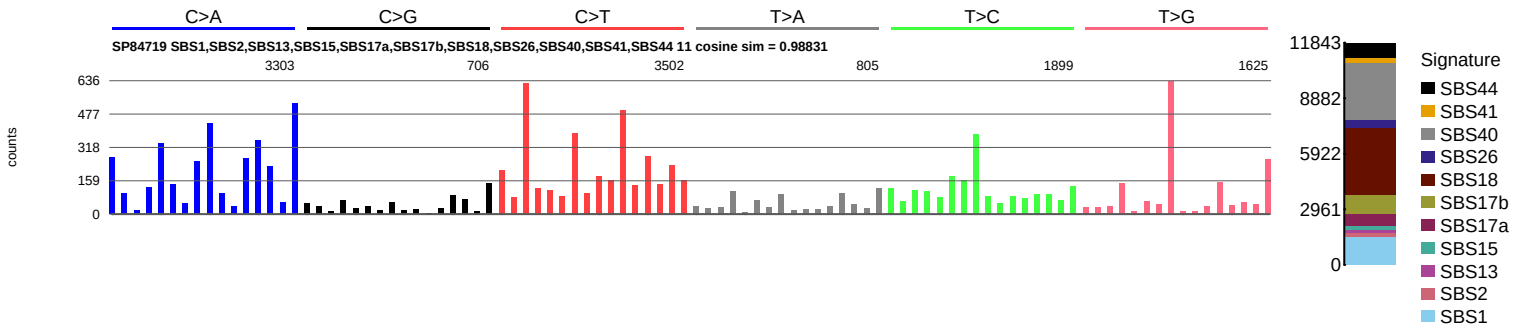
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98832
3272
685
3686
702
2010
1490
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98831
3303
706
3502
805
1899
1625
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
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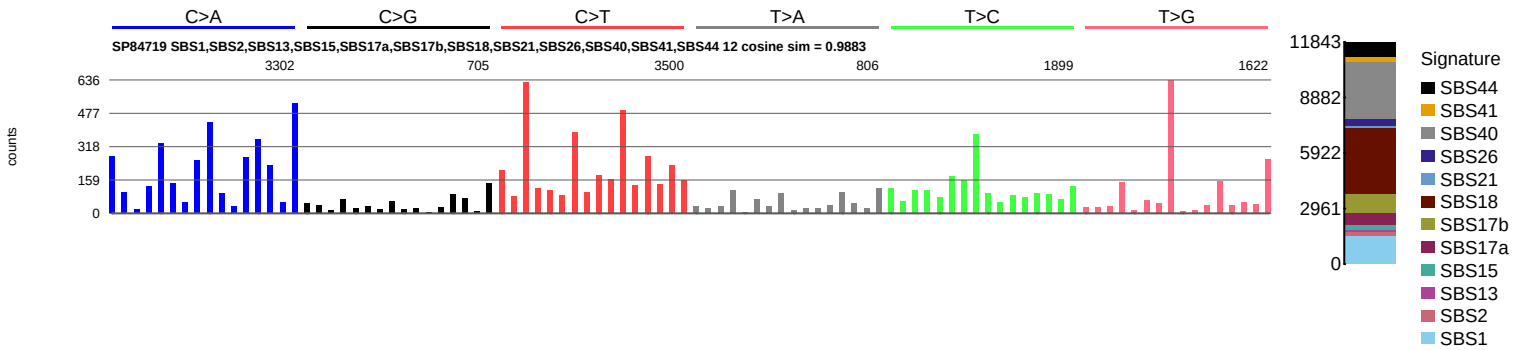
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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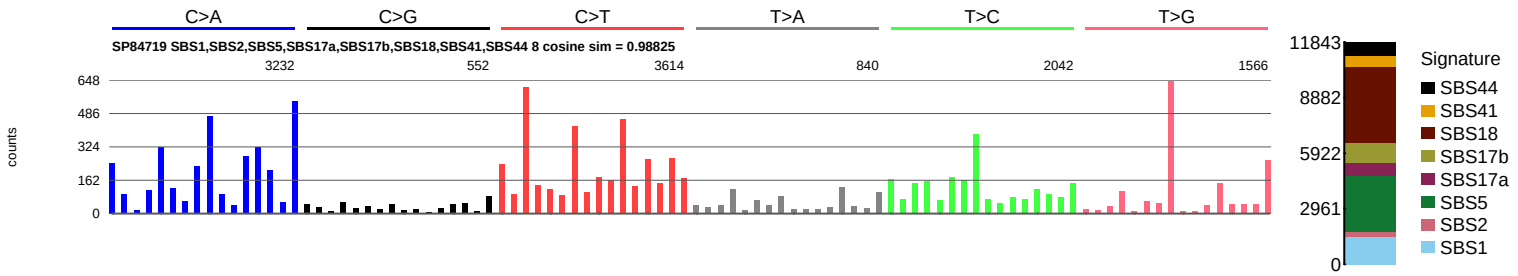
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.9883
3302
705
3500
806
1899
1622
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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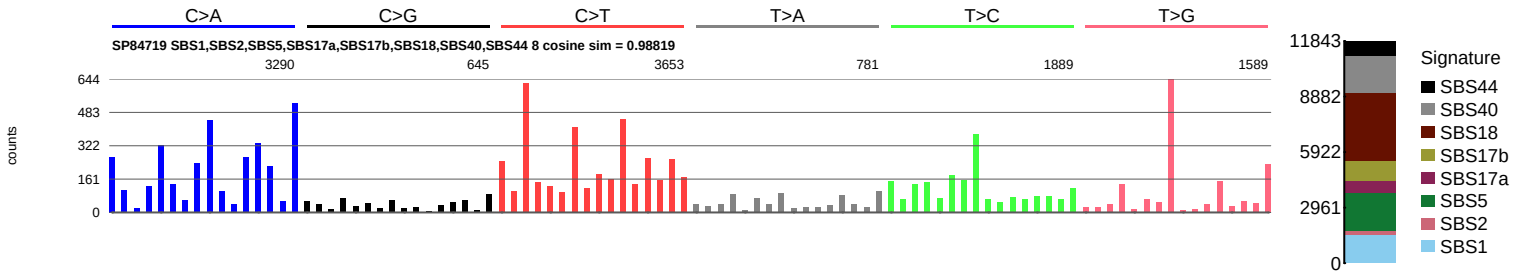
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98825
3232
552
3614
840
2042
1566
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98819
3290
645
3653
781
1889
1589
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
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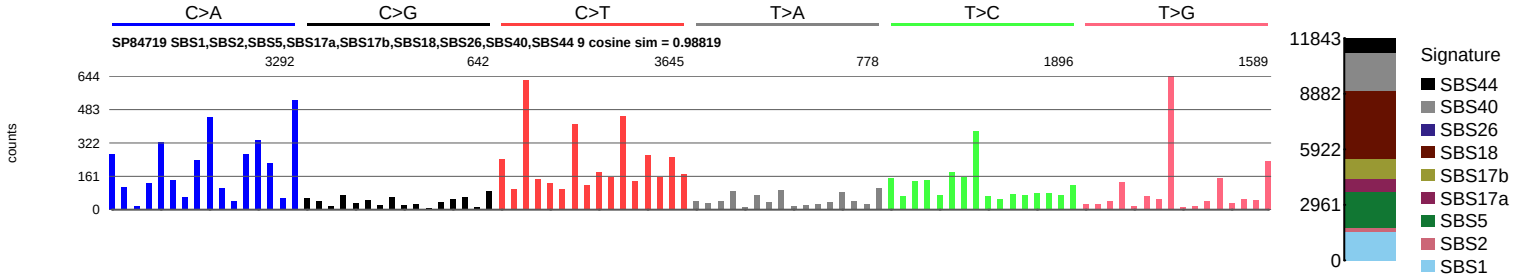
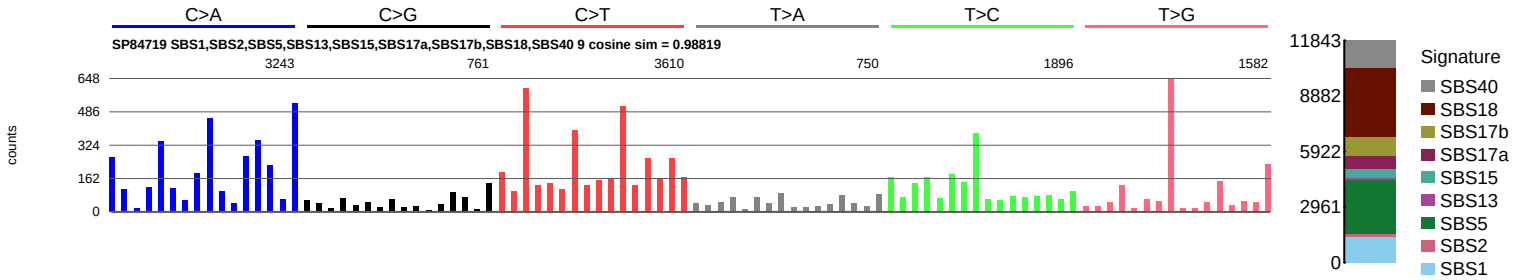
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98819
3292
642
3645
778
1896
1589
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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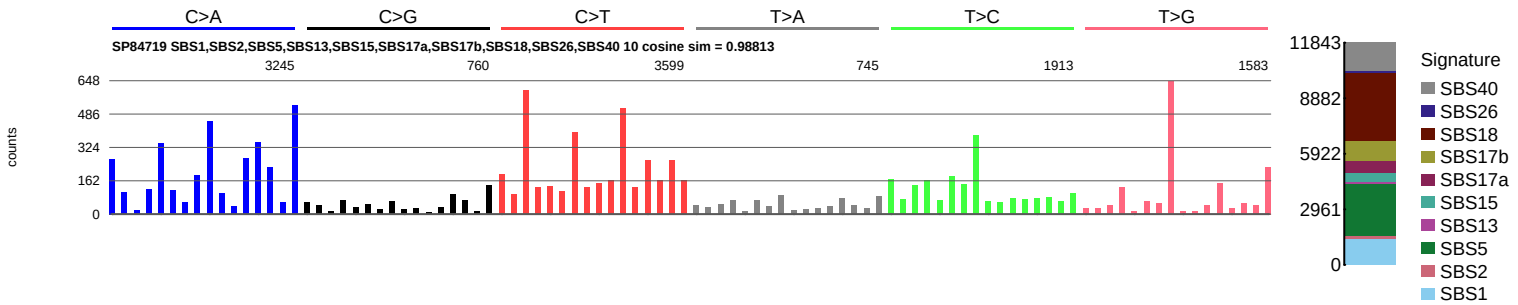
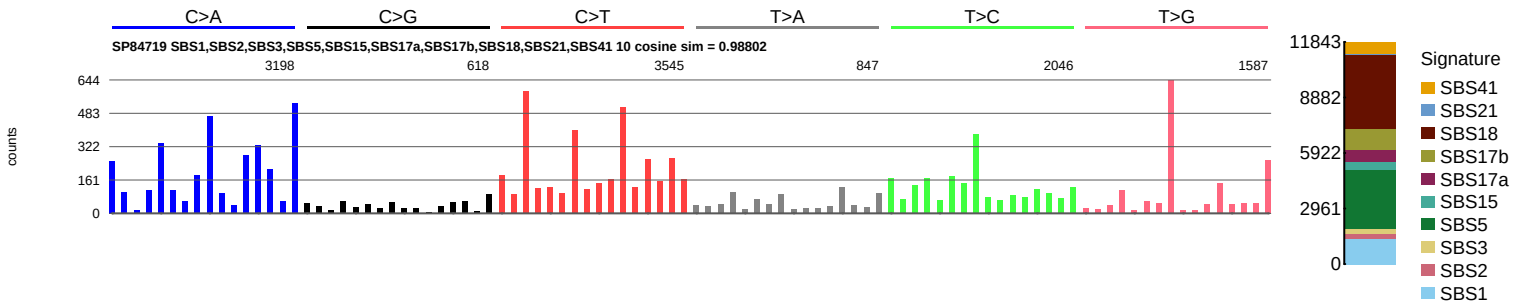
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98819
3243
761
3610
750
1896
1582
648
486
324
162
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98813
3245
760
3599
745
1913
1583
648
486
324
162
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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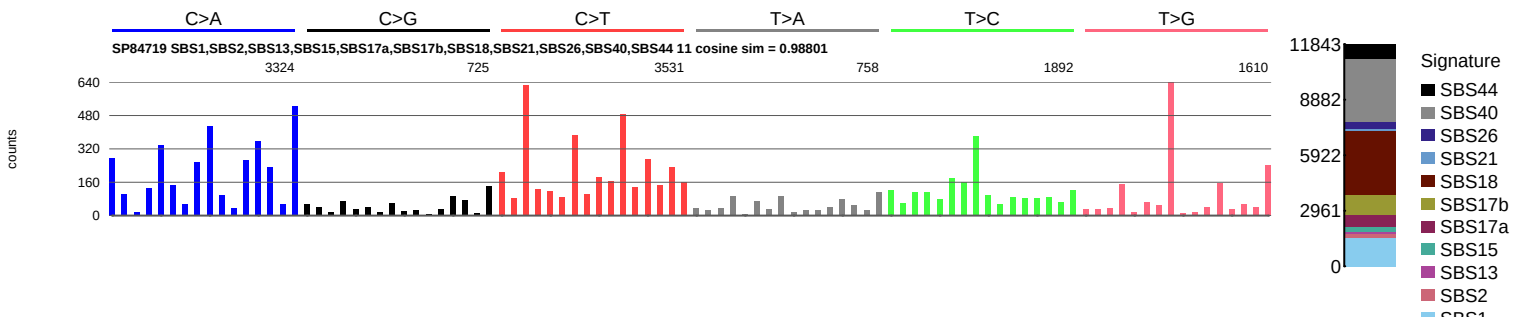
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98802
3198
618
3545
847
2046
1587
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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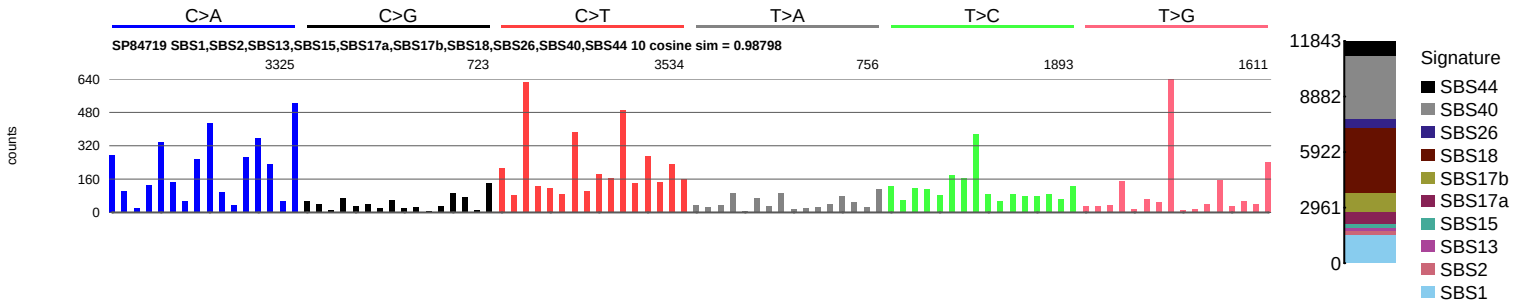
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98801
3324
725
3531
758
1892
1610
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2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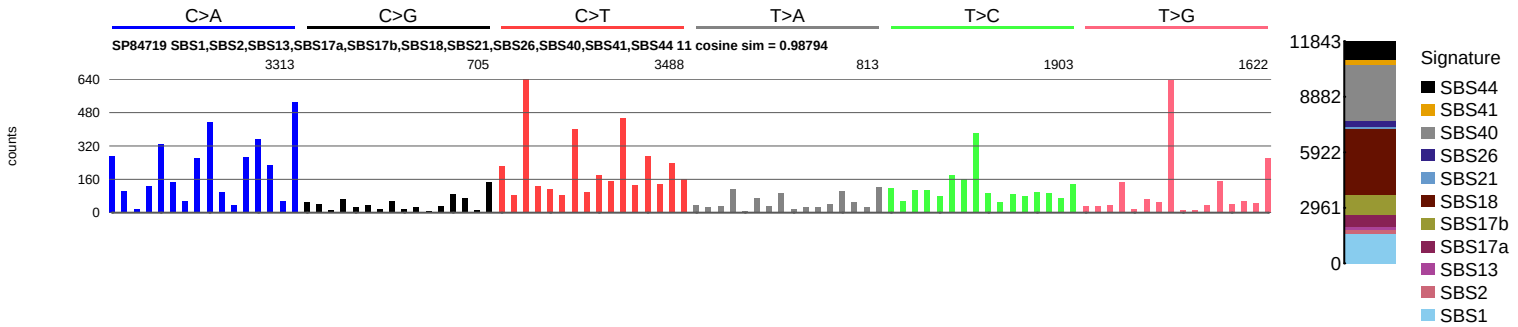
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98798
3325
723
3534
756
1893
1611
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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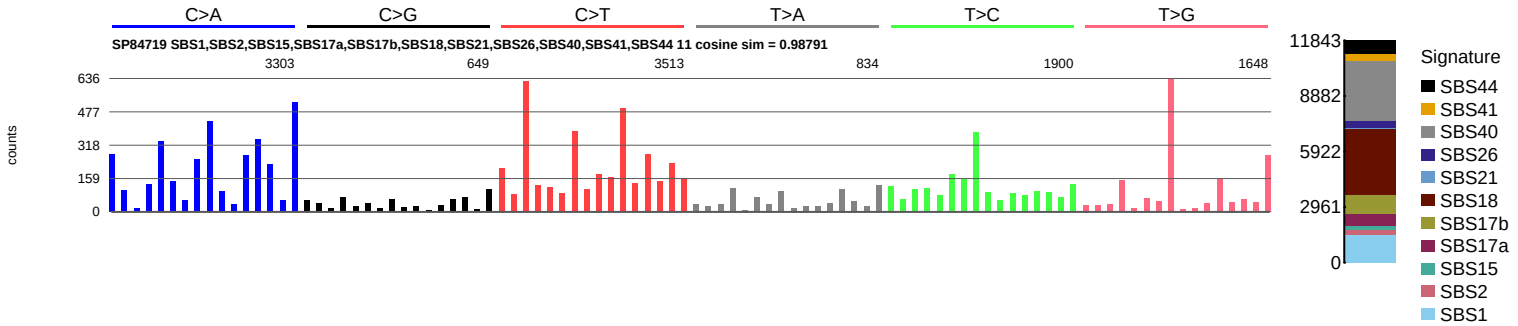
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98794
3313
705
3488
813
1903
1622
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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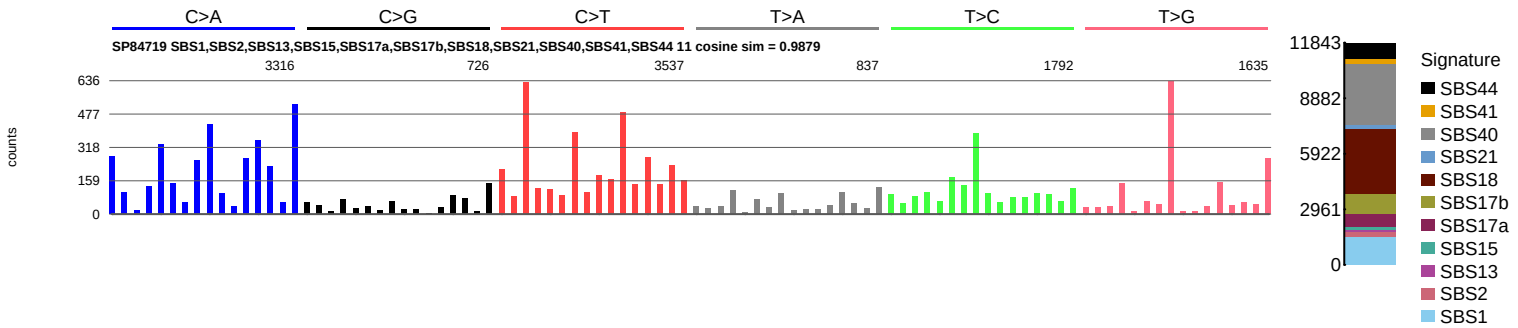
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98791
3303
649
3513
834
1900
1648
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9879
3316
726
3537
837
1792
1635
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
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SBS40
SBS21
SBS18
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SBS17a
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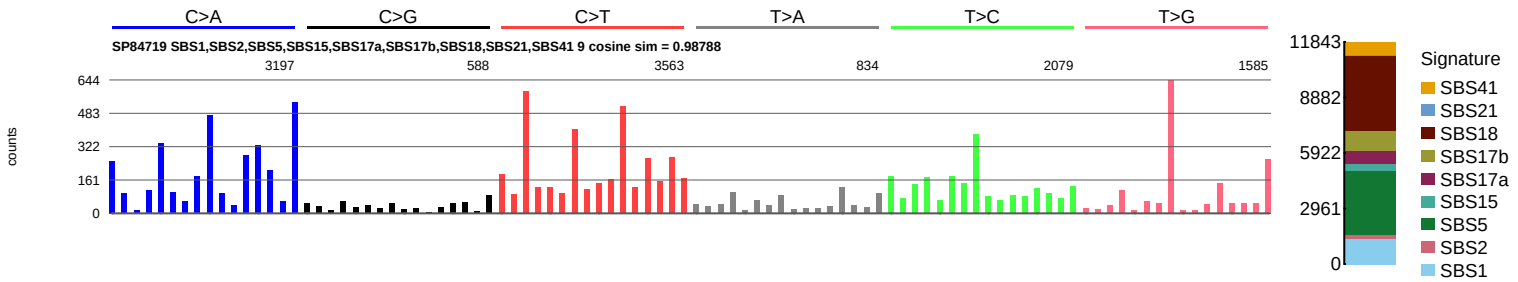
SBS2
SBS1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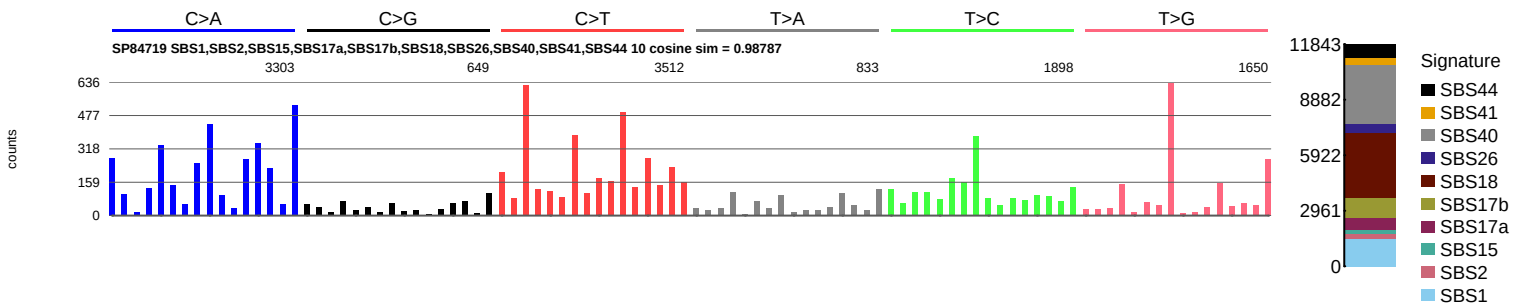
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98788
3197
588
3563
834
2079
1585
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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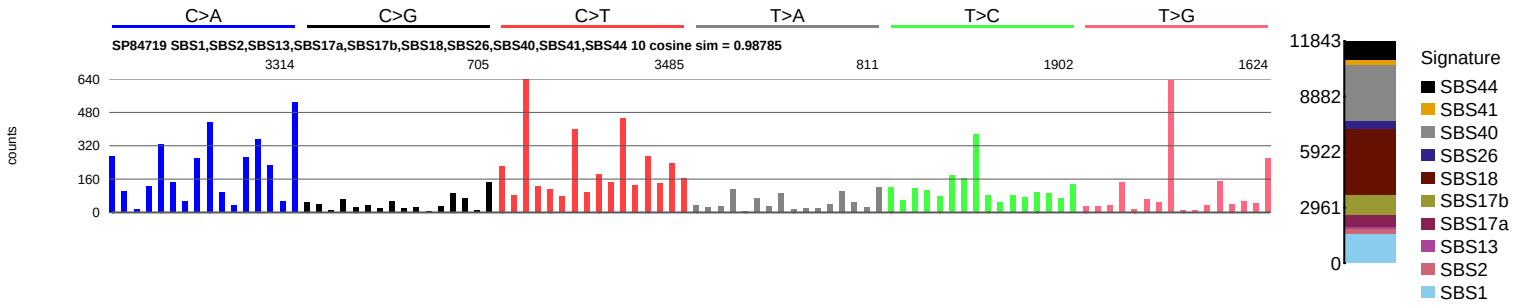
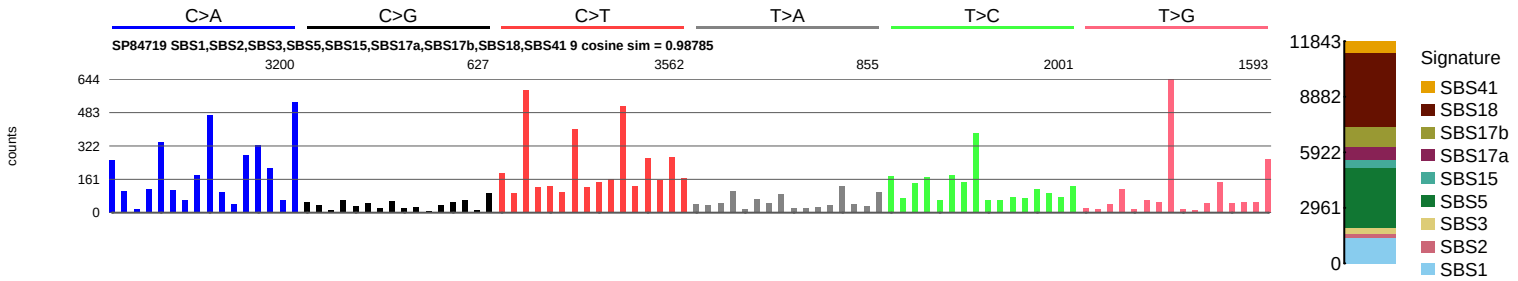
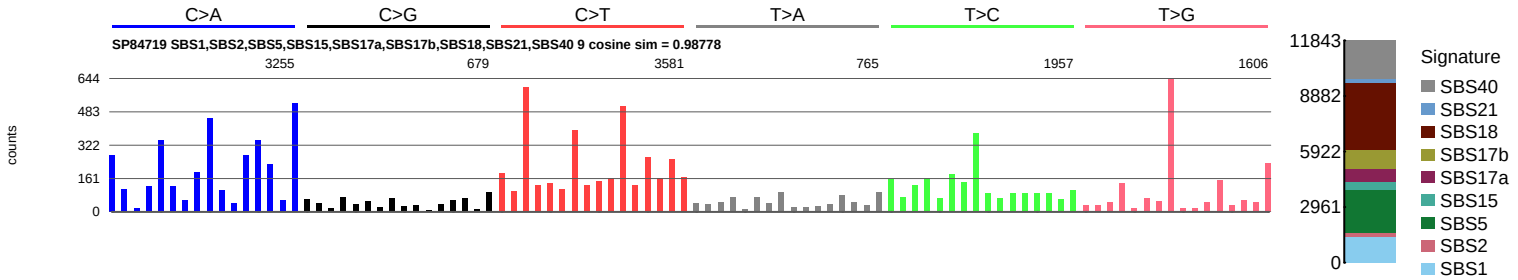
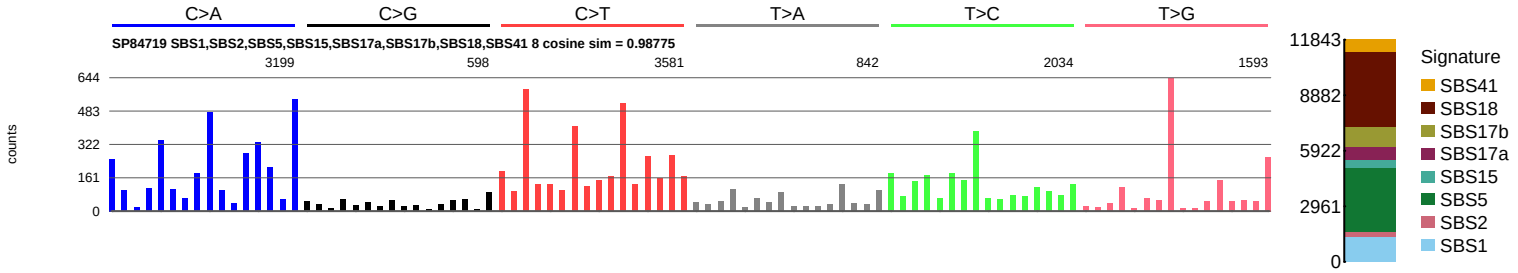
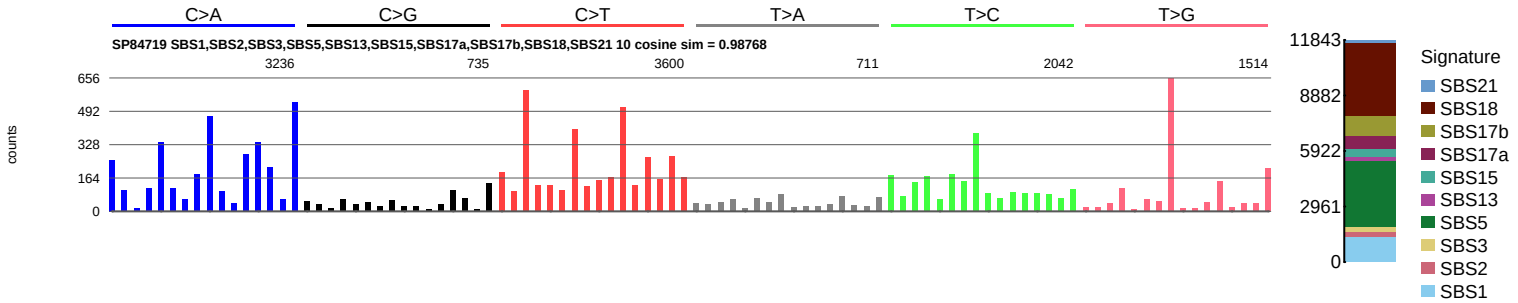
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98787
3303
649
3512
833
1898
1650
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

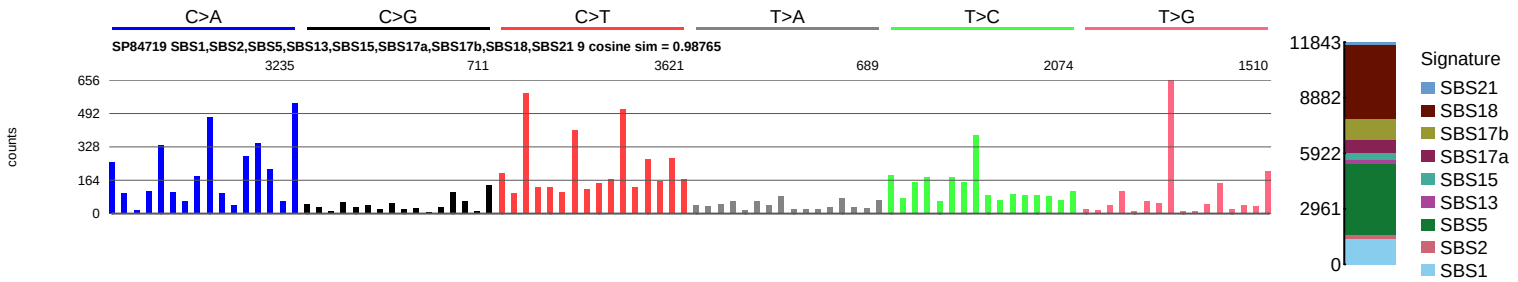

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98785
3314
705
3485
811
1902
1624
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98785
3200
627
3562
855
2001
1593
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98778
3255
679
3581
765
1957
1606
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98775
3199
598
3581
842
2034
1593
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98768
3236
735
3600
711
2042
1514
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98765
3235
711
3621
689
2074
1510
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
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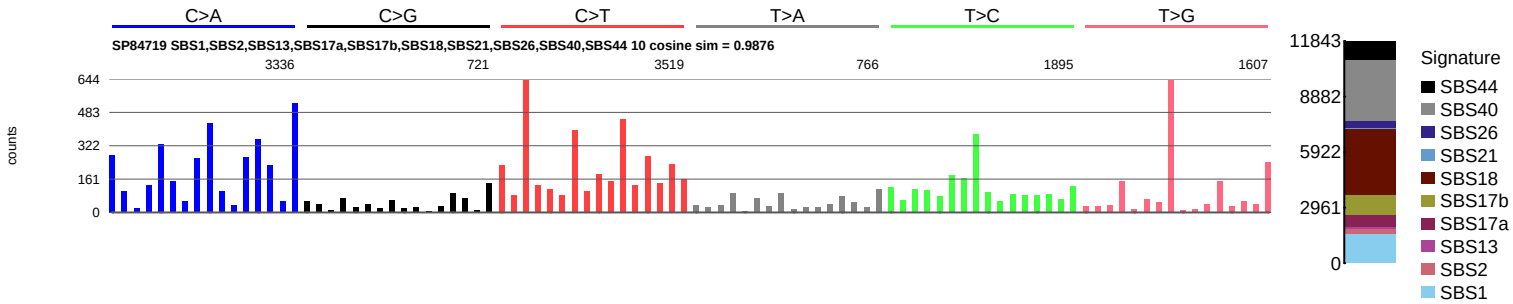
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SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.9876
3336
721
3519
766
1895
1607
counts
644
483
322
161
0
11843
8882
5922
2961
0
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SBS44
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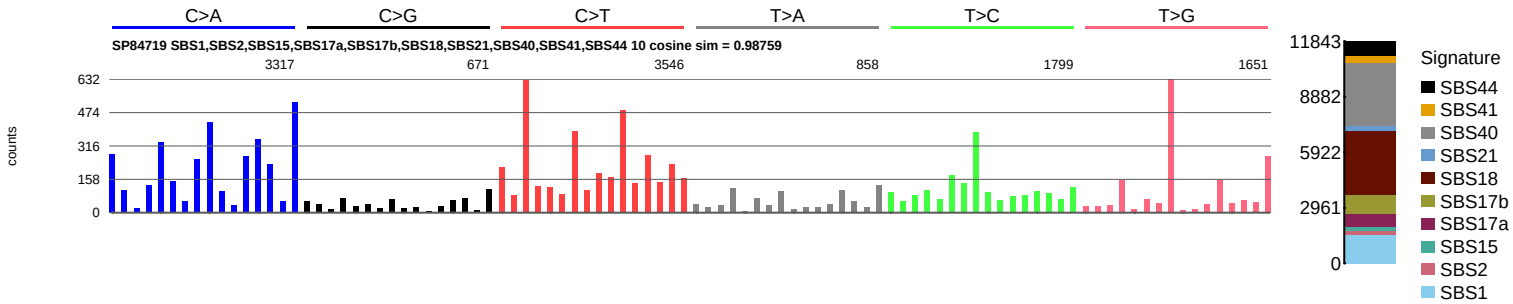
SBS2
SBS1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SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98759
3317
671
3546
858
1799
1651
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
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SBS17a
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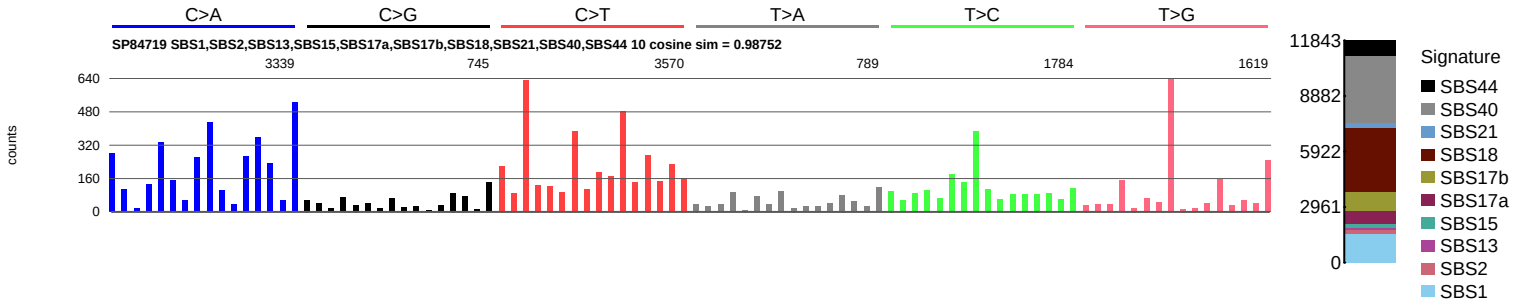
SBS2
SBS1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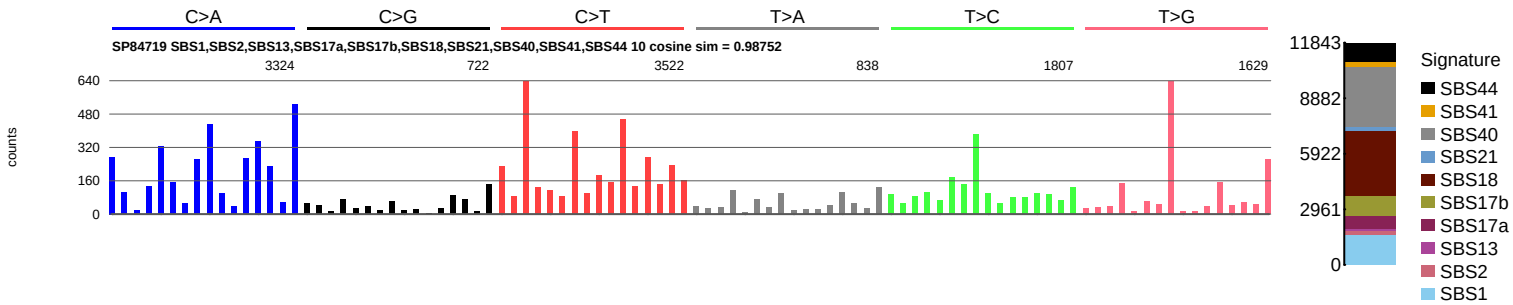
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98752
3339
745
3570
789
1784
1619
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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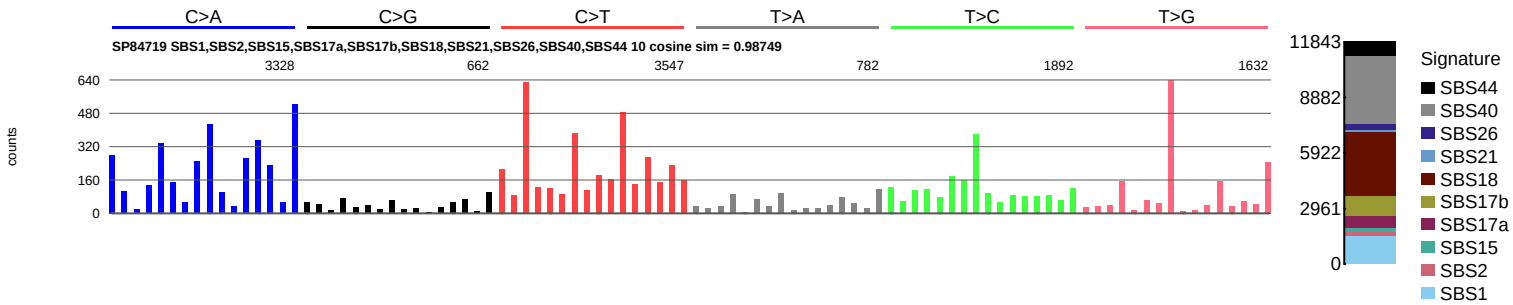
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98752
3324
722
3522
838
1807
1629
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
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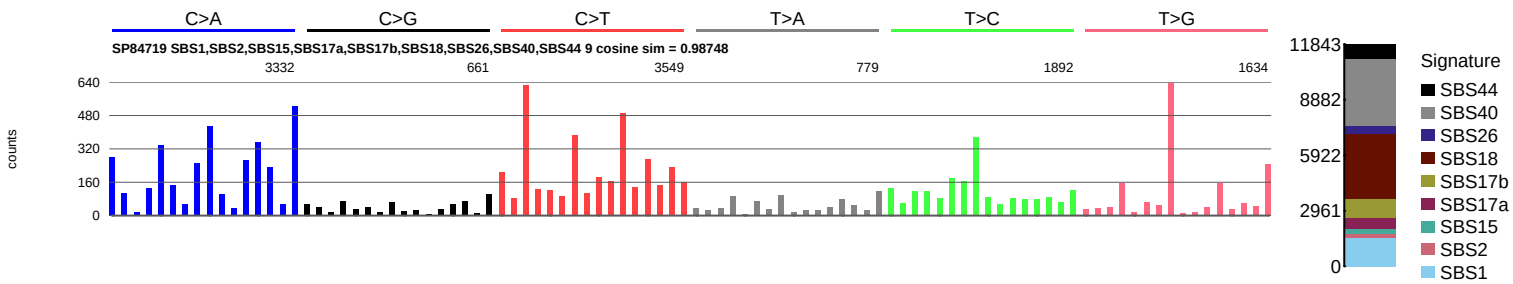
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98749
3328
662
3547
782
1892
1632
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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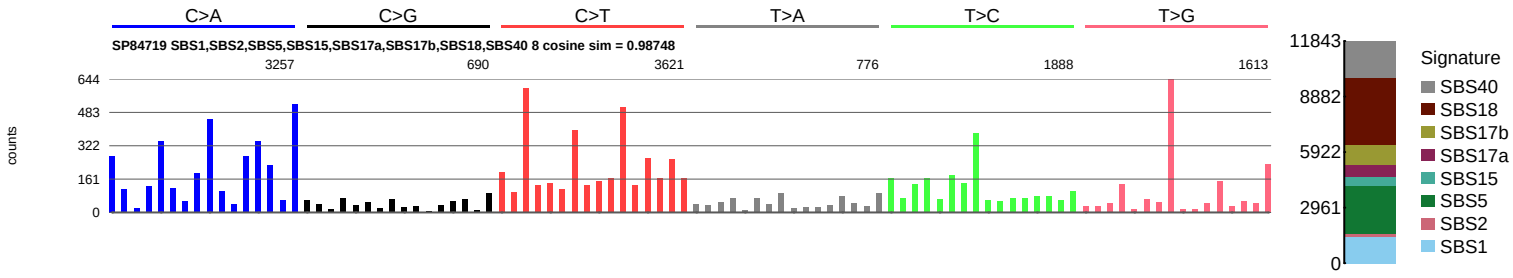
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98748
3332
661
3549
779
1892
1634
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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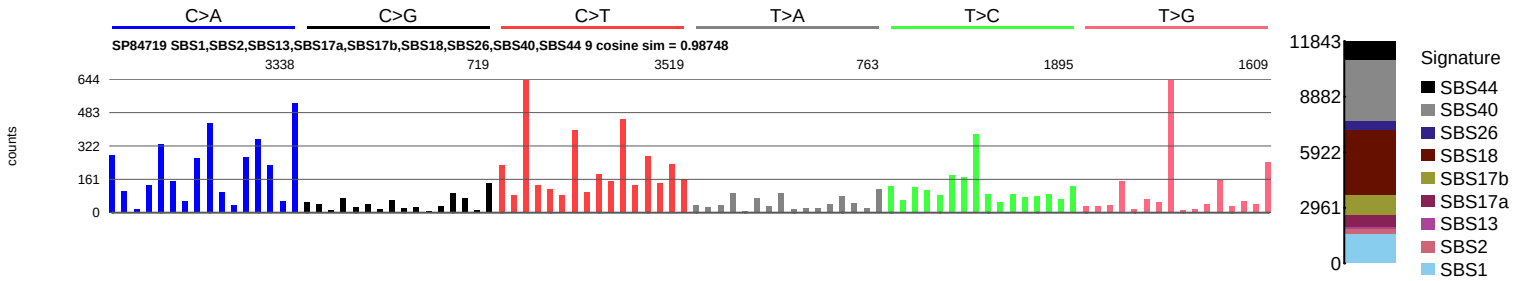
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98748
3257
690
3621
776
1888
1613
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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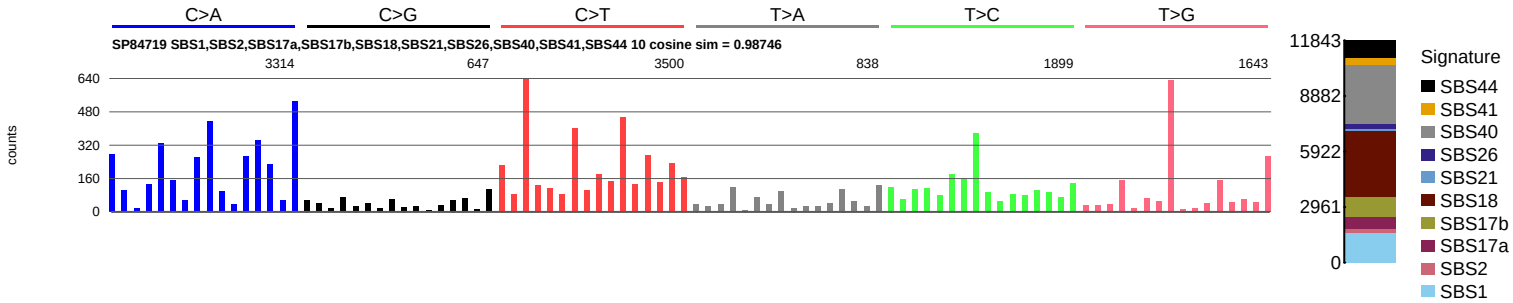
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98748
3338
719
3519
763
1895
1609
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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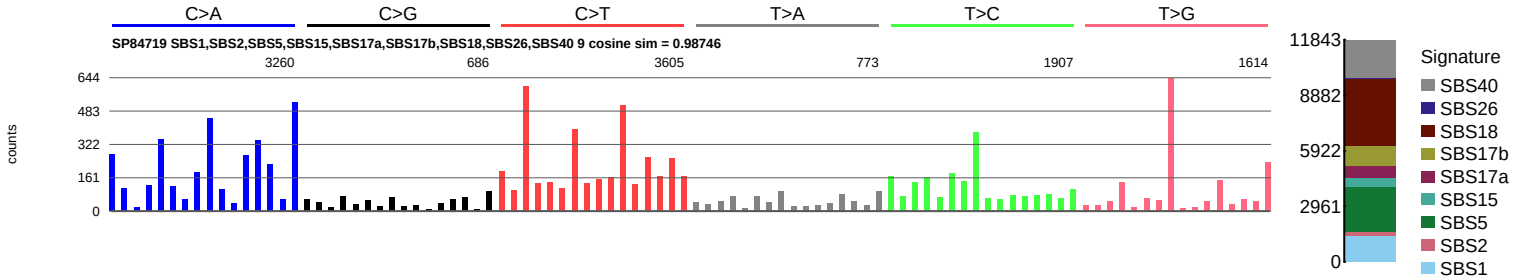
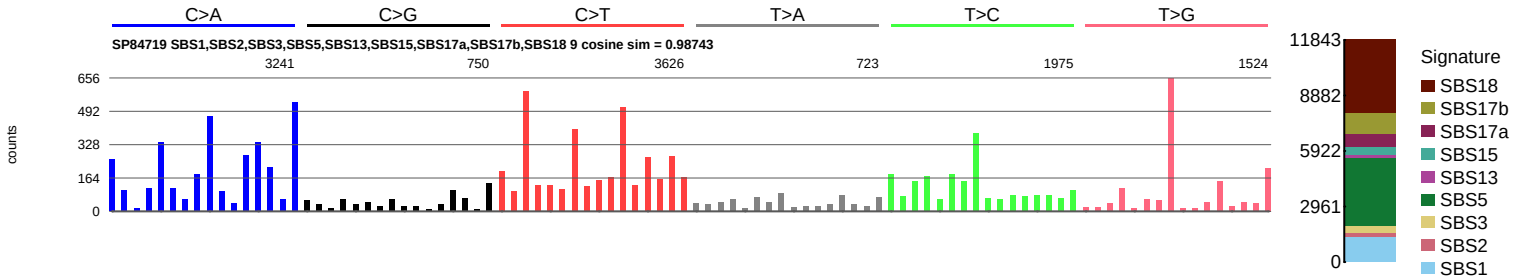
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98746
3314
647
3500
838
1899
1643
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98746
3260
686
3605
773
1907
1614
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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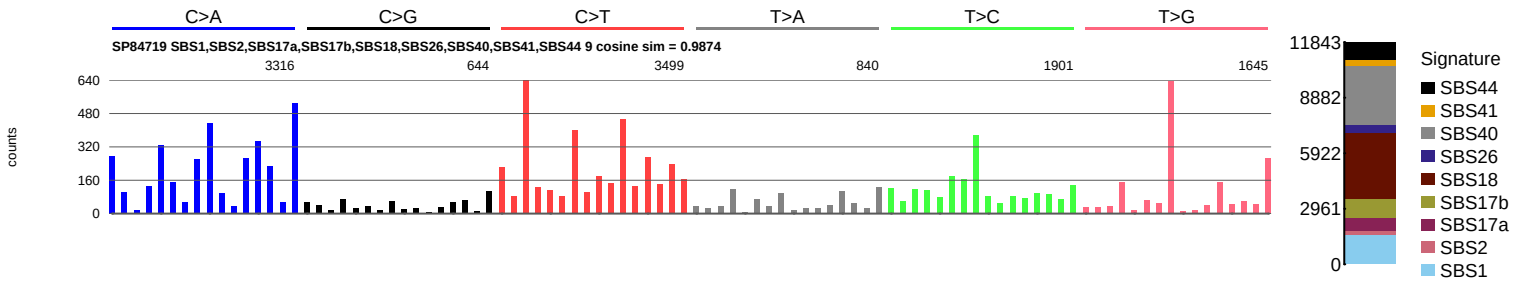
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98743
3241
750
3626
723
1975
1524
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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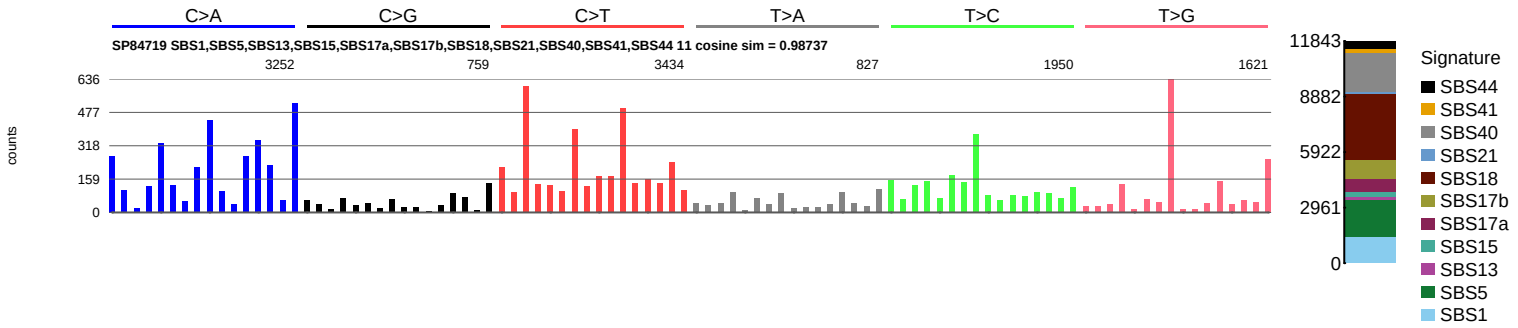
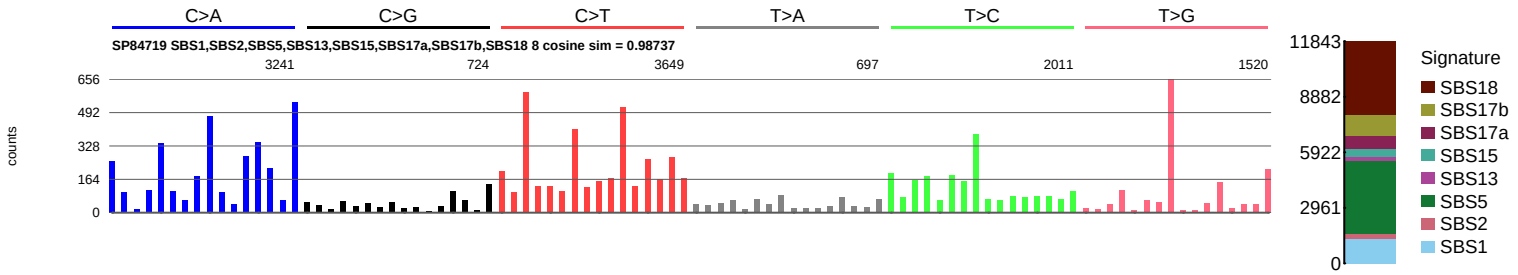
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9874
3316
644
3499
840
1901
1645
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98737
3252
759
3434
827
1950
1621
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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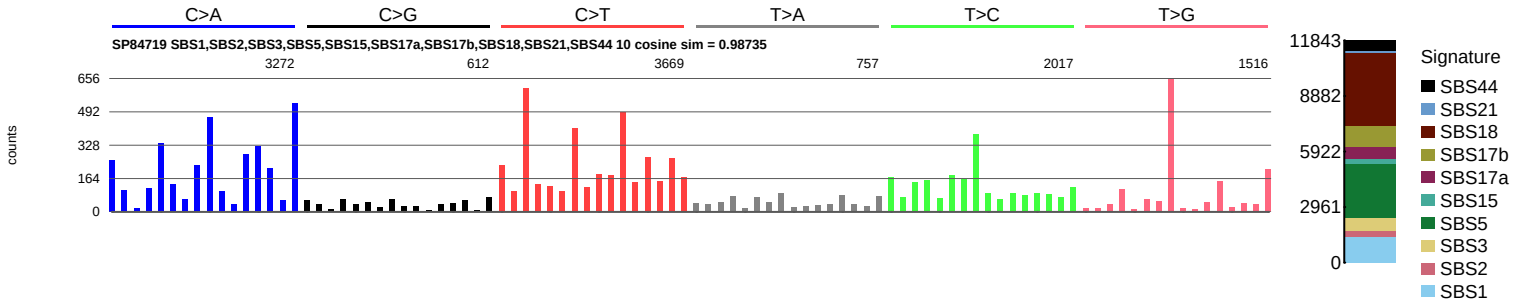
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98737
3241
724
3649
697
2011
1520
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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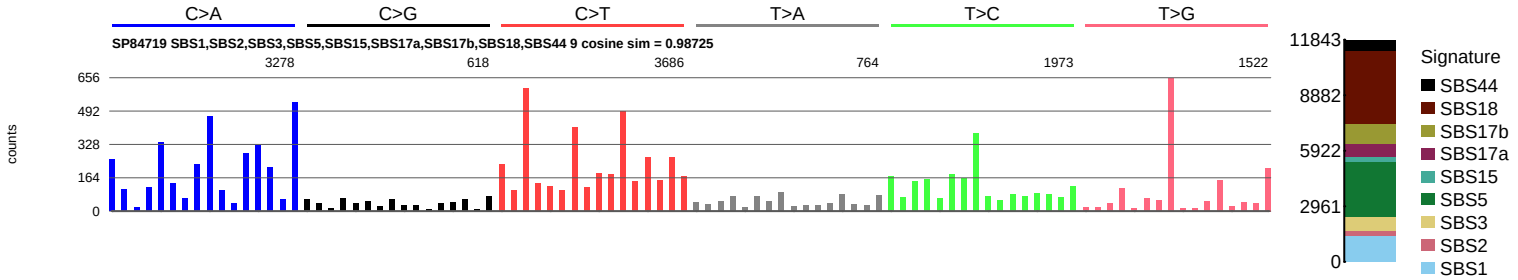
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98735
3272
612
3669
757
2017
1516
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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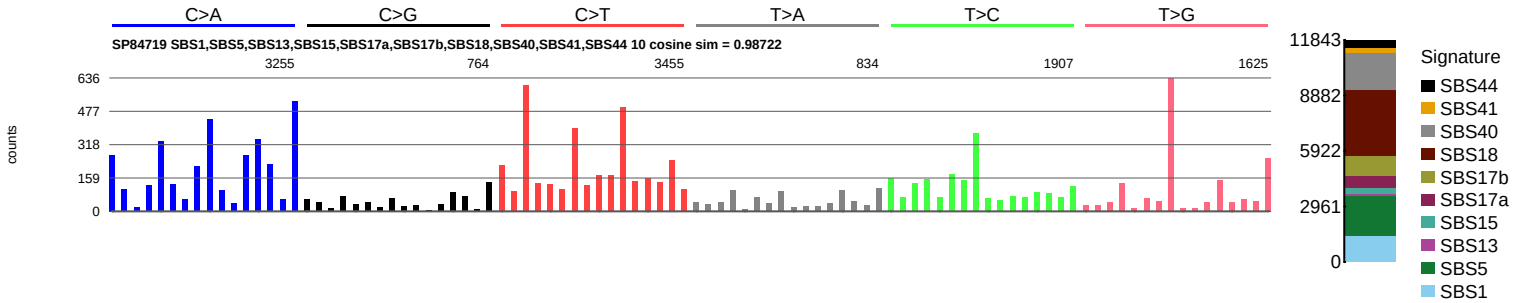
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98725
3278
618
3686
764
1973
1522
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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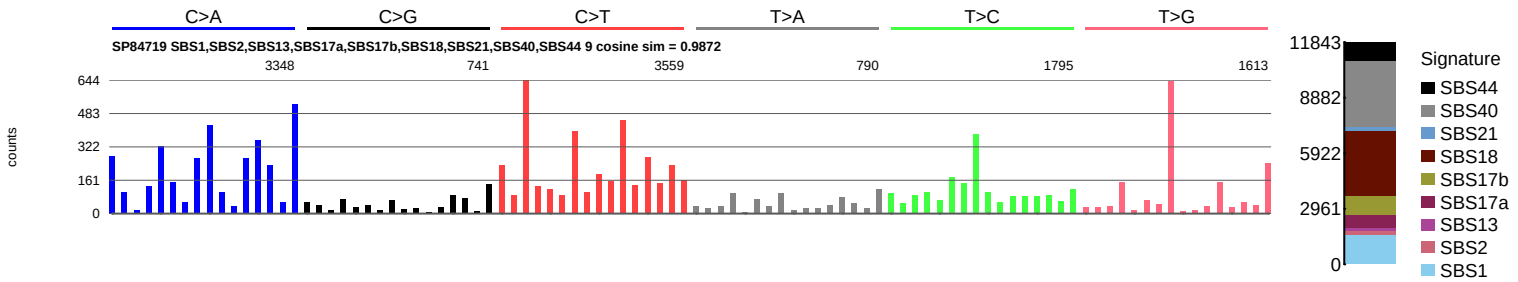
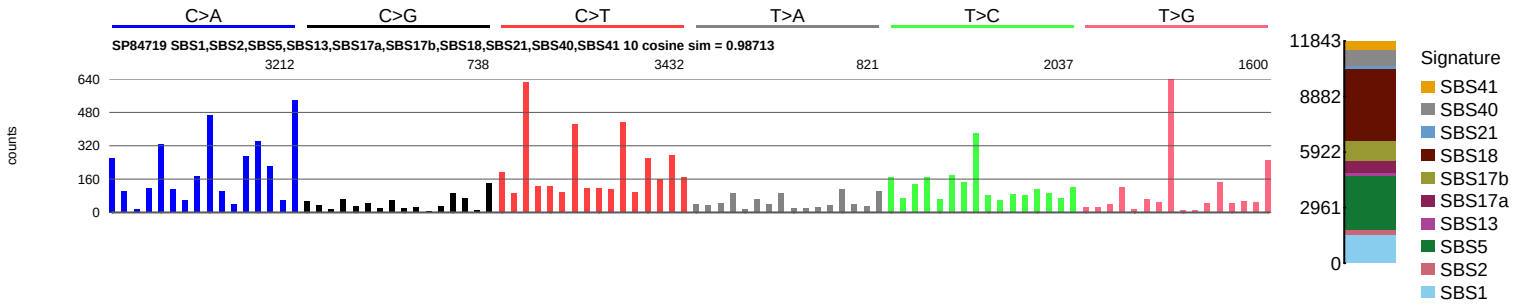
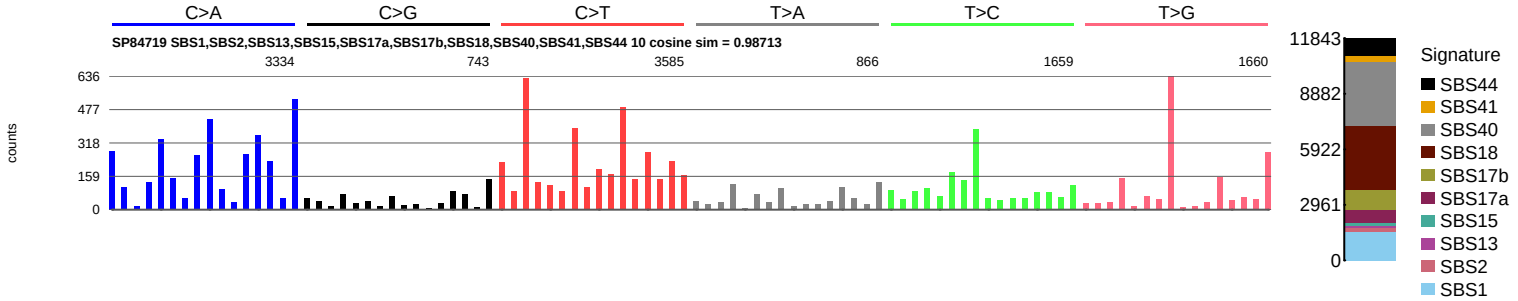
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98722
3255
764
3455
834
1907
1625
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9872
3348
741
3559
790
1795
1613
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98713
3212
738
3432
821
2037
1600
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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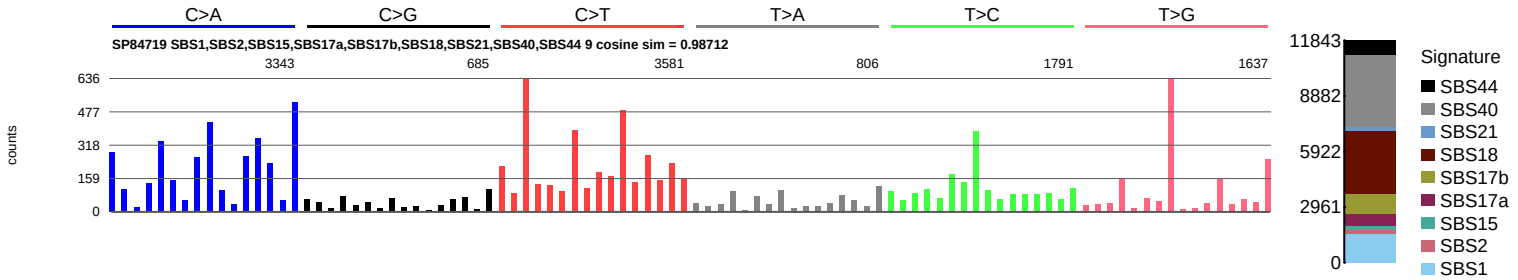
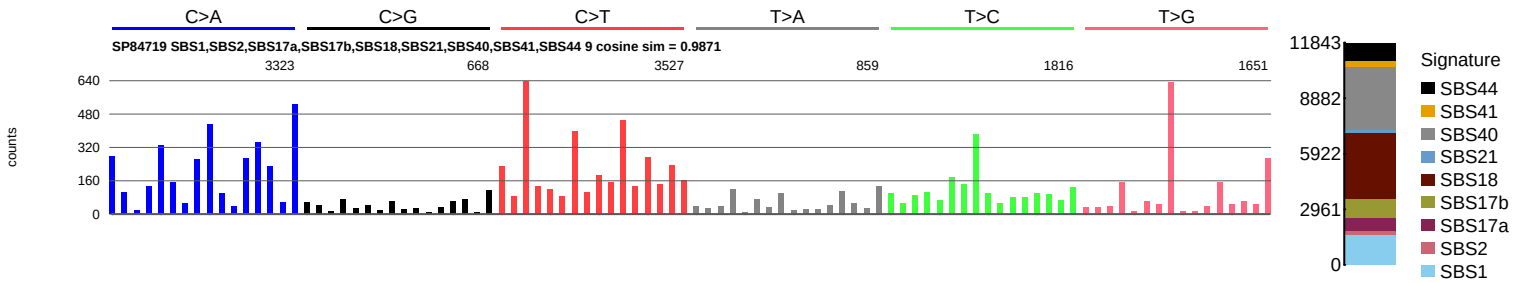
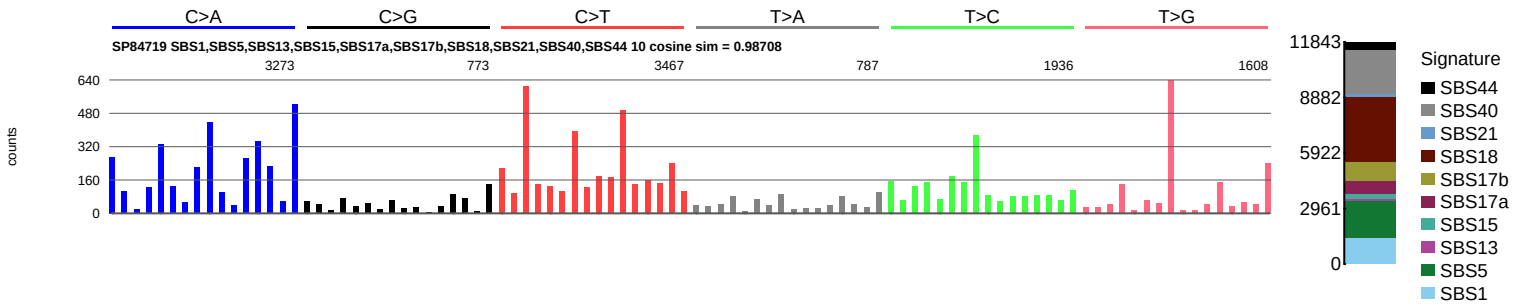
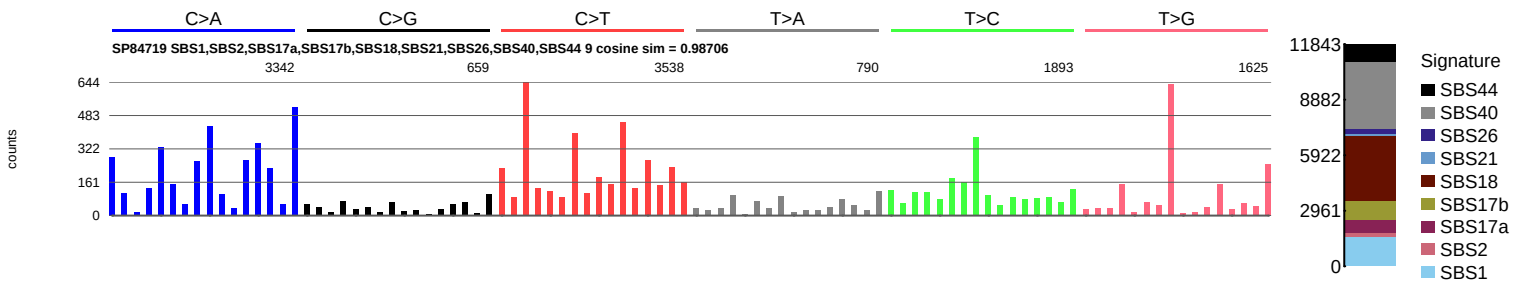
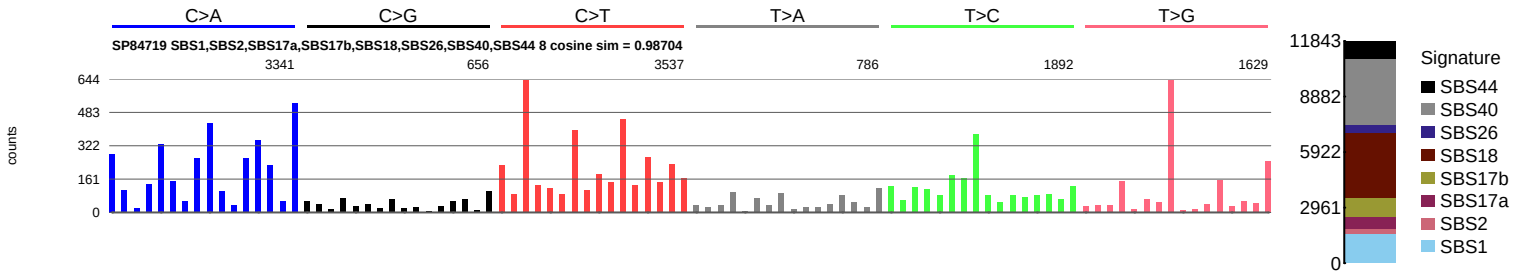
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98713
3334
743
3585
866
1659
1660
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98712
3343
685
3581
806
1791
1637
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9871
3323
668
3527
859
1816
1651
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98708
3273
773
3467
787
1936
1608
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98706
3342
659
3538
790
1893
1625
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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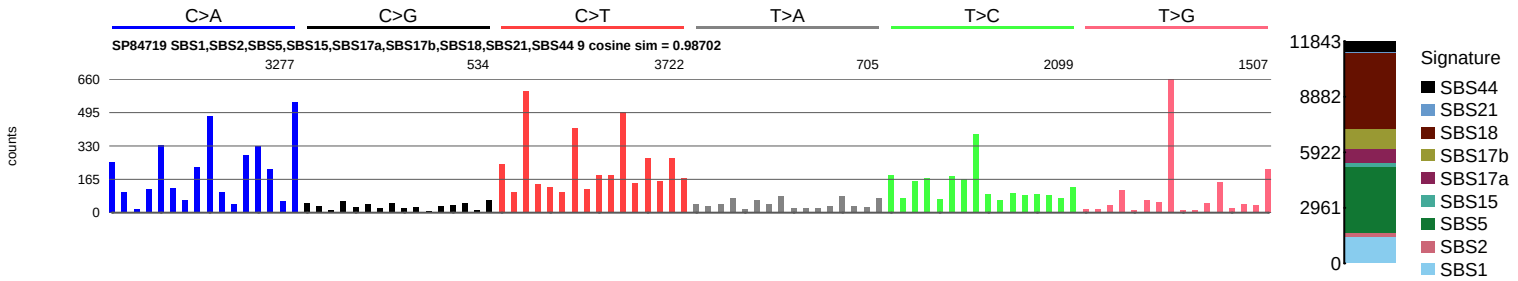
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98704
3341
656
3537
786
1892
1629
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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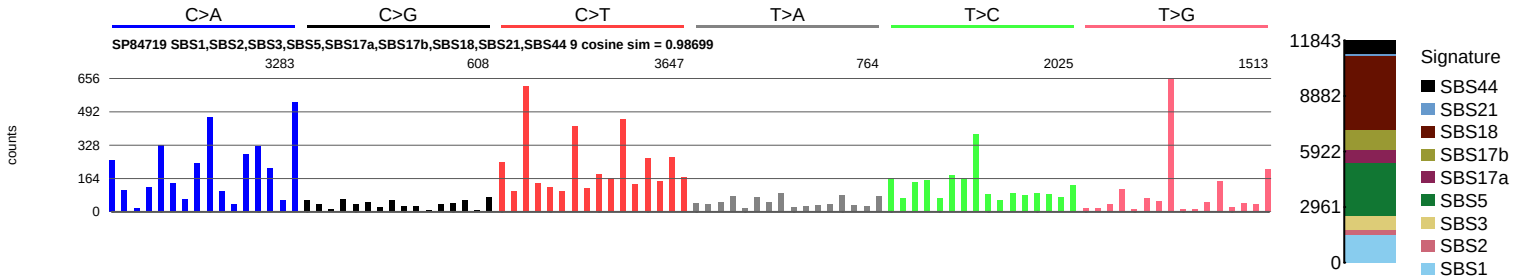
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98702
3277
534
3722
705
2099
1507
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>T
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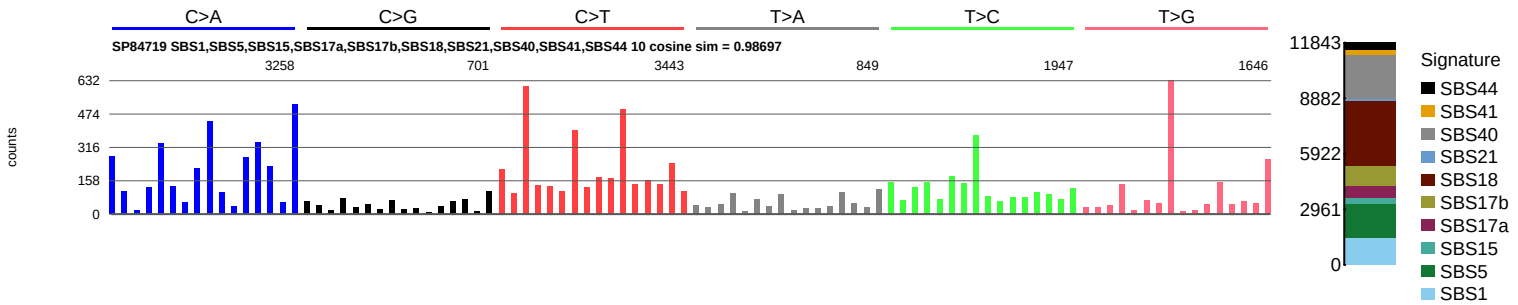
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98699
3283
608
3647
764
2025
1513
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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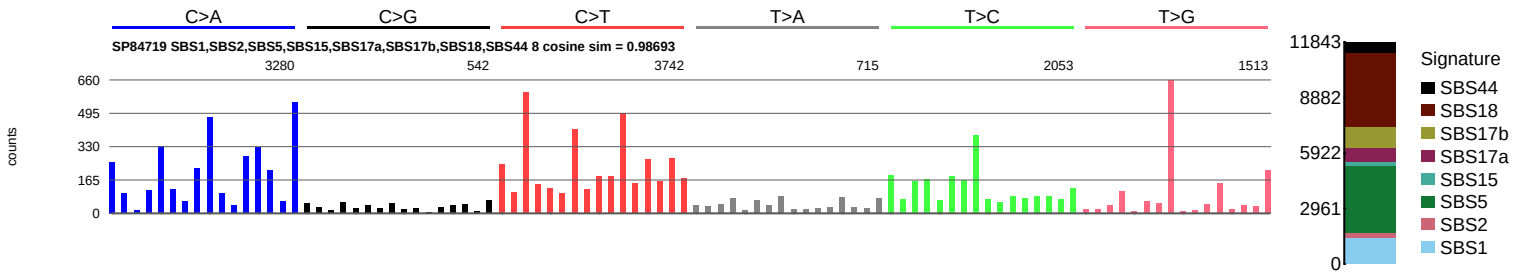
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98697
3258
701
3443
849
1947
1646
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98693
3280
542
3742
715
2053
1513
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
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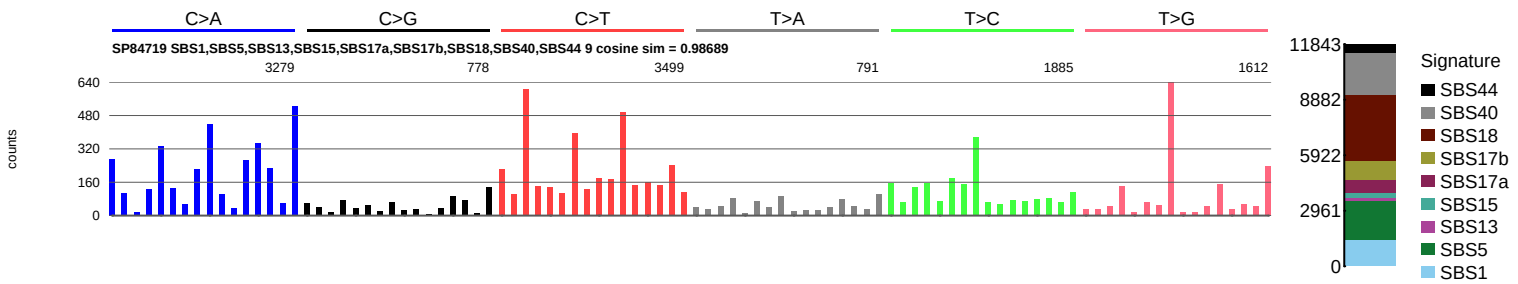
SBS5
SBS2
SBS1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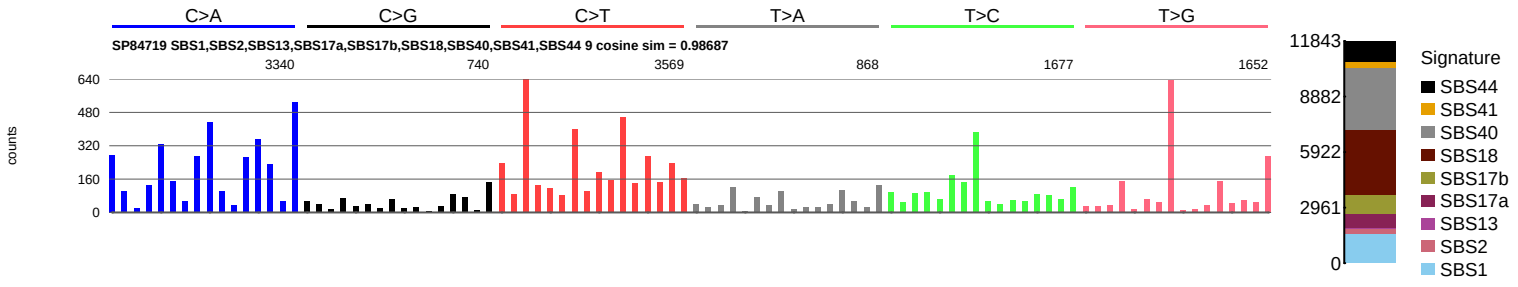
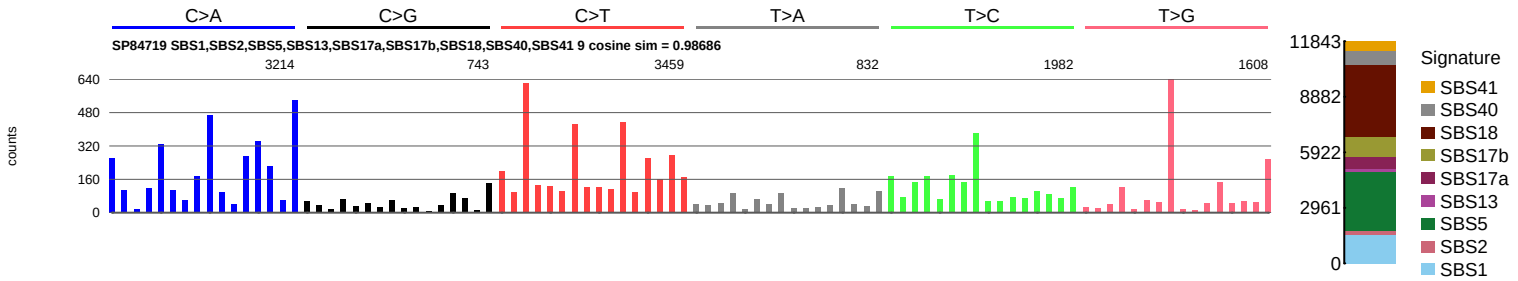
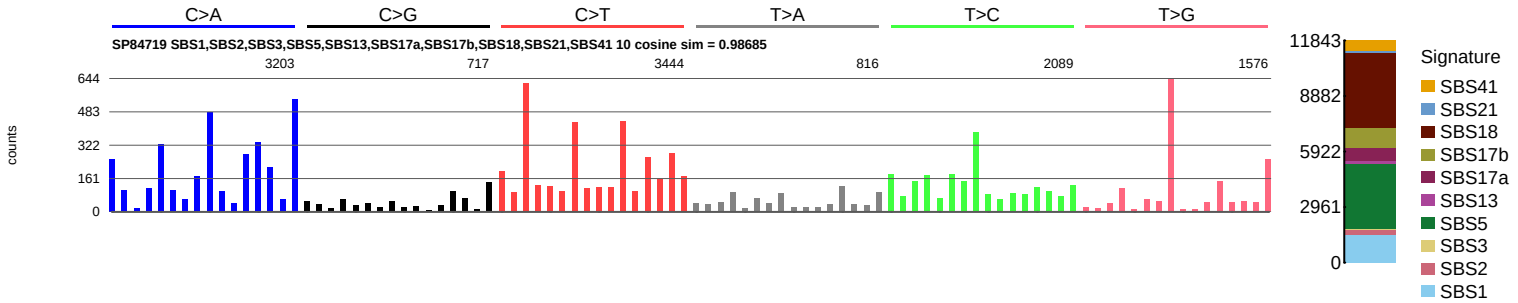
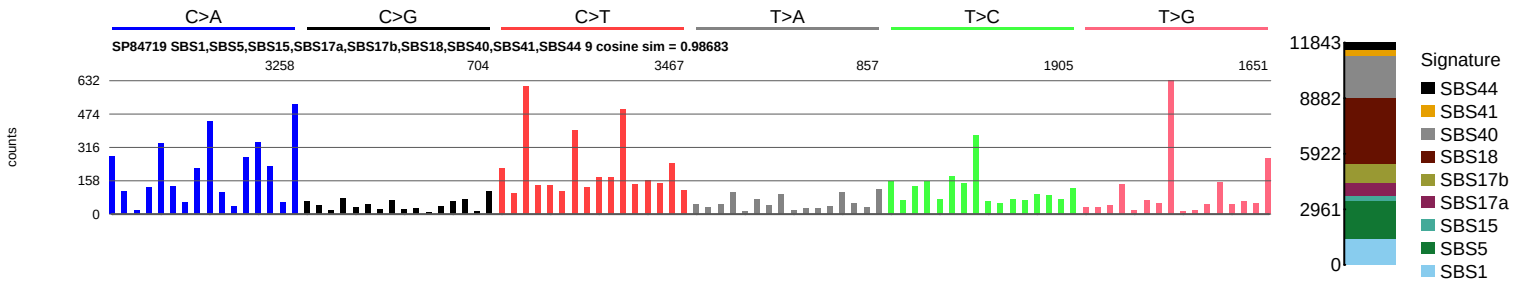
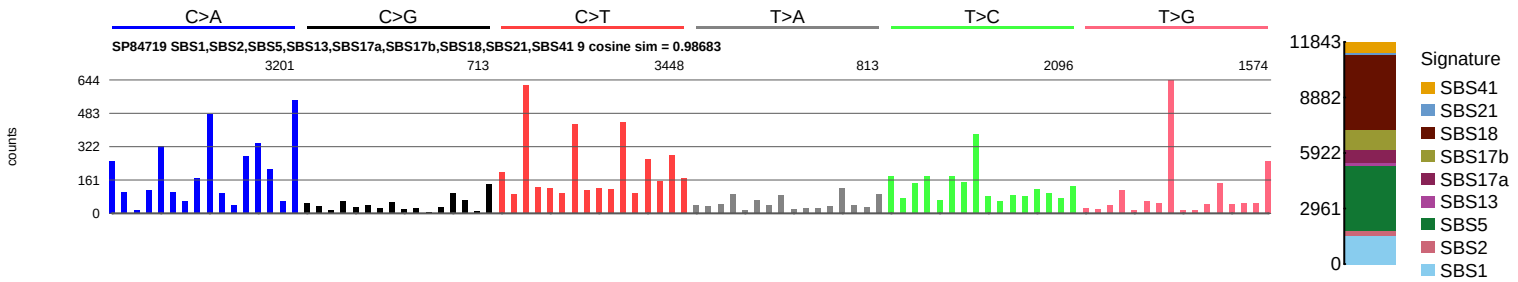
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98689
3279
778
3499
791
1885
1612
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

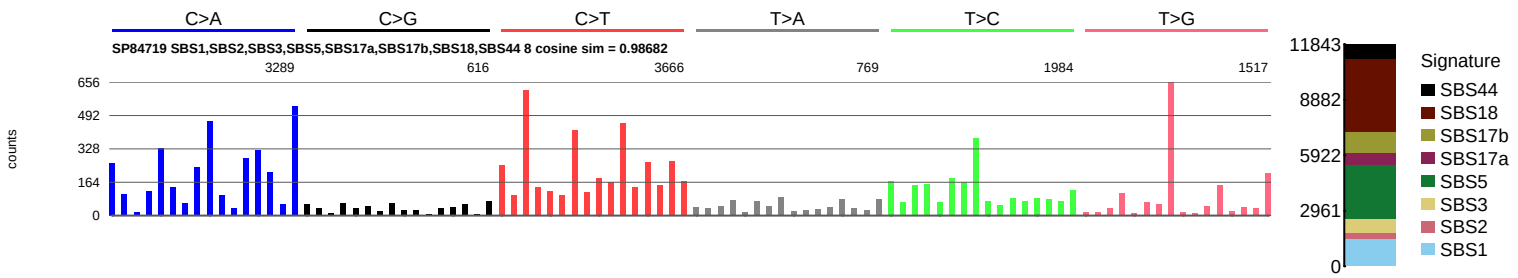

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98687
3340
740
3569
868
1677
1652
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

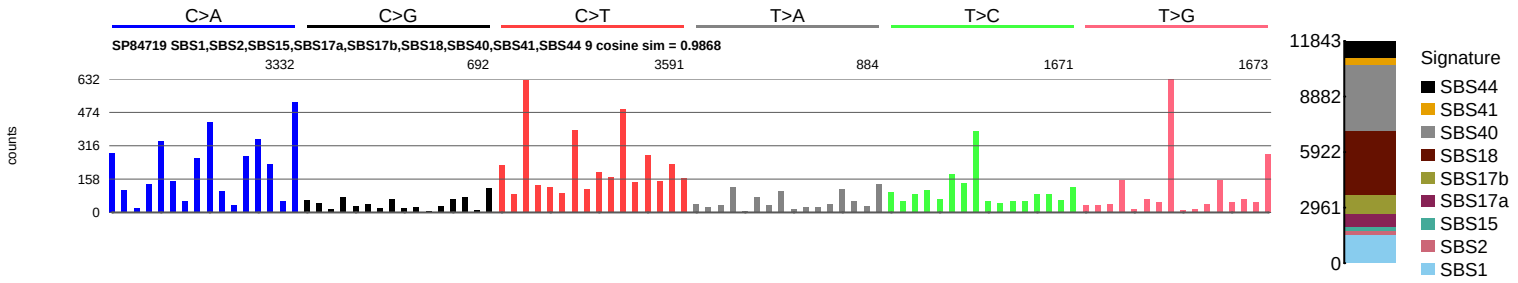

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98686
3214
743
3459
832
1982
1608
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

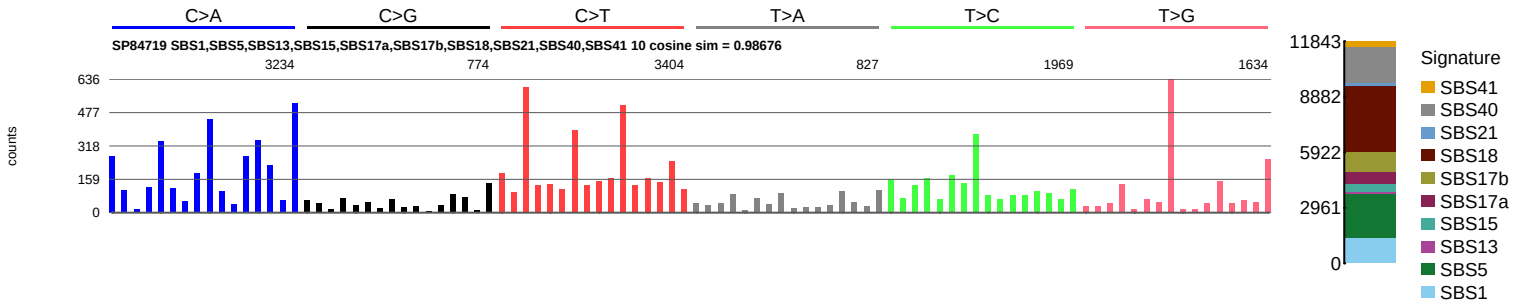

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98685
3203
717
3444
816
2089
1576
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

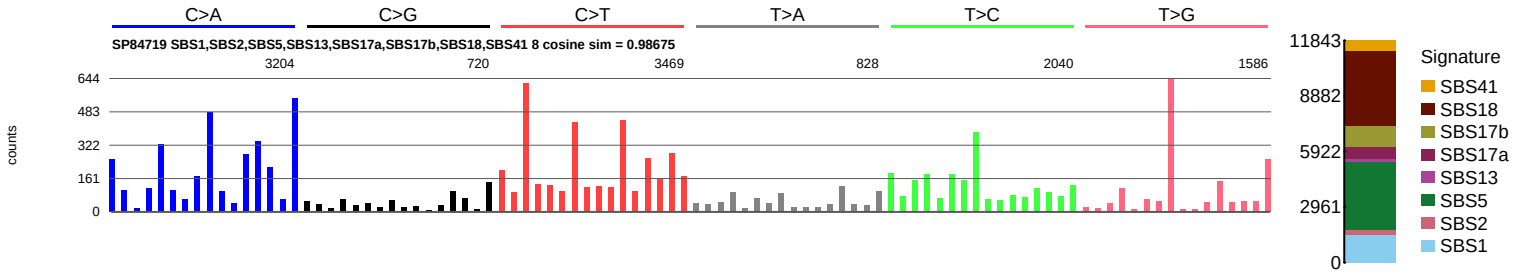

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98683
3258
704
3467
857
1905
1651
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

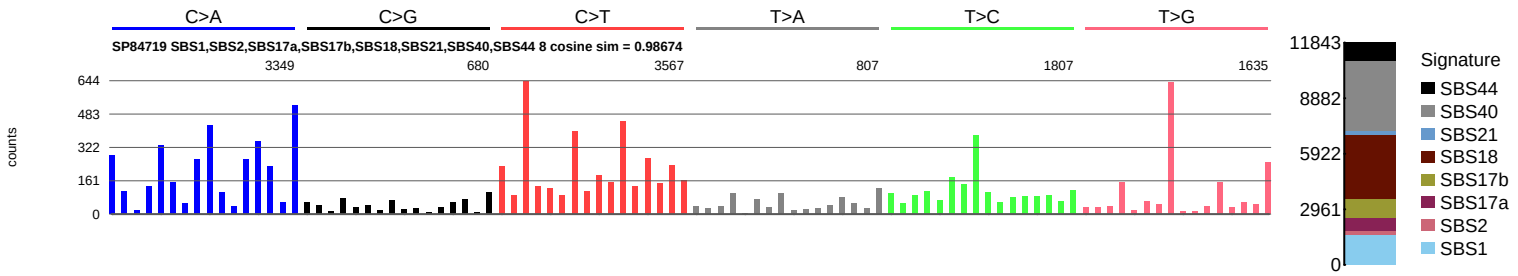

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98683
3201
713
3448
813
2096
1574
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

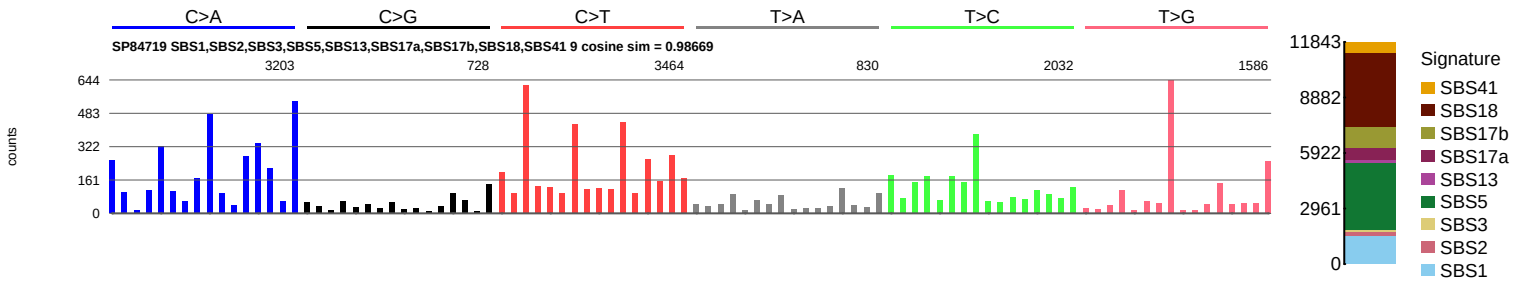

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98682
3289
616
3666
769
1984
1517
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

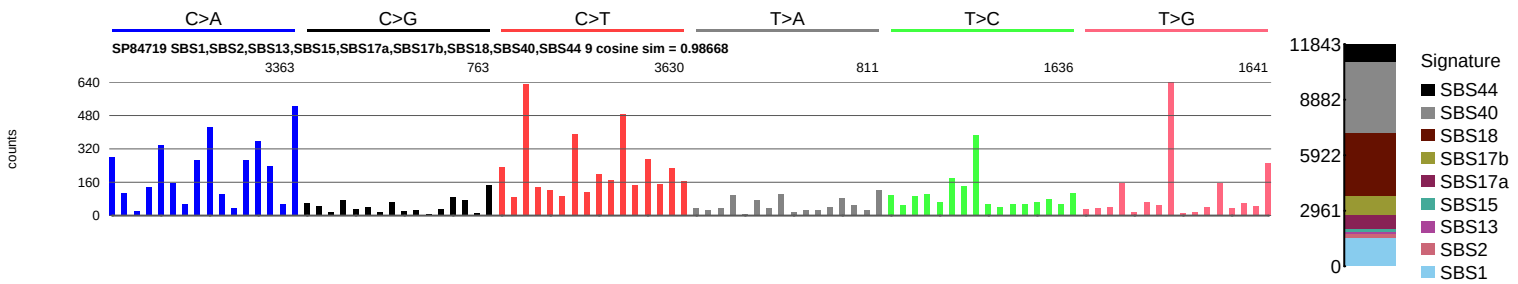

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.9868
3332
692
3591
884
1671
1673
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

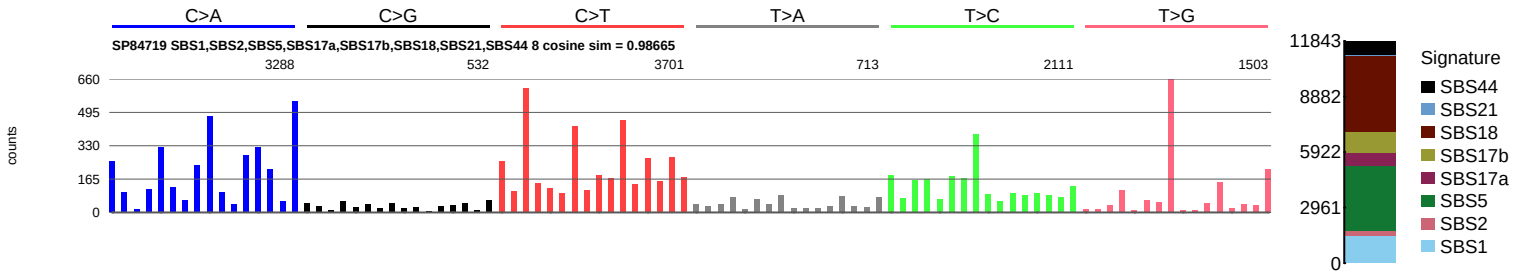

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98676
3234
774
3404
827
1969
1634
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

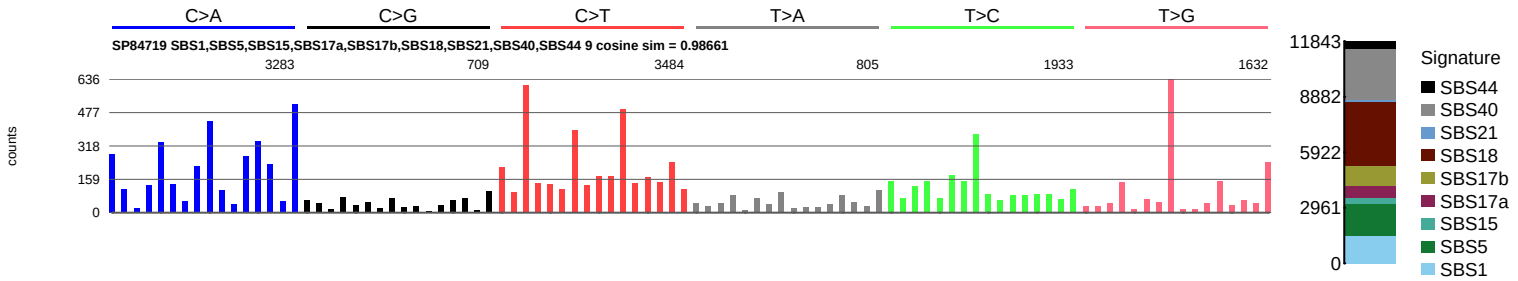

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98675
3204
720
3469
828
2040
1586
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

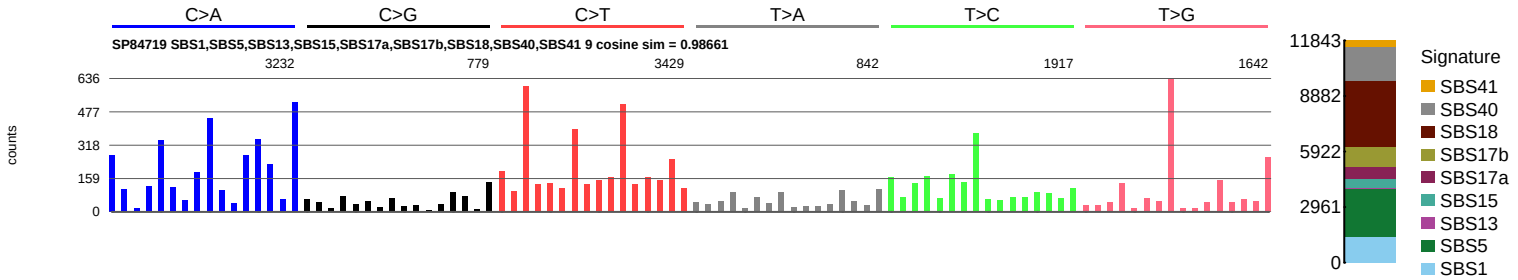

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98674
3349
680
3567
807
1807
1635
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

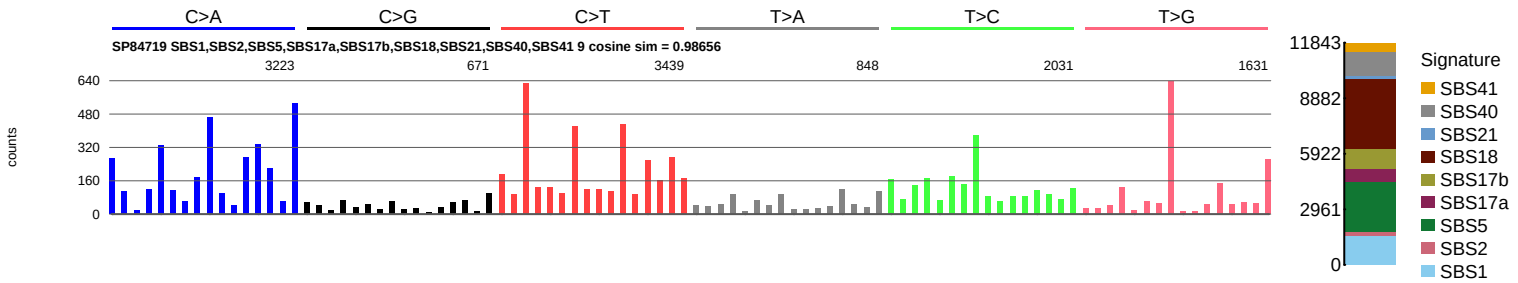

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98669
3203
728
3464
830
2032
1586
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98668
3363
763
3630
811
1636
1641
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98665
3288
532
3701
713
2111
1503
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98661
3283
709
3484
805
1933
1632
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98661
3232
779
3429
842
1917
1642
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98656
3223
671
3439
848
2031
1631
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
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SBS17b
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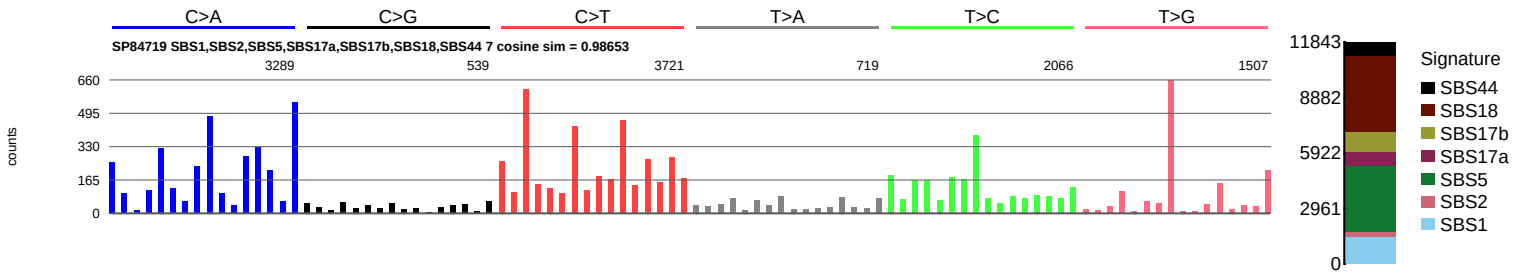
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98653
3289
539
3721
719
2066
1507
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
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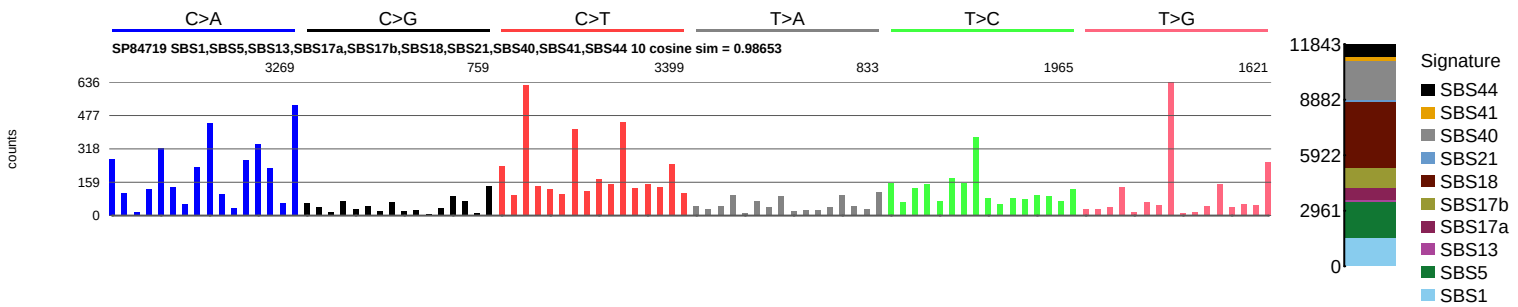
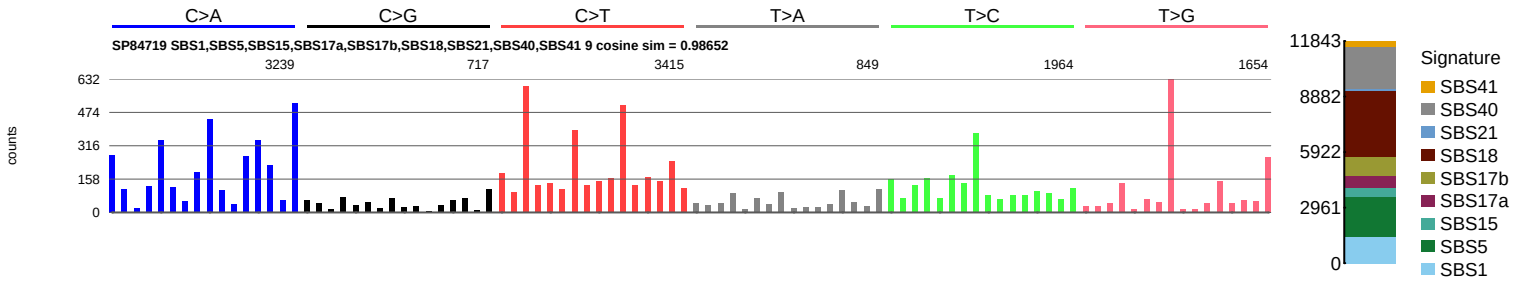
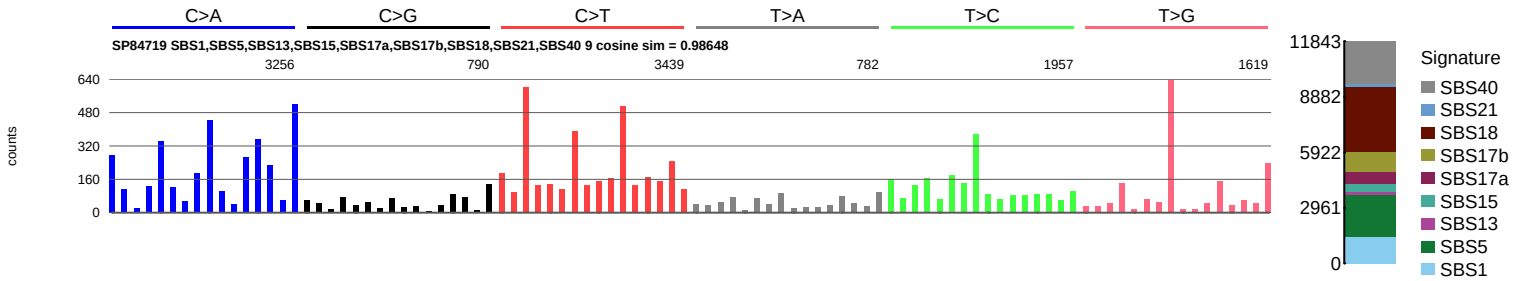
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98653
3269
759
3399
833
1965
1621
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98652
3239
717
3415
849
1964
1654
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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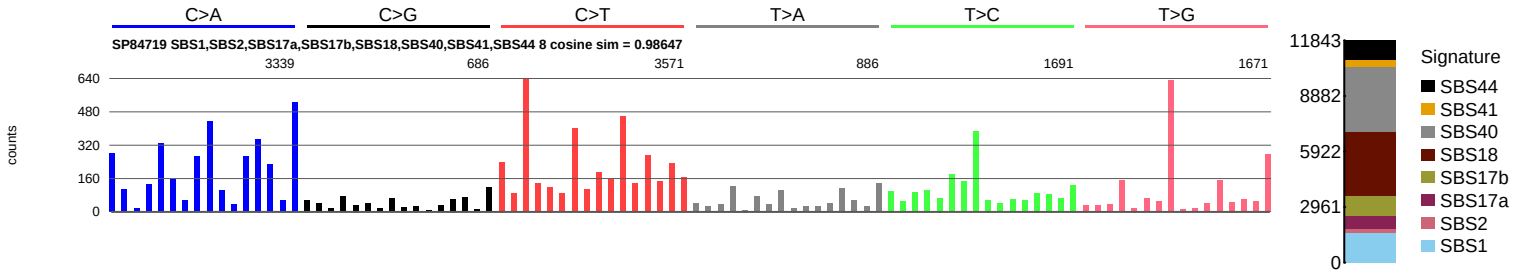
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98648
3256
790
3439
782
1957
1619
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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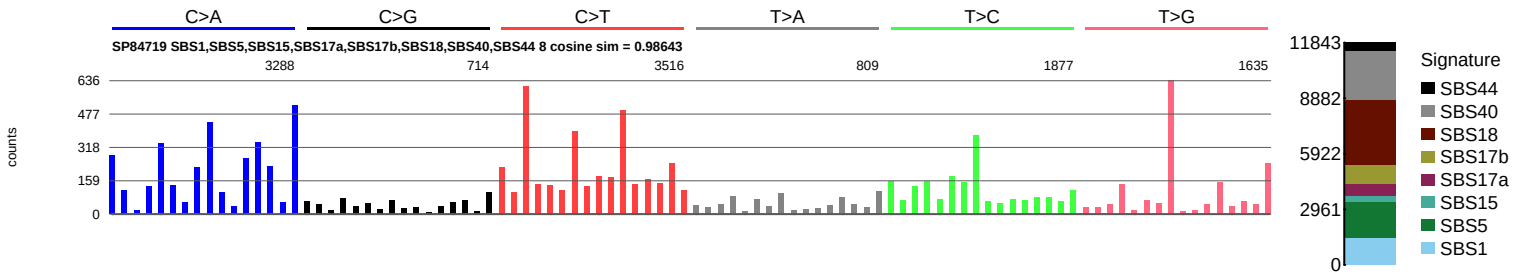
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98647
3339
686
3571
886
1691
1671
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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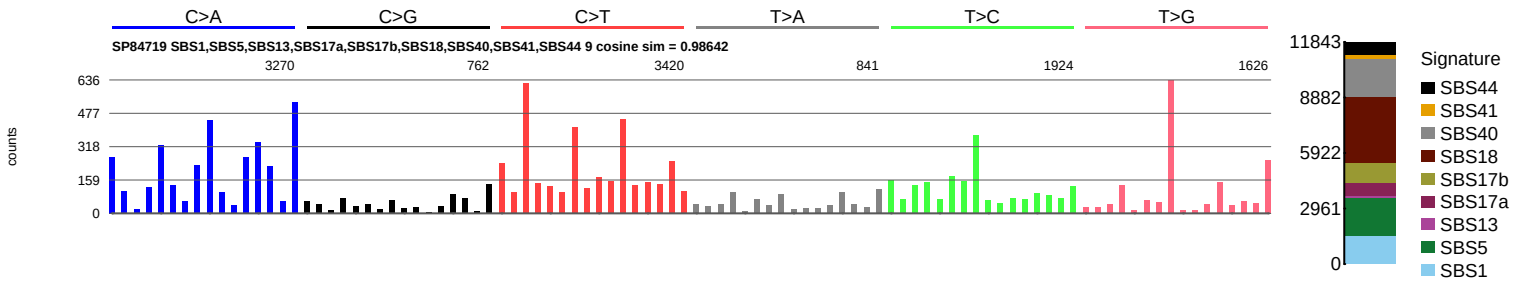
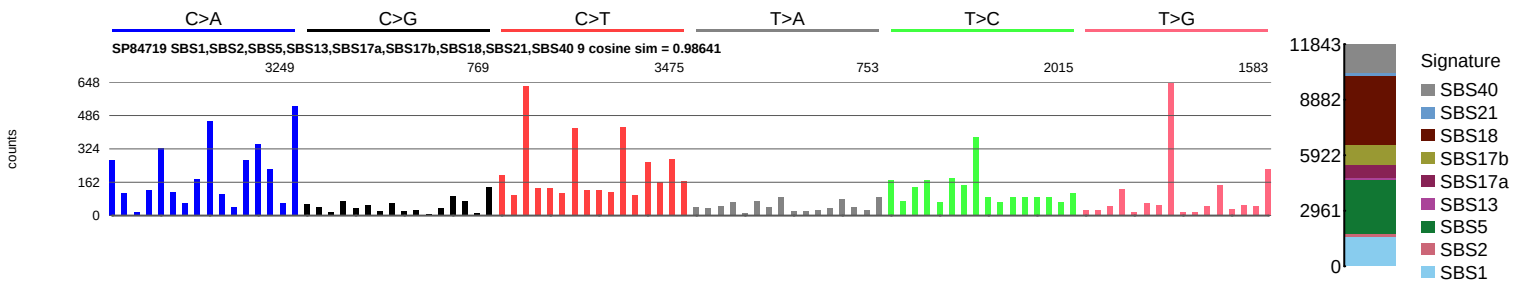
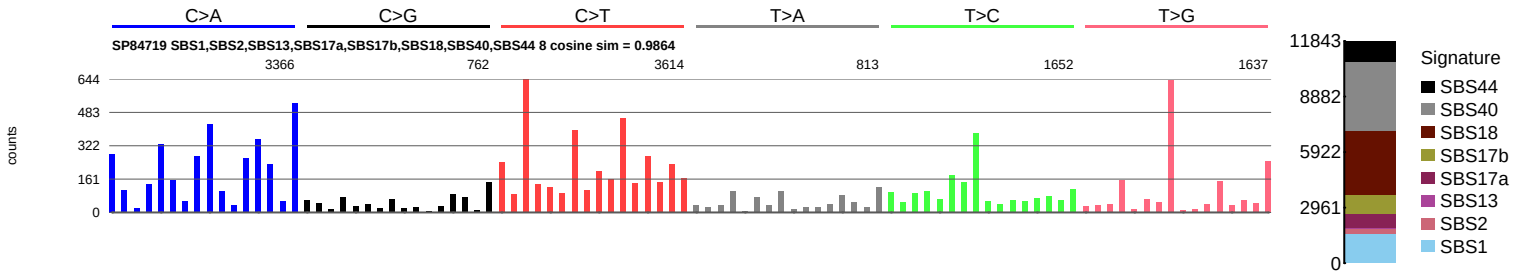
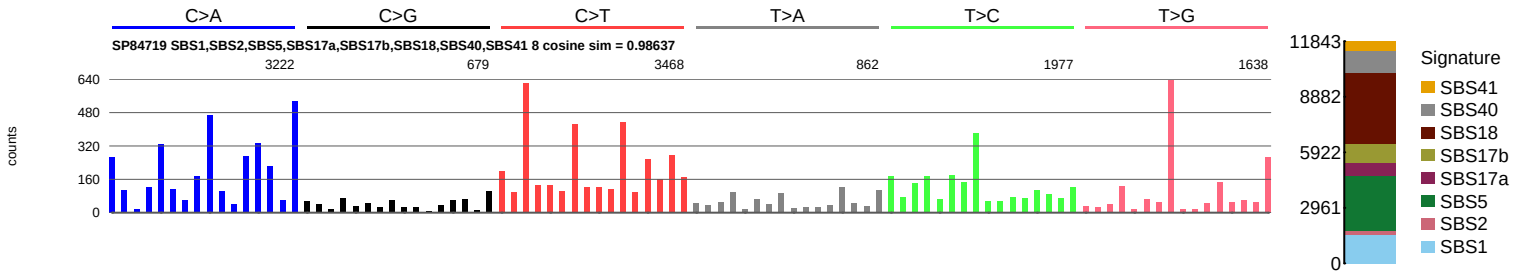
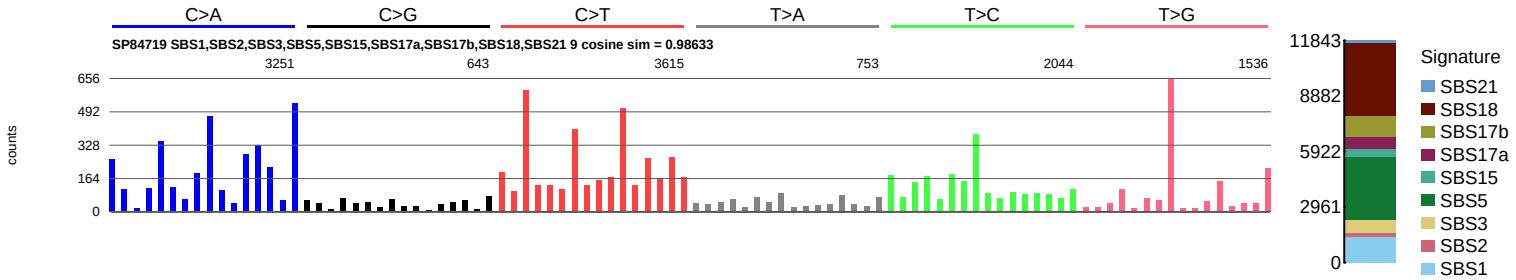
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98643
3288
714
3516
809
1877
1635
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

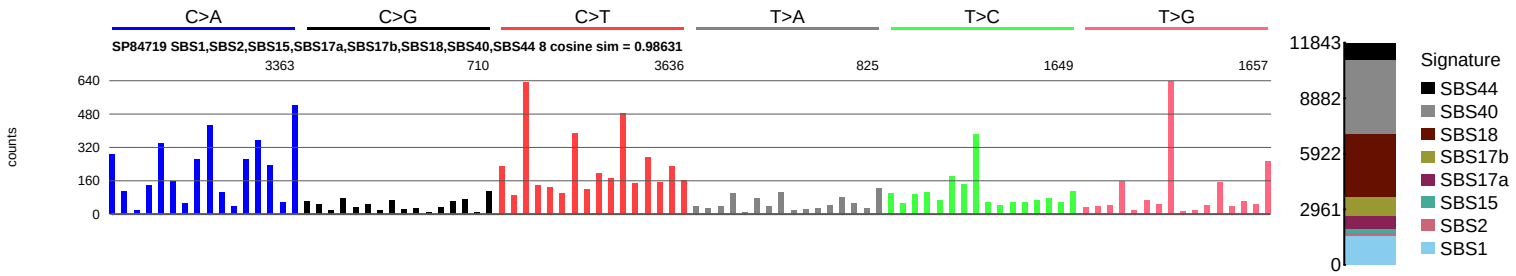

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98642
3270
762
3420
841
1924
1626
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

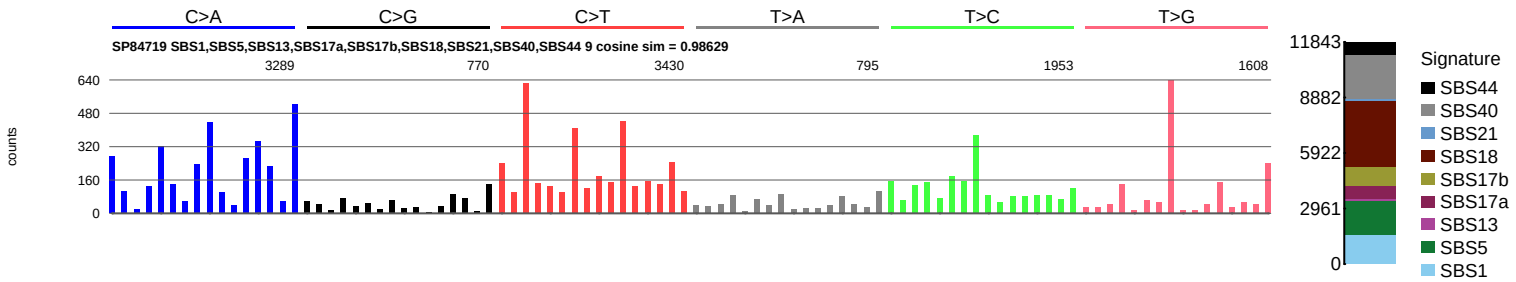

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98641
3249
769
3475
753
2015
1583
648
486
324
162
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

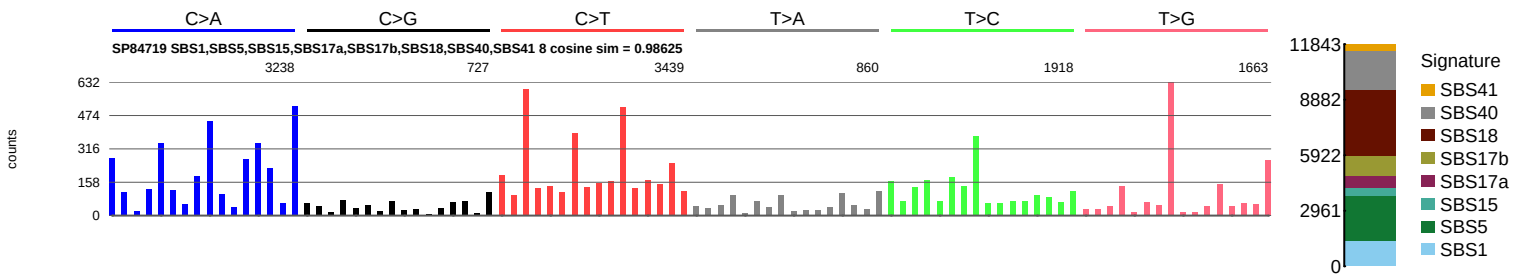

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9864
3366
762
3614
813
1652
1637
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

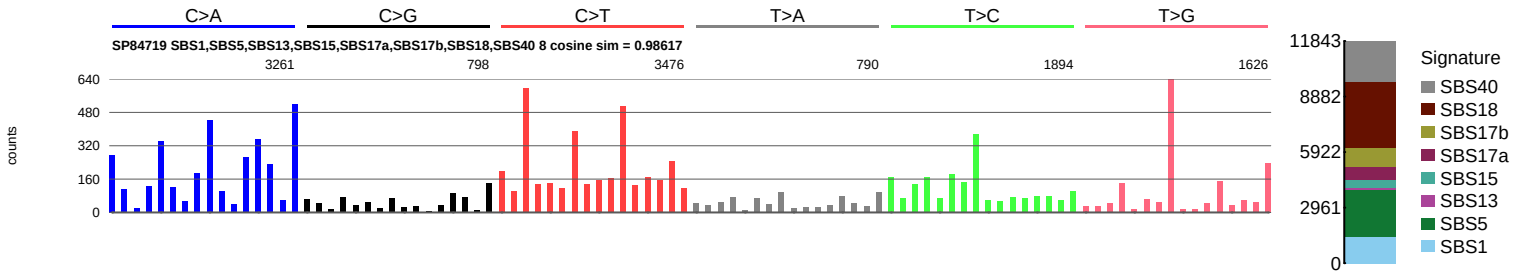

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98637
3222
679
3468
862
1977
1638
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

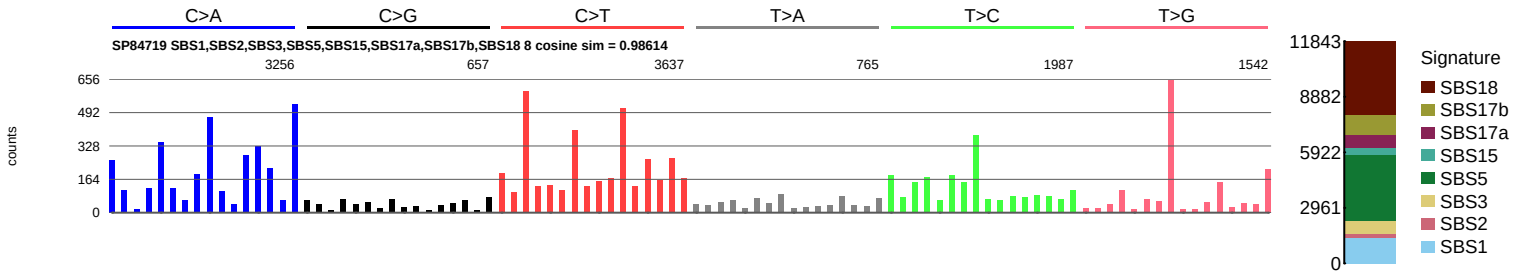

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98633
3251
643
3615
753
2044
1536
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

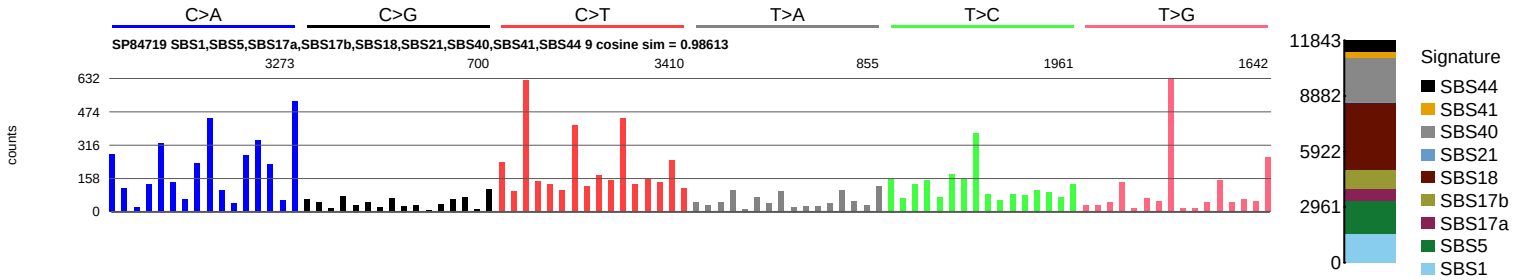

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98631
3363
710
3636
825
1649
1657
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

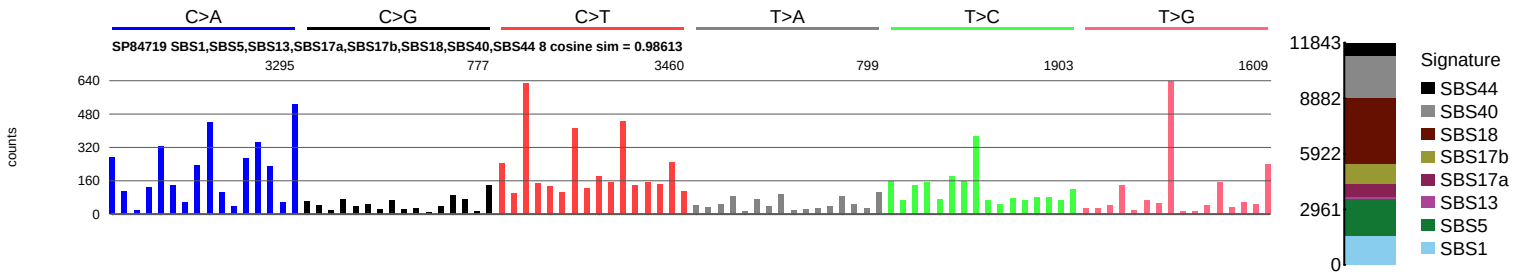

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98629
3289
770
3430
795
1953
1608
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

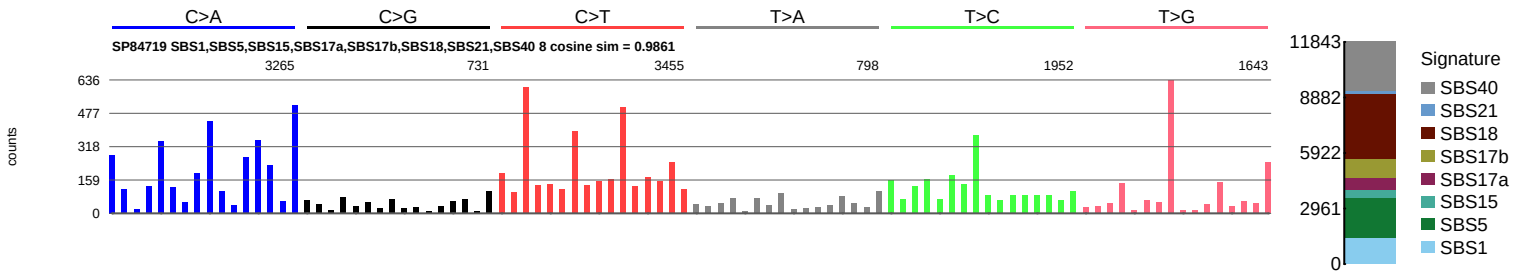

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98625
3238
727
3439
860
1918
1663
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

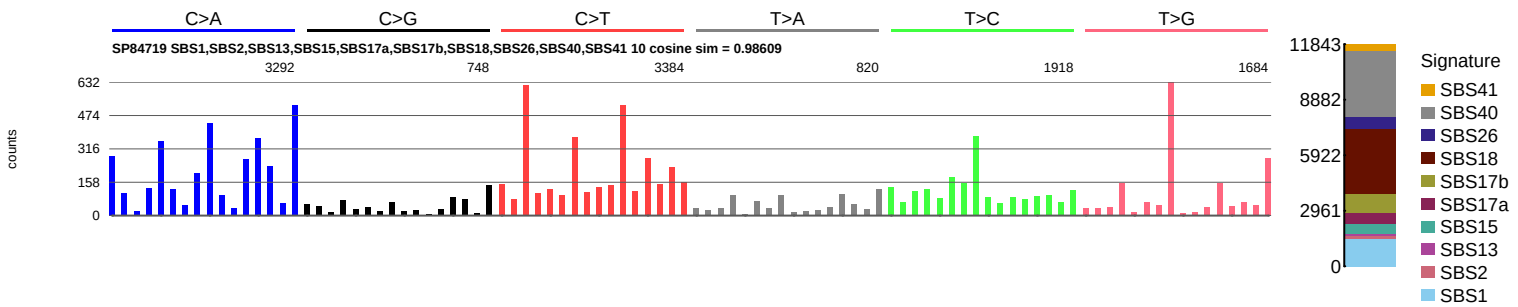

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98617
3261
798
3476
790
1894
1626
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

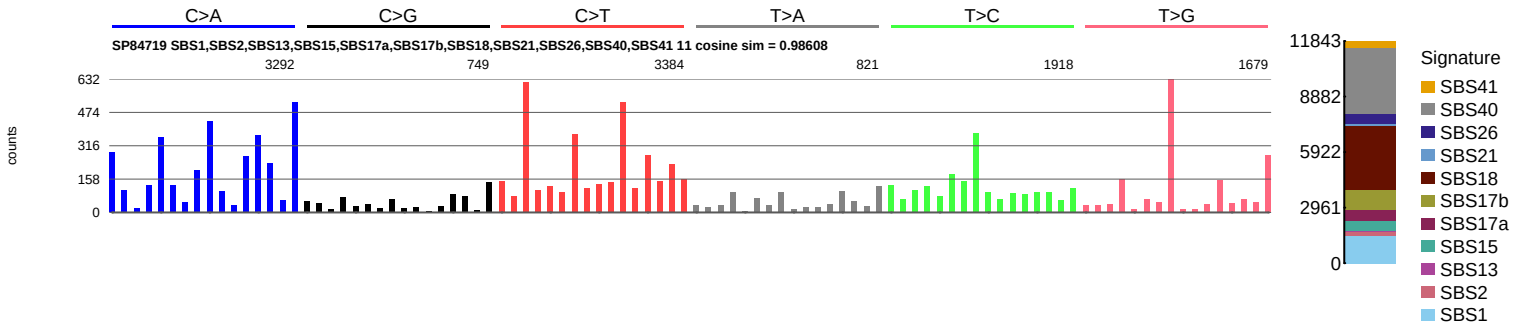

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98614
3256
657
3637
765
1987
1542
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

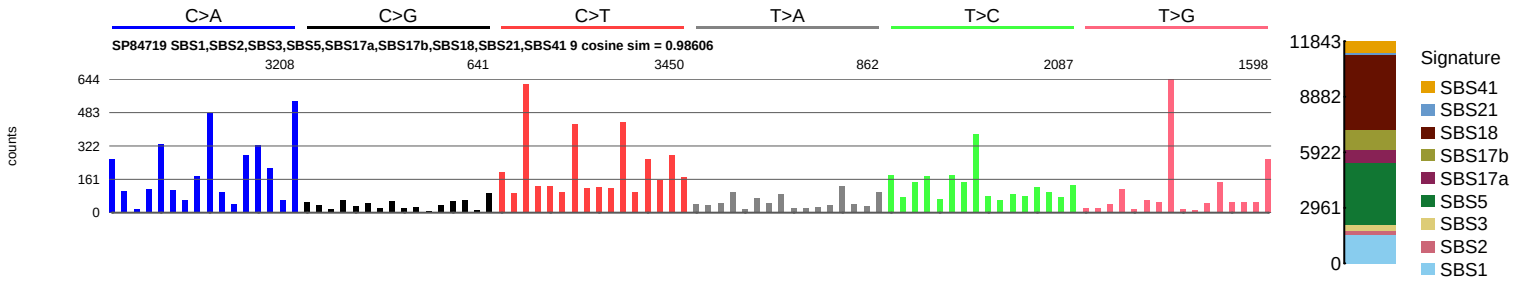

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98613
3273
700
3410
855
1961
1642
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

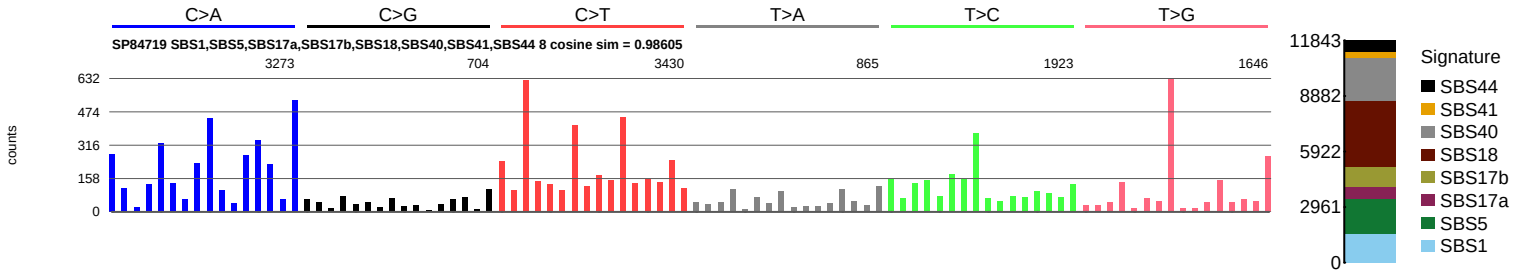

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98613
3295
777
3460
799
1903
1609
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

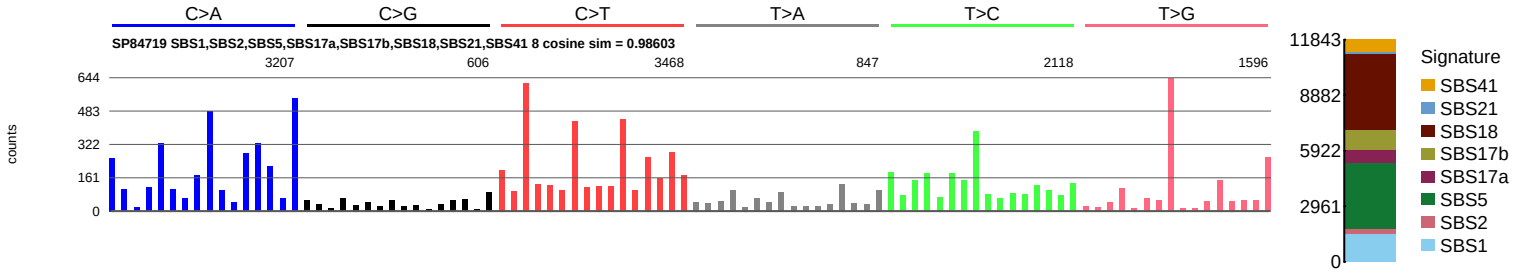

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.9861
3265
731
3455
798
1952
1643
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

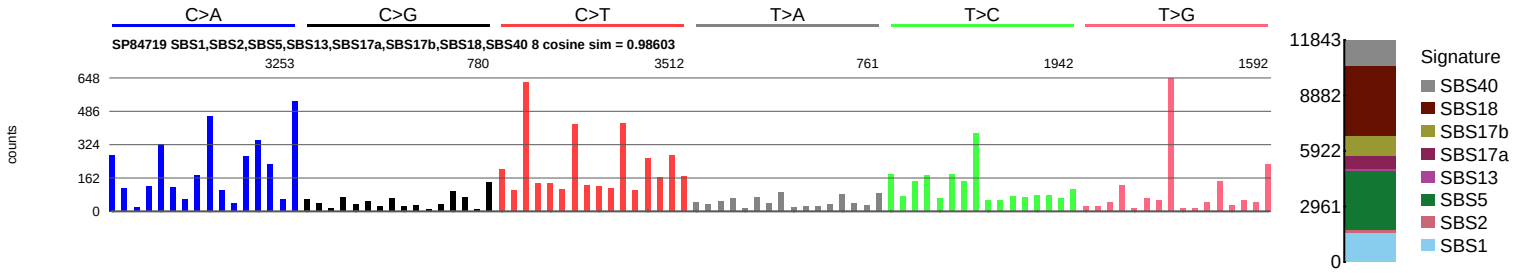

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98609
3292
748
3384
820
1918
1684
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

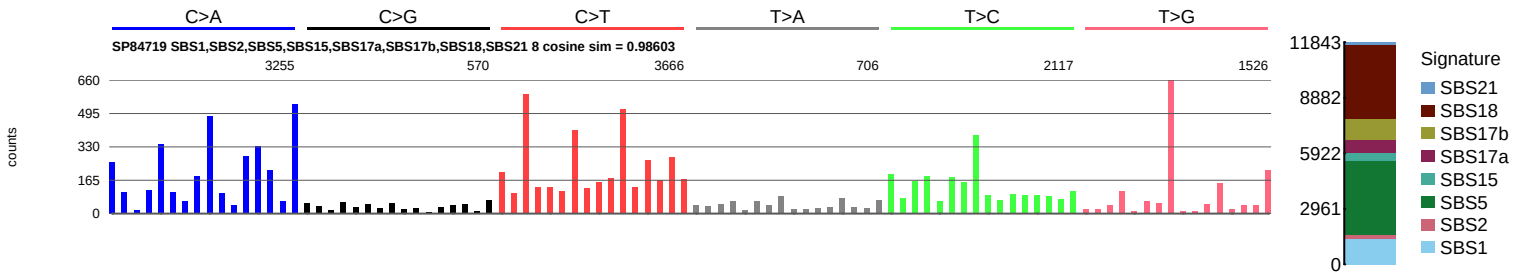

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98608
3292
749
3384
821
1918
1679
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

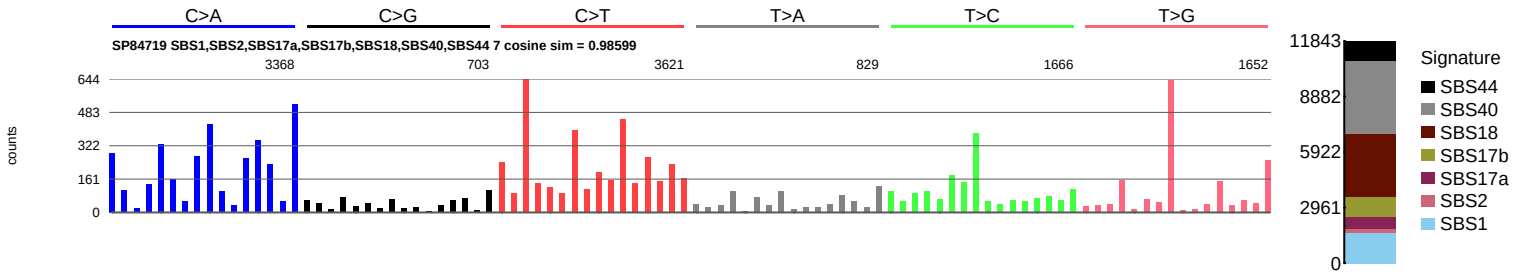

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98606
3208
641
3450
862
2087
1598
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98605
3273
704
3430
865
1923
1646
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98603
3207
606
3468
847
2118
1596
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98603
3253
780
3512
761
1942
1592
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98603
3255
570
3666
706
2117
1526
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98599
3368
703
3621
829
1666
1652
counts
644
483
322
161
0
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8882
5922
2961
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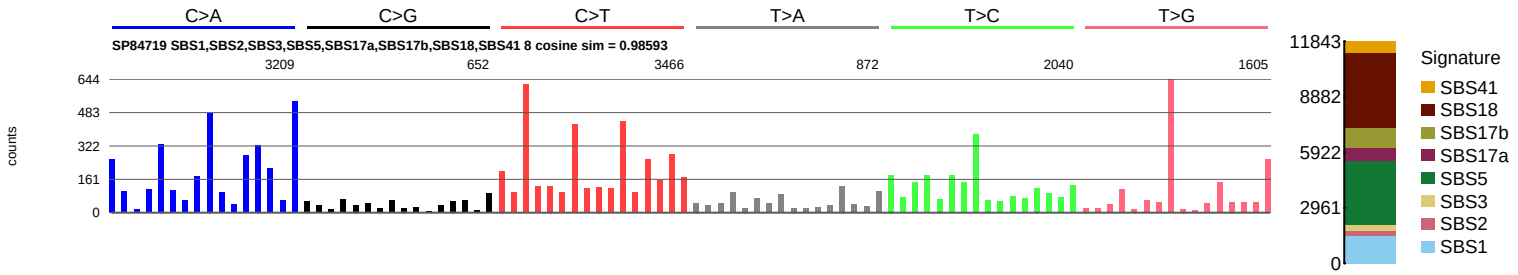
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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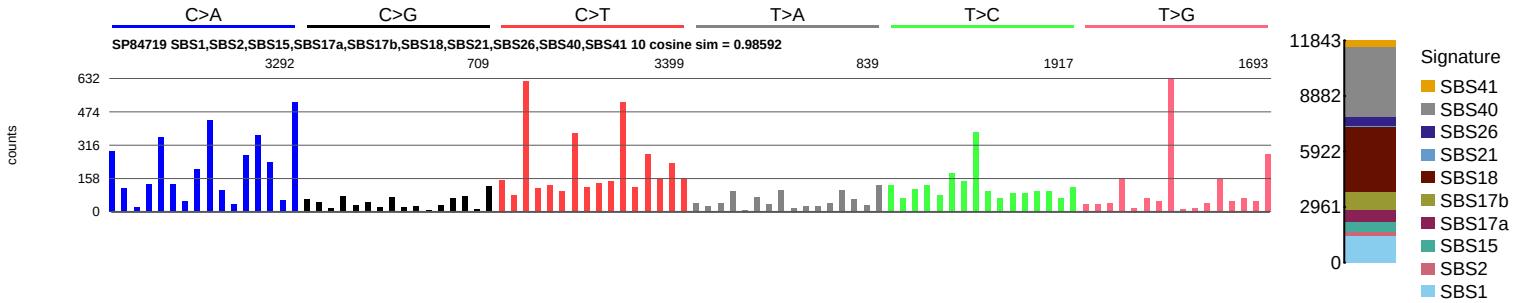
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98593
3209
652
3466
872
2040
1605
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>T
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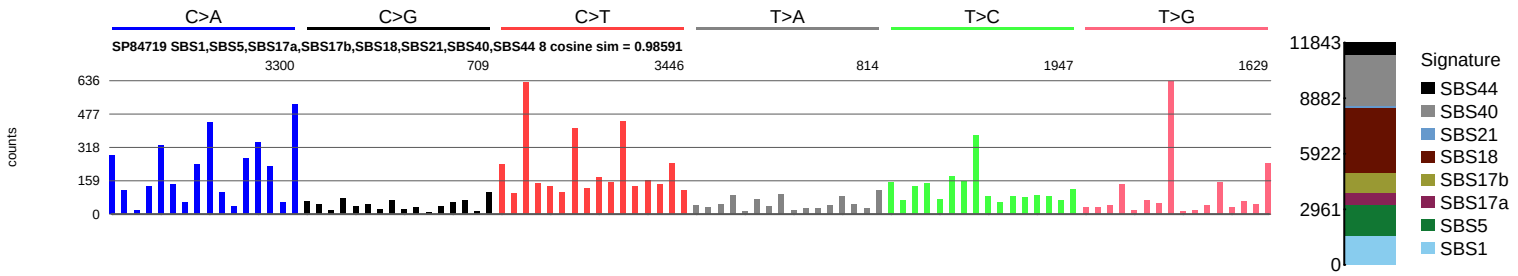
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98592
3292
709
3399
839
1917
1693
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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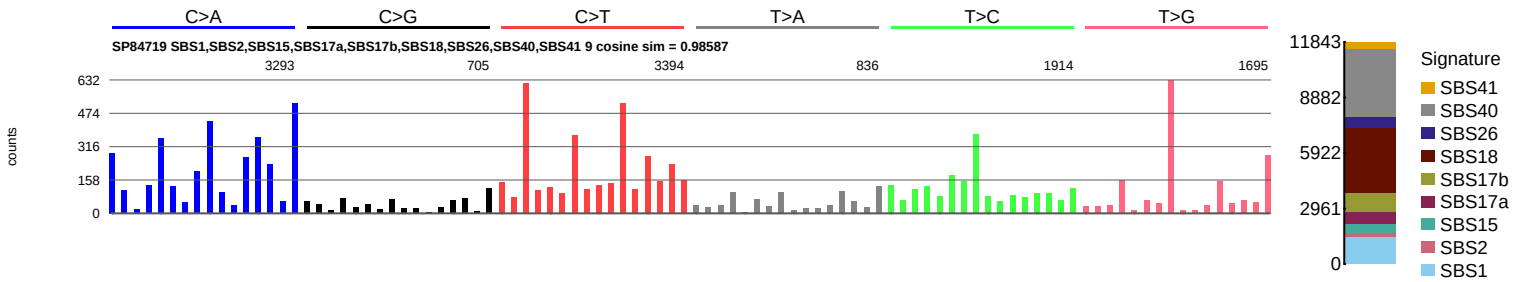
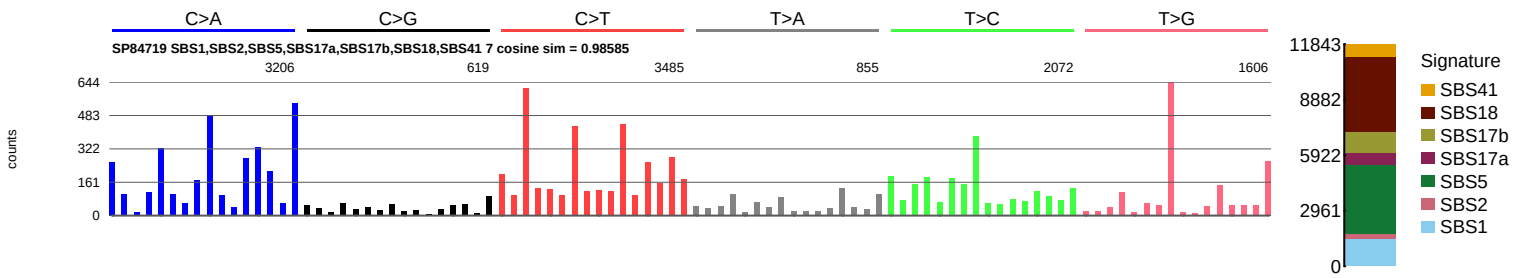
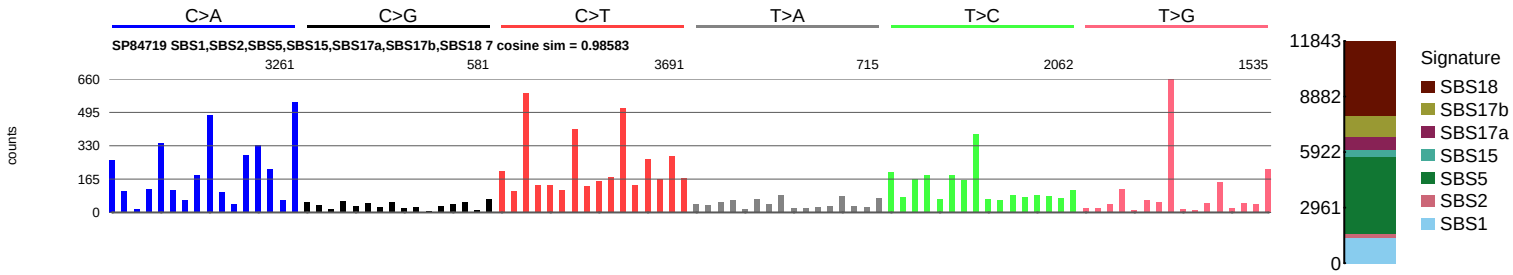
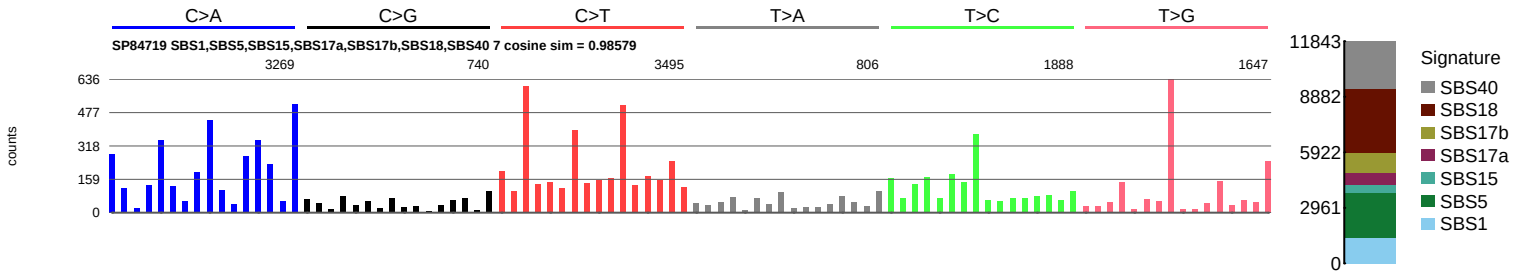
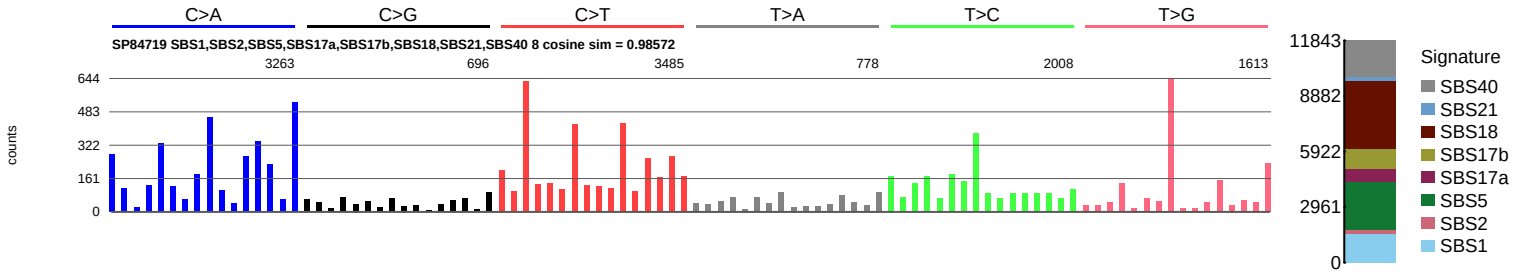
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98591
3300
709
3446
814
1947
1629
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

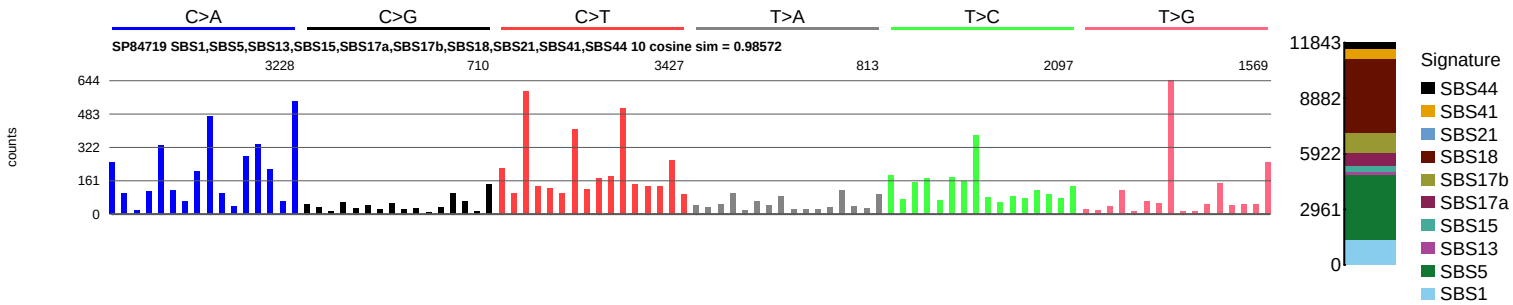

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98587
3293
705
3394
836
1914
1695
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

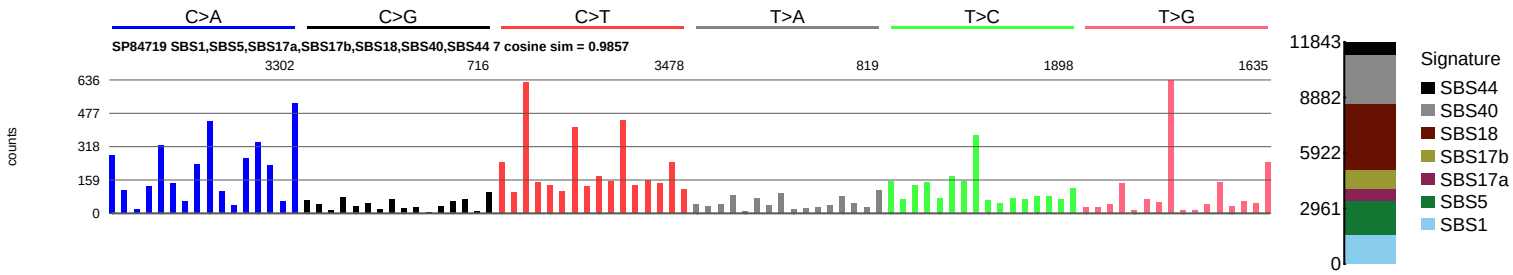

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98585
3206
619
3485
855
2072
1606
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

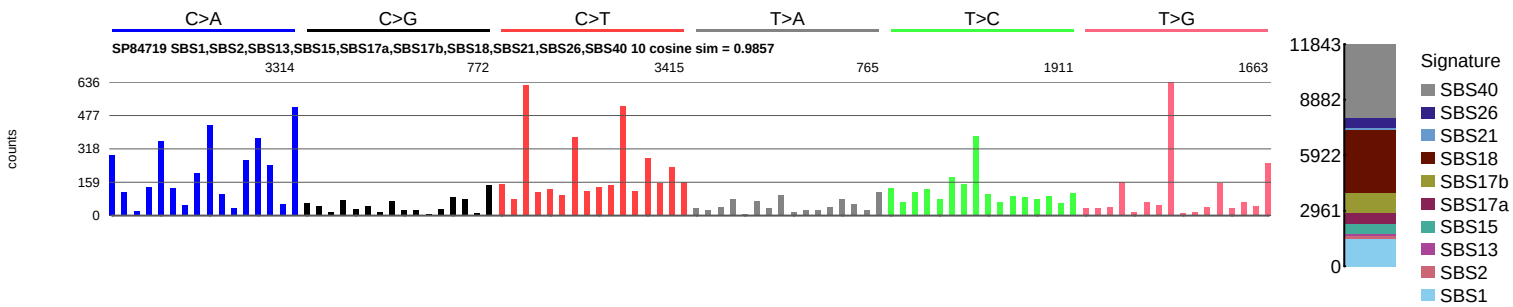

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98583
3261
581
3691
715
2062
1535
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

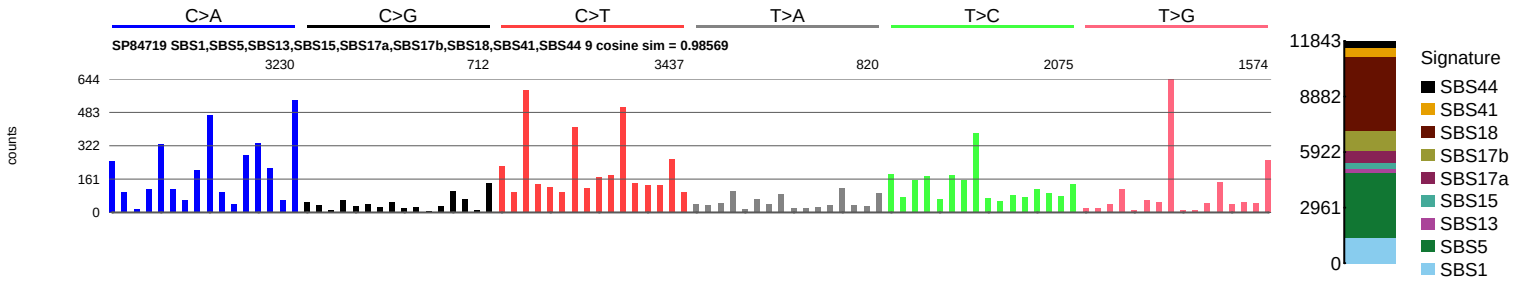

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98579
3269
740
3495
806
1888
1647
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

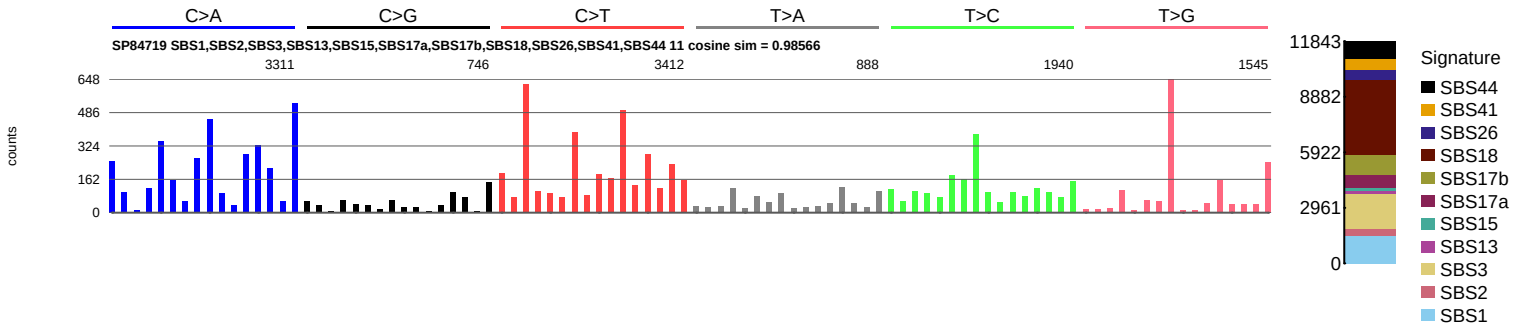

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98572
3263
696
3485
778
2008
1613
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

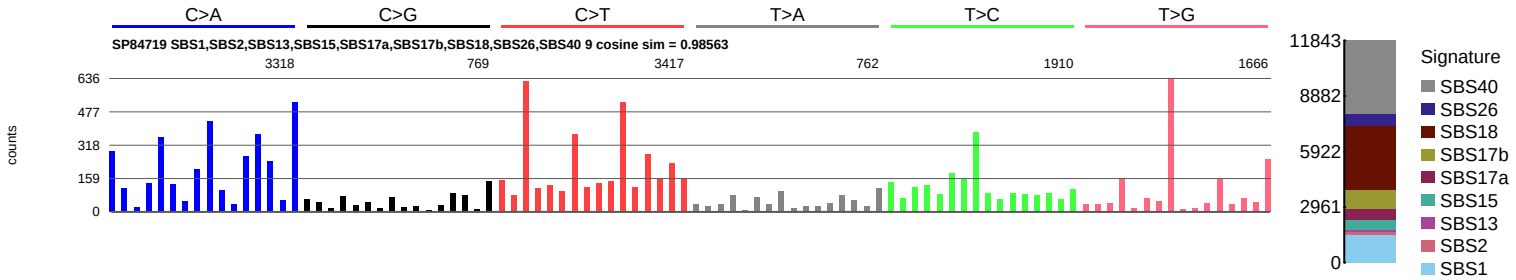

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98572
3228
710
3427
813
2097
1569
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

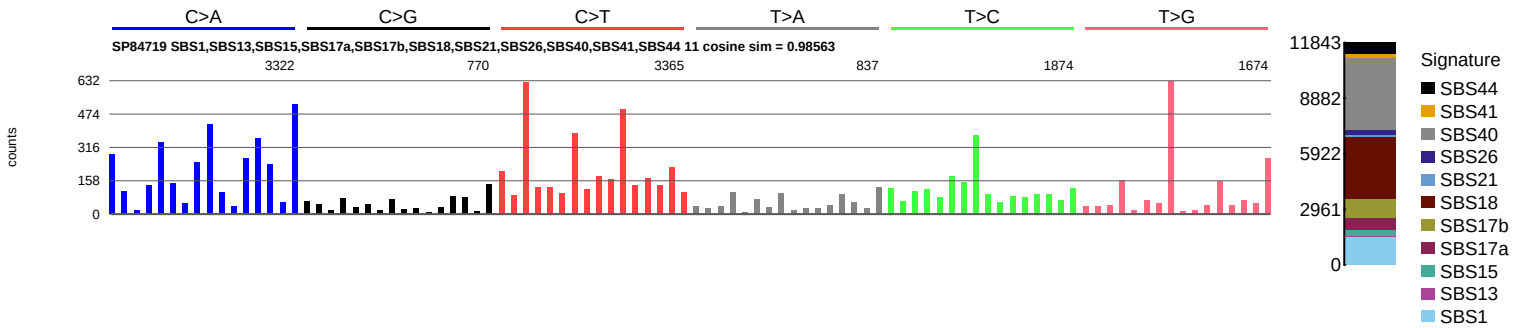

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9857
3302
716
3478
819
1898
1635
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

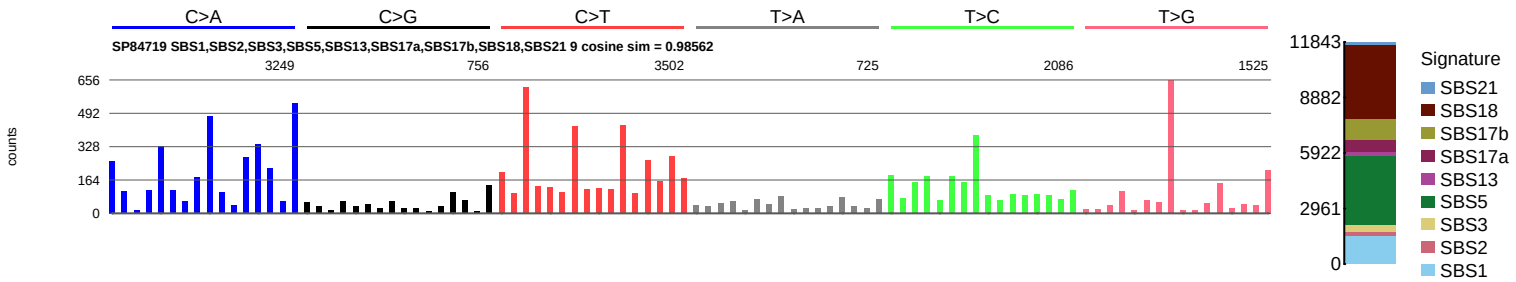

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.9857
3314
772
3415
765
1911
1663
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

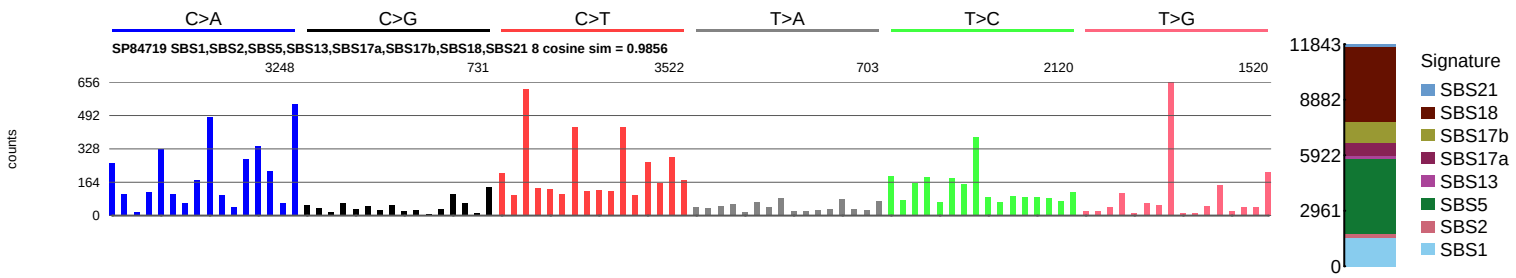

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98569
3230
712
3437
820
2075
1574
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

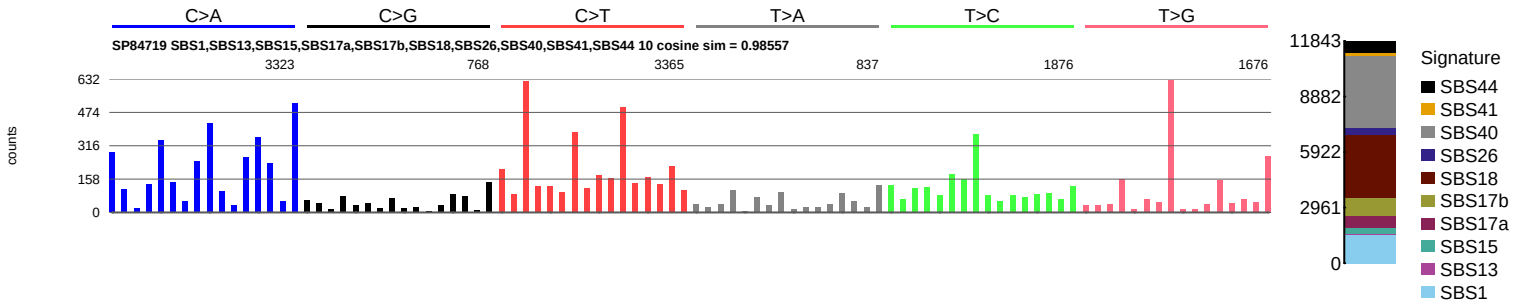

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98566
3311
746
3412
888
1940
1545
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

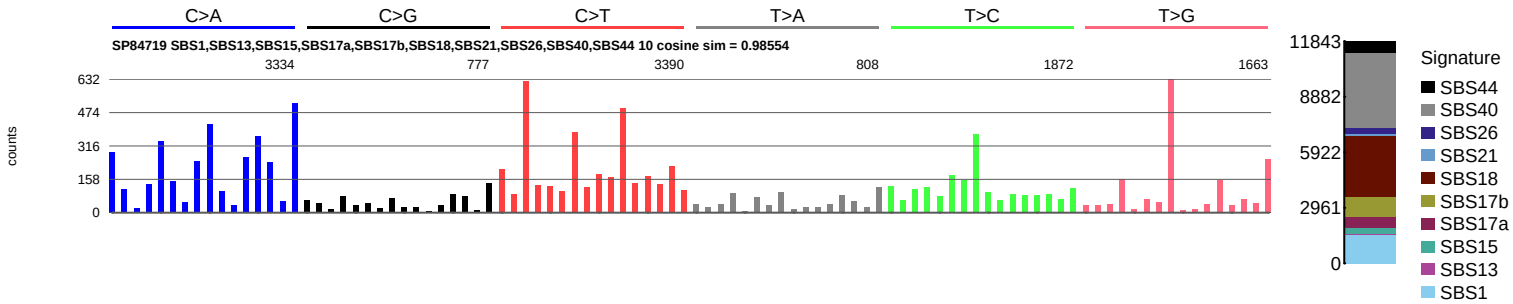

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98563
3318
769
3417
762
1910
1666
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

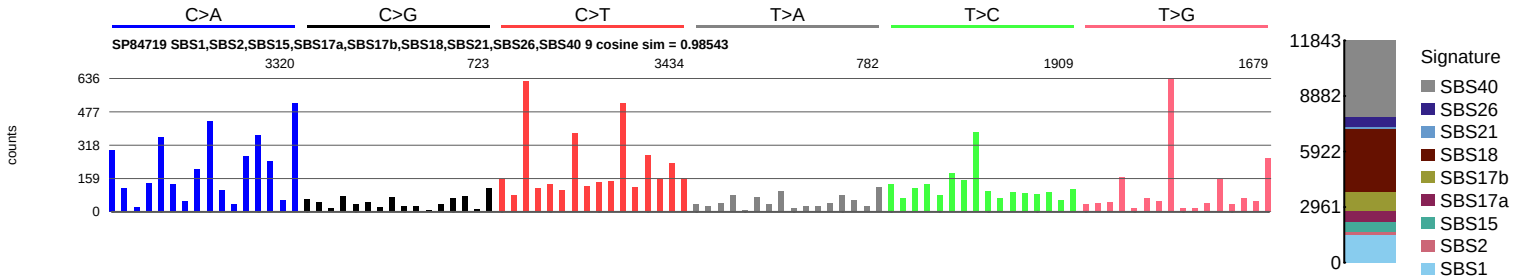

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98563
3322
770
3365
837
1874
1674
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

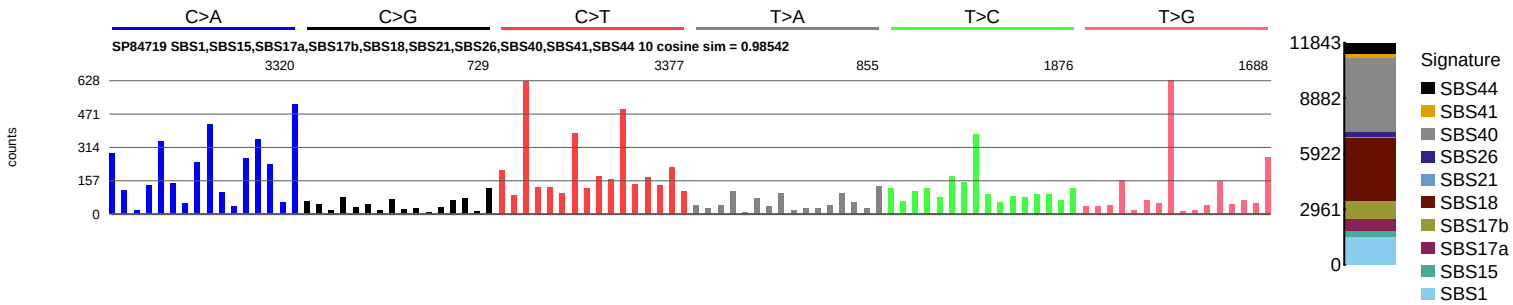

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98562
3249
756
3502
725
2086
1525
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

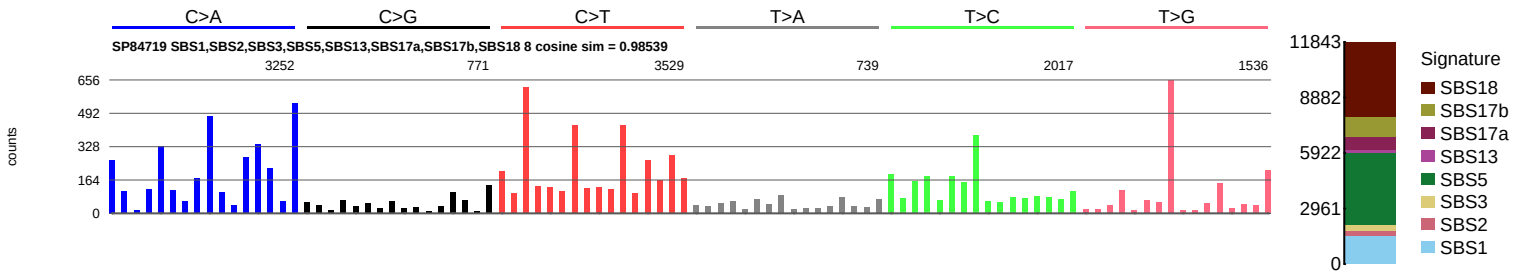

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.9856
3248
731
3522
703
2120
1520
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98557
3323
768
3365
837
1876
1676
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98554
3334
777
3390
808
1872
1663
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98543
3320
723
3434
782
1909
1679
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98542
3320
729
3377
855
1876
1688
counts
628
471
314
157
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98539
3252
771
3529
739
2017
1536
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
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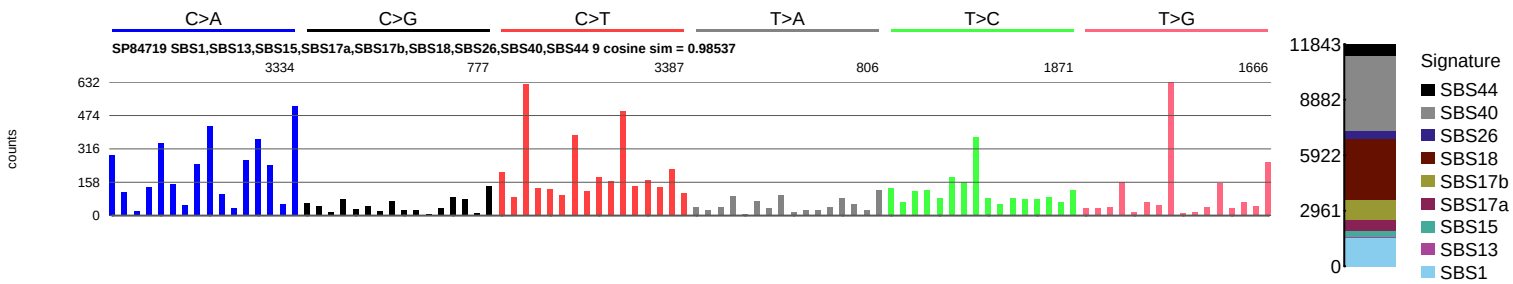
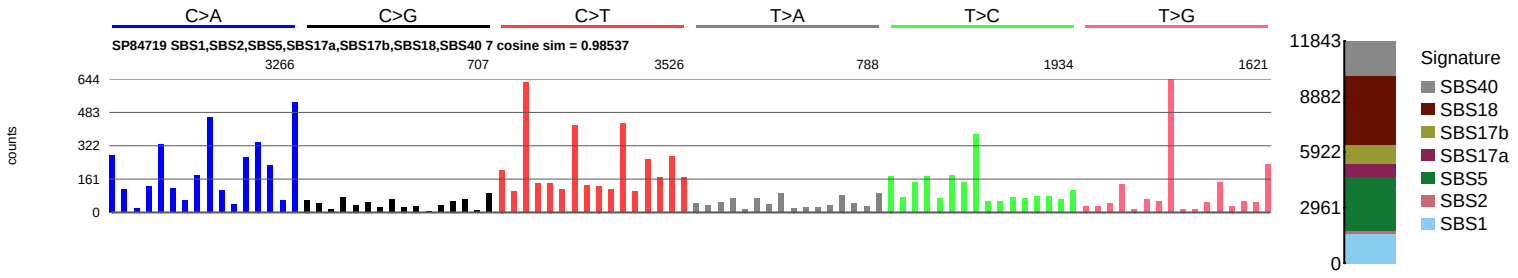
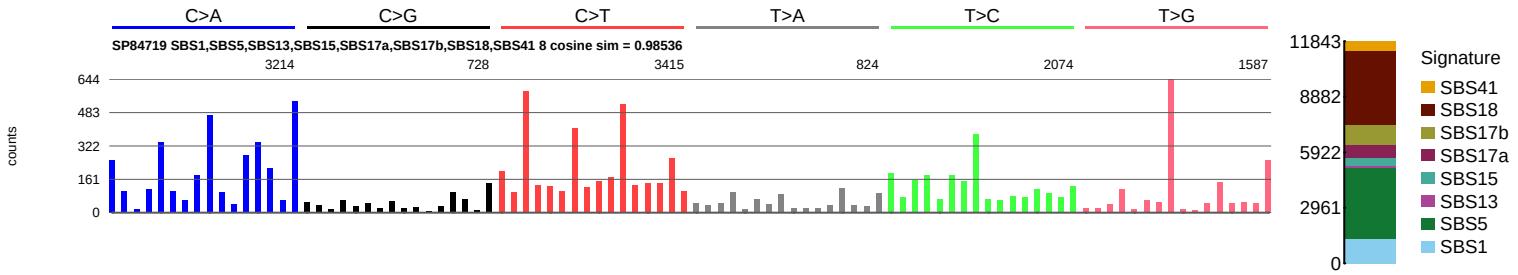
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98537
3334
777
3387
806
1871
1666
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98537
3266
707
3526
788
1934
1621
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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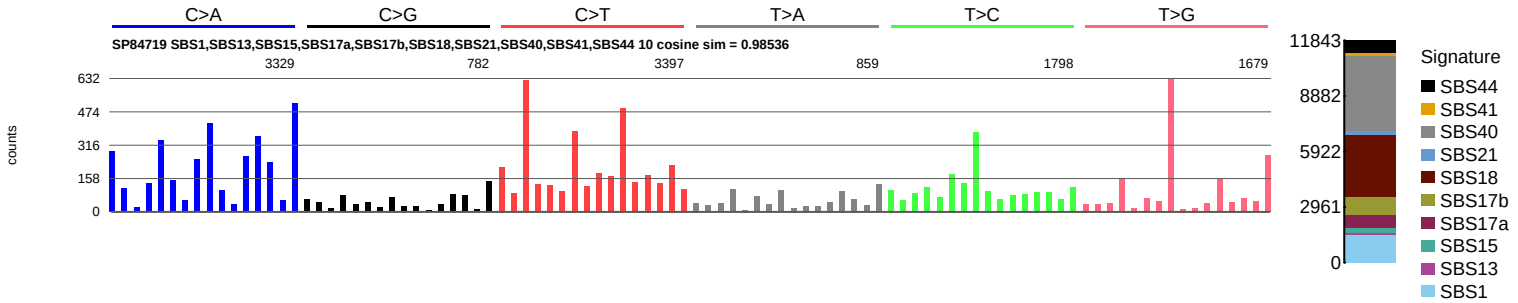
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98536
3214
728
3415
824
2074
1587
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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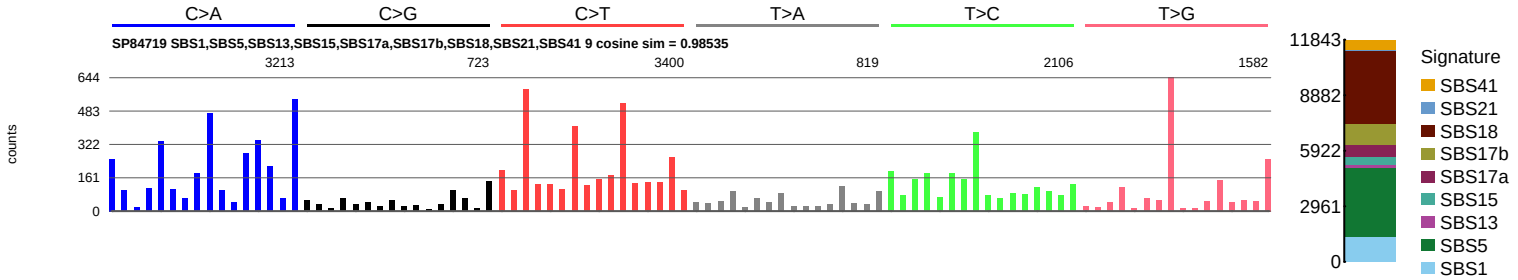
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98536
3329
782
3397
859
1798
1679
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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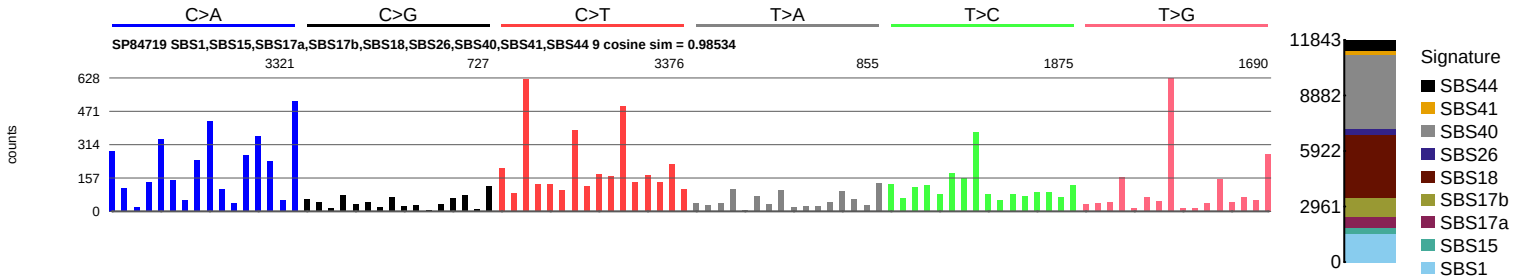
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98535
3213
723
3400
819
2106
1582
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
C>T
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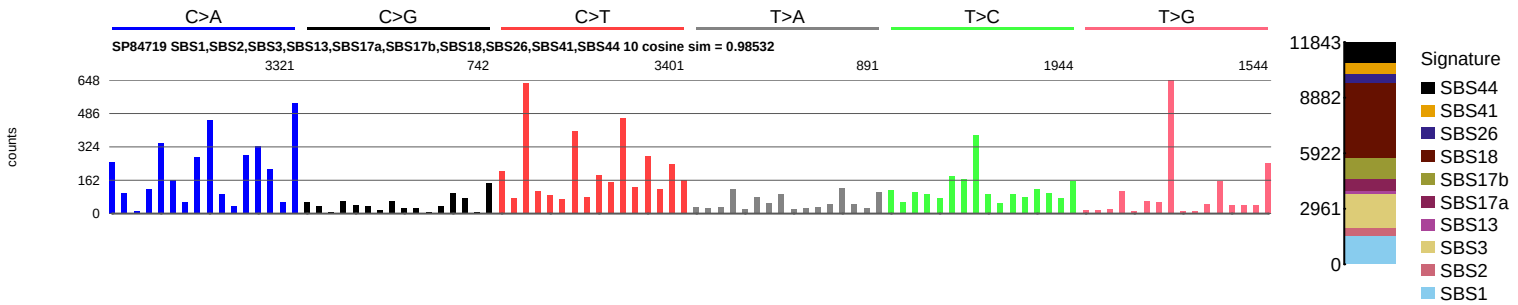
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98534
3321
727
3376
855
1875
1690
counts
628
471
314
157
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
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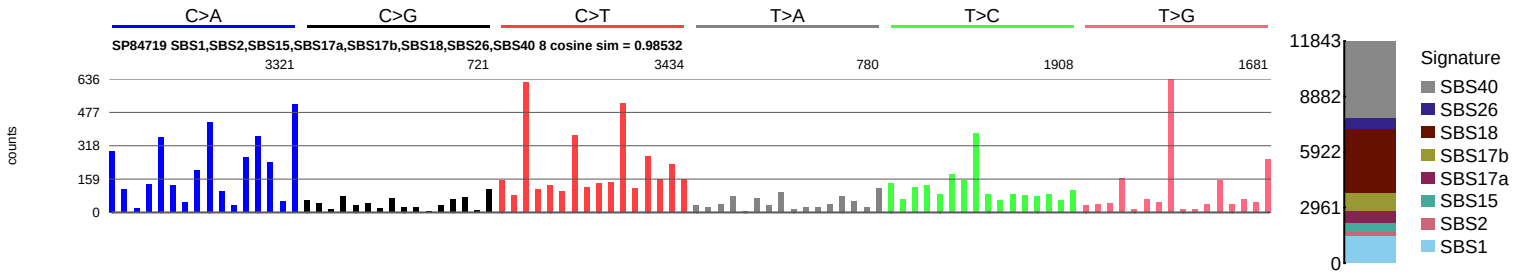
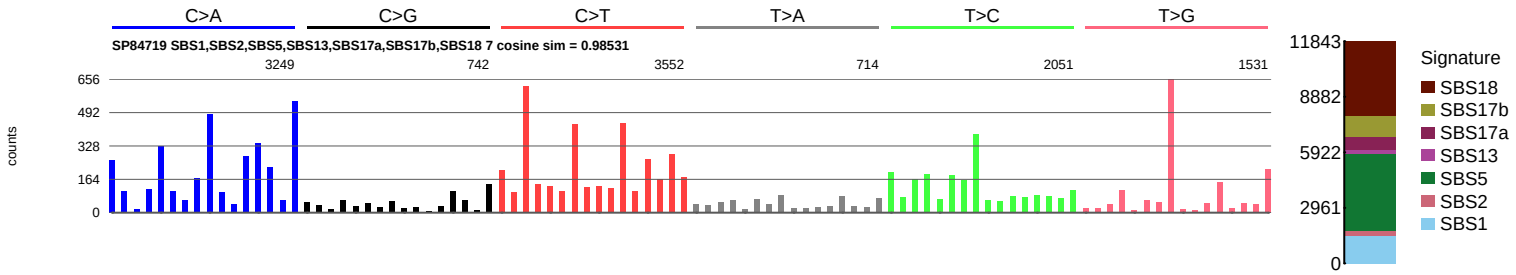
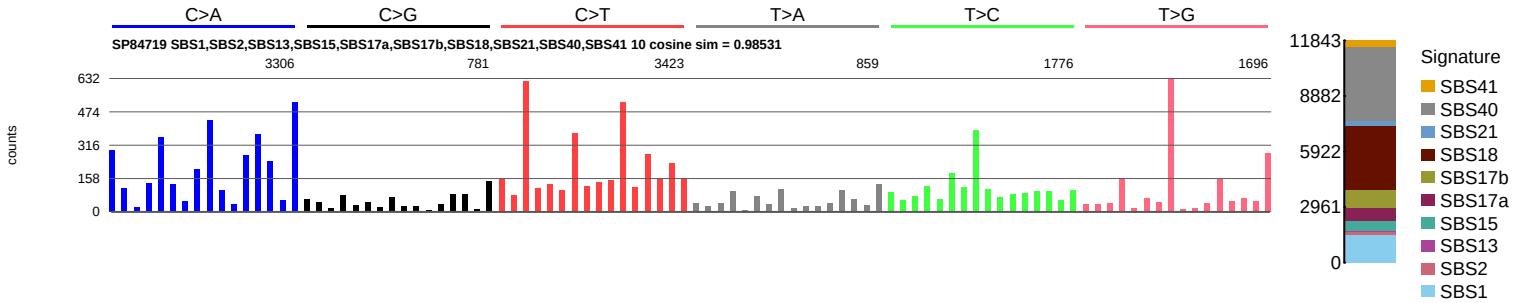
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98532
3321
742
3401
891
1944
1544
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98532
3321
721
3434
780
1908
1681
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98531
3249
742
3552
714
2051
1531
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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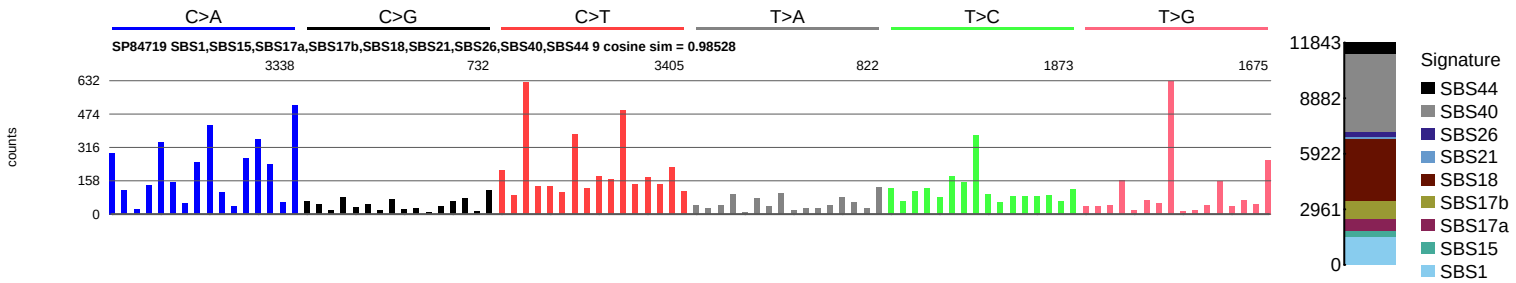
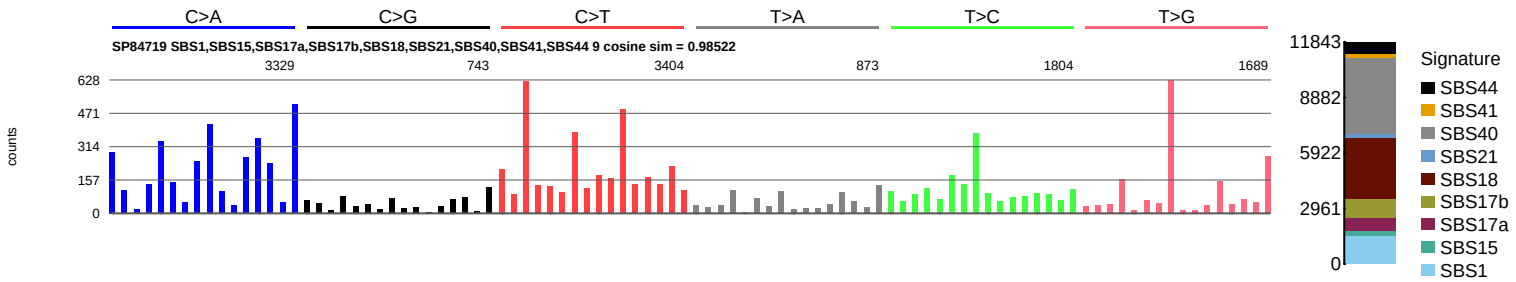
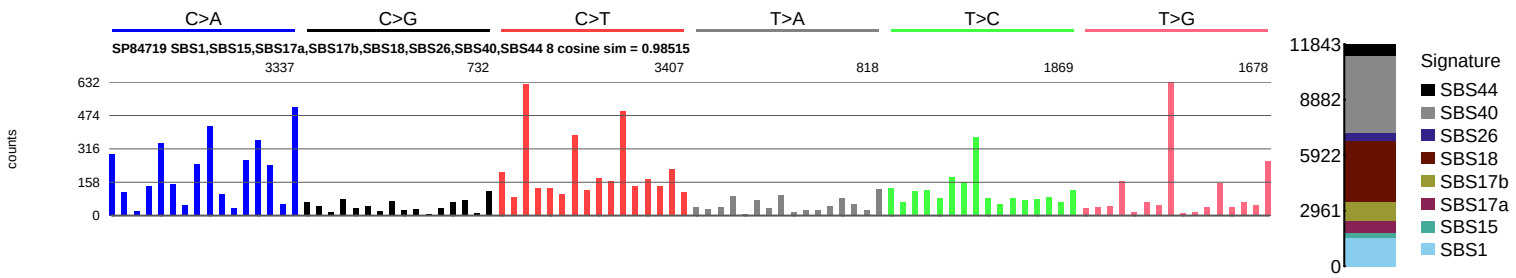
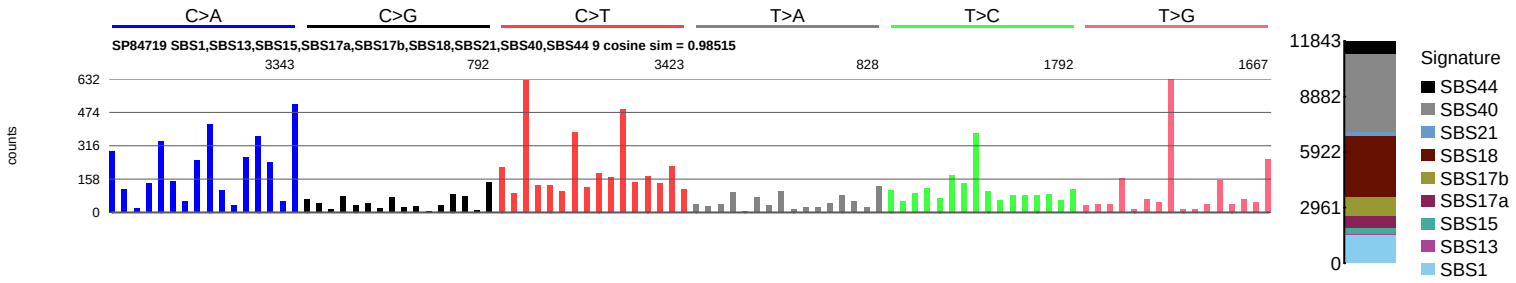
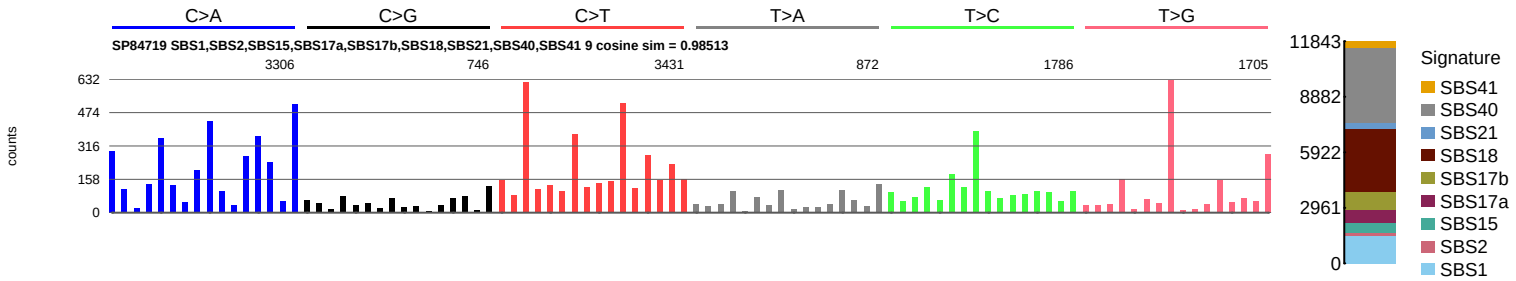
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98531
3306
781
3423
859
1776
1696
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

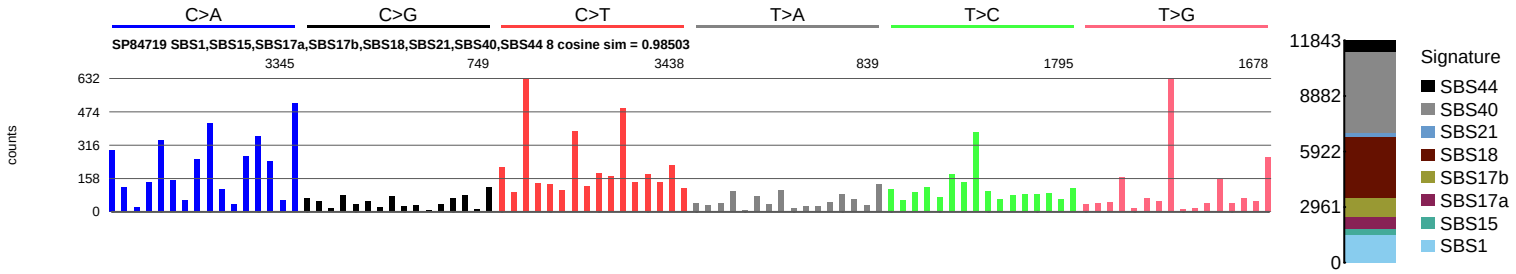

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98528
3338
732
3405
822
1873
1675
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

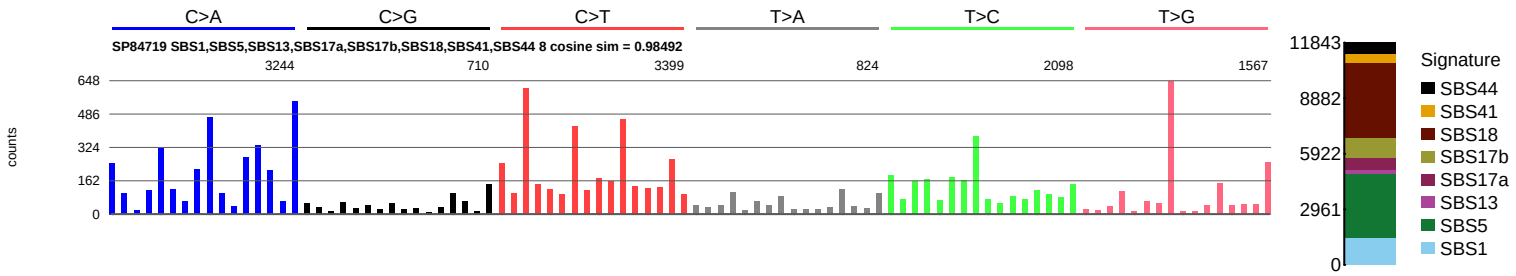

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98522
3329
743
3404
873
1804
1689
628
471
314
157
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98515
3337
732
3407
818
1869
1678
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98515
3343
792
3423
828
1792
1667
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98513
3306
746
3431
872
1786
1705
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98503
3345
749
3438
839
1795
1678
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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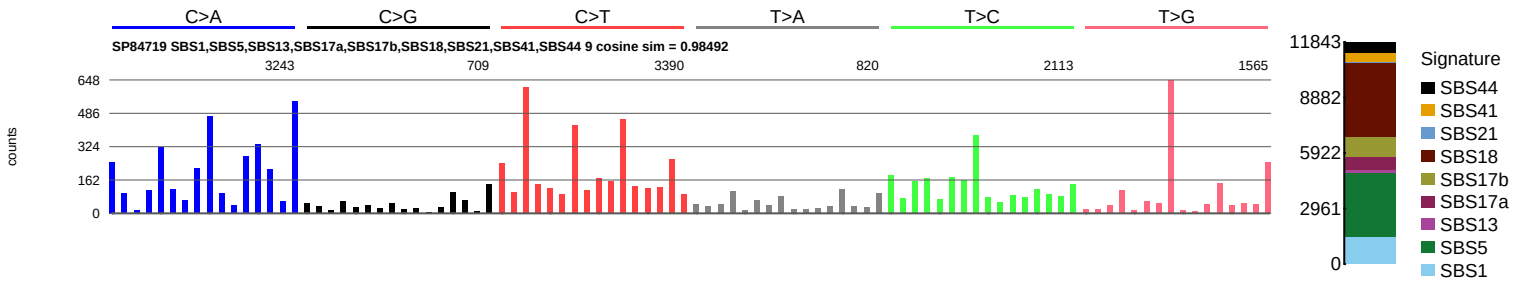
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98492
3244
710
3399
824
2098
1567
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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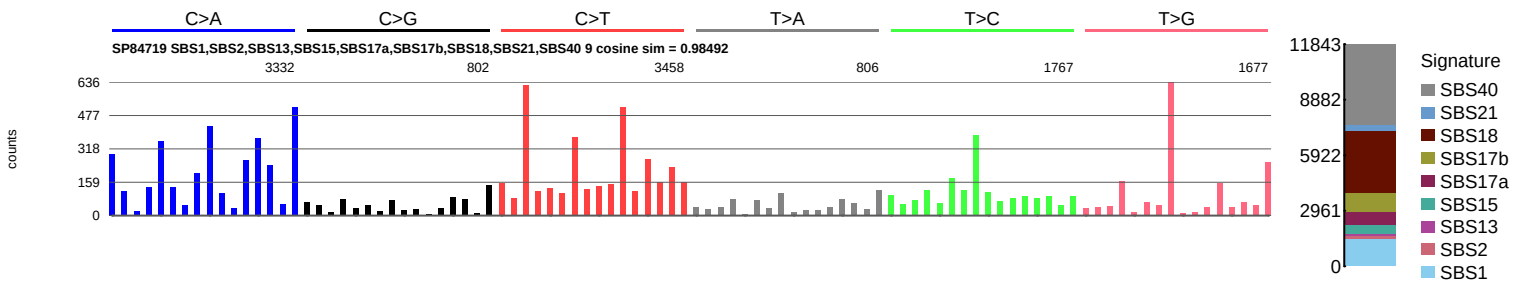
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98492
3243
709
3390
820
2113
1565
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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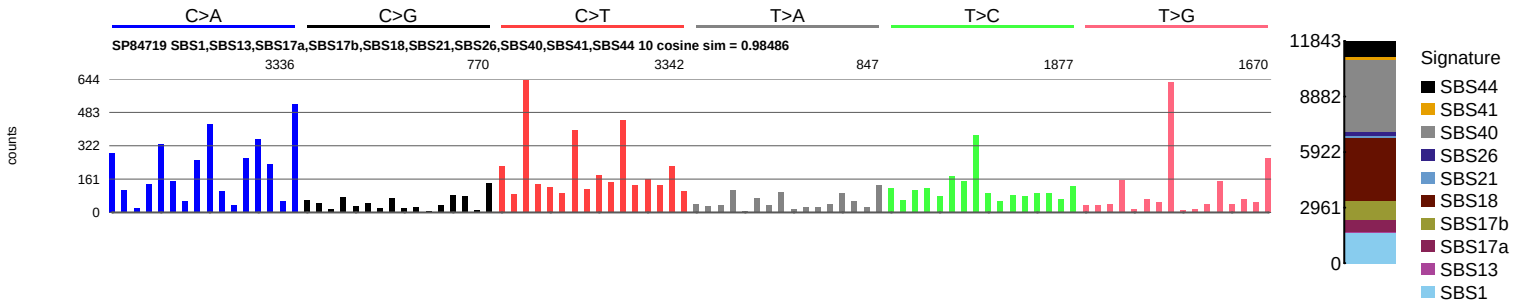
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98492
3332
802
3458
806
1767
1677
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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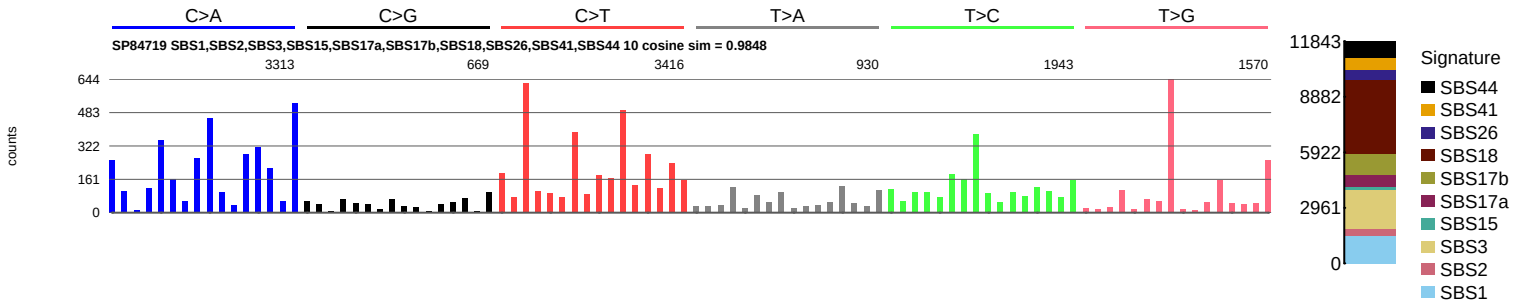
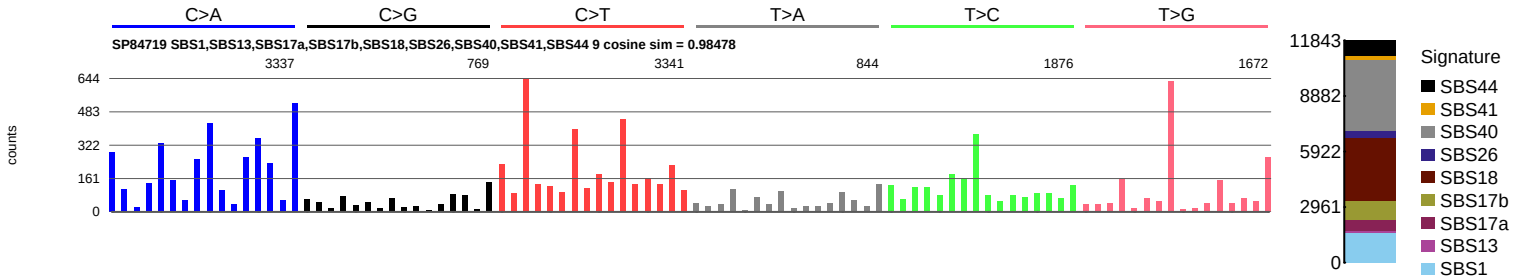
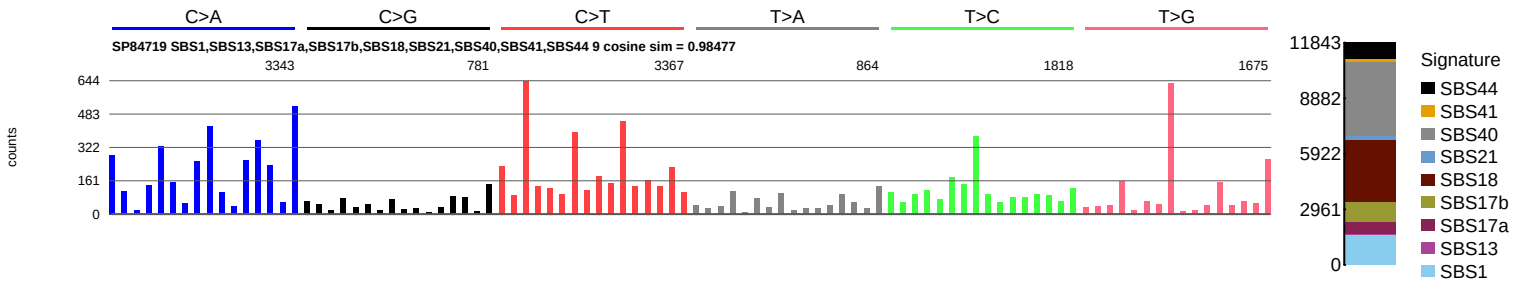
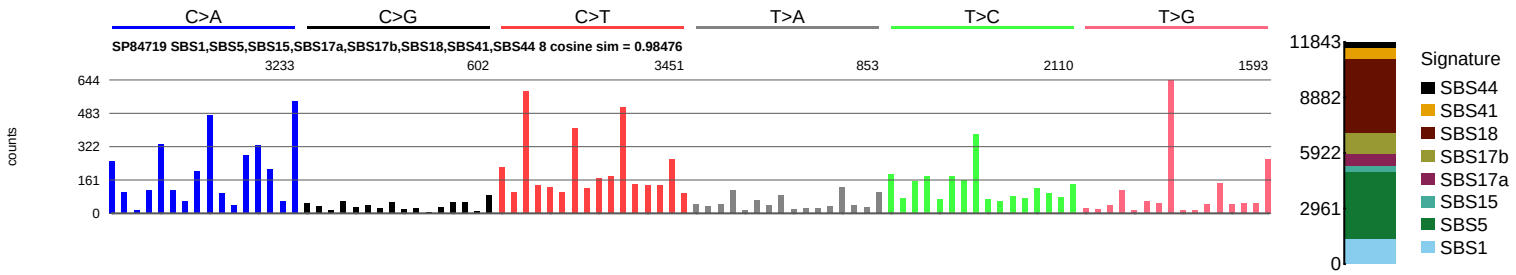
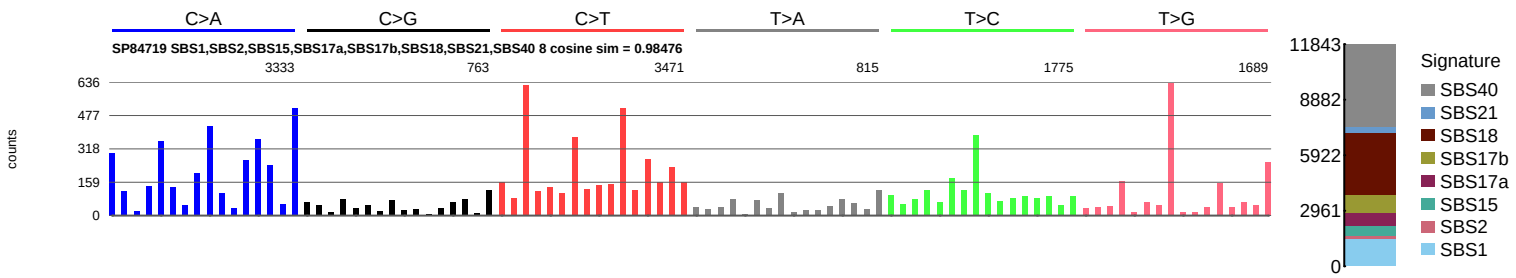
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98486
3336
770
3342
847
1877
1670
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

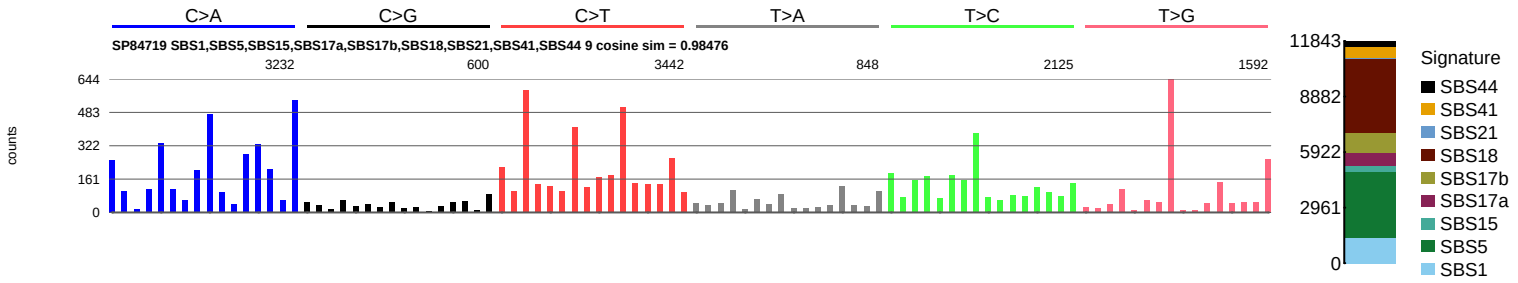

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.9848
3313
669
3416
930
1943
1570
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

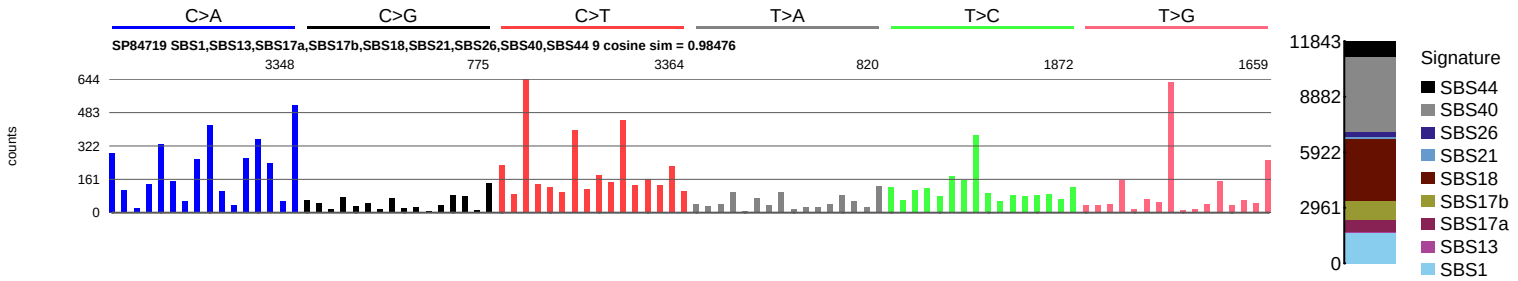

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98478
3337
769
3341
844
1876
1672
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98477
3343
781
3367
864
1818
1675
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98476
3233
602
3451
853
2110
1593
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98476
3333
763
3471
815
1775
1689
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98476
3232
600
3442
848
2125
1592
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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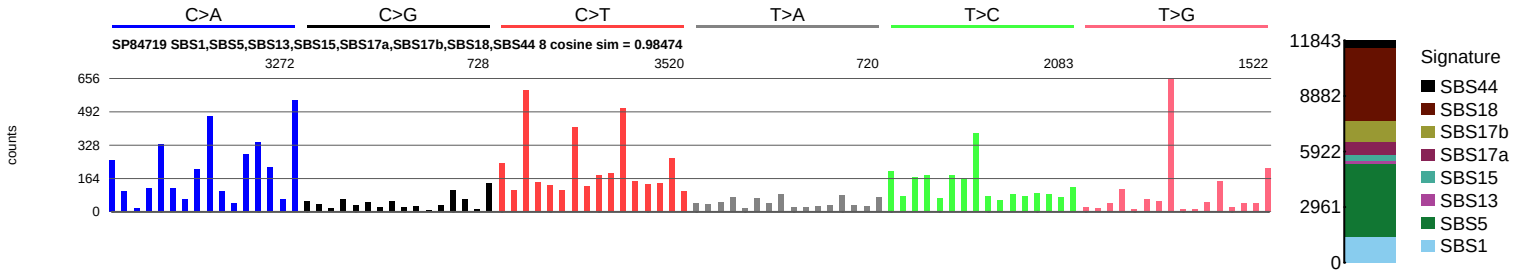
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98476
3348
775
3364
820
1872
1659
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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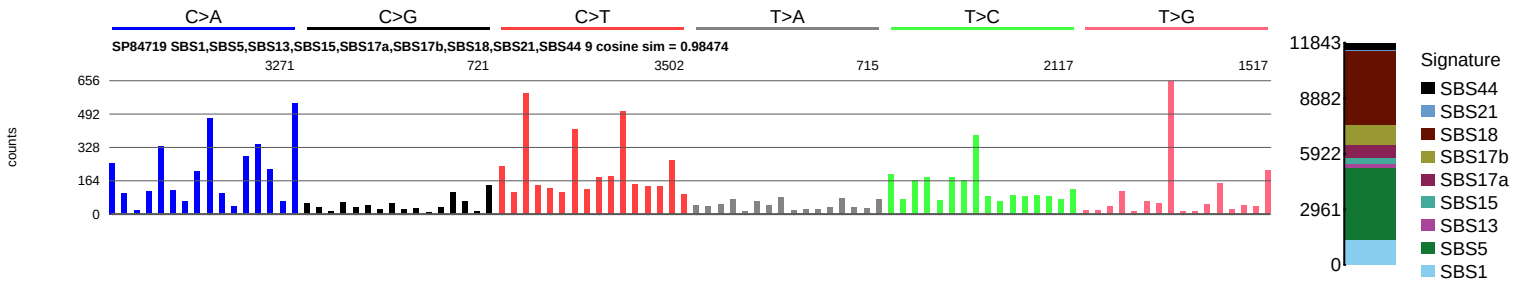
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98474
3272
728
3520
720
2083
1522
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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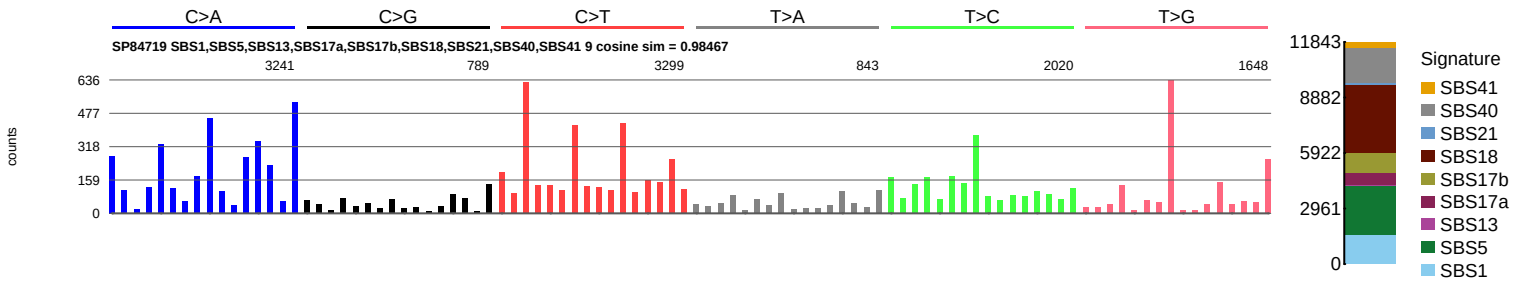
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98474
3271
721
3502
715
2117
1517
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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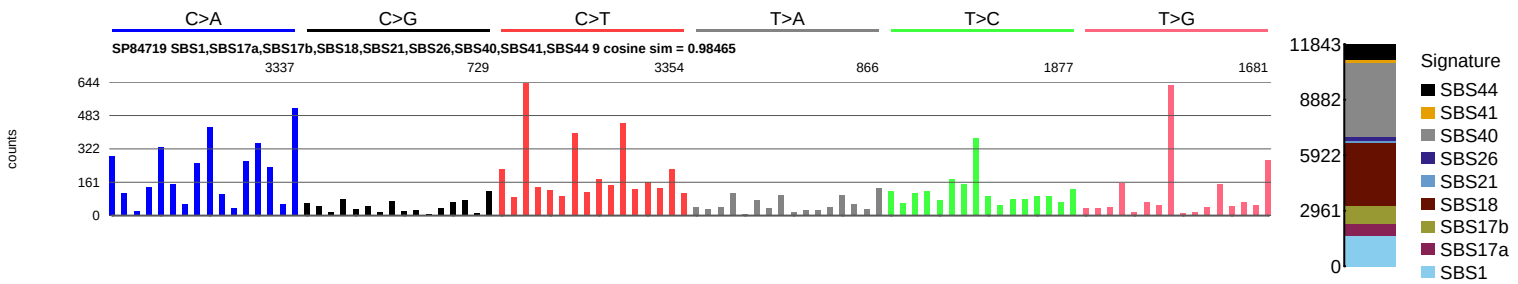
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98467
3241
789
3299
843
2020
1648
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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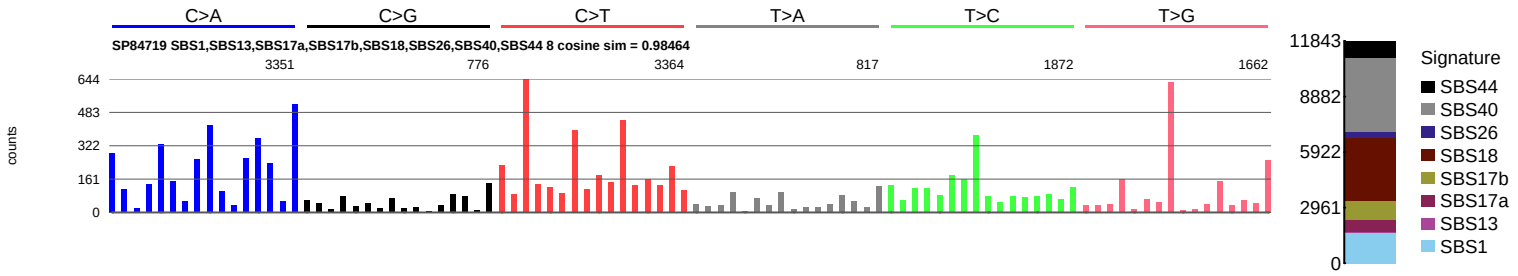
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98465
3337
729
3354
866
1877
1681
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1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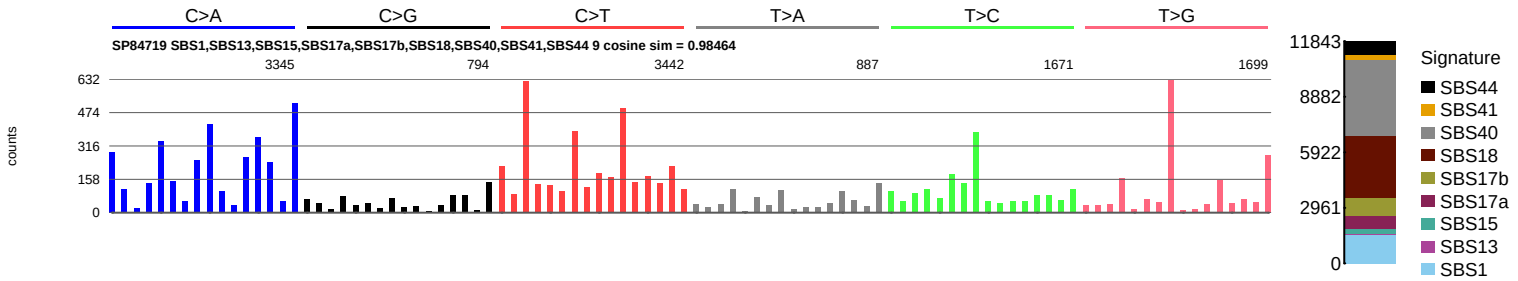
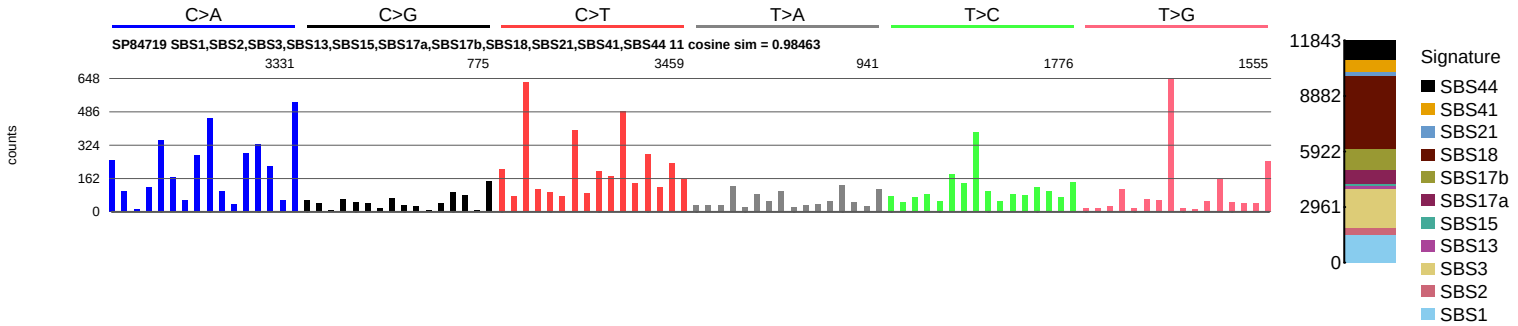
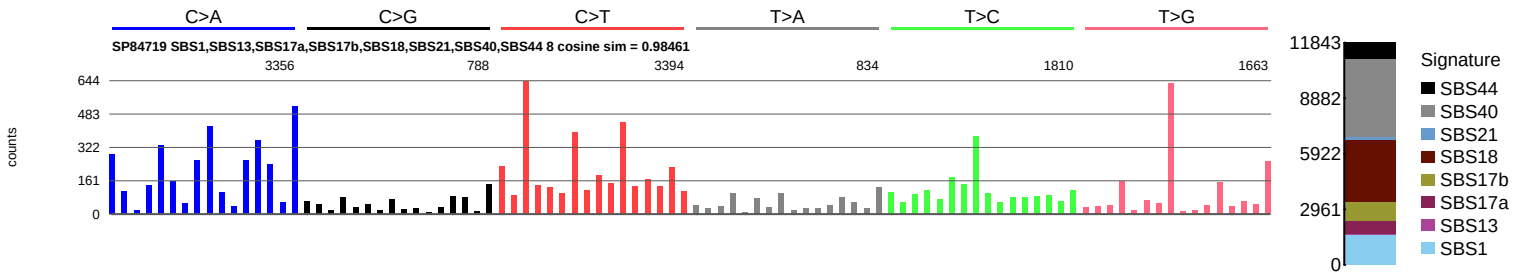
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98464
3351
776
3364
817
1872
1662
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98464
3345
794
3442
887
1671
1699
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98463
3331
775
3459
941
1776
1555
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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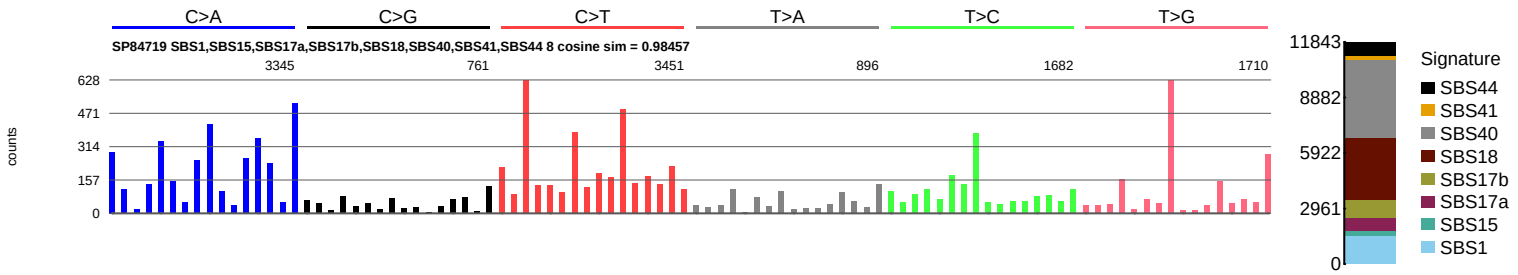
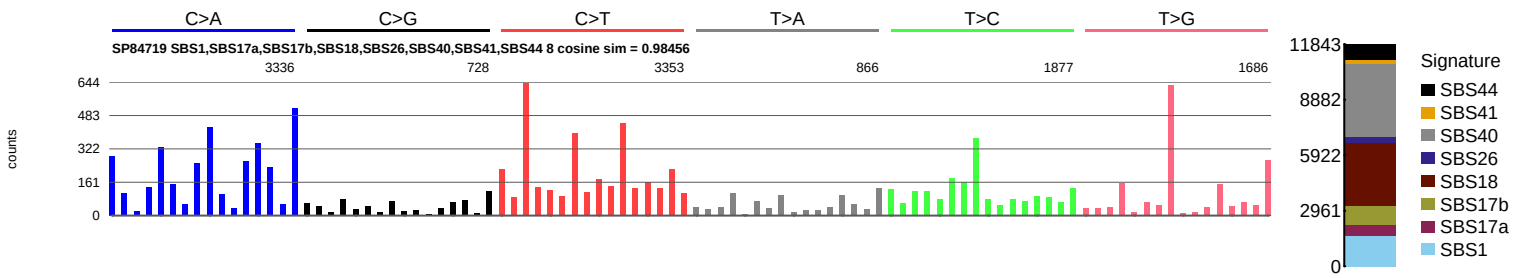
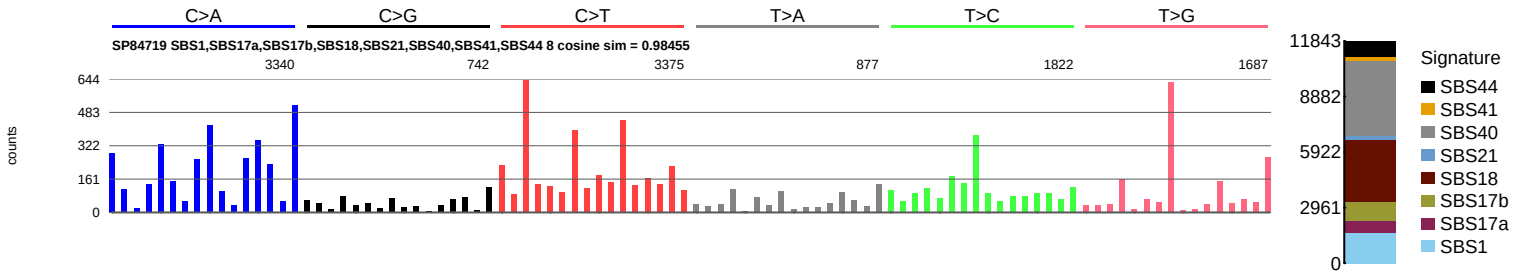
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98461
3356
788
3394
834
1810
1663
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98457
3345
761
3451
896
1682
1710
counts
628
471
314
157
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98456
3336
728
3353
866
1877
1686
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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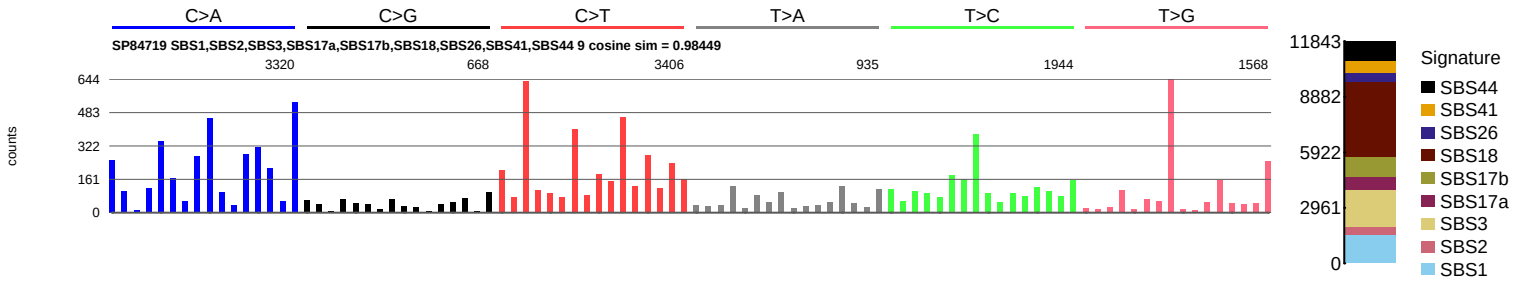
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98455
3340
742
3375
877
1822
1687
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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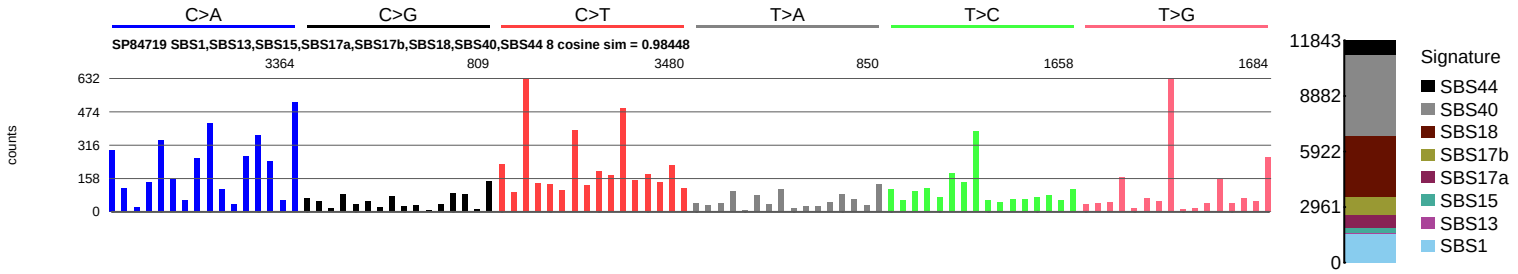
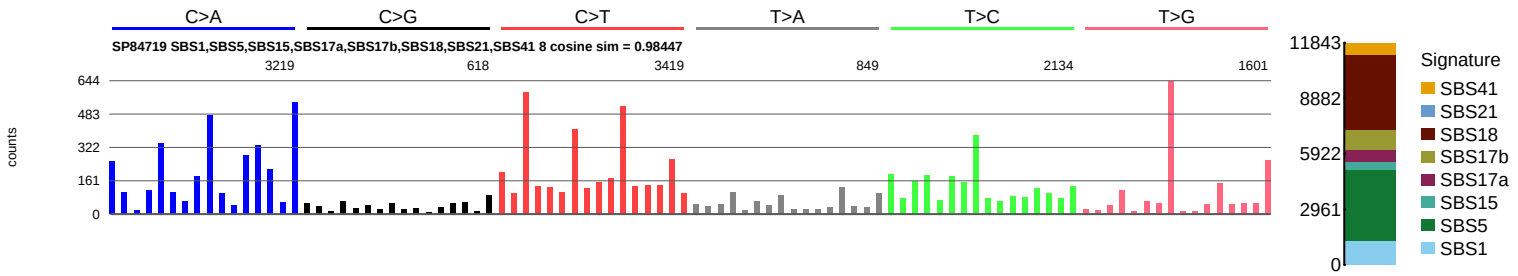
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98449
3320
668
3406
935
1944
1568
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98448
3364
809
3480
850
1658
1684
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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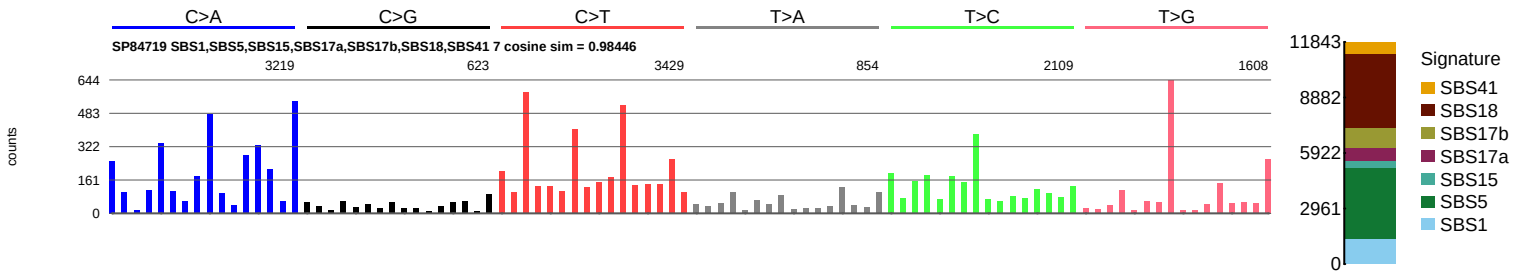
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98447
3219
618
3419
849
2134
1601
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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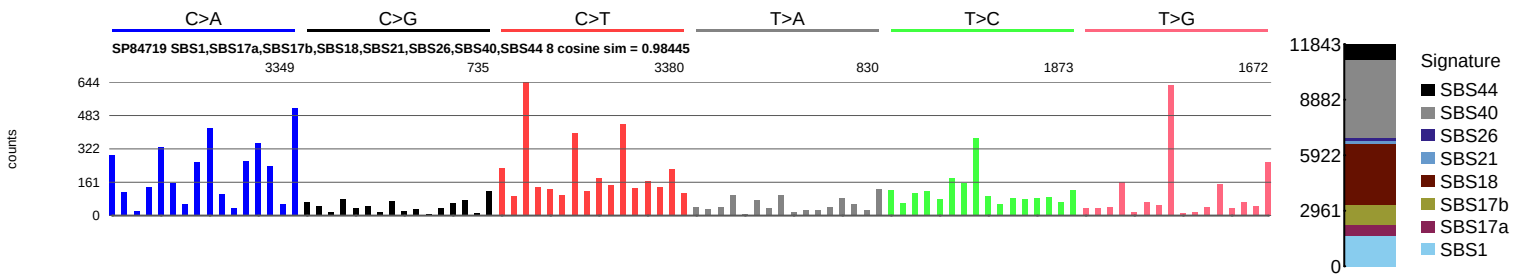
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98446
3219
623
3429
854
2109
1608
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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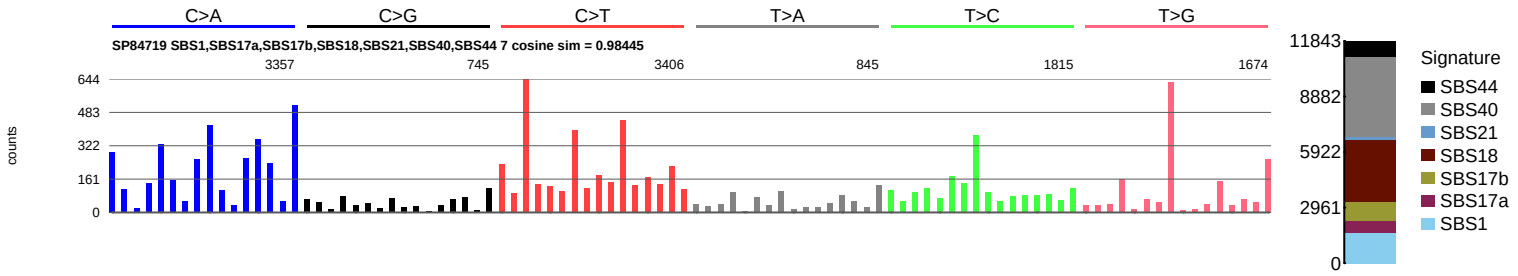
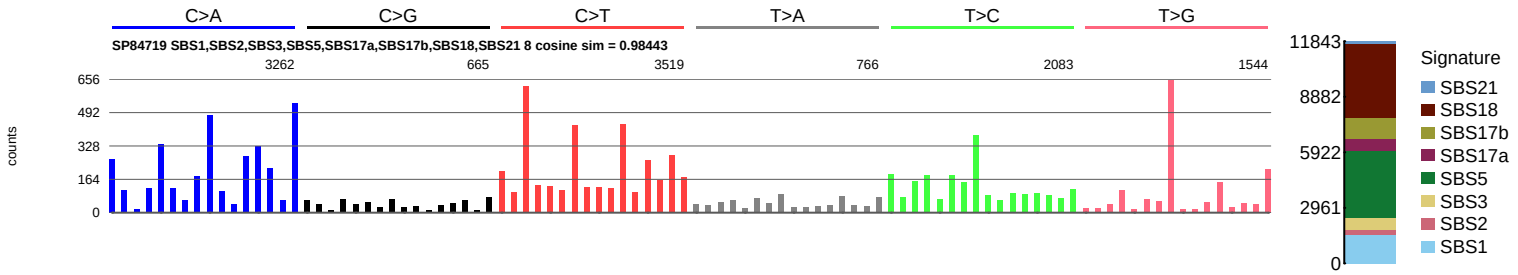
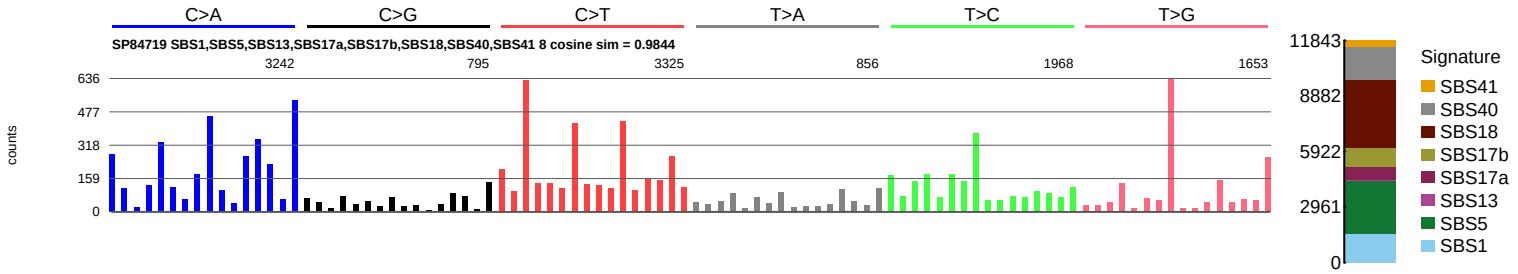
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98445
3349
735
3380
830
1873
1672
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98445
3357
745
3406
845
1815
1674
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98443
3262
665
3519
766
2083
1544
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9844
3242
795
3325
856
1968
1653
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS41
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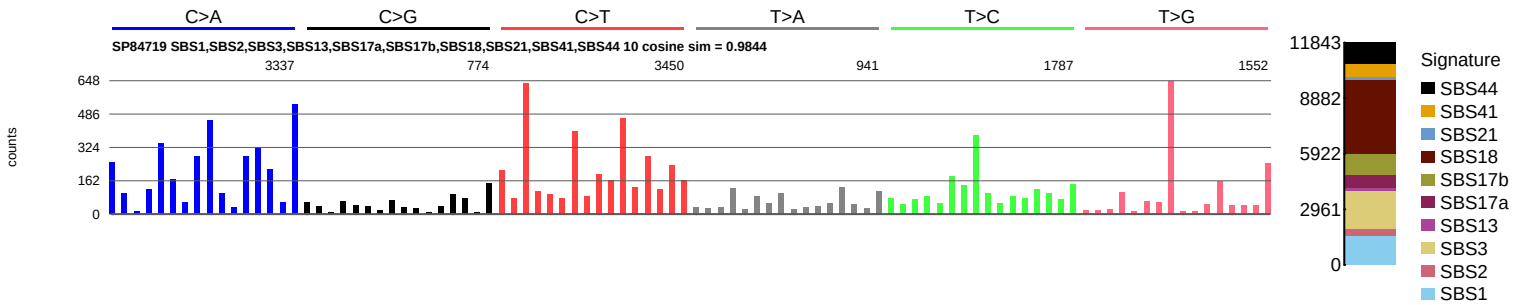
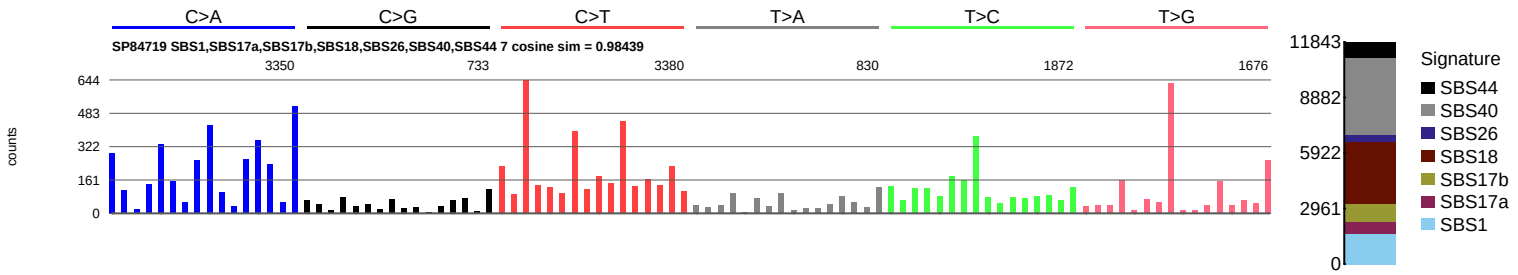
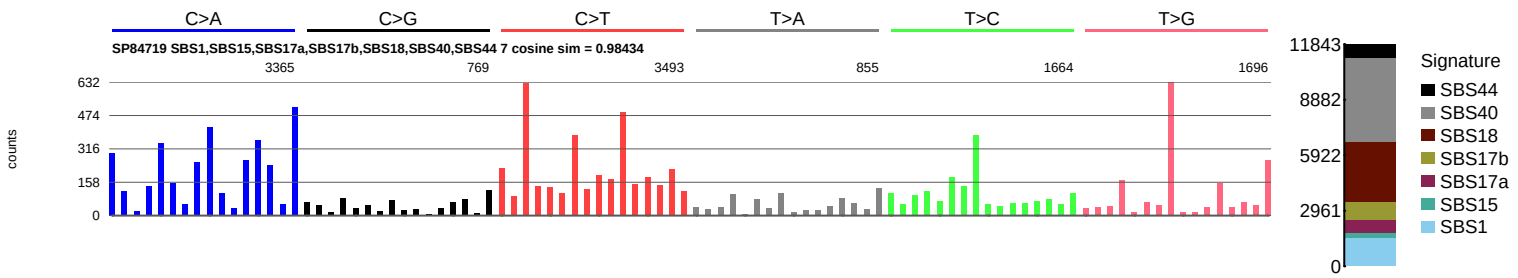
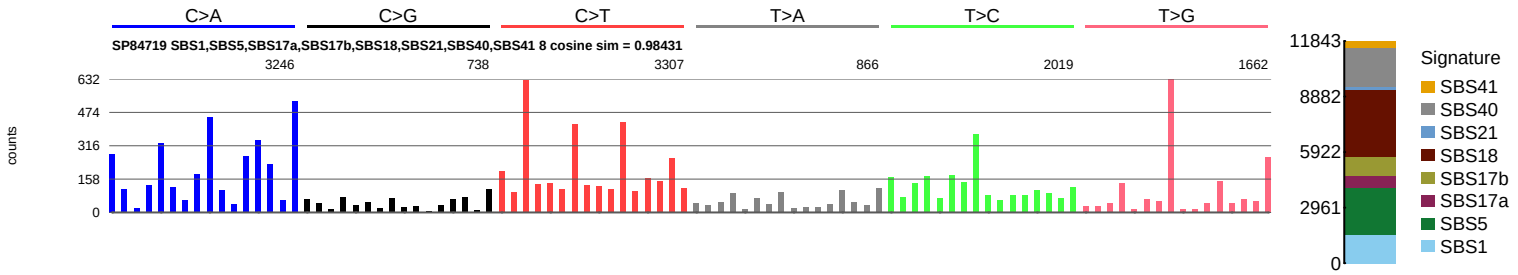
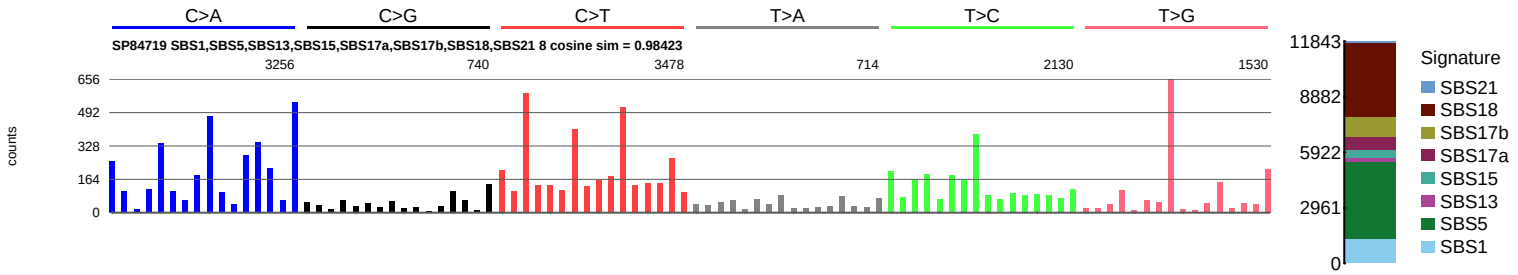
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9844
3337
774
3450
941
1787
1552
648
486
324
162
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98439
3350
733
3380
830
1872
1676
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98434
3365
769
3493
855
1664
1696
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98431
3246
738
3307
866
2019
1662
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98423
3256
740
3478
714
2130
1530
656
492
328
164
0
counts
11843
8882
5922
2961
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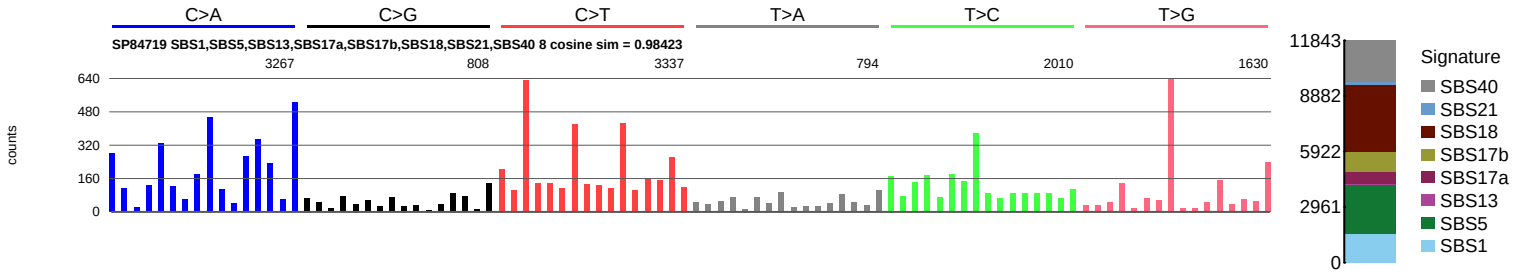
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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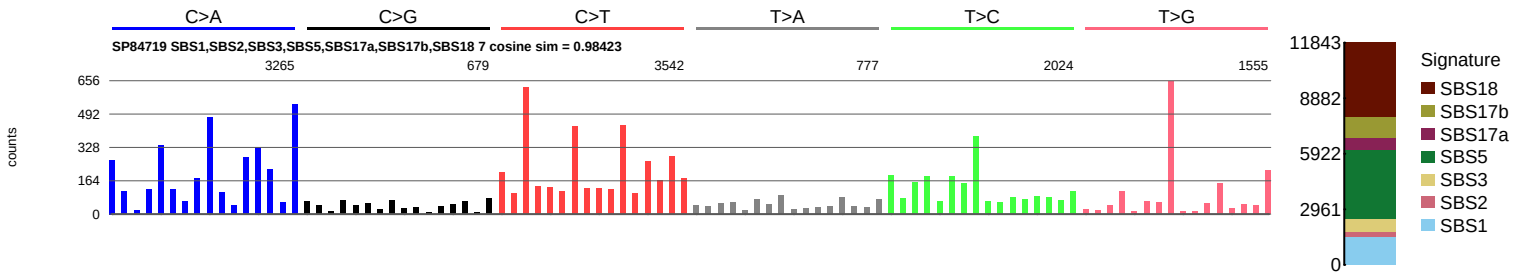
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98423
3267
808
3337
794
2010
1630
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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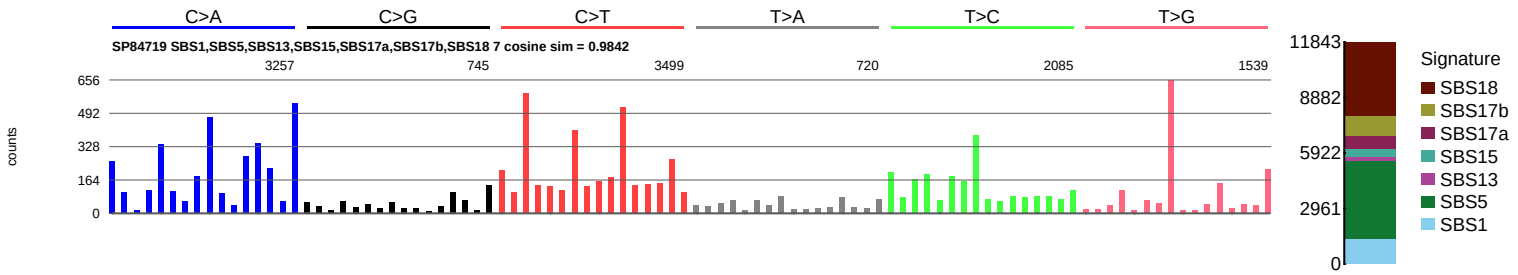
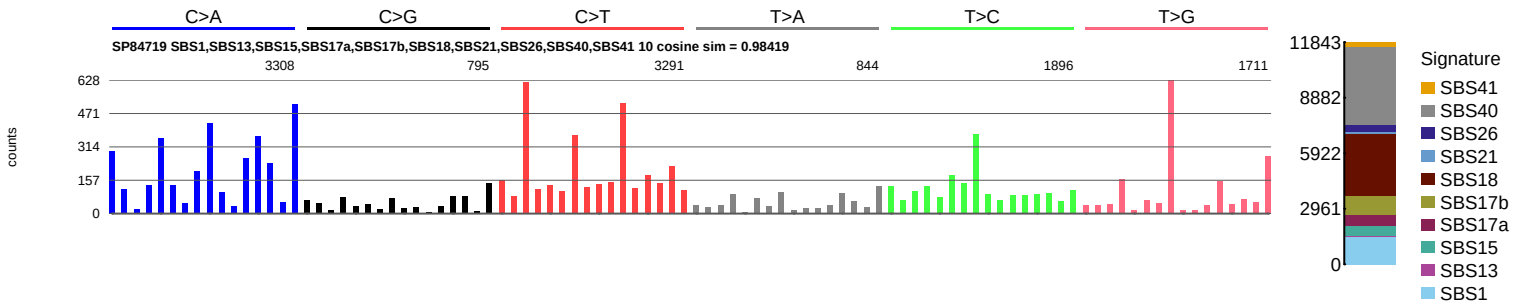
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98423
3265
679
3542
777
2024
1555
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.9842
3257
745
3499
720
2085
1539
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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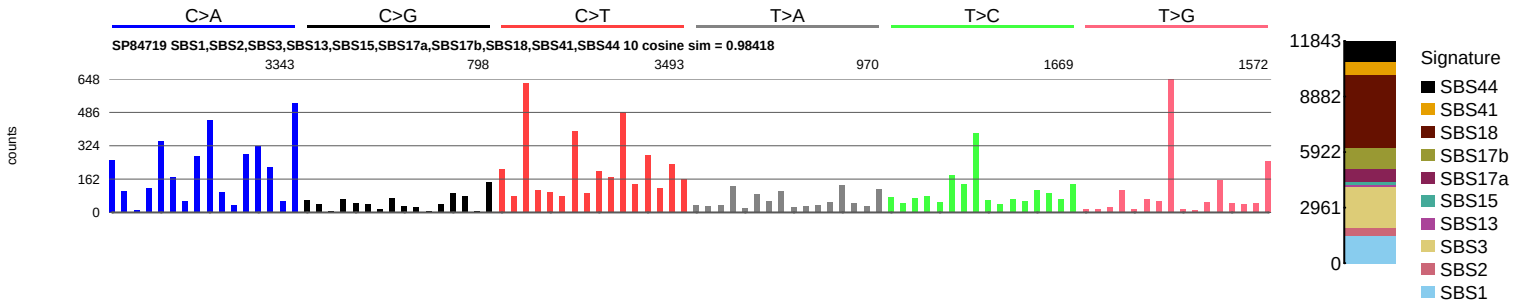
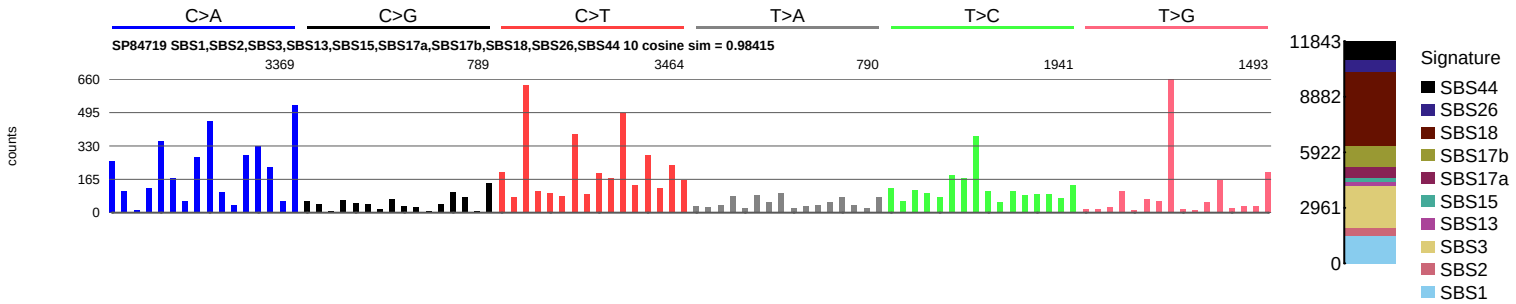
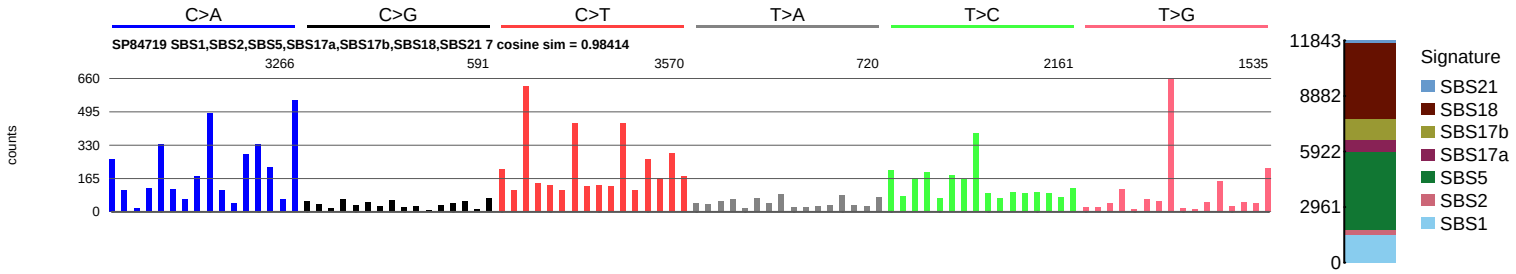
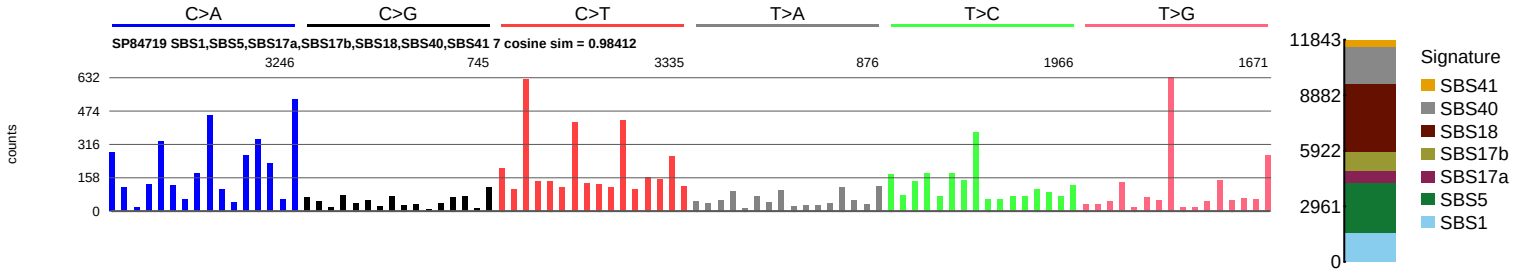
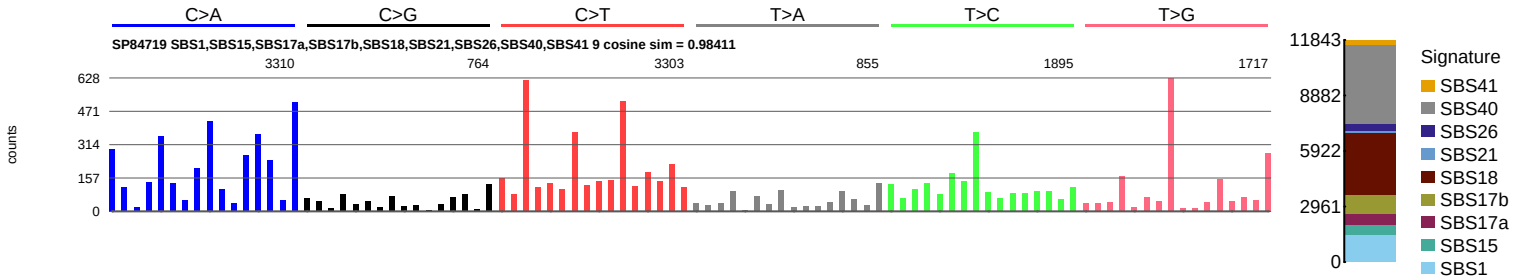
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98419
3308
795
3291
844
1896
1711
628
471
314
157
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

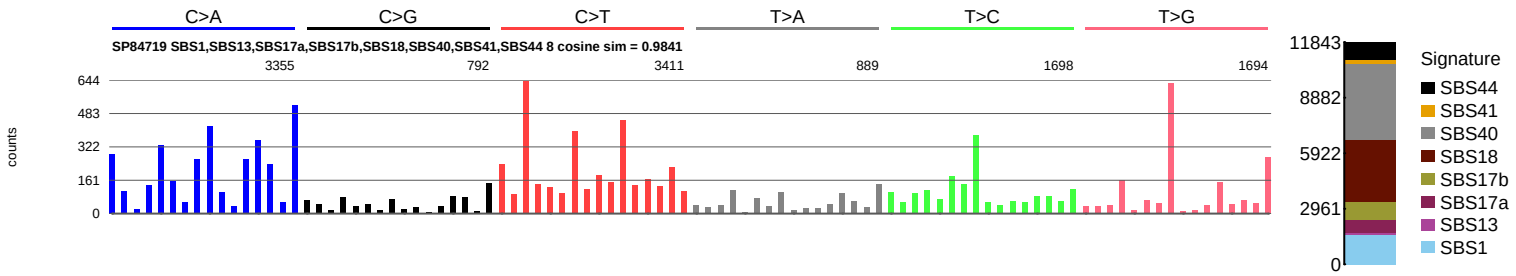

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98418
3343
798
3493
970
1669
1572
648
486
324
162
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

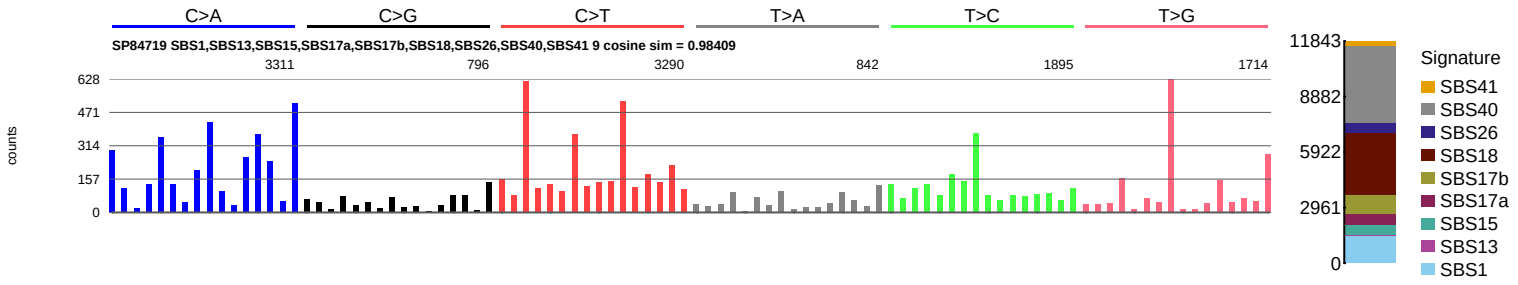

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98415
3369
789
3464
790
1941
1493
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

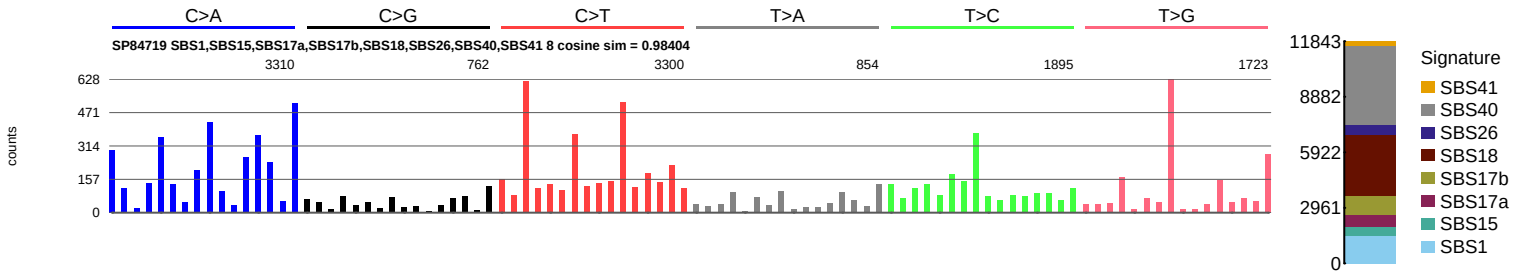

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98414
3266
591
3570
720
2161
1535
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

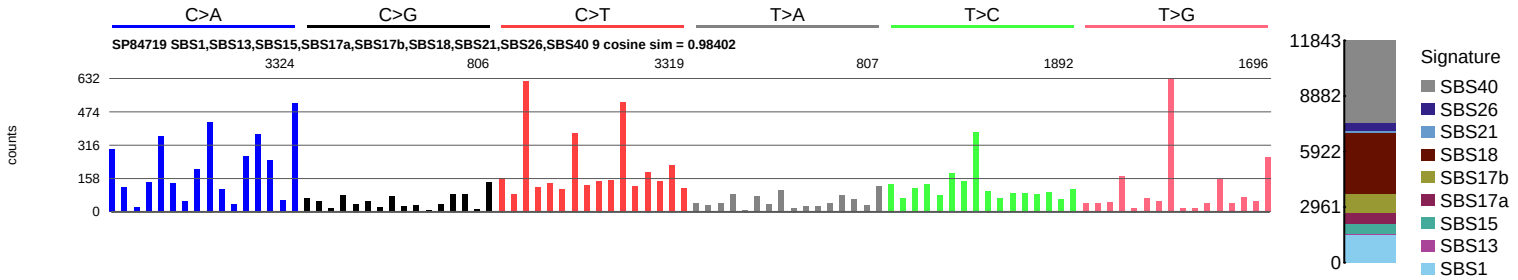

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98412
3246
745
3335
876
1966
1671
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

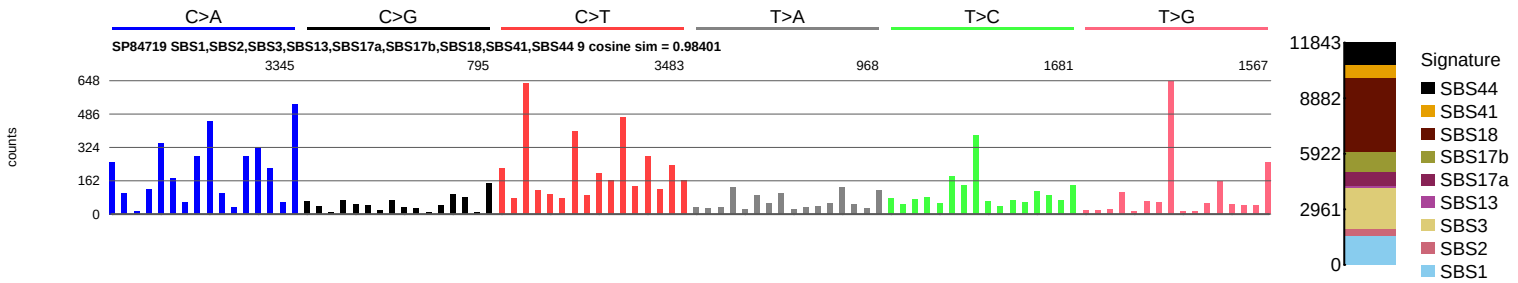

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98411
3310
764
3303
855
1895
1717
628
471
314
157
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

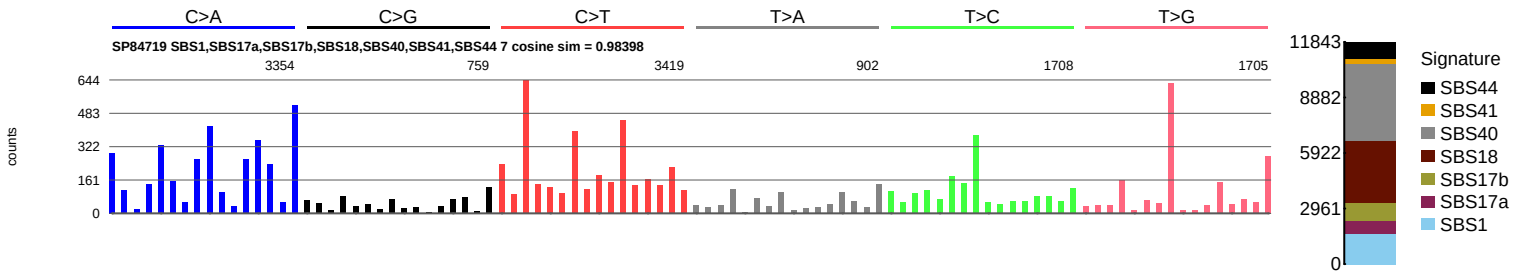

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9841
3355
792
3411
889
1698
1694
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1

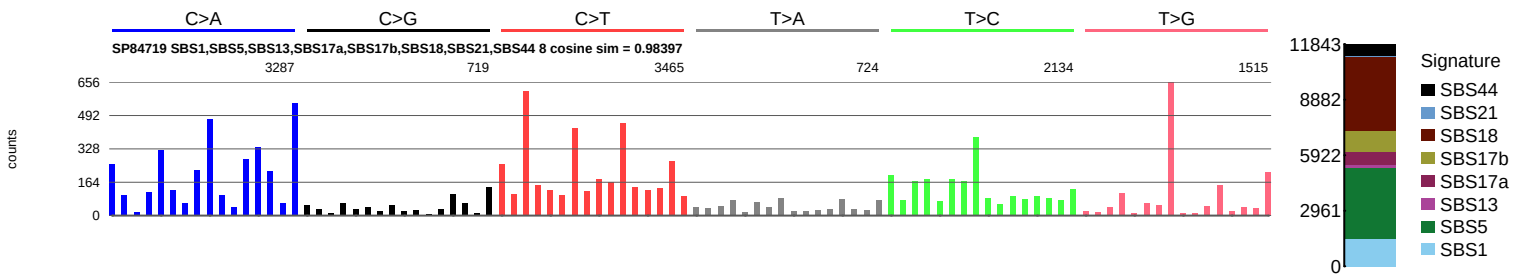

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98409
3311
796
3290
842
1895
1714
counts
628
471
314
157
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

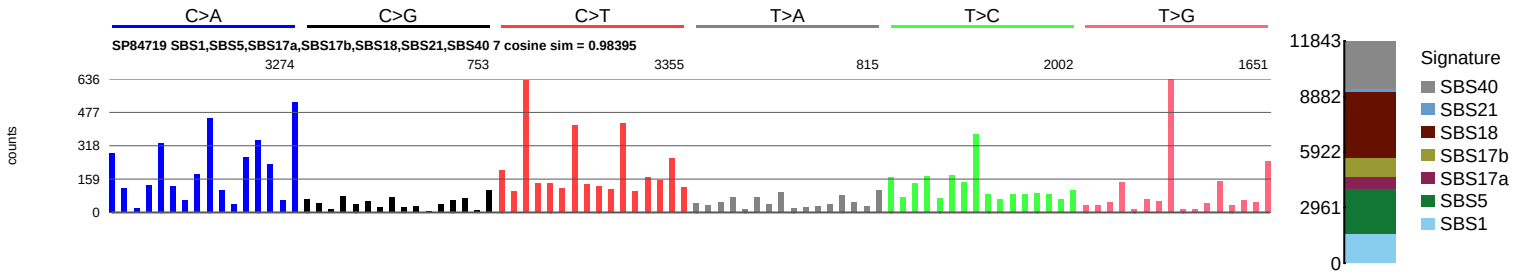

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98404
3310
762
3300
854
1895
1723
counts
628
471
314
157
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

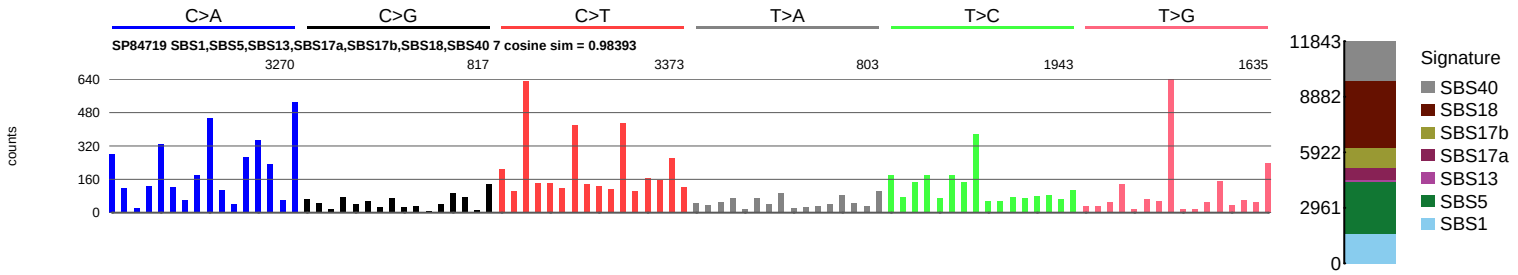

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98402
3324
806
3319
807
1892
1696
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

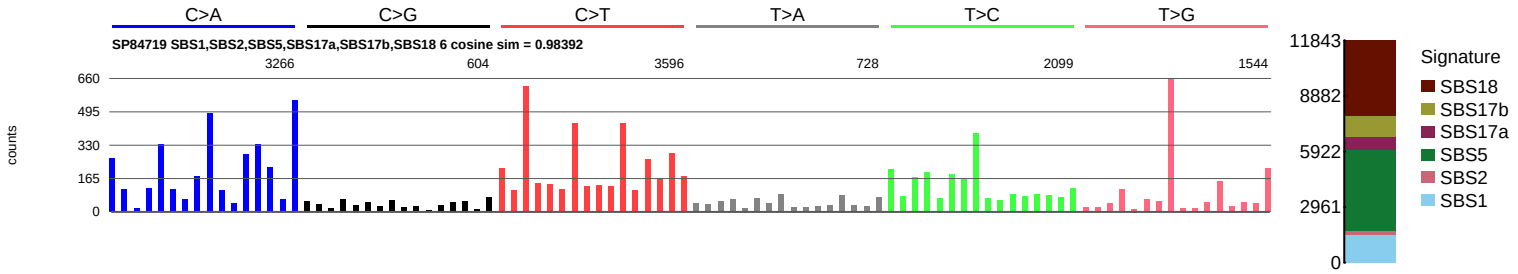

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98401
3345
795
3483
968
1681
1567
counts
648
486
324
162
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

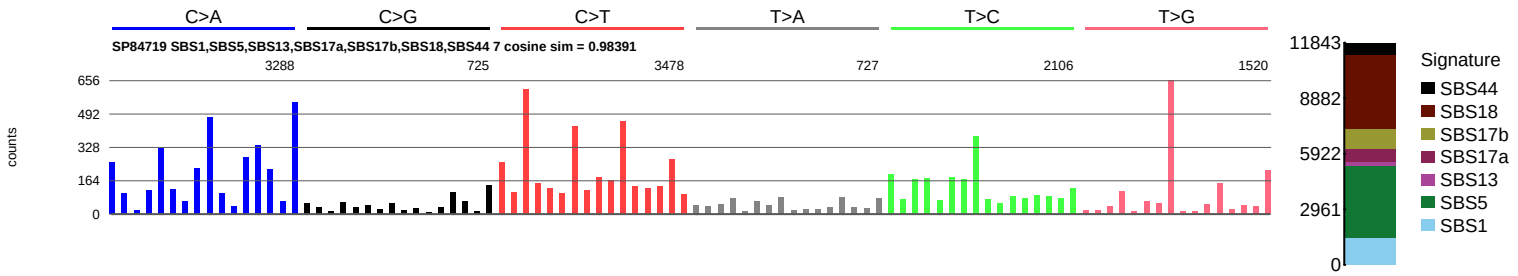

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98398
3354
759
3419
902
1708
1705
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1

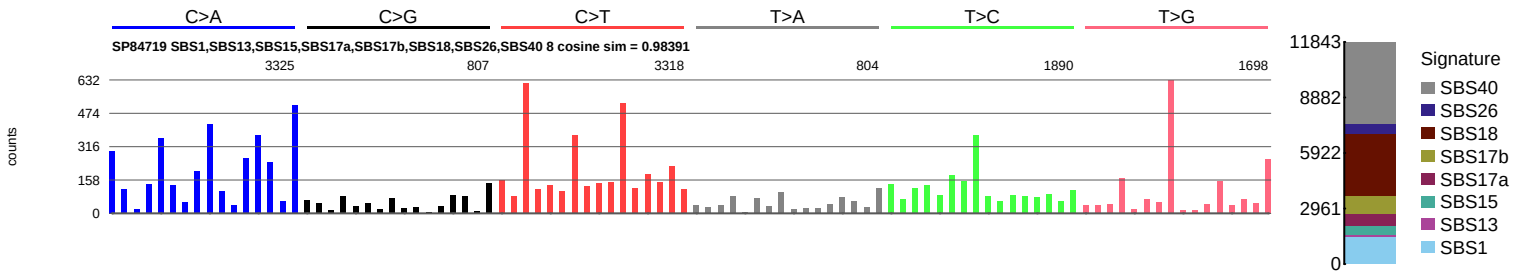

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98397
3287
719
3465
724
2134
1515
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

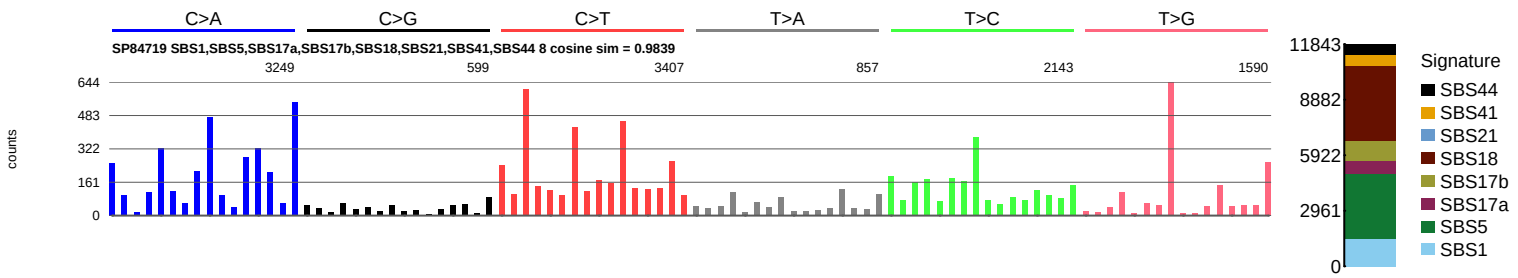

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98395
3274
753
3355
815
2002
1651
636
477
318
159
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

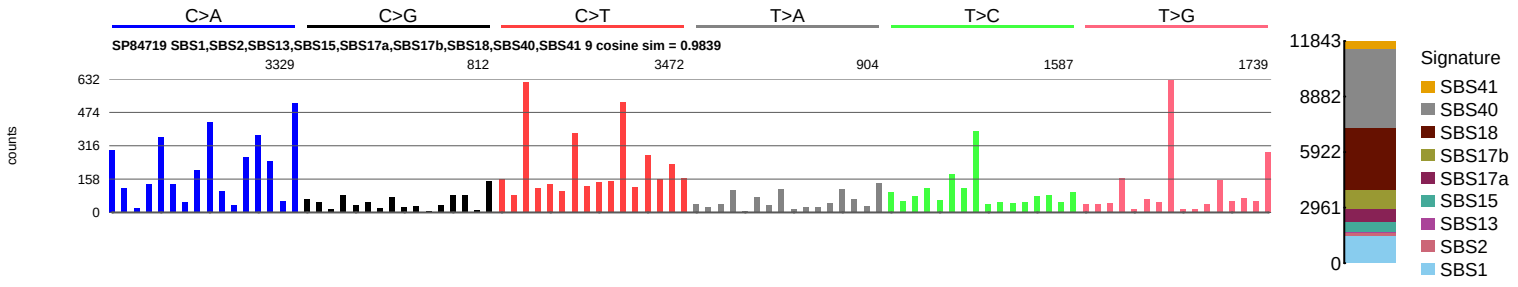

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98393
3270
817
3373
803
1943
1635
640
480
320
160
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

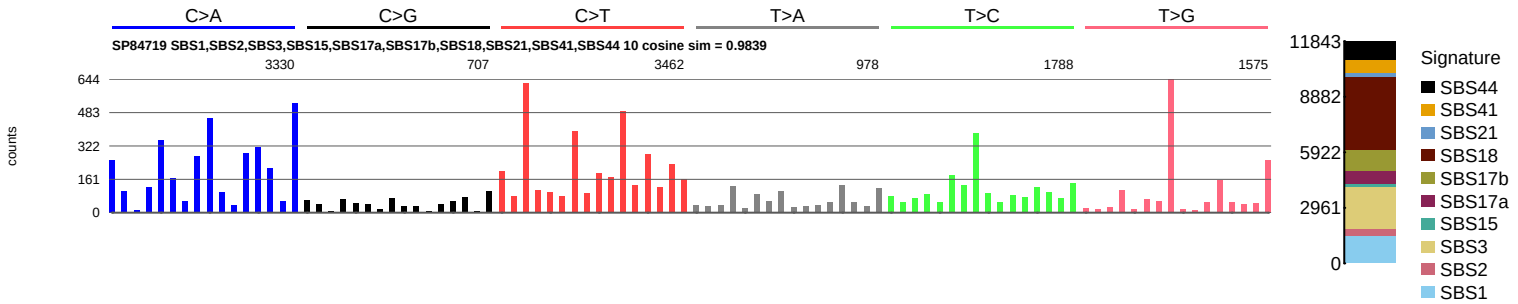

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98392
3266
604
3596
728
2099
1544
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

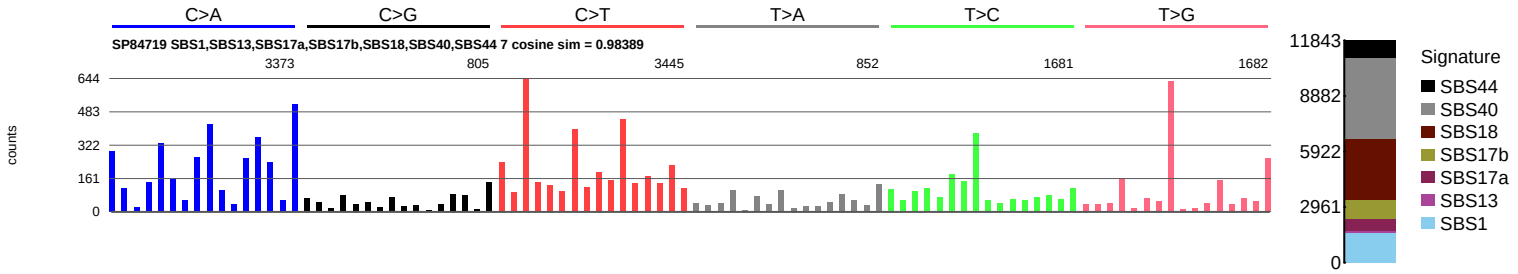

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98391
3288
725
3478
727
2106
1520
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98391
3325
807
3318
804
1890
1698
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.9839
3249
599
3407
857
2143
1590
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.9839
3329
812
3472
904
1587
1739
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9839
3330
707
3462
978
1788
1575
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98389
3373
805
3445
852
1681
1682
644
483
322
161
0
counts
11843
8882
5922
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SBS44
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SBS17b
SBS17a
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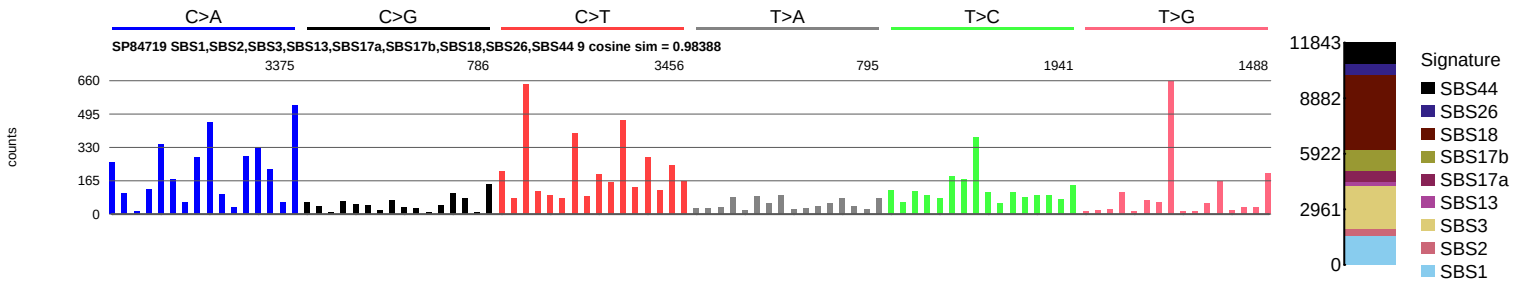
SBS1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SP84719 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98388
3375
786
3456
795
1941
1488
660
495
330
165
0
counts
11843
8882
5922
2961
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Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
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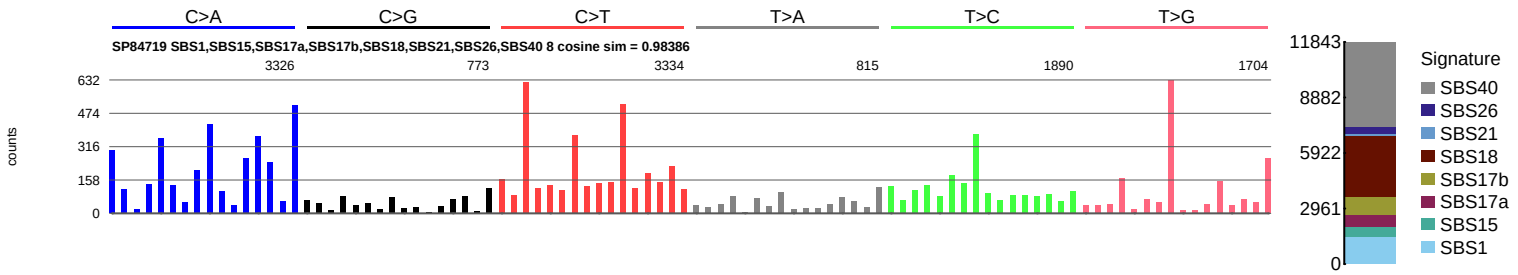
SBS2
SBS1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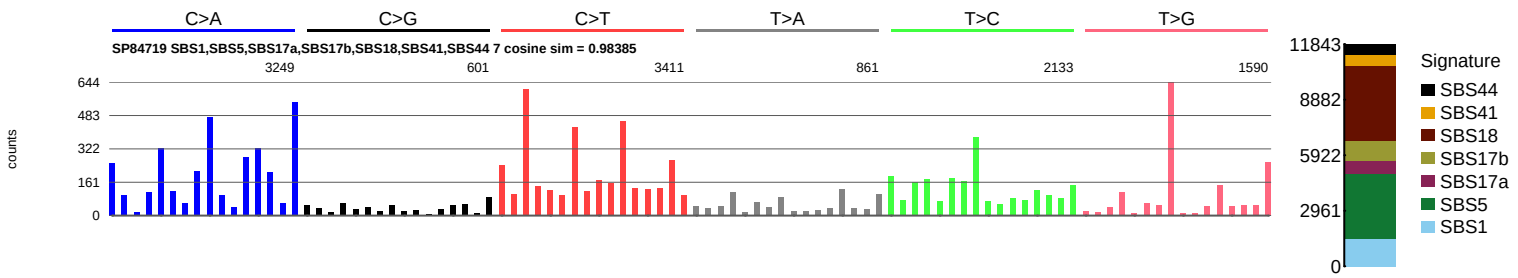
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98386
3326
773
3334
815
1890
1704
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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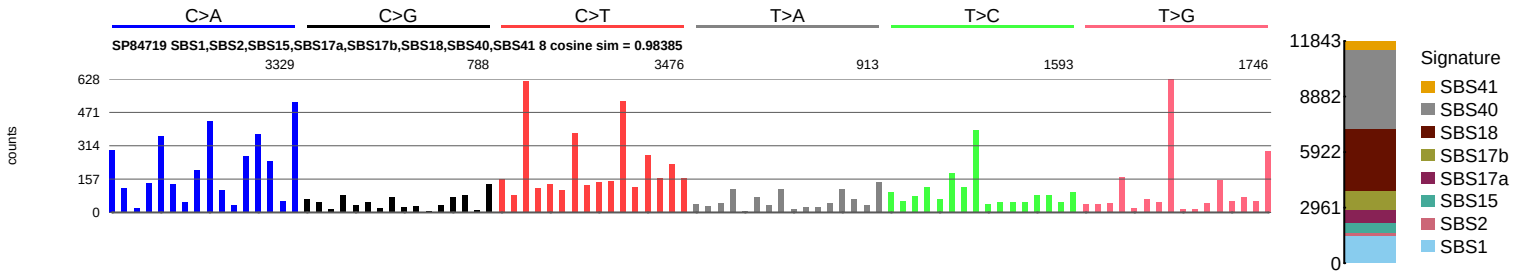
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98385
3249
601
3411
861
2133
1590
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1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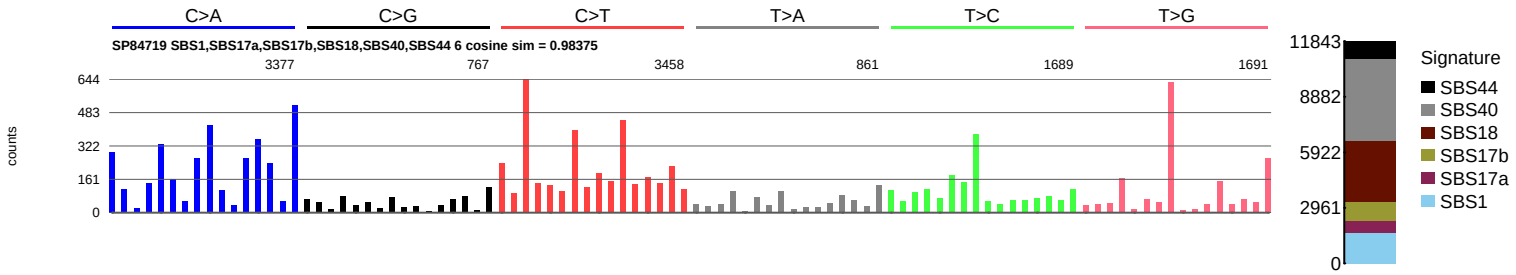
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98385
3329
788
3476
913
1593
1746
counts
628
471
314
157
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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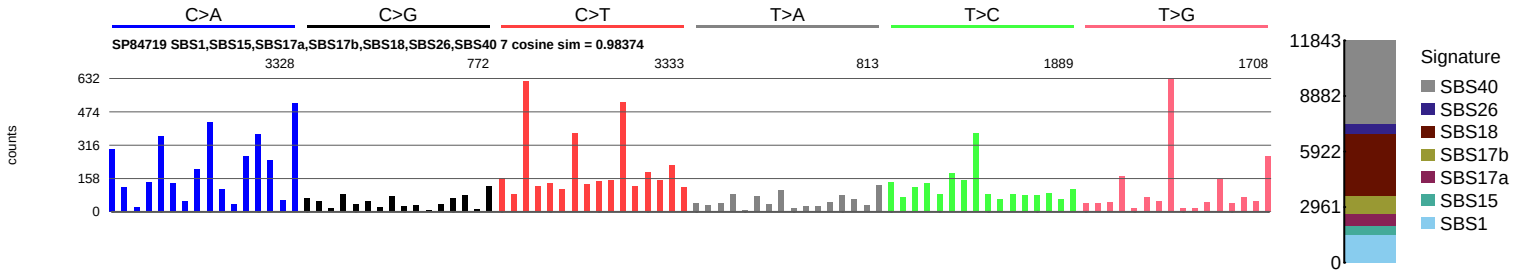
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98375
3377
767
3458
861
1689
1691
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS1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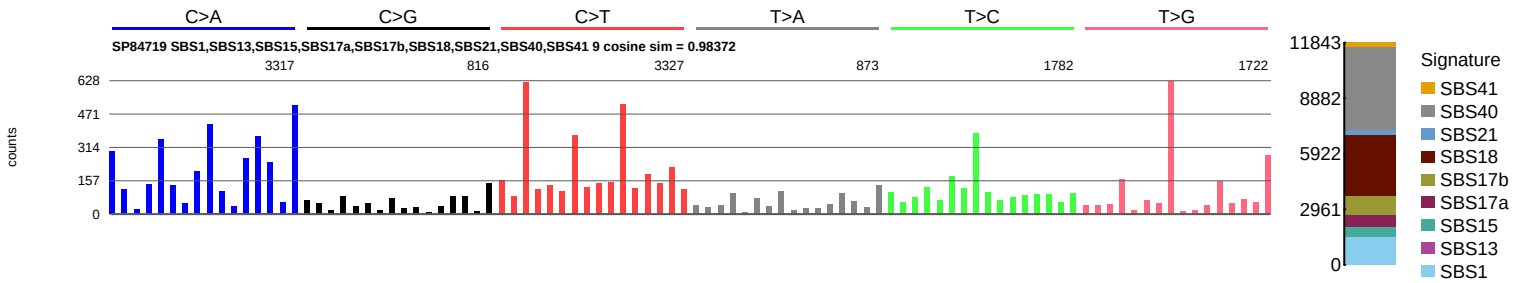
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98374
3328
772
3333
813
1889
1708
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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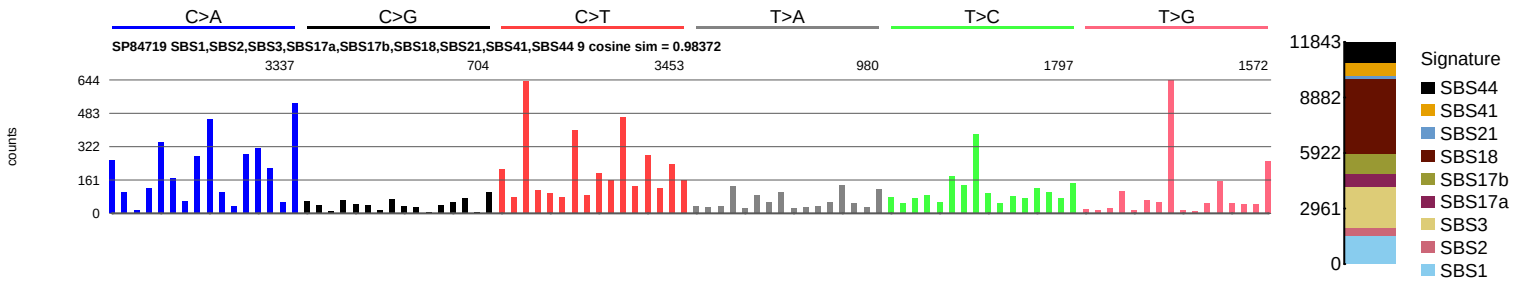
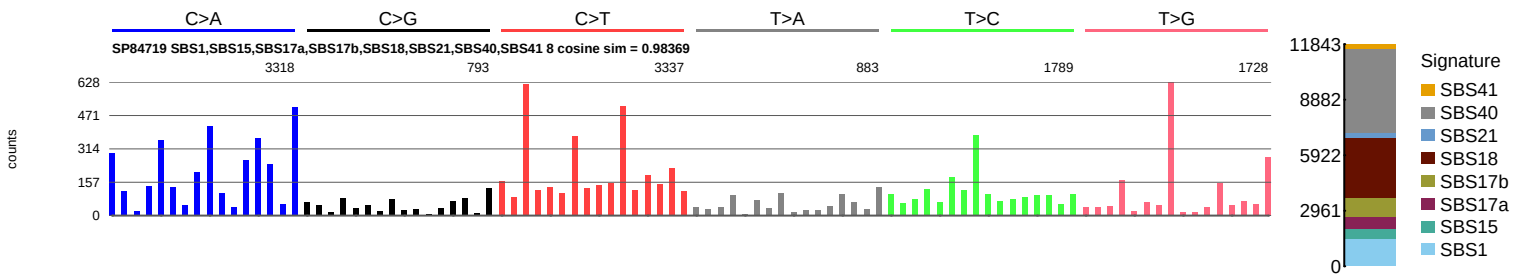
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98372
3317
816
3327
873
1782
1722
counts
628
471
314
157
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98372
3337
704
3453
980
1797
1572
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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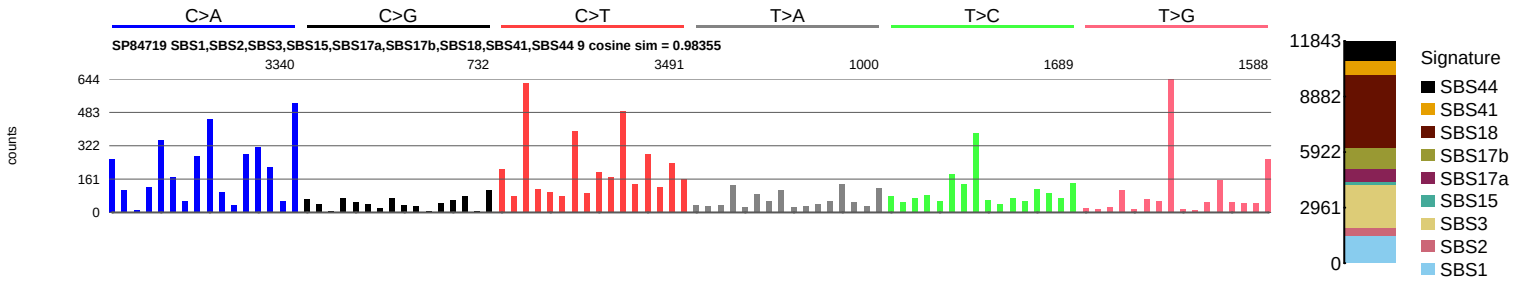
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98369
3318
793
3337
883
1789
1728
counts
628
471
314
157
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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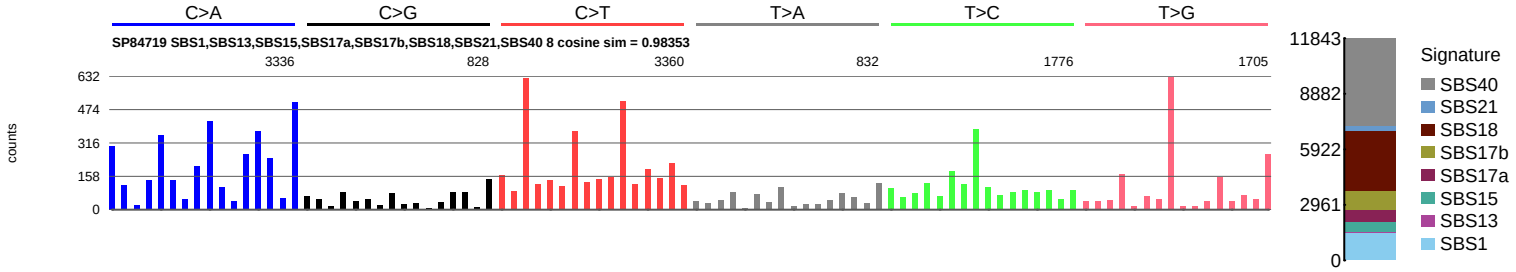
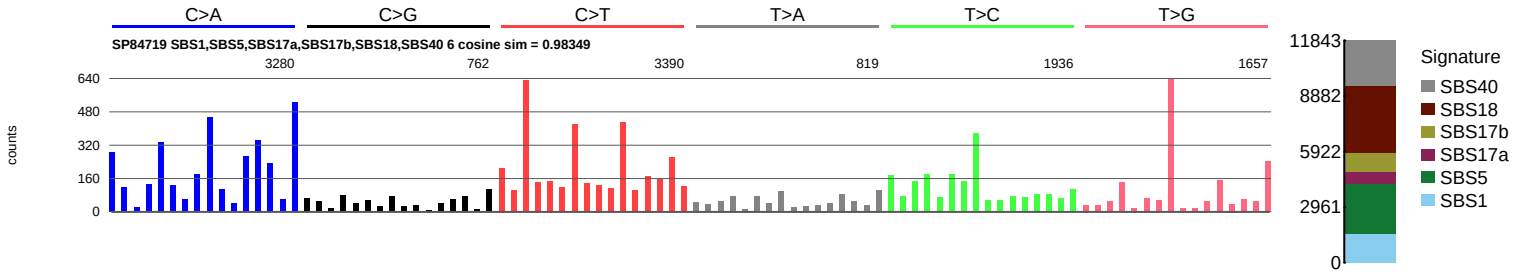
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98355
3340
732
3491
1000
1689
1588
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98353
3336
828
3360
832
1776
1705
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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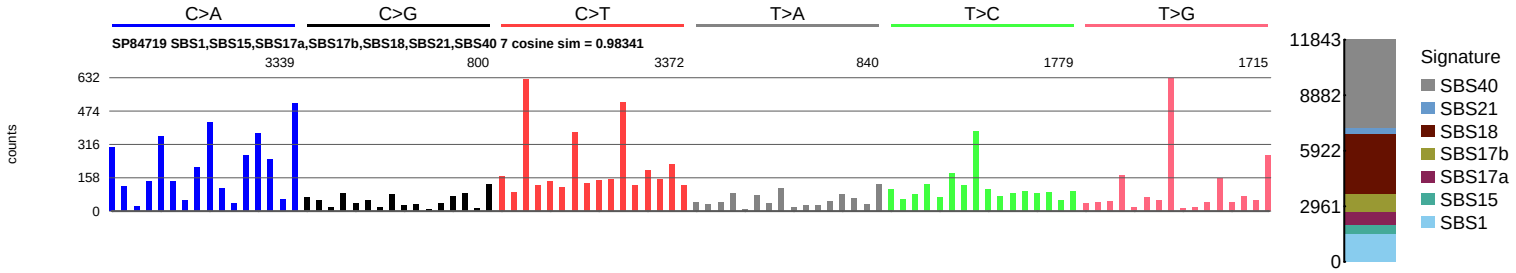
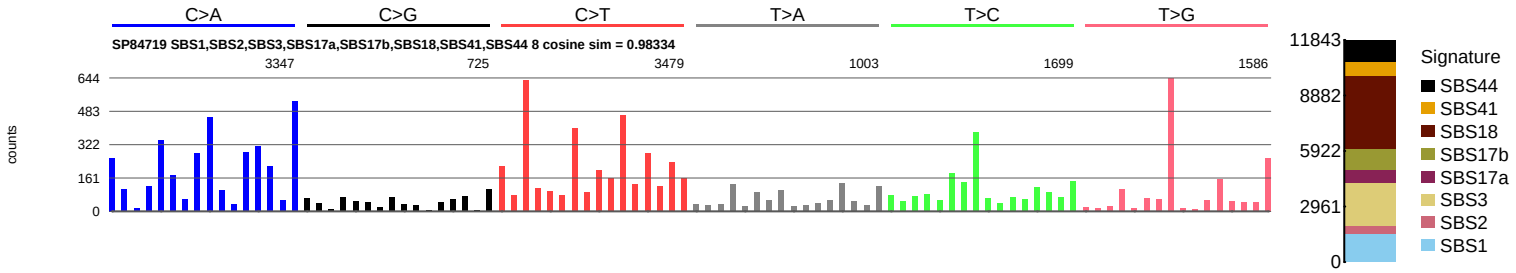
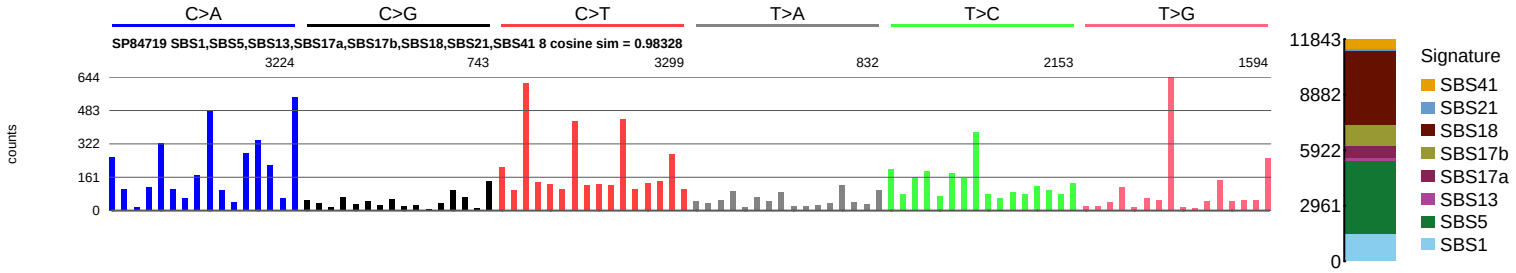
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98349
3280
762
3390
819
1936
1657
counts
640
480
320
160
0
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98341
3339
800
3372
840
1779
1715
counts
632
474
316
158
0
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98334
3347
725
3479
1003
1699
1586
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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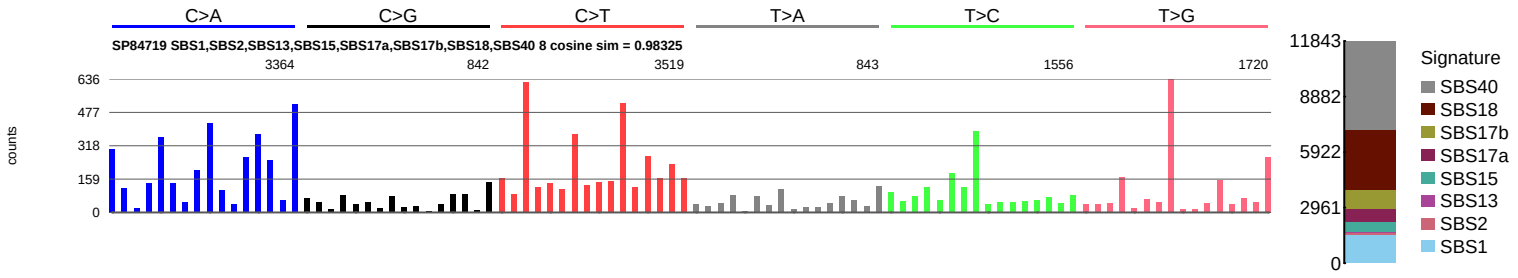
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98328
3224
743
3299
832
2153
1594
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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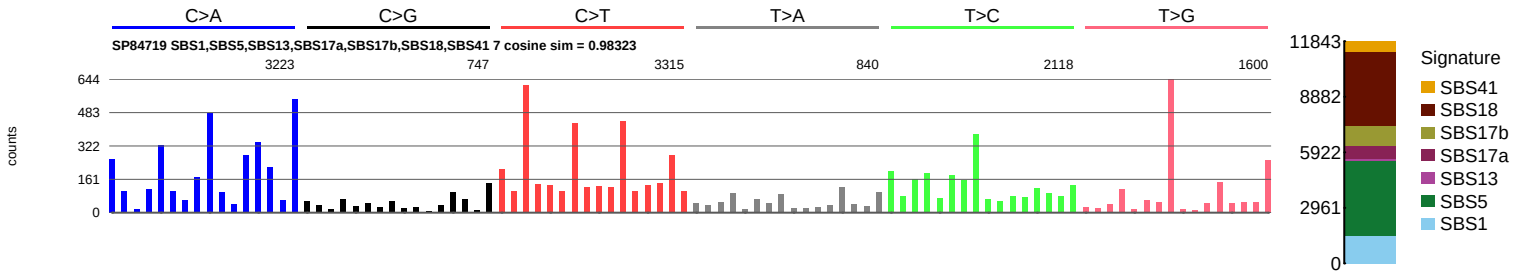
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98325
3364
842
3519
843
1556
1720
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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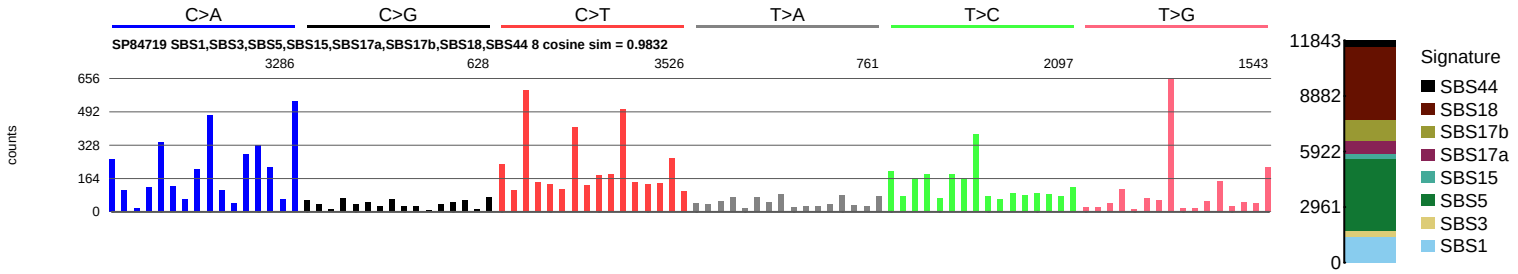
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98323
3223
747
3315
840
2118
1600
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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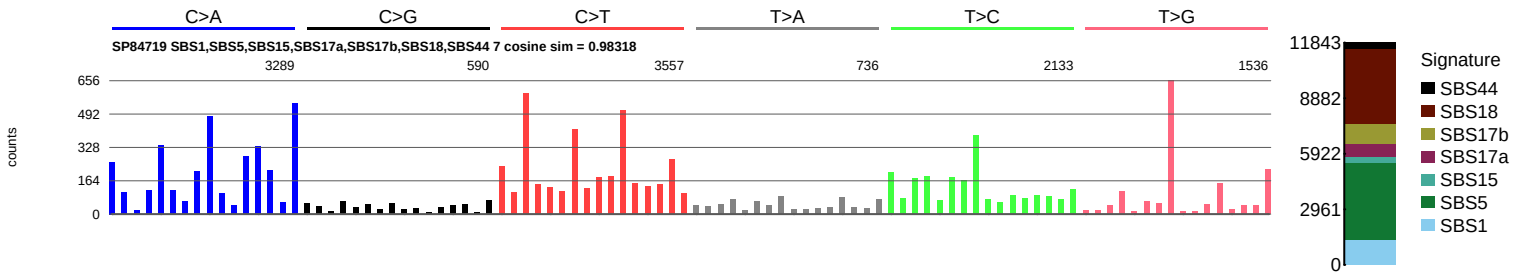
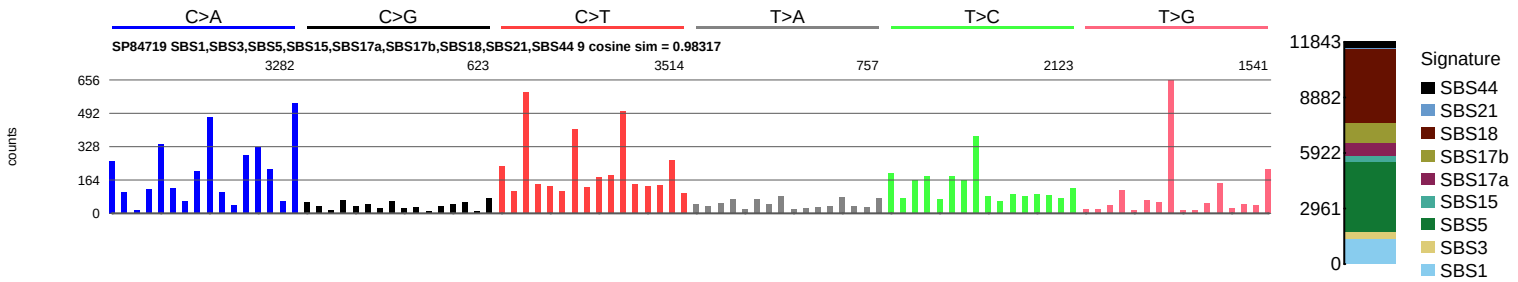
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.9832
3286
628
3526
761
2097
1543
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98318
3289
590
3557
736
2133
1536
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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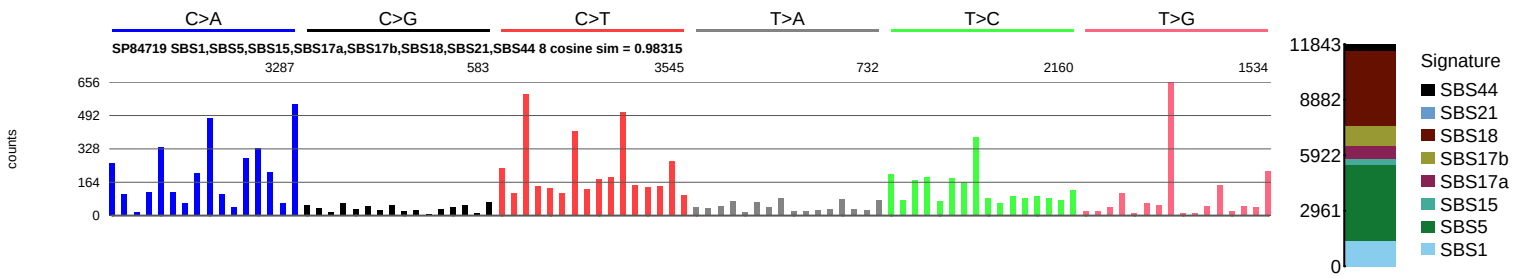
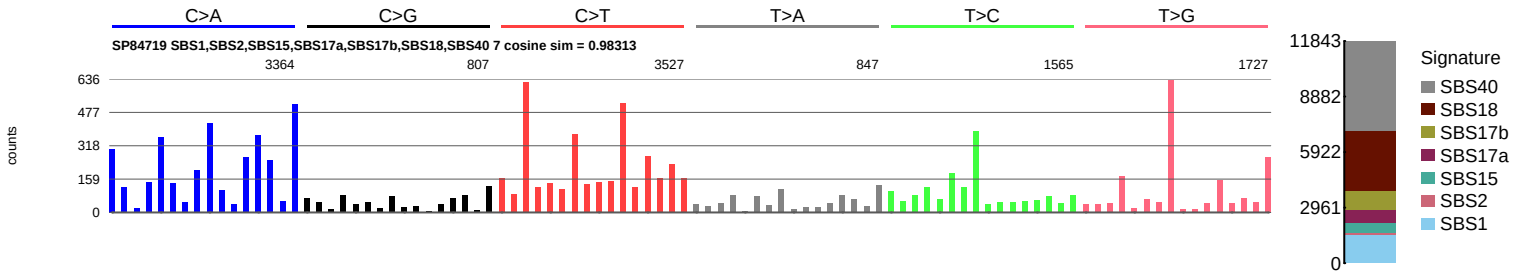
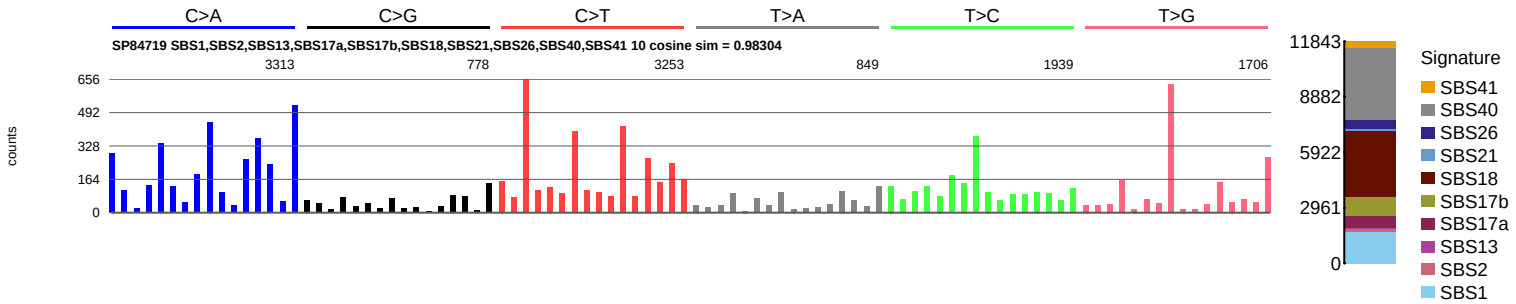
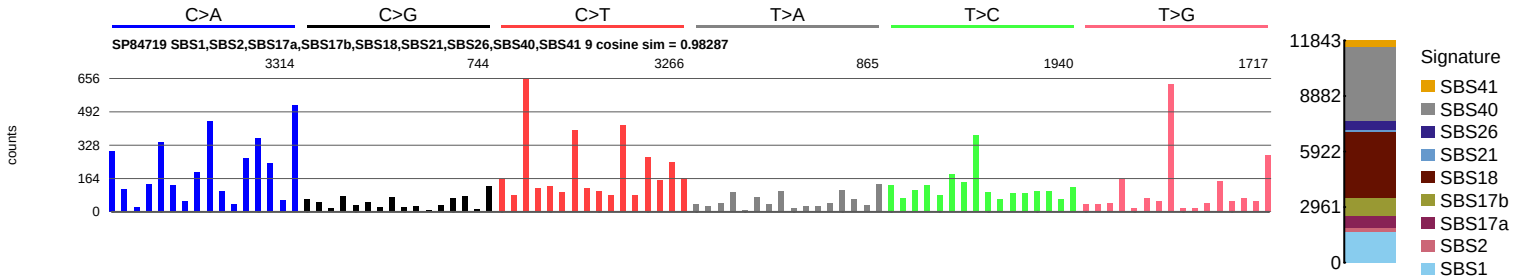
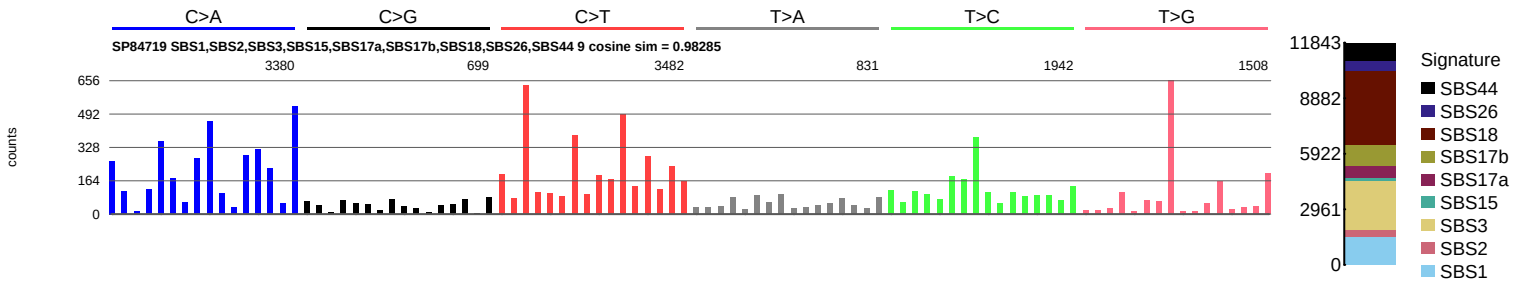
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98317
3282
623
3514
757
2123
1541
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

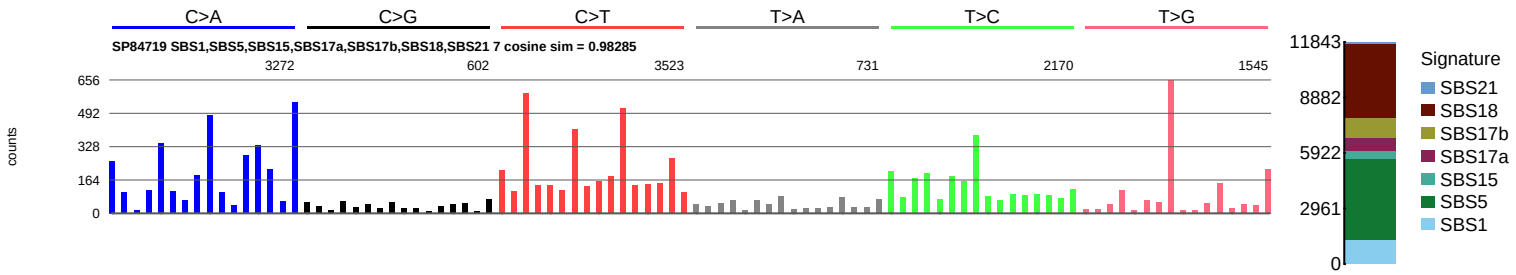

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98315
3287
583
3545
732
2160
1534
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98313
3364
807
3527
847
1565
1727
counts
636
477
318
159
0
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98304
3313
778
3253
849
1939
1706
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98287
3314
744
3266
865
1940
1717
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98285
3380
699
3482
831
1942
1508
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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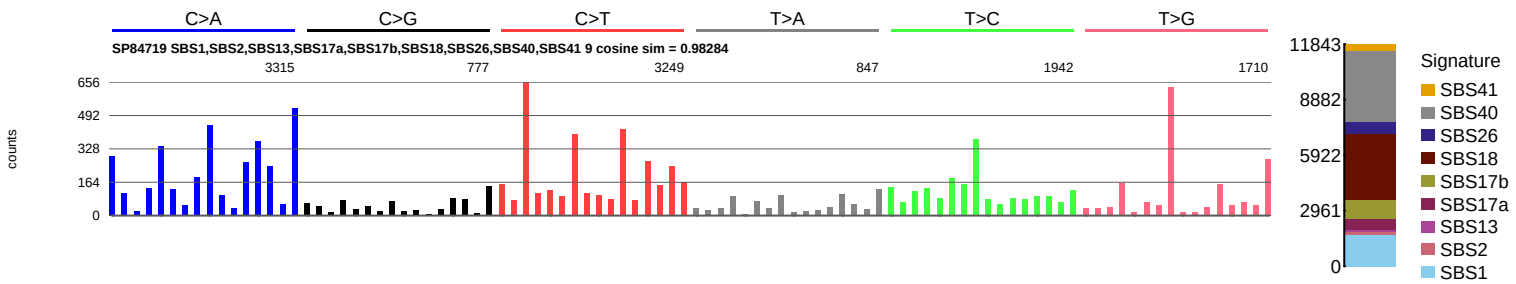
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98285
3272
602
3523
731
2170
1545
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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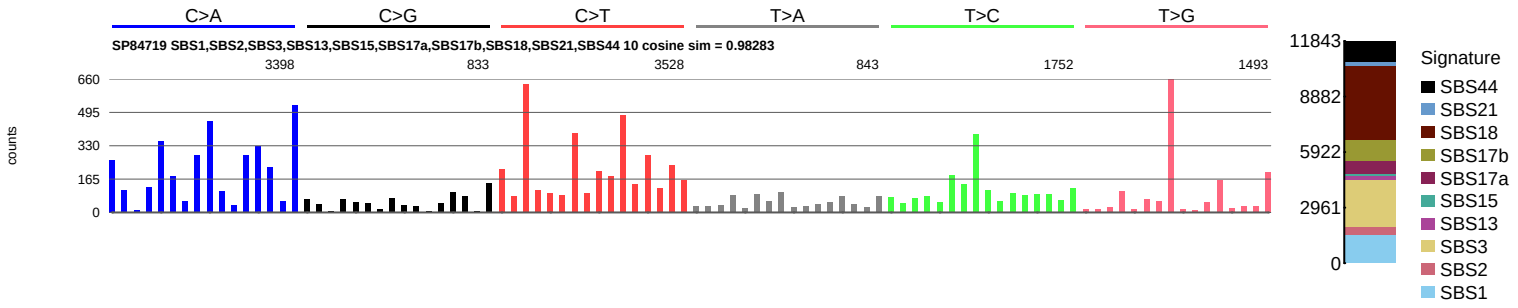
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98284
3315
777
3249
847
1942
1710
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
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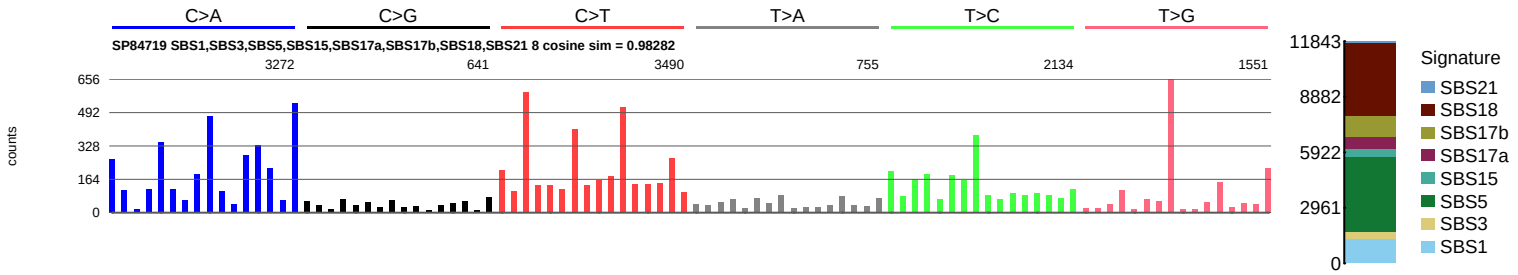
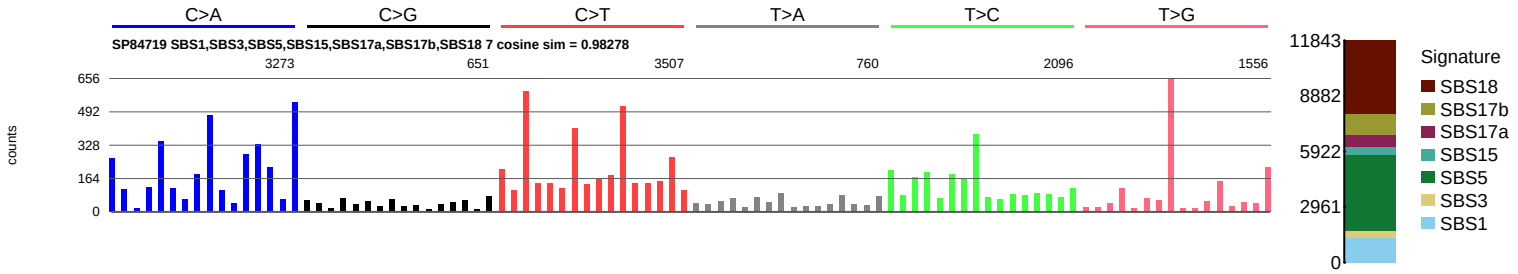
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98283
3398
833
3528
843
1752
1493
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98282
3272
641
3490
755
2134
1551
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98278
3273
651
3507
760
2096
1556
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
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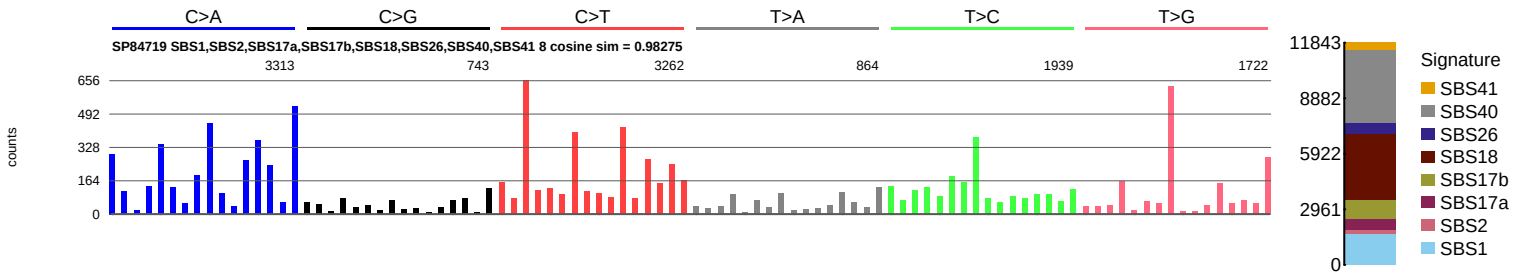
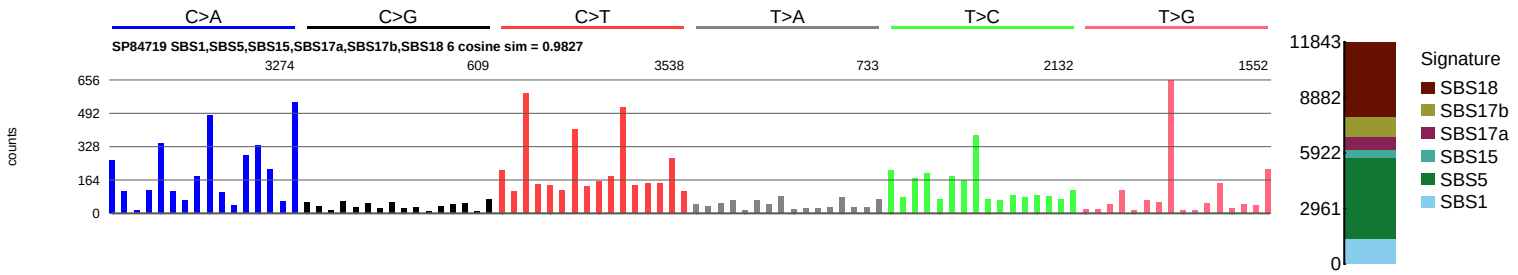
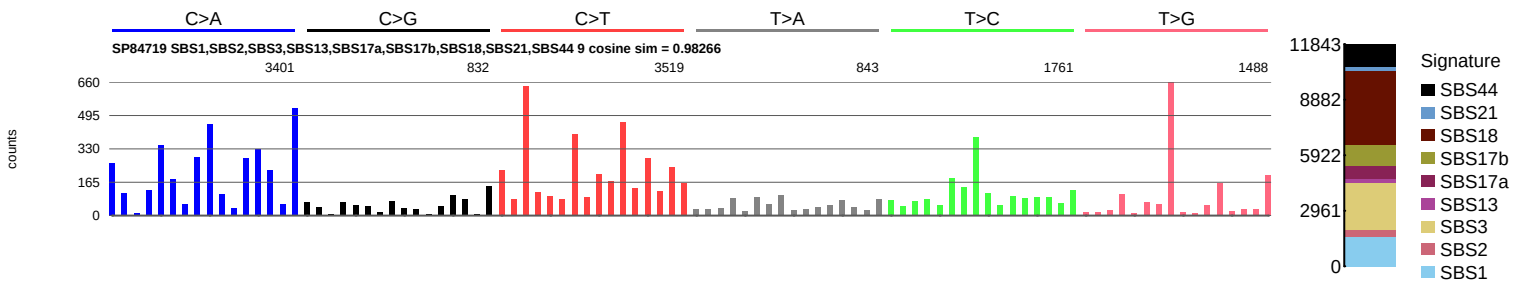
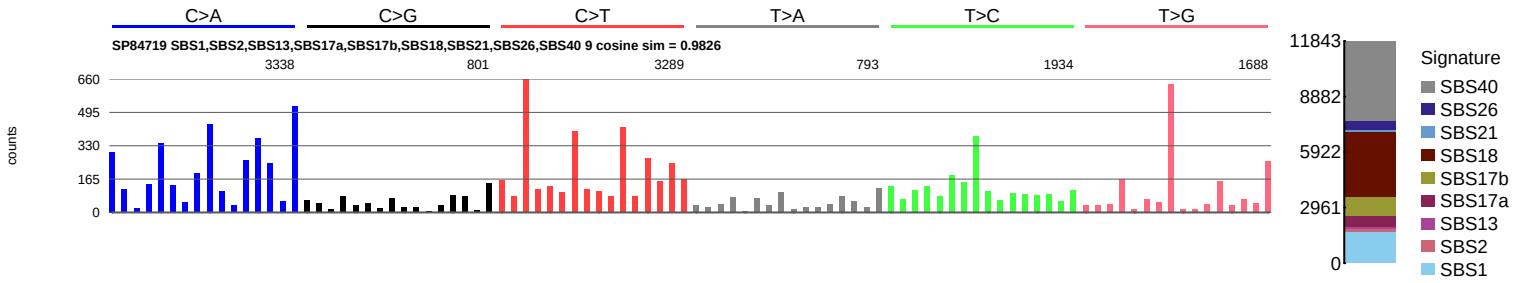
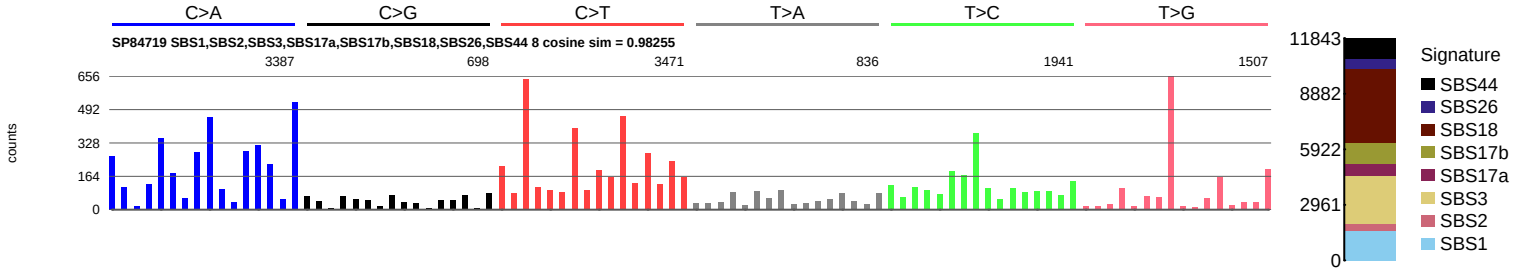
SBS5
SBS3
SBS1

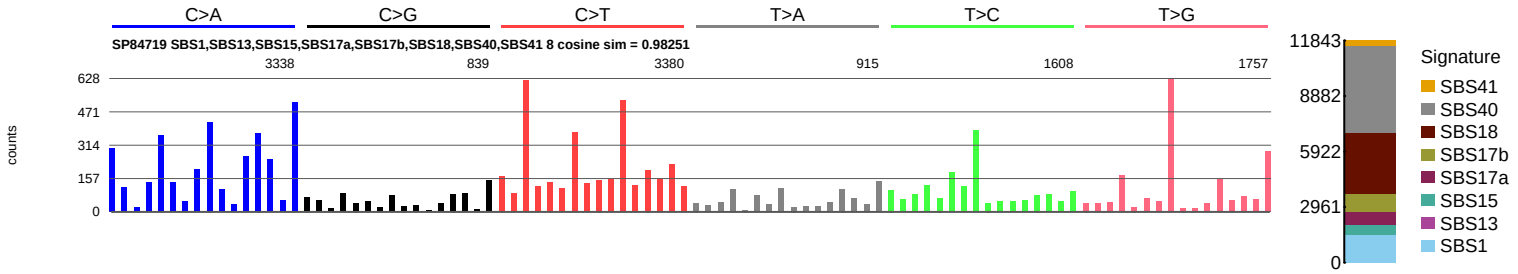

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98275
3313
743
3262
864
1939
1722
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

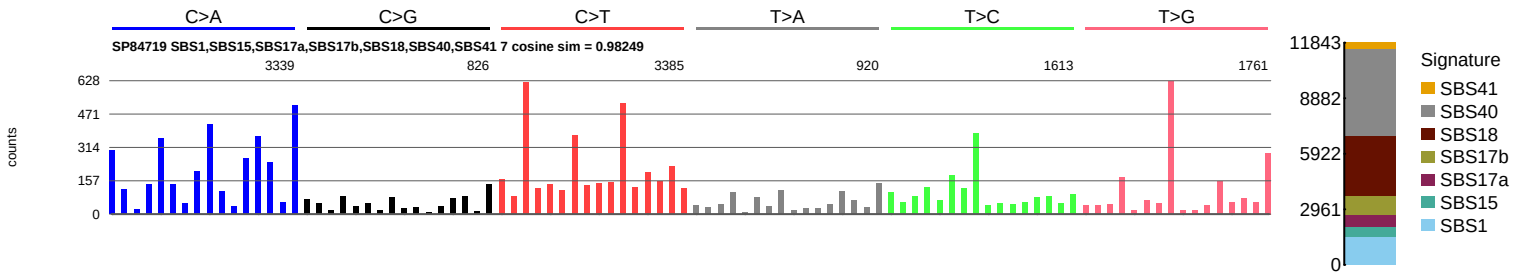

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.9827
3274
609
3538
733
2132
1552
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

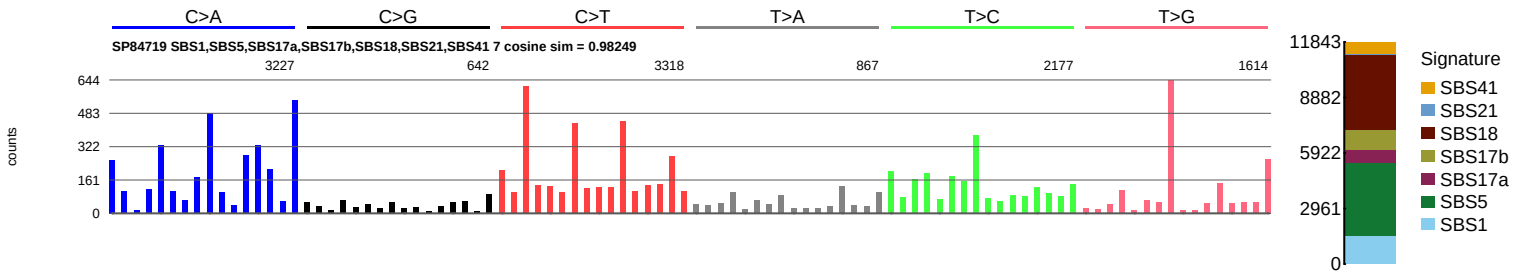

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98266
3401
832
3519
843
1761
1488
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

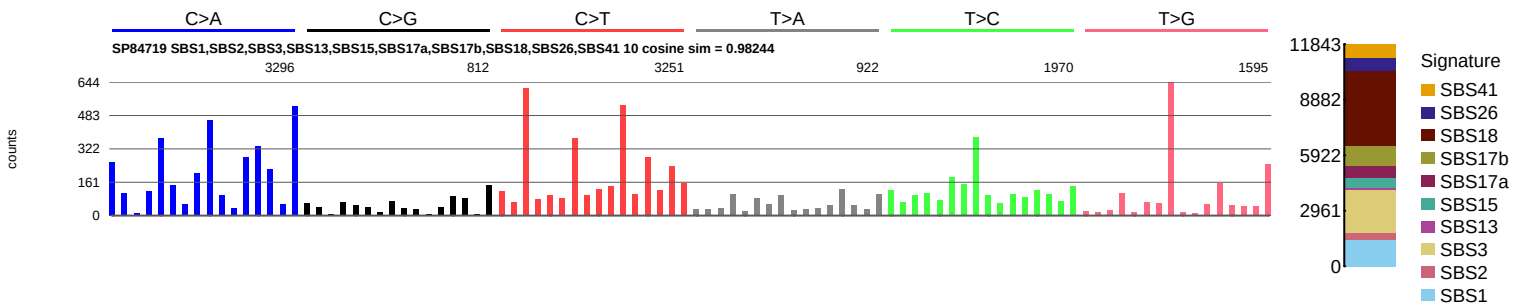

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.9826
3338
801
3289
793
1934
1688
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

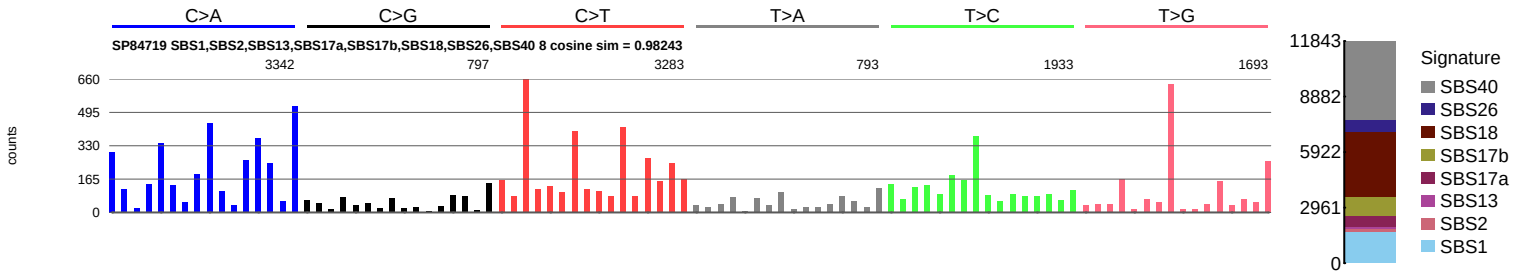

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98255
3387
698
3471
836
1941
1507
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

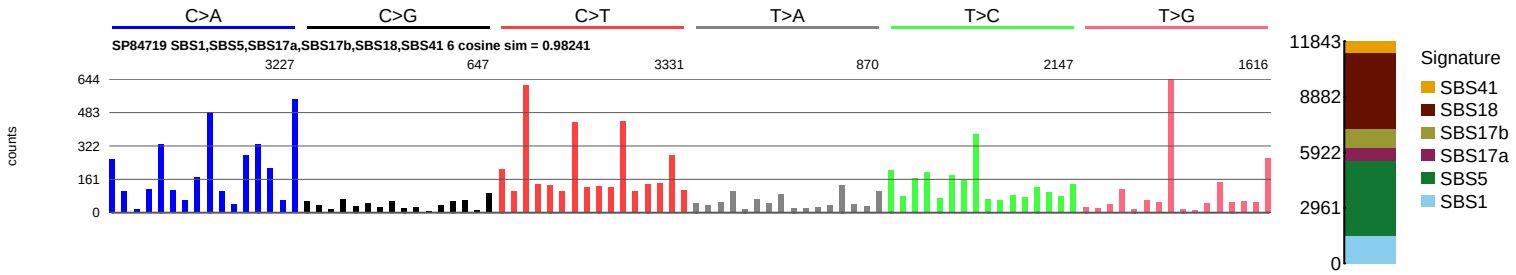

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98251
3338
839
3380
915
1608
1757
628
471
314
157
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

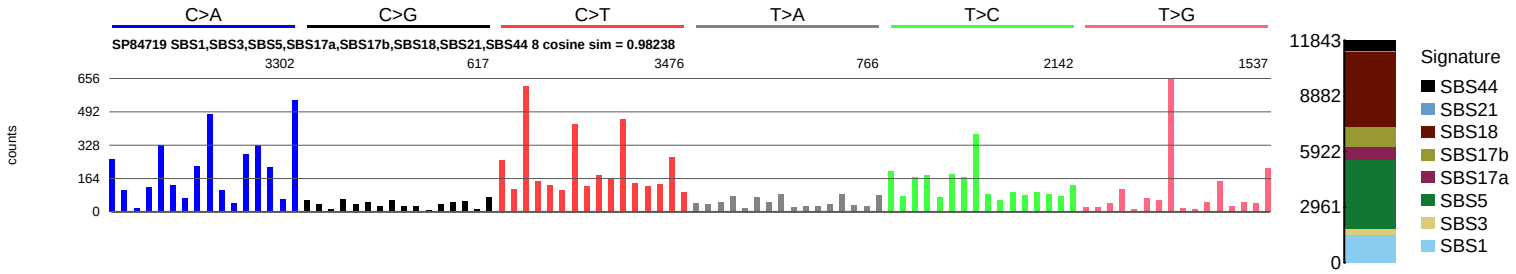

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98249
3339
826
3385
920
1613
1761
628
471
314
157
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

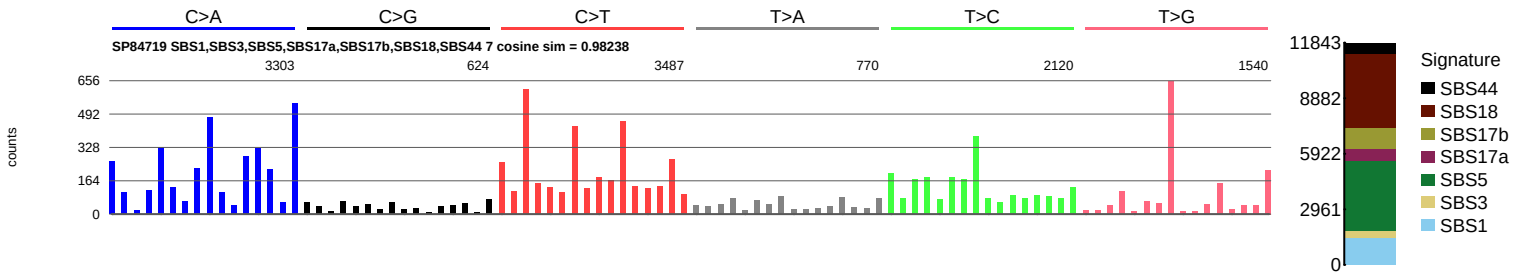

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98249
3227
642
3318
867
2177
1614
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

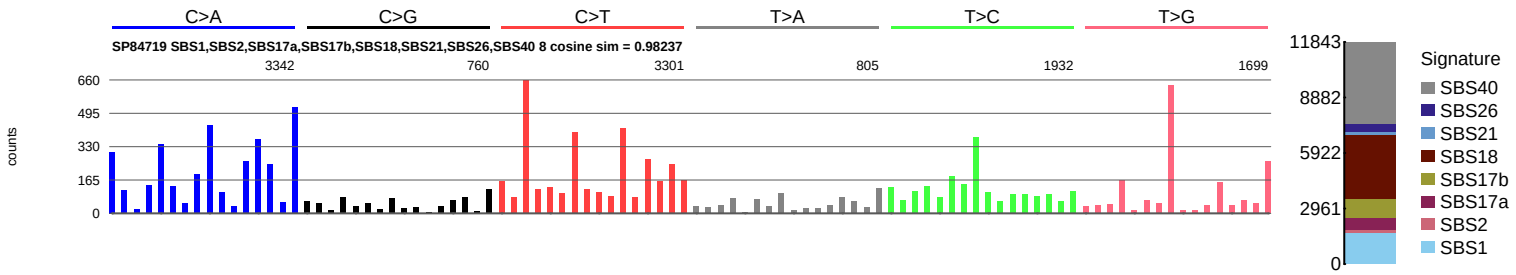

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98244
3296
812
3251
922
1970
1595
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

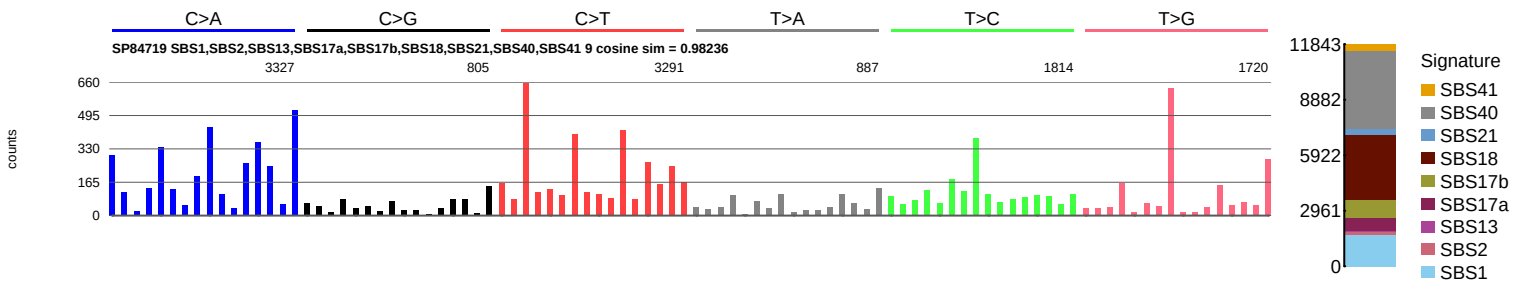

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98243
3342
797
3283
793
1933
1693
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

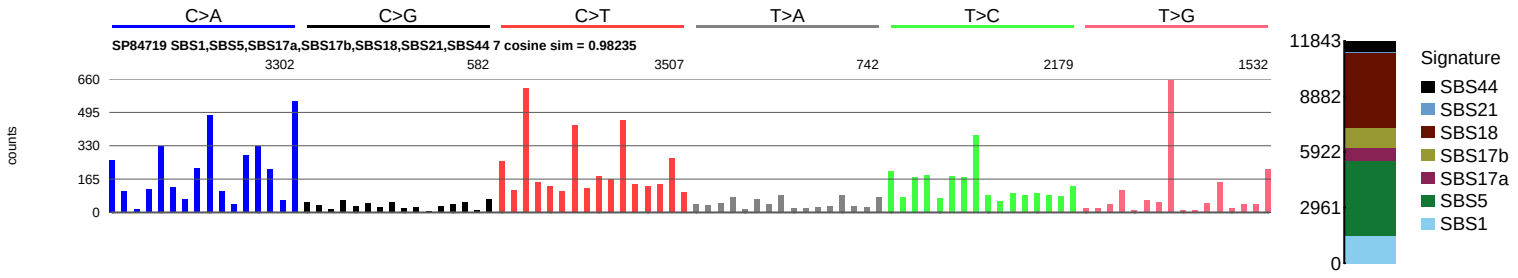

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98241
3227
647
3331
870
2147
1616
counts
644
483
322
161
0
11843
8882
5922
2961
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1

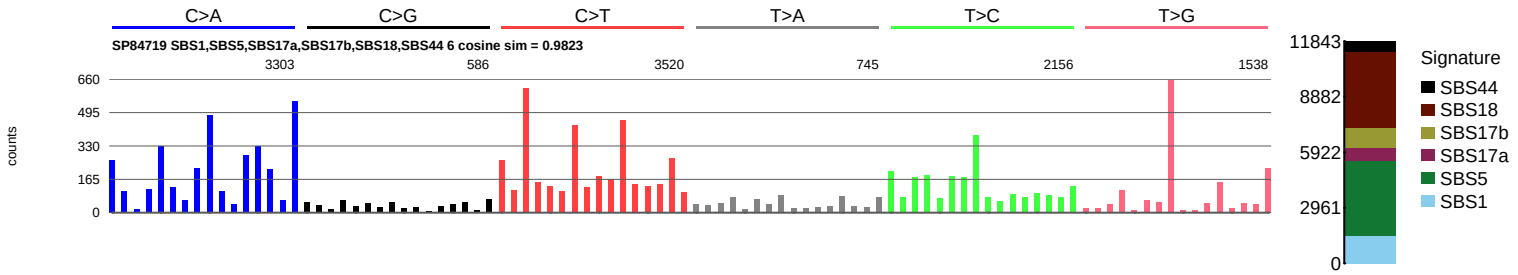

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98238
3302
617
3476
766
2142
1537
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

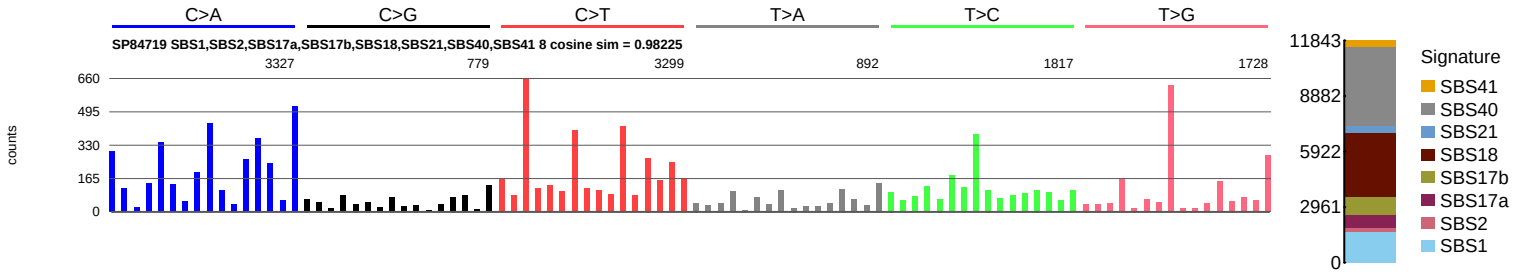

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98238
3303
624
3487
770
2120
1540
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

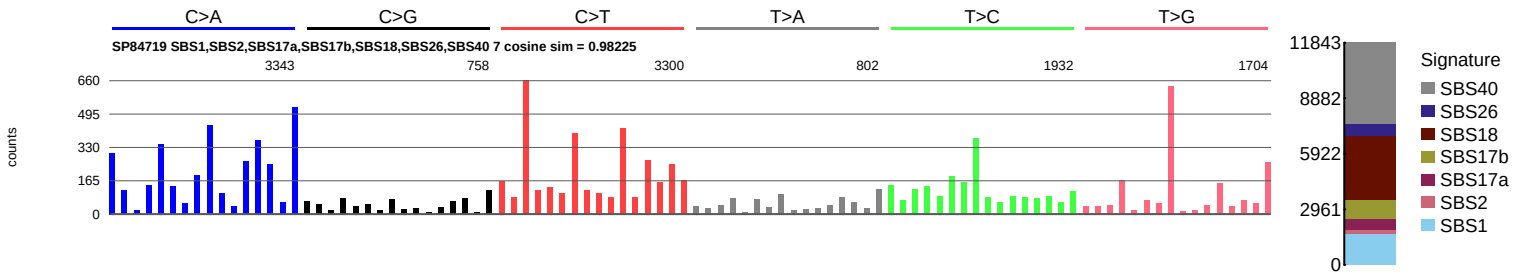

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98237
3342
760
3301
805
1932
1699
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

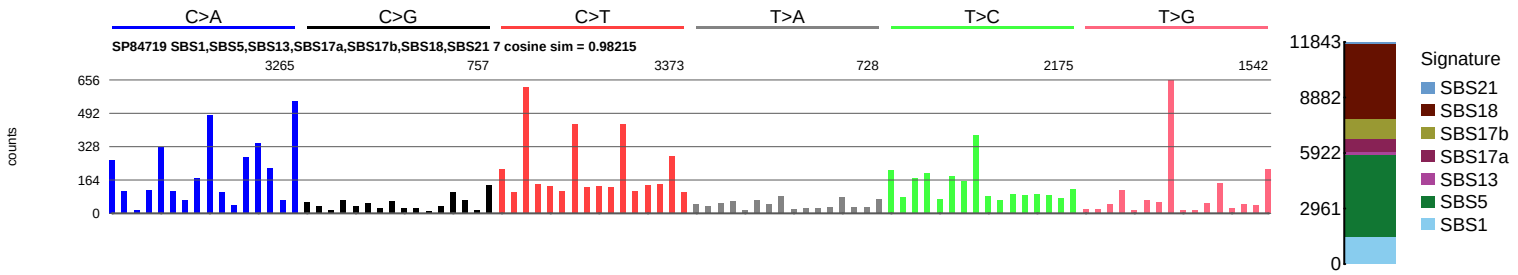

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98236
3327
805
3291
887
1814
1720
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

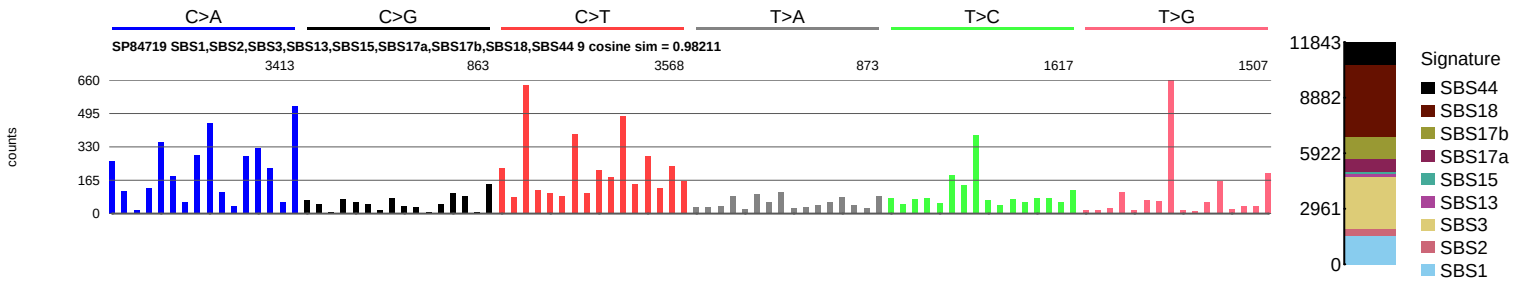

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98235
3302
582
3507
742
2179
1532
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.9823
3303
586
3520
745
2156
1538
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98225
3327
779
3299
892
1817
1728
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98225
3343
758
3300
802
1932
1704
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98215
3265
757
3373
728
2175
1542
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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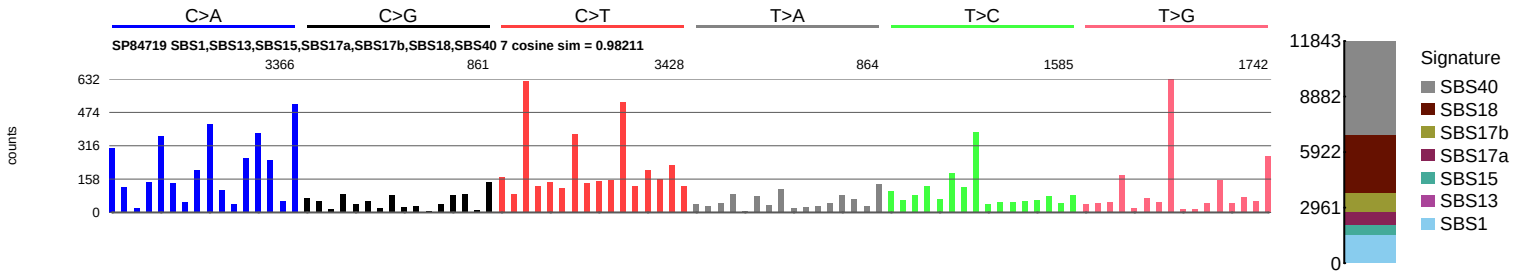
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98211
3413
863
3568
873
1617
1507
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
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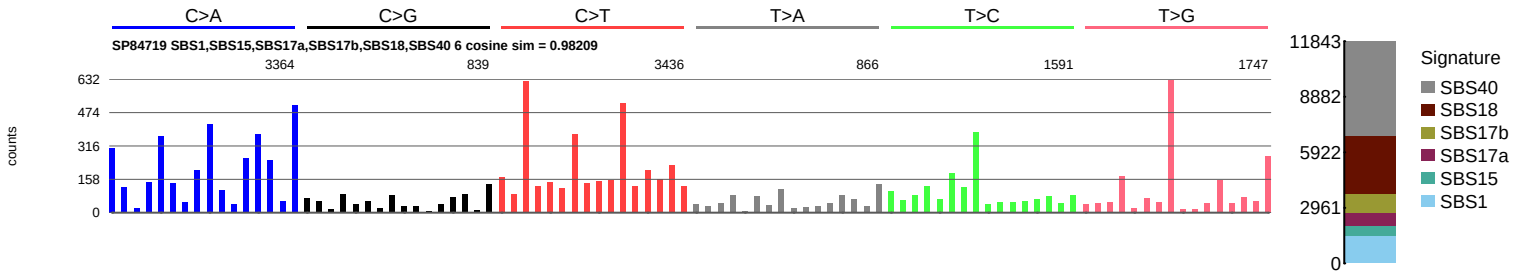
T>C
T>G
SP84719 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98211
3366
861
3428
864
1585
1742
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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T>A
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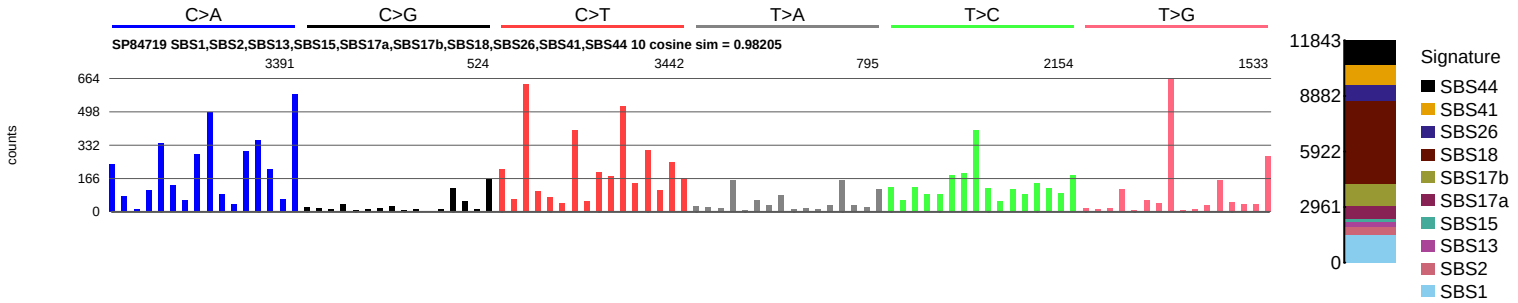
T>G
SP84719 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98209
3364
839
3436
866
1591
1747
632
474
316
158
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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T>A
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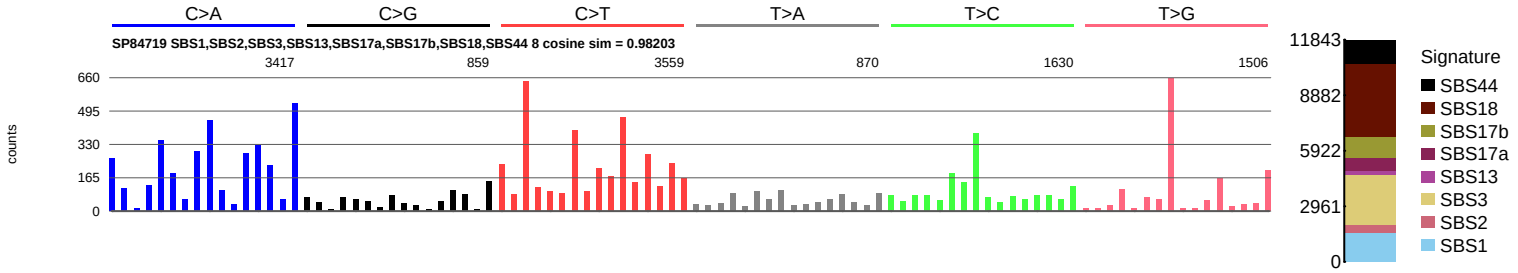
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98205
3391
524
3442
795
2154
1533
664
498
332
166
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98203
3417
859
3559
870
1630
1506
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS44
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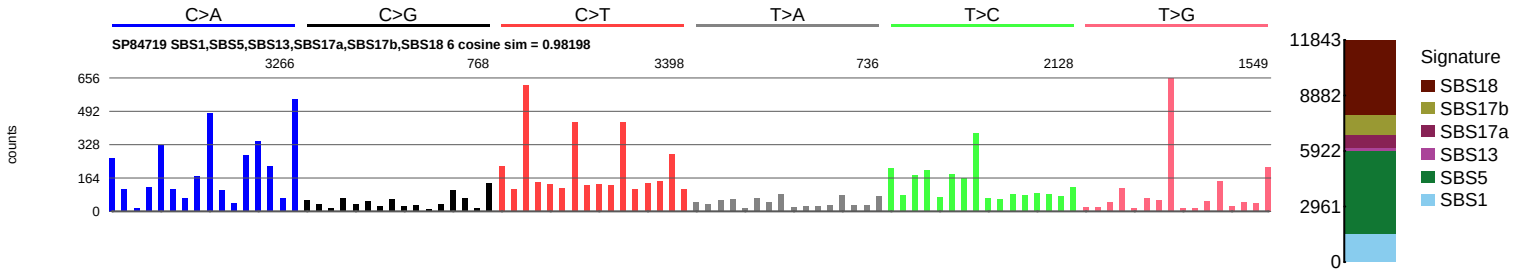
SBS2
SBS1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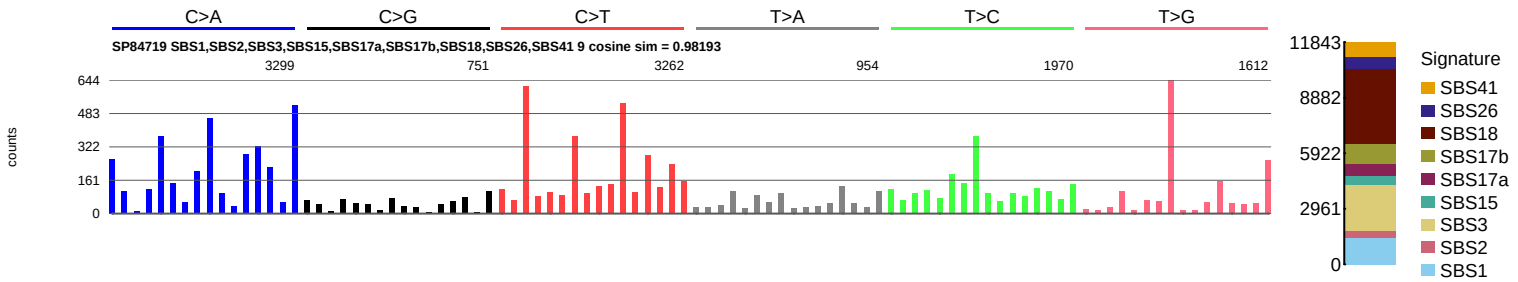
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98198
3266
768
3398
736
2128
1549
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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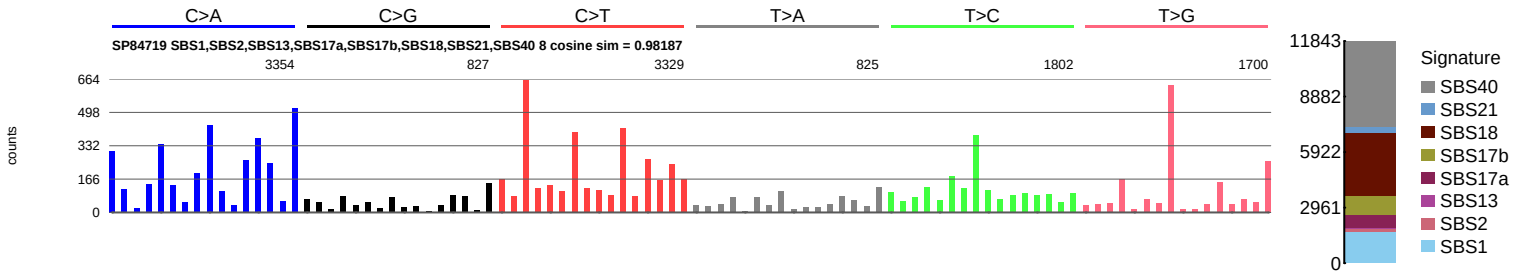
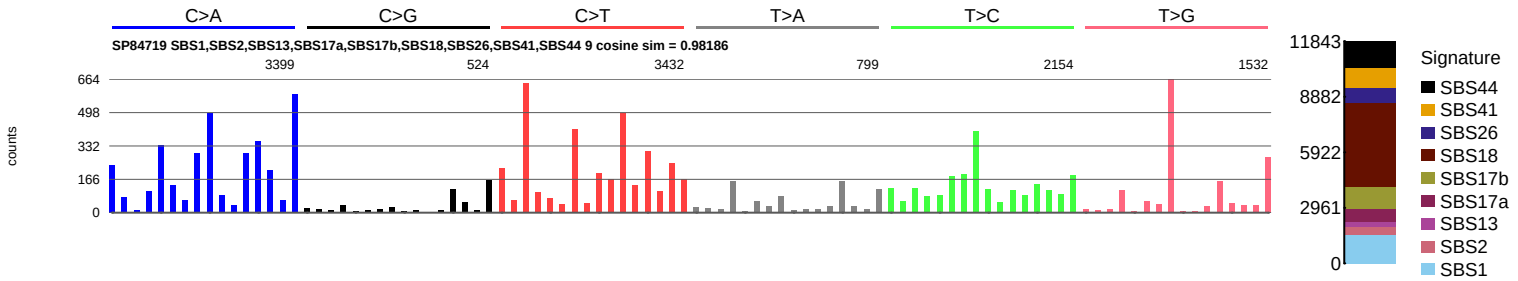
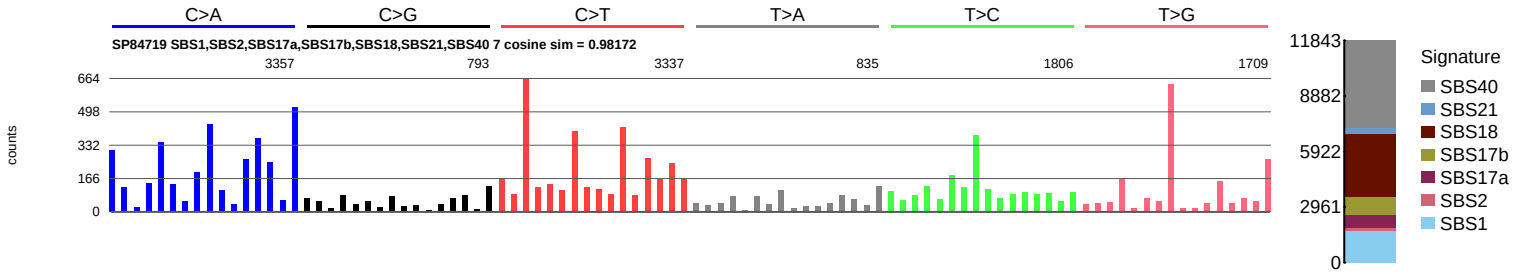
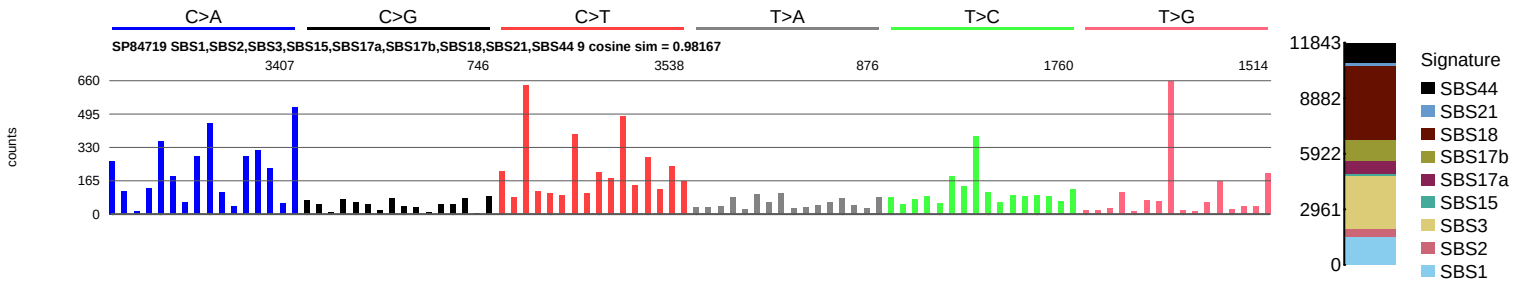
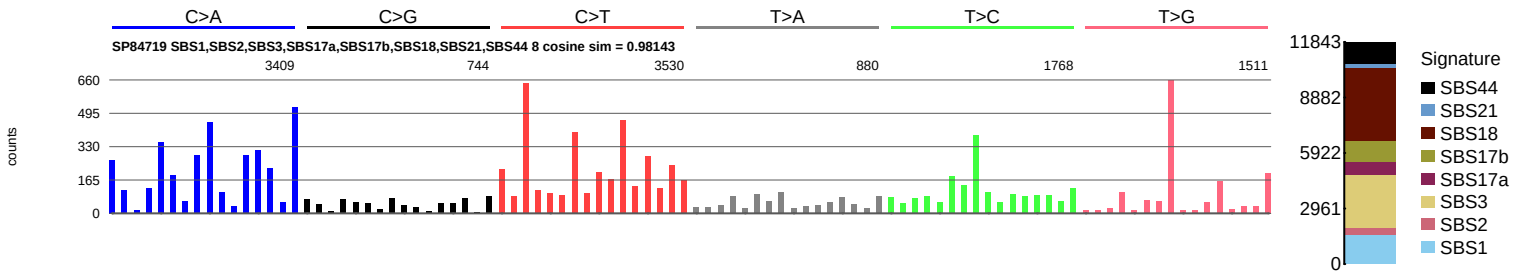
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98193
3299
751
3262
954
1970
1612
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

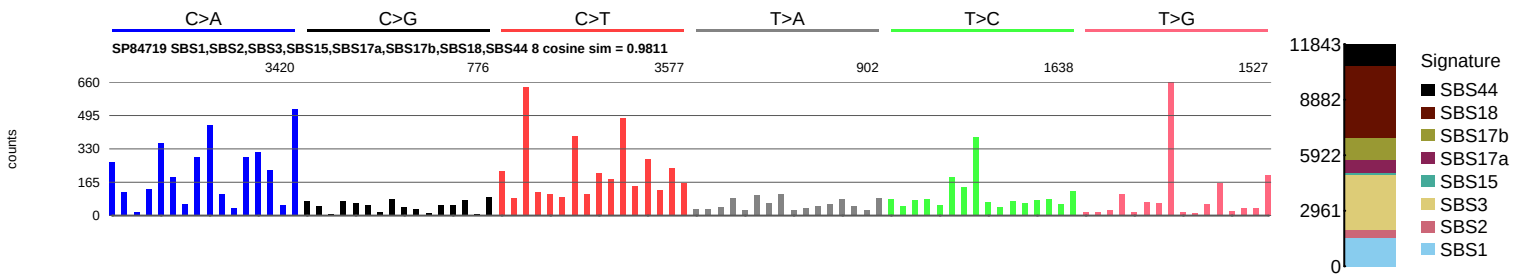

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98187
3354
827
3329
825
1802
1700
counts
664
498
332
166
0
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

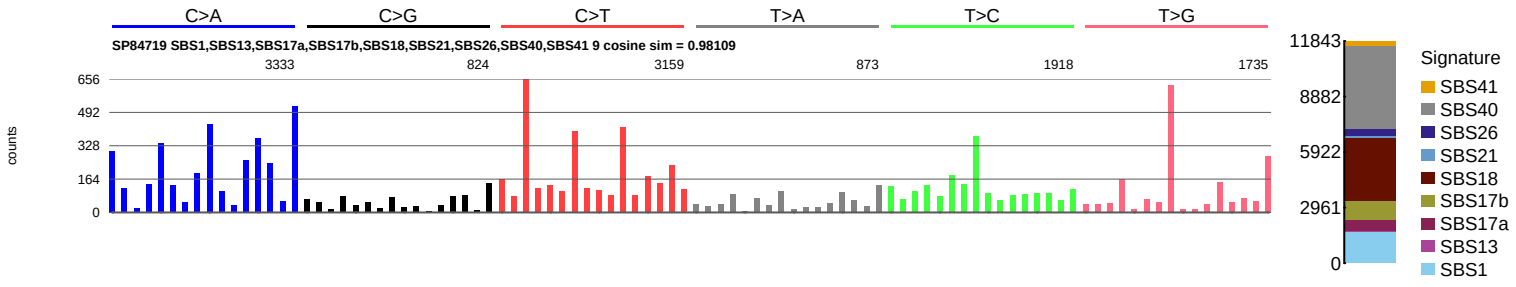

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98186
3399
524
3432
799
2154
1532
counts
664
498
332
166
0
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

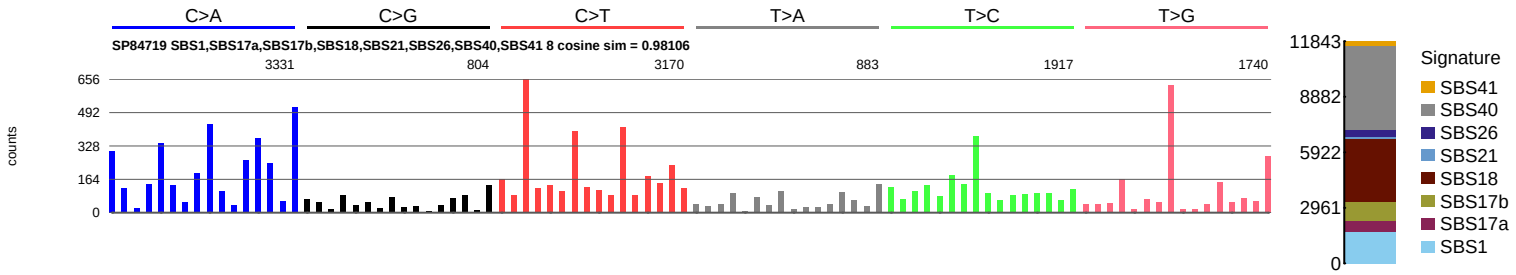

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98172
3357
793
3337
835
1806
1709
counts
664
498
332
166
0
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

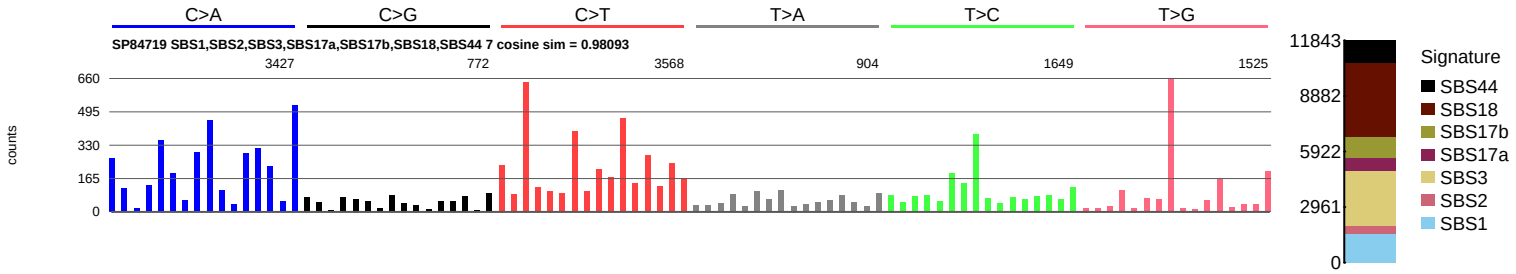

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98167
3407
746
3538
876
1760
1514
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

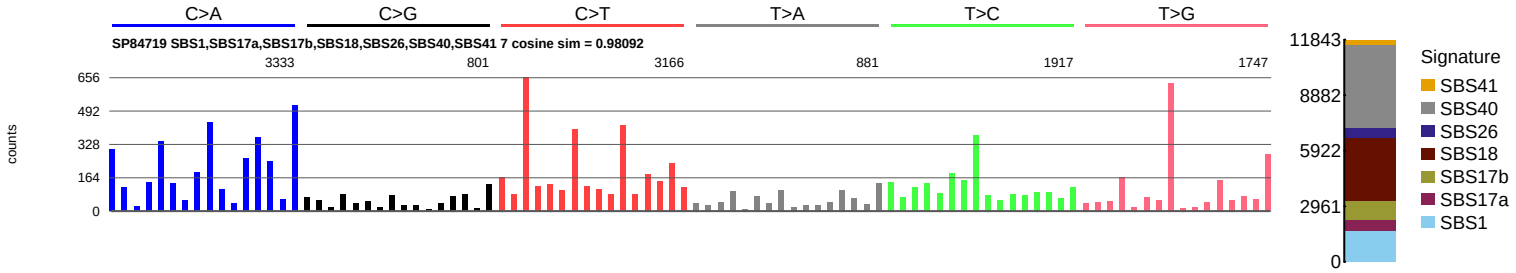

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98143
3409
744
3530
880
1768
1511
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

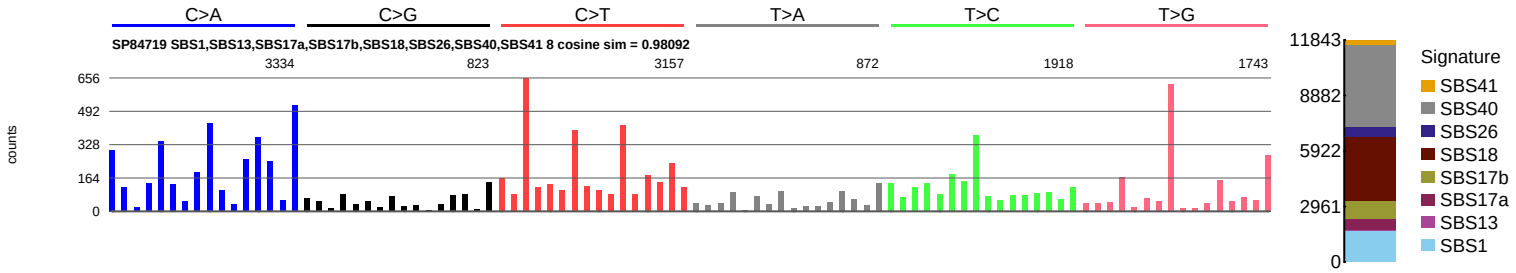

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.9811
3420
776
3577
902
1638
1527
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

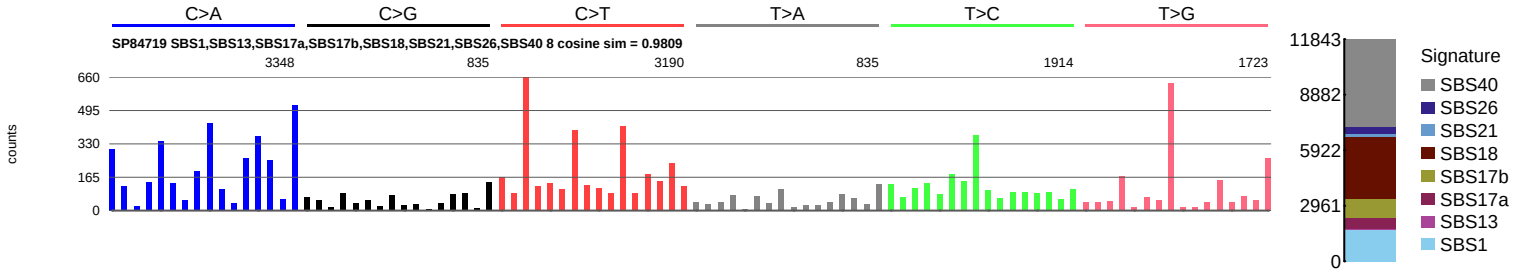

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98109
3333
824
3159
873
1918
1735
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98106
3331
804
3170
883
1917
1740
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98093
3427
772
3568
904
1649
1525
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98092
3333
801
3166
881
1917
1747
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98092
3334
823
3157
872
1918
1743
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1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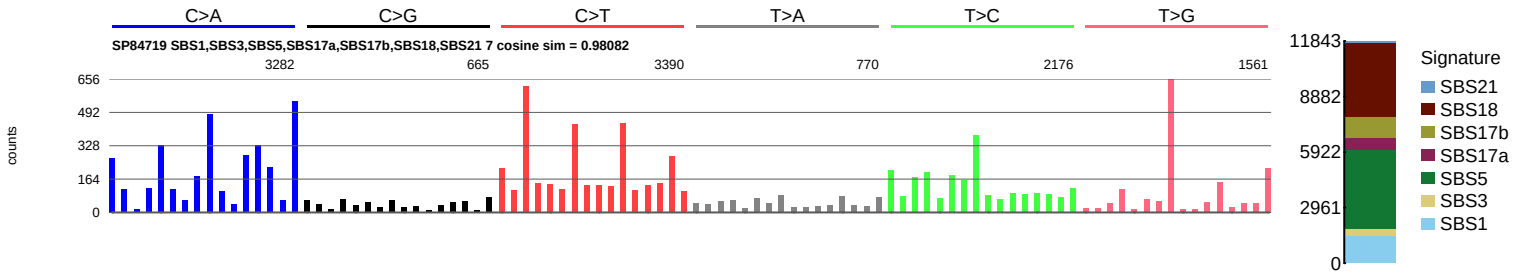
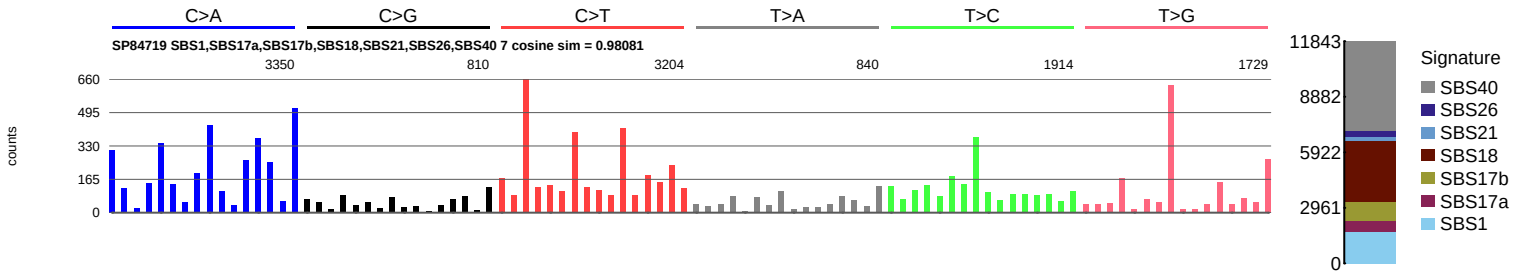
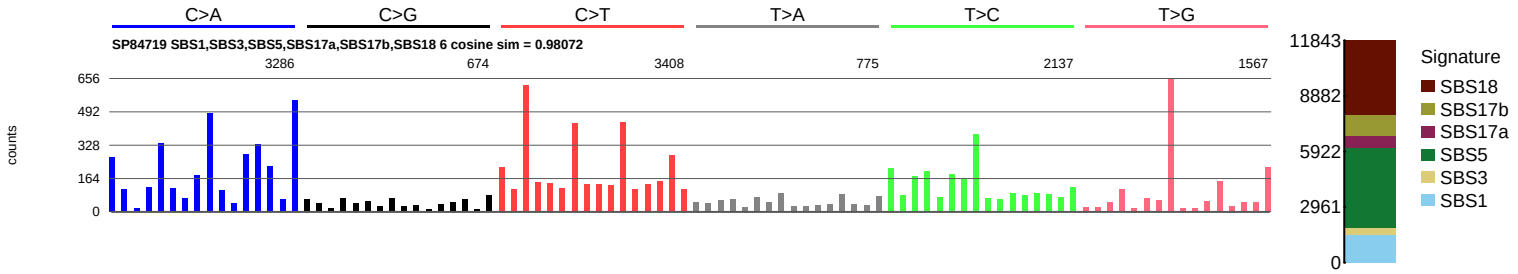
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.9809
3348
835
3190
835
1914
1723
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98082
3282
665
3390
770
2176
1561
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98081
3350
810
3204
840
1914
1729
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98072
3286
674
3408
775
2137
1567
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
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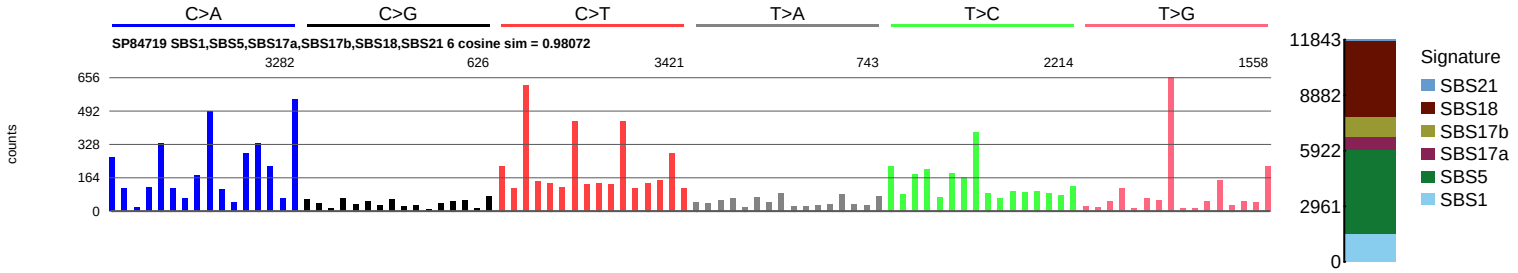
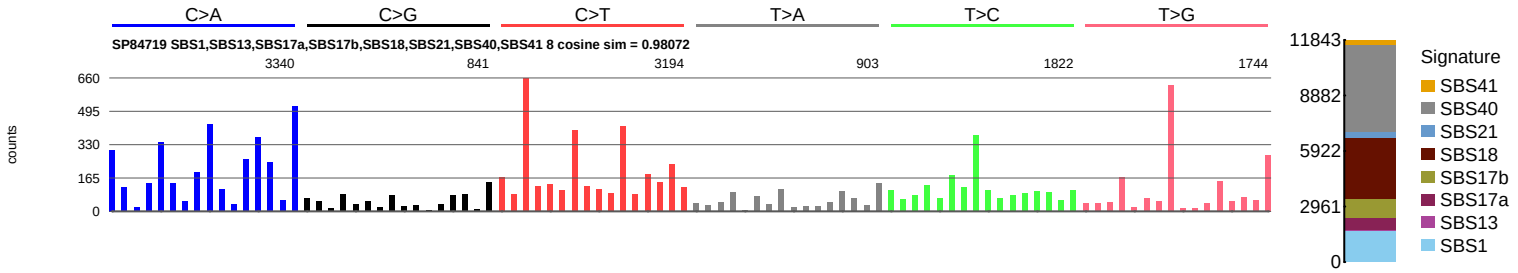
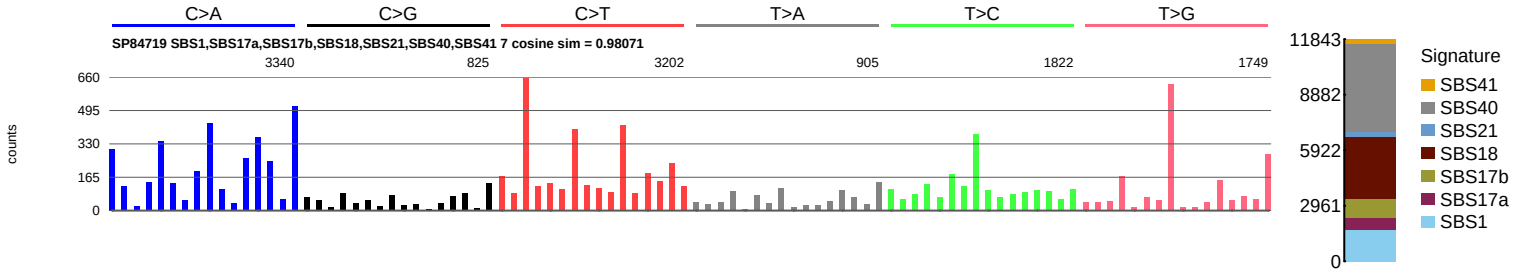
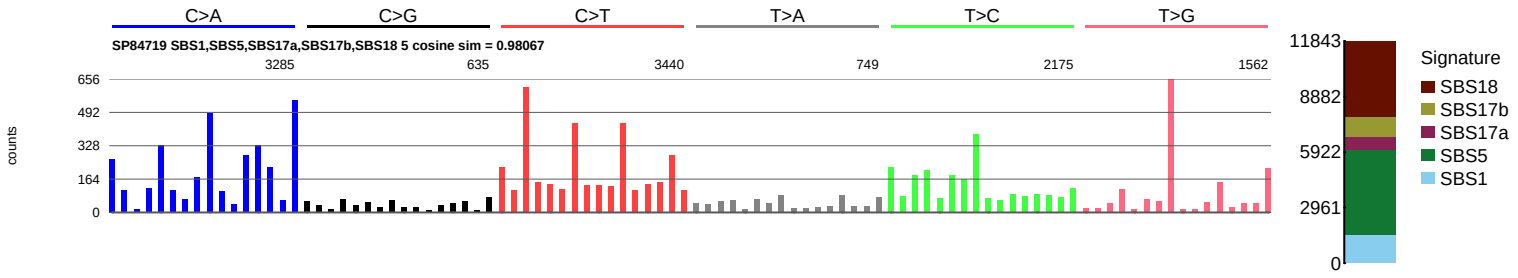
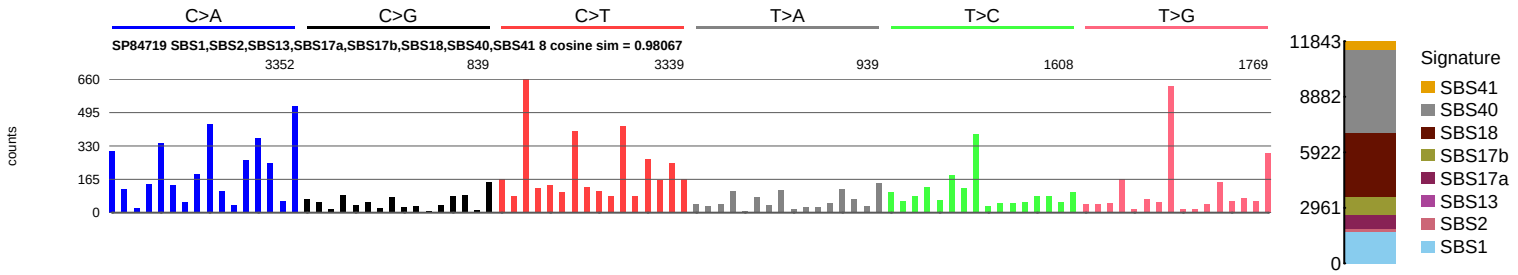
SBS5
SBS3
SBS1

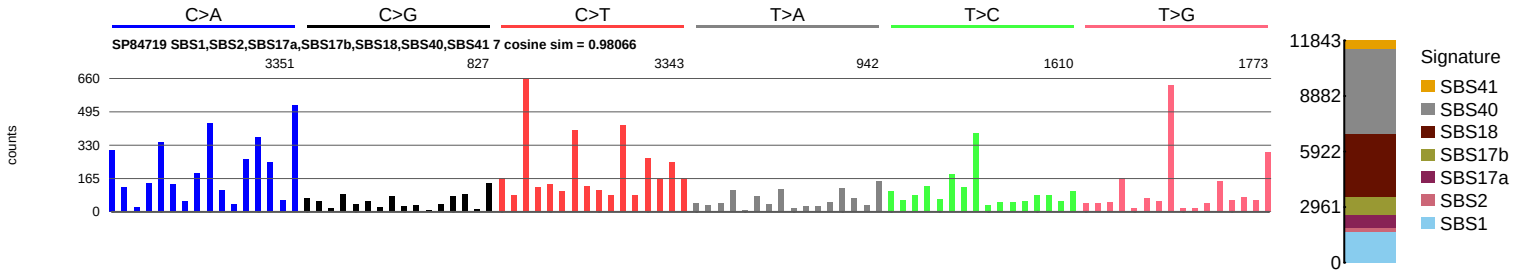

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98072
3282
626
3421
743
2214
1558
counts
656
492
328
164
0
11843
8882
5922
2961
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

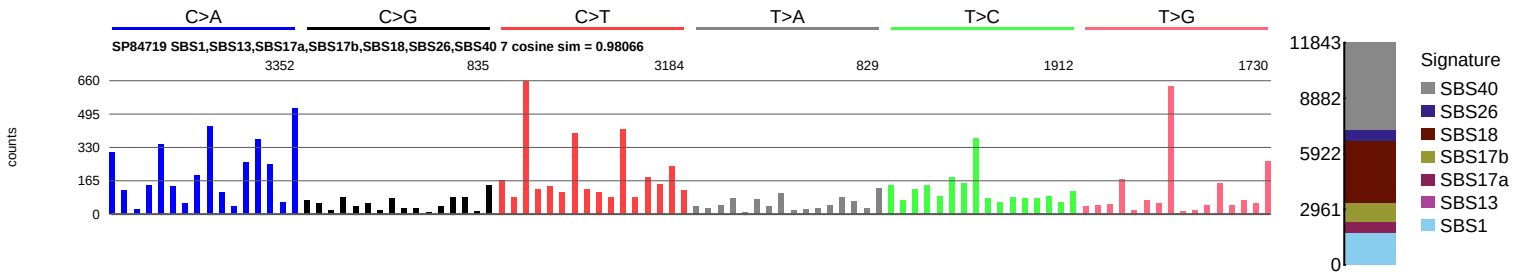

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98072
3340
841
3194
903
1822
1744
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

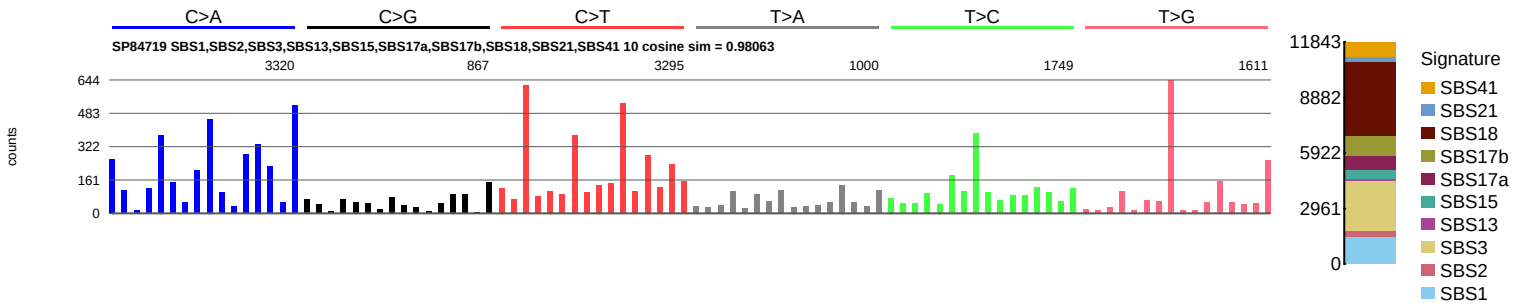

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98071
3340
825
3202
905
1822
1749
counts
660
495
330
165
0
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

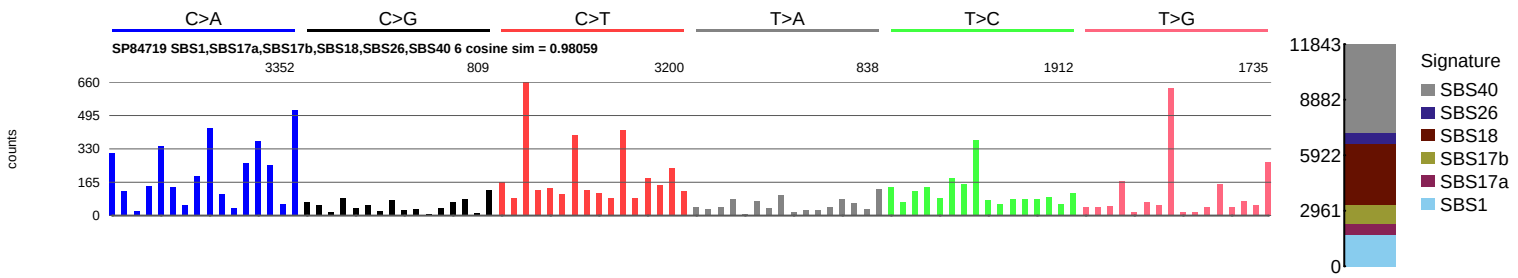

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98067
3285
635
3440
749
2175
1562
656
492
328
164
0
counts
11843
8882
5922
2961
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1

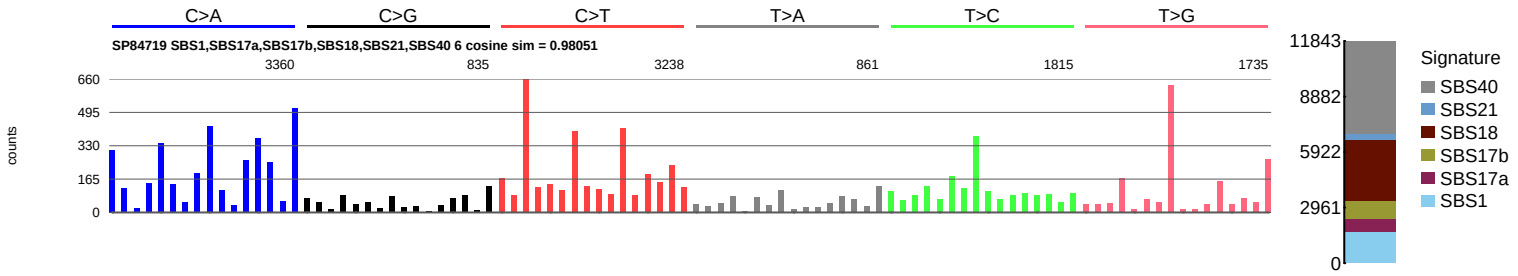

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98067
3352
839
3339
939
1608
1769
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

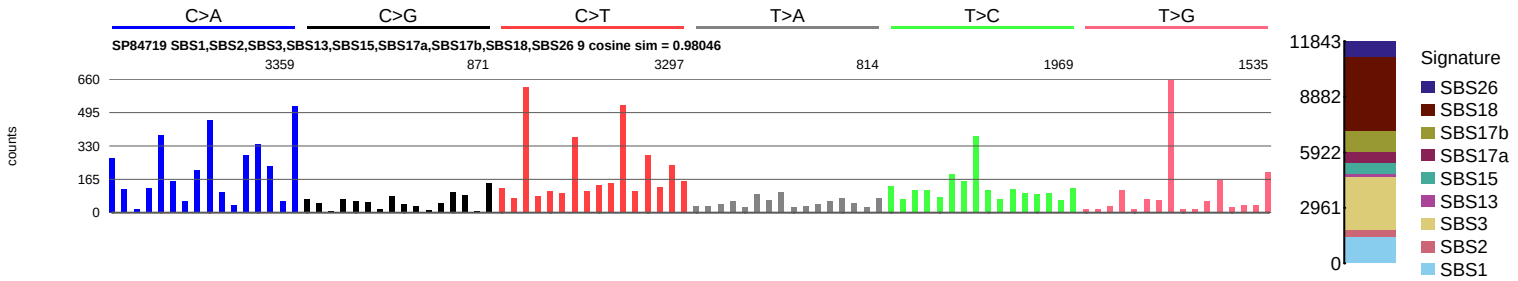

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98066
3351
827
3343
942
1610
1773
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

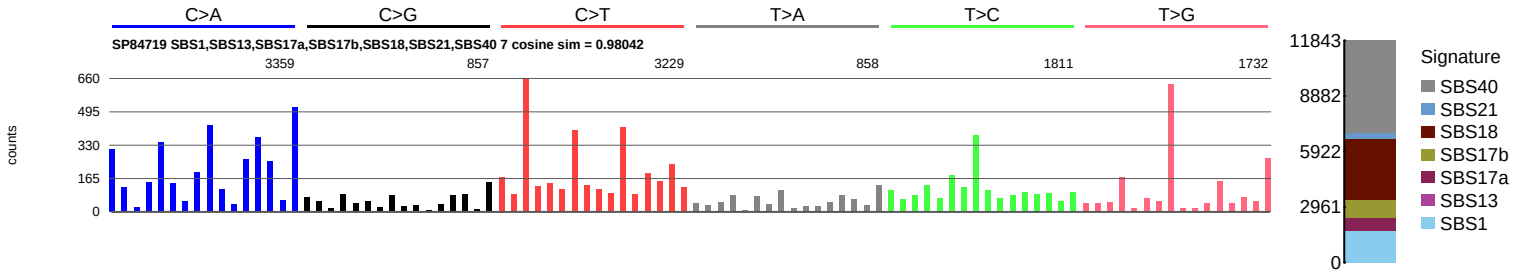

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98066
3352
835
3184
829
1912
1730
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

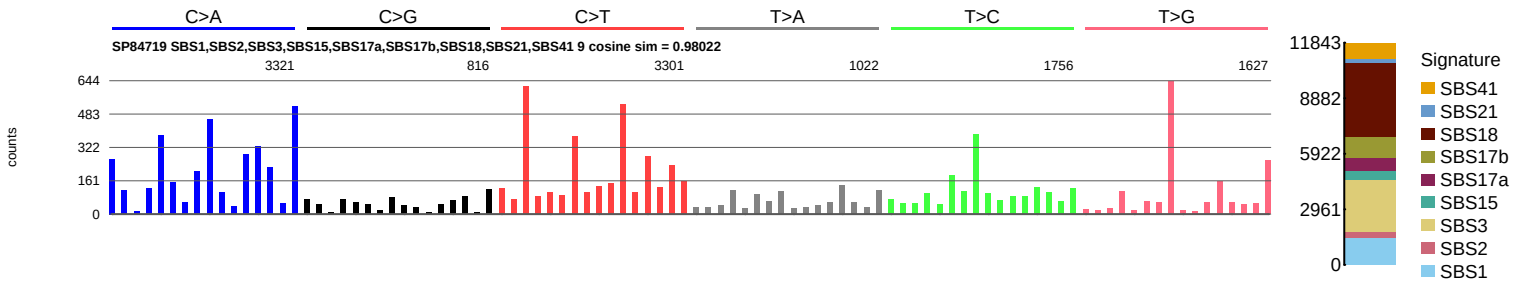

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98063
3320
867
3295
1000
1749
1611
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

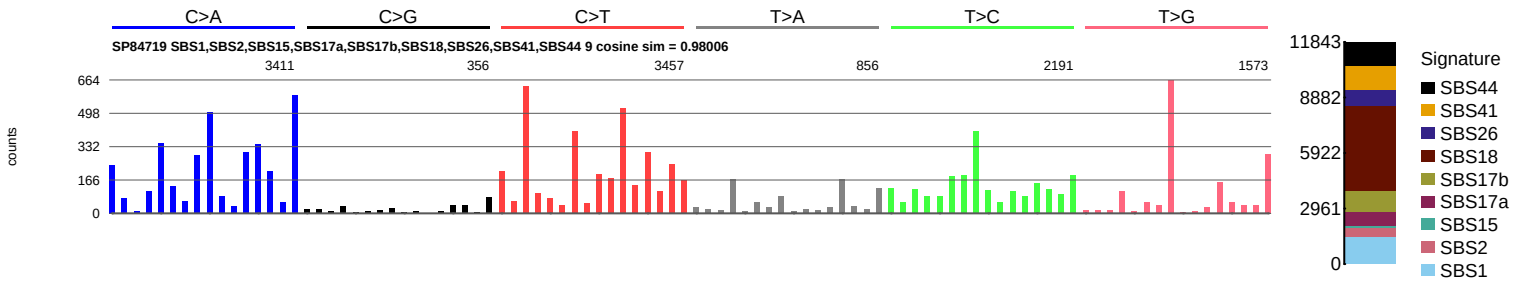

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98059
3352
809
3200
838
1912
1735
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

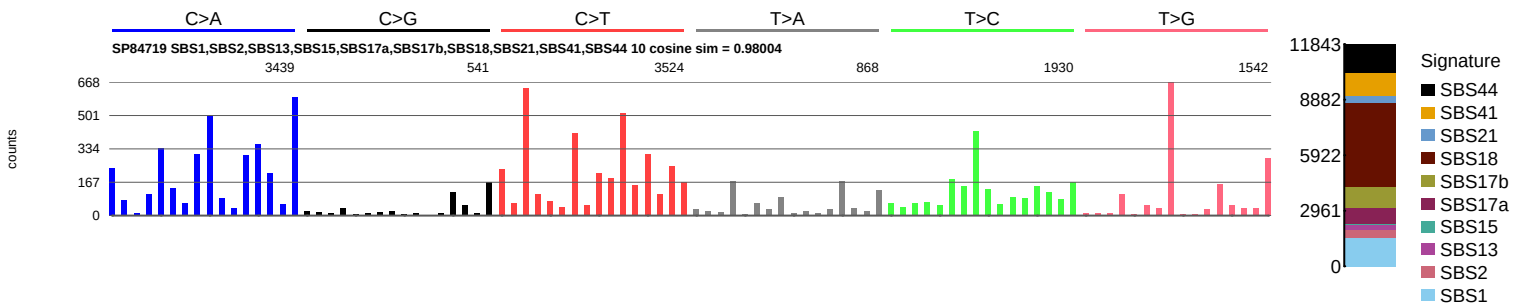

C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98051
3360
835
3238
861
1815
1735
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98046
3359
871
3297
814
1969
1535
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98042
3359
857
3229
858
1811
1732
660
495
330
165
0
counts
11843
8882
5922
2961
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98022
3321
816
3301
1022
1756
1627
644
483
322
161
0
counts
11843
8882
5922
2961
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98006
3411
356
3457
856
2191
1573
664
498
332
166
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84719 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98004
3439
541
3524
868
1930
1542
668
501
334
167
0
counts
11843
8882
5922
2961
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1